TYPE II REGULARITY FOR NAVIER–STOKES

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ABSTRACT. We establish a local retained-branch exclusion criterion for the three-dimensional Navier–Stokes equations near a possible singular point and record the named exit rows together with their local closures. From a positive Caffarelli–Kohn–Nirenberg concentration sequence we select renormalized local windows around a retained core and analyze the resulting alternatives by compactness, pressure reconstruction, and local energy estimates. The argument combines a compact single-core exclusion criterion, a repaired-gauge representation for nondegenerate concentration cores, local Calderón–Zygmund pressure control, a Caccioppoli estimate on compact cylinders, multibubble and cascade reductions, and scale-collapse cost estimates. The retained compact scale-collapse branch is closed by local state-space routing into the scale-rigid discharge, so no additional stationary or self-similar classification theorem is used implicitly. Thin negative drift is closed as an independent terminal row by diffuse occupation-state routing. Branches that leave the retained state through selected compactness-package failure are now routed to more specific local rows. Modulation failure, exterior concentration, retained active-core loss, and noncanonical carrier/cost failure are not treated as hidden inputs; they are listed as named exits and closed by their own local closure theorems.

CONTENTS

1. Introduction

The three-dimensional Navier–Stokes regularity problem is governed by the interaction between local energy control, scale-critical concentration, and the possible formation of singular structures at arbitrarily small parabolic scales. Since Leray's construction of global weak solutions [Ler34], the theory of suitable weak solutions and the partial regularity theorem of Scheffer and Caffarelli–Kohn–Nirenberg have provided the basic local language for this interaction [Sch76, CKN82, Lin98]. In that theory, singularity formation is detected by scale-invariant velocity-pressure quantities, and the critical L^3 scale plays a central role alongside the local energy inequality, pressure reconstruction, and compactness. The analytic background also includes the classical local and critical-space theory for Navier–Stokes [Lad69, Tem77, CF88, Kat84, LR02], the $L_{3,\infty}$ regularity theorem of Escauriaza–Seregin–Šverák [ESS03], and the rigidity theorem of Nečas–Růžička–Šverák for Leray self-similar profiles [NRŠ96]. These classical rigidity results are cited as background; the retained Type II branch is closed by the local state-space reductions below, not by importing an external stationary or self-similar classification theorem.

This paper studies the local Type II alternative near a possible singular point. Starting from a positive Caffarelli–Kohn–Nirenberg concentration sequence, we select renormalized local windows around a retained core and analyze all branches that can arise from the scale-center dynamics. The result in the present version is a retained-state exclusion theorem together with an explicit ledger of named exits and a closure theorem for each remaining ledger row. Pressure reconstruction, repaired-gauge compactness, terminal profile decoupling, cost comparison, and local state-space closure are proved on retained branches or used to route a first failure to one of those named exits. No additional stationary, ancient, or self-similar classification theorem is invoked implicitly.

The proof begins with local concentration and the Type I/Type II separation. For a non-degenerate retained core, the solution is rewritten in local scale-translation variables. The repaired-gauge representation produces a renormalized Navier–Stokes equation with controlled modulation terms, local pressure reconstruction modulo functions of time, and compact-window suitable estimates. Within this representation, a compact single-core branch with a retained positive velocity core and vanishing localized dissipation is impossible: local compactness gives a spatially constant limit, the zero constant contradicts retained velocity by strong L^3 convergence, and a nonzero constant gives a bounded selected-window limit, which exits the retained local Type II row. Pressure-only harmonic concentration is routed to the pressure/exterior rows rather than closed by the single-core velocity argument.

The remaining sections remove the ways in which a Type II branch could avoid that compact single-core mechanism. Caccioppoli estimates convert finite outer velocity-pressure control into inner windowed H^1 control, so rough-core loss is returned to an explicit compact-cylinder alternative. Multibubble and cascade arguments separate active profiles, same-point cascades, and gauge-degenerate branches from a single retained core. Scale-collapse branches are treated by localized cost identities: finite scale-collapsing cost, and the corresponding finite absolute cost, are incompatible with a nonvanishing retained core, while thick autonomous windows lead only to the stated modulation, compactness, or retained compact scale-collapse state; the latter is routed into the scale-rigid discharge rather than into a new Liouville assumption. The diffuse thin-drift boundary face is handled by normalized negative-drift occupation blocks: persistent windows return to the thick-window ledger, while genuinely diffuse blocks make the negative scale term a vanishing local perturbation and route the branch to another local row or to a closed retained row. The terminal profile and corrected monotonicity sections then turn these local reductions into a branchwise classification: retained states are eliminated, while first failures outside the retained state are recorded as named exits.

The conclusion should therefore be read in this local state-space sense. In the retained branch, a positive local Type II concentration sequence cannot survive the repaired-gauge, compactness, multibubble, rough-core, compact-cost, and retained scale-collapse reductions. A failure of one of the local analytic rows is not left as an unstated input; it is recorded as a named state-space exit. In the final assembly, the remaining named exits are then closed by their own local routing arguments, giving the stated no-local-Type-II conclusion without importing global critical estimates.

2. Local Single-Core Criterion

2.1. **Setup and definitions.** For $R > 0$, write

$$Q_R = B_R \times (-R^2, 0).$$

For a suitable pair (v, q) on Q_R, define

$$A_v(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |v(y,s)|^2 \, dy,$$

$$E_v(R) = R^{-1} \int_{Q_R} |\nabla v|^2 \, dy \, ds,$$

$$C_v(R) = R^{-2} \int_{Q_R} |v|^3 \, dy \, ds,$$

$$D_q(R) = R^{-2} \int_{-R^2}^{0} \int_{B_R} |q(y,s) - (q)_{B_R}(s)|^{3/2} \, dy \, ds.$$

All pressure quantities are understood modulo functions of time.

Definition 2.1 (Bounded selected-window limit)**.** Let (V_n, P_n) be a sequence of represented local rescalings on a compact cylinder Q_r. A selected-window subsequence has a bounded selected-window limit on Q_r if, after passing to a subsequence,

$$V_n \to V_* \qquad \text{strongly in } L^3(Q_r),$$

where $V_* \in L^\infty(Q_r)$. The limit is nonzero if $V_* \not\equiv 0$ on Q_r.

Definition 2.2 (Local Type II sequence)**.** A represented sequence is called a local Type II sequence if it has positive local Caffarelli–Kohn–Nirenberg concentration on some fixed compact cylinder and no selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder in the sense of definition 2.1.

Definition 2.3 (Compact single-core vanishing-dissipation branch)**.** Fix radii $0 < r_* < R_*$, constants $M < \infty$, $\eta_0 > 0$, and a sequence of represented local rescalings (V_n, P_n, a_n, b_n) on Q_{R_*}. The sequence is a compact single-core vanishing-dissipation branch if:

(i) each (V_n, P_n) is suitable on Q_{R_*} and solves the represented Navier–Stokes equation

$$\partial_s V_n + (V_n \cdot \nabla) V_n + \nabla P_n = \nu \Delta V_n + a_n(s)(V_n + y \cdot \nabla V_n) + b_n(s) \cdot \nabla V_n, \qquad \nabla \cdot V_n = 0; \tag{1}$$

(ii) the compact-cylinder suitable estimates are uniformly bounded:

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M;$$

(iii) the selected core has retained positive velocity concentration on Q_{r_*}:

$$C_{V_n}(r_*) \ge \eta_0 \qquad \text{for all } n;$$

Pressure-only concentration is not included in this compact single-core row; it is routed to the harmonic/exterior-pressure alternative.

(iv) the localized scale and center are nondegenerate for the selected core, so the representation theorem gives the variables in (1);

(v) the modulation coefficients are uniformly bounded on the selected cylinder:

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M;$$

(vi) the branch has vanishing localized dissipation in the sense of definition 2.4 below.

Definition 2.4 (Vanishing localized dissipation)**.** Let $r_* < \rho < R_*$. Selected windows are first translated and rescaled to the fixed cylinders $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. The localized dissipation on Q_ρ is

$$\mathcal{D}_n(\rho) = \int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds.$$

The branch has vanishing localized dissipation if, after passing to a selected-window subsequence, there is $\rho \in (r_*, R_*)$ such that

$$\mathcal{D}_n(\rho) \to 0.$$

2.2. **Local analytic inputs.** The compact single-core argument is based on the following local analytic statements.

Theorem 2.5 (Local concentration dichotomy)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then, for every $x_0 \in \Sigma(T)$, using the pressure-normalized quantities C, D, there are $\eta > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n.$$

Moreover, at each singular point exactly one of the following alternatives holds: a local Type I scale-invariant bound holds, or the point lies in the local Type II alternative, meaning that no such local Type I bound holds.

Proof. By the Caffarelli–Kohn–Nirenberg epsilon-regularity criterion, there is a universal $\varepsilon_{\mathrm{CKN}} > 0$ such that if

$$C((x_0, T), r) + D((x_0, T), r) < \varepsilon_{\mathrm{CKN}}$$

for all sufficiently small r, then u is locally bounded in a smaller backward cylinder ending at (x_0, T). Since $x_0 \in \Sigma(T)$, this regularity conclusion is impossible. Hence for every k there exists $0 < r_k < 1/k$ with

$$C((x_0, T), r_k) + D((x_0, T), r_k) \geq \varepsilon_{\mathrm{CKN}}.$$

After passing to a decreasing subsequence and setting $\eta = \varepsilon_{\mathrm{CKN}}$, the displayed concentration sequence follows. The Type I/Type II split is the dichotomy between existence and failure of the local Type I scale-invariant bound at the same point. ☐

Proposition 2.6 (Passage to local Type II sequences)**.** *Let (x_0, T) be a concentration point in the local Type II alternative of theorem 2.5. After choosing the positive concentration scales given there and applying the local scale-translation selection used in the present section to a nondegenerate compact single core, the represented selected windows form a sequence with positive local CKN concentration on a fixed compact cylinder. In the selected-window Type II branch, no subsequence has a nonzero bounded selected-window limit on any compact cylinder. Hence the represented selected windows are a local Type II sequence in the sense of definition 2.2.*

Proof. The local concentration dichotomy supplies the positive CKN concentration sequence and the exclusion of the local Type I alternative at the chosen concentration point. The scale-translation selection rewrites those shrinking cylinders as fixed selected windows. The last condition is the selected-window formulation of the Type II branch considered here: any branch with a nonzero bounded selected-window limit belongs to the bounded-window alternative, not to definition 2.2. ☐

Proposition 2.7 (Local representation input)**.** *For a nondegenerate selected single core, the local scale-translation construction gives represented variables (V_n, P_n, a_n, b_n) satisfying (1) on compact cylinders. Failure of the localized nondegeneracy condition is a multibubble, cascade, or gauge-degenerate alternative, not a compact single-core branch.*

Proposition 2.8 (Local pressure decomposition)**.** *For every $B_r \Subset B_R$, the represented pressure decomposes as*

$$P_n = P_{\mathrm{loc},n} + H_n, \qquad -\Delta P_{\mathrm{loc},n} = \partial_i \partial_j (\zeta V_{n,i} V_{n,j}),$$

where $\zeta \in C_c^\infty(B_R)$ equals one on B_r, and H_n is harmonic in B_r. The local part satisfies the Calderón–Zygmund bound [CZ52, Ste70]

$$\|P_{\mathrm{loc},n}\|_{L^{3/2}(B_R)} \le C \|V_n\|_{L^3(B_R)}^2$$

for a.e. time. The harmonic part is controlled only by the local pressure norm, modulo spatial means; no vanishing of the harmonic part follows from $V_n \to 0$ without an additional pressure-row hypothesis.

Lemma 2.9 (Localization of pressure oscillation bounds)**.** *Let $0 < r < R$, and let $q \in L^{3/2}(Q_R)$. Then*

$$D_q(r) \le C \left(\frac{R}{r}\right)^2 D_q(R),$$

where C is universal.

Proof. For a.e. $s \in (-r^2, 0)$, Jensen's inequality gives

$$|(q)_{B_r}(s) - (q)_{B_R}(s)|^{3/2} \le \frac{1}{|B_r|} \int_{B_r} |q(y,s) - (q)_{B_R}(s)|^{3/2}\, dy.$$

Hence

$$\int_{B_r} |q - (q)_{B_r}(s)|^{3/2}\, dy \le C \int_{B_r} |q - (q)_{B_R}(s)|^{3/2}\, dy.$$

Integrating over $(-r^2, 0)$ and using $Q_r \subset Q_R$,

$$\begin{aligned} D_q(r) &\le C r^{-2} \int_{-r^2}^{0} \int_{B_r} |q - (q)_{B_R}(s)|^{3/2}\, dy\, ds \\ &\le C r^{-2} \int_{-R^2}^{0} \int_{B_R} |q - (q)_{B_R}(s)|^{3/2}\, dy\, ds = C \left(\frac{R}{r}\right)^2 D_q(R). \end{aligned}$$

□

2.3. Compactness and localized pressure compactness.

Proposition 2.10 (Compactness on selected windows)**.** *Let (V_n, P_n) be suitable on Q_{R_*}, solve* (1)*, and satisfy*

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M,$$

with

$$\|a_n\|_{L^\infty(-R_*^2, 0)} + \|b_n\|_{L^\infty(-R_*^2, 0)} \le M.$$

Then for every $r < R_*$, after passing to a subsequence there are (V, P) such that

$$\begin{aligned} V_n &\overset{*}{\rightharpoonup} V \quad \text{in } L^\infty(-r^2, 0; L^2(B_r)), \\ V_n &\rightharpoonup V \quad \text{in } L^2(-r^2, 0; H^1(B_r)), \\ V_n &\to V \quad \text{strongly in } L^3(Q_r), \end{aligned}$$

and

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \text{in } L^{3/2}(Q_r).$$

After passing to a further subsequence, a_n and b_n converge weak- in L^∞ to bounded coefficients, and the limit satisfies the corresponding represented equation distributionally on smaller cylinders.*

Proof. The bounds on A and E give weak-* compactness in $L^\infty L^2$ and weak compactness in $L^2 H^1$ on smaller cylinders. Fix $r < r_1 < R_*$, and choose m so large that

$$H_0^m(B_{r_1}) \hookrightarrow W_0^{1,\infty}(B_{r_1}).$$

We claim that $\partial_s V_n$ is uniformly bounded in

$$L^1(-r_1^2, 0; H^{-m}(B_{r_1})).$$

Let $\varphi \in H_0^m(B_{r_1})$. Pairing (1) with φ, integrating by parts in space where needed, and using the embedding above gives

$$|\langle (V_n \cdot \nabla) V_n, \varphi \rangle| = \left| \int_{B_{r_1}} V_{n,i} V_{n,j} \partial_j \varphi_i \, dy \right| \le C \|V_n\|^2_{L^3(B_{r_1})} \|\varphi\|_{H^m},$$

$$|\langle \nu \Delta V_n, \varphi \rangle| \le C \|\nabla V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

$$|\langle \nabla P_n, \varphi \rangle| = \left| \int_{B_{r_1}} (P_n - (P_n)_{B_{r_1}}) \nabla \cdot \varphi \, dy \right| \le C \|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(B_{r_1})} \|\varphi\|_{H^m}.$$

The modulation terms obey

$$|\langle a_n V_n, \varphi \rangle| \le C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

$$|\langle a_n \, y \cdot \nabla V_n, \varphi \rangle| = \left| \int_{B_{r_1}} a_n V_{n,i} \partial_j (y_j \varphi_i) \, dy \right| \le C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

and

$$|\langle b_n \cdot \nabla V_n, \varphi \rangle| = \left| \int_{B_{r_1}} V_{n,i} b_{n,j} \partial_j \varphi_i \, dy \right| \le C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

where the constant uses r_1 and the uniform L^∞ bounds for a_n, b_n. Integrating these estimates in time gives a uniform bound because

$$\int_{-r_1^2}^0 \|V_n(s)\|^2_{L^3(B_{r_1})} \, ds \le r_1^{2/3} \|V_n\|^2_{L^3(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|\nabla V_n(s)\|_{L^2(B_{r_1})} \, ds \le r_1 \|\nabla V_n\|_{L^2(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|P_n(s) - (P_n)_{B_{r_1}}(s)\|_{L^{3/2}(B_{r_1})} \, ds \le r_1^{2/3} \|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(Q_{r_1})},$$

and

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^2(B_{r_1})} \, ds \le r_1^2 \|V_n\|_{L^\infty(-r_1^2, 0; L^2(B_{r_1}))}.$$

The uniform C, E, D, and A bounds therefore prove the claimed time-derivative bound; the D-bound on Q_{r_1} follows from the bound on Q_{R_*} by lemma 2.9.

The compact embedding $H^1(B_{r_1}) \Subset L^2(B_r)$, together with Aubin–Lions compactness [Aub63, LM72, Sim87], gives

$$V_n \to V \qquad \text{strongly in } L^2(Q_r)$$

after passing to a subsequence. The same $L^\infty L^2 \cap L^2 H^1$ bounds give the parabolic interpolation estimate

$$\|V_n\|_{L^{10/3}(Q_r)} \le C(r, M).$$

Indeed, for a.e. s, the Sobolev and Gagliardo–Nirenberg inequalities give

$$\|V_n(s)\|_{L^{10/3}(B_r)} \le C \|V_n(s)\|^{2/5}_{L^2(B_r)} \|\nabla V_n(s)\|^{3/5}_{L^2(B_r)}.$$

Raising to the power 10/3, integrating in time, and using the A and E bounds yields

$$\int_{-r^2}^{0} \|V_n(s)\|_{L^{10/3}(B_r)}^{10/3}\, ds \leq C \left(\operatorname*{ess\,sup}_{-r^2<s<0} \|V_n(s)\|_{L^2(B_r)}^2 \right)^{2/3} \int_{Q_r} |\nabla V_n|^2 \, dy\, ds \leq C(r, M).$$

The same estimate applies to V by weak lower semicontinuity, and therefore to $V_n - V$ by the triangle inequality. Interpolating between L^2 and $L^{10/3}$, one obtains

$$\|V_n - V\|_{L^3(Q_r)} \leq \|V_n - V\|_{L^2(Q_r)}^{1/6} \|V_n - V\|_{L^{10/3}(Q_r)}^{5/6}.$$

The second factor is uniformly bounded and the first tends to zero, so

$$V_n \to V \qquad \text{strongly in } L^3(Q_r).$$

The pressure compactness follows from the uniform D-bound: after subtracting spatial means,

$$P_n - (P_n)_{B_r} \rightharpoonup P \qquad \text{in } L^{3/2}(Q_r).$$

We pass to the limit distributionally using the preceding convergences. It remains to justify the modulation terms. For every compactly supported test field φ,

$$\int a_n(s)\, y \cdot \nabla V_n \cdot \varphi = -\int a_n(s)\, V_{n,i}\, \partial_j(y_j \varphi_i),$$

where summation over repeated spatial indices is understood. Equivalently,

$$\partial_j(y_j\varphi_i) = 3\varphi_i + y_j \partial_j \varphi_i.$$

Hence this term passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of a_n: if $F_n \to F$ in L^1 and $a_n \rightharpoonup^* a$ in L^∞, then

$$\int a_n F_n - \int aF = \int a_n(F_n - F) + \int (a_n - a)F \to 0.$$

The term $a_n V_n$ is immediate from the same argument, while

$$\int b_n(s) \cdot \nabla V_n \cdot \varphi = -\int V_{n,i}\, b_{n,j}(s)\, \partial_j \varphi_i$$

passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of b_n. These passages give the limiting represented equation in the sense of distributions. □

Remark 2.11 (Use of bounded modulation). The uniform L^∞ bounds on a_n and b_n enter the compactness argument in exactly two places. First, they give the negative Sobolev time-derivative bound for the three lower-order modulation terms in (1). Second, after subsequential weak-* convergence of a_n, b_n, they allow the modulation terms to pass to the distributional limit once $V_n \to V$ strongly in L^2_{loc}.

Lemma 2.12 (Stability of the localized pressure part)**.** *Let $V_n \to V$ strongly in $L^3(Q_R)$, and use the pressure decomposition of proposition 2.8. Then for every $r < R$,*

$$P_{\text{loc},n} \to P_{\text{loc}} \quad \textit{strongly in } L^{3/2}(Q_r),$$

where $-\Delta P_{\text{loc}} = \partial_i\partial_j(\zeta V_i V_j)$. If $V \equiv 0$ on Q_R, then

$$P_{\text{loc},n} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

This lemma does not assert convergence of the full mean-normalized pressure: the harmonic component may contain local parasitic pressure modes independent of V_n.

Proof. Strong convergence in L^3 gives

$$V_{n,i}V_{n,j} \to V_iV_j \quad \text{strongly in } L^{3/2}(Q_R).$$

Indeed,

$$\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)} \le \big(\|V_n\|_{L^3(Q_R)} + \|V\|_{L^3(Q_R)}\big)\|V_n - V\|_{L^3(Q_R)}.$$

Calderón–Zygmund estimates for the compactly supported localized source yield strong convergence of the localized pressures in $L^{3/2}$ on smaller balls:

$$\|P_{\mathrm{loc},n} - P_{\mathrm{loc}}\|_{L^{3/2}(Q_r)} \le C_{r,R}\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)}.$$

If $V \equiv 0$, then $P_{\mathrm{loc}} = 0$, and the first part gives the displayed convergence of $P_{\mathrm{loc},n}$. The harmonic part H_n is not controlled by Calderón–Zygmund from the velocity source and may persist even when $V_n \to 0$; it is therefore handled only in the pressure/exterior rows, not in the compact single-core zero-velocity contradiction. □

Lemma 2.13 (Zero-dissipation limit)**.** *If*

$$V_n \rightharpoonup V \quad \text{in } L^2(I; H^1(B_\rho))$$

and

$$\int_I\int_{B_\rho} |\nabla V_n|^2\, dy\, ds \to 0,$$

then $\nabla V = 0$ a.e. on $B_\rho \times I$. Hence for a.e. time, $V(\cdot, s)$ is spatially constant on B_ρ.

Proof. Weak lower semicontinuity gives

$$\int_I\int_{B_\rho} |\nabla V|^2\, dy\, ds \le \liminf_{n\to\infty} \int_I\int_{B_\rho} |\nabla V_n|^2\, dy\, ds = 0.$$

Thus $\nabla V = 0$ a.e. on $B_\rho \times I$. By Fubini, for a.e. $s \in I$ one has $\nabla V(\cdot, s) = 0$ in $L^2(B_\rho)$. Since B_ρ is connected, every $H^1(B_\rho)$ function with zero weak gradient is a.e. equal to a constant. Hence $V(\cdot, s)$ is spatially constant for a.e. s. □

Lemma 2.14 (Spatially constant limits)**.** *Let V be a limit on Q_ρ and suppose $\nabla V = 0$ a.e. there. Then $V(y, s) = c(s)$ on Q_ρ, with $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the approximating selected-window subsequence has a nonzero bounded selected-window limit.*

Proof. The spatial constancy follows from lemma 2.13. Since $V(\cdot, s) = c(s)$ on B_ρ, the local $L^\infty_s L^2_y$ bound gives

$$|c(s)|^2|B_\rho| \le \operatorname*{ess\,sup}_{-\rho^2<s<0} \int_{B_\rho} |V(y, s)|^2\, dy,$$

so $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the strong L^3 convergence of proposition 2.10 gives a subsequential limit to a nonzero bounded function on Q_ρ, which is precisely a nonzero bounded selected-window limit in the sense of definition 2.1. □

2.4. Contradiction lemmas.

Lemma 2.15 (Persistence of retained velocity core)**.** *For a compact single-core branch in the sense of definition 2.3, there are fixed $r_* > 0$ and $\eta_0 > 0$ such that along the selected-window subsequence,*

$$C_{V_n}(r_*) \ge \eta_0 \qquad \textit{for all } n.$$

Proof. The retained velocity lower bound is part of the definition of the selected compact core and is preserved by passing to subsequences of the selected windows. □

Lemma 2.16 (Zero constant contradicts retained velocity)**.** *Let $r_* < \rho$. Suppose the compactness limit in proposition 2.10 is spatially constant on Q_ρ and is identically zero on Q_{r_*}. Then*

$$C_{V_n}(r_*) \to 0.$$

Consequently this case contradicts lemma 2.15.

Proof. Strong $L^3(Q_{r_*})$ convergence to zero gives

$$C_{V_n}(r_*) = r_*^{-2} \int_{Q_{r_*}} |V_n|^3 \, dy \, ds = r_*^{-2} \|V_n\|_{L^3(Q_{r_*})}^3 \to 0.$$

This contradicts the retained velocity lower bound

$$C_{V_n}(r_*) \geq \eta_0.$$

No assertion is made about $D_{P_n}(r_*)$: a pressure-only harmonic mode may remain and is not part of the retained compact single-core velocity row. □

Lemma 2.17 (Nonzero constant exits the retained Type II row)**.** *If the compactness limit is spatially constant and nonzero on a compact subcylinder, then the original sequence is not a local Type II sequence.*

Proof. By lemma 2.14, the selected-window subsequence has a nonzero bounded selected-window limit. By definition 2.2, such a sequence is assigned to the bounded-window alternative rather than the retained local Type II row. □

2.5. Main theorem.

Theorem 2.18 (Local single-core Type II criterion)**.** *Under the analytic inputs propositions 2.6 to 2.8, a compact single-core vanishing-dissipation branch in the sense of definition 2.3 cannot be a local Type II sequence.*

Proof. By the definition of vanishing localized dissipation, after passing to a selected-window subsequence there is a fixed radius $\rho \in (r_*, R_*)$ such that

$$\int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds \to 0$$

and the positive core bound on Q_{r_*} remains valid along this subsequence. Apply compactness on the fixed compact cylinder Q_ρ and pass to a subsequence. The compactness proposition gives

$$V_n \rightharpoonup V \quad \text{in } L^2(-\rho^2, 0; H^1(B_\rho))$$

and strong convergence in L^3 on every smaller cylinder. Since the localized dissipation tends to zero, the zero-dissipation lemma gives

$$\nabla V = 0 \quad \text{a.e. on } Q_\rho.$$

Thus $V(y, s) = c(s)$ on Q_ρ, and in particular on the selected core cylinder Q_{r_*}.

If this spatially constant limit is zero on the core cylinder, the zero-limit contradiction lemma gives

$$C_{V_n}(r_*) \to 0,$$

contradicting persistence of the retained velocity core.

If the spatially constant limit is not identically zero on Q_{r_*}, the nonzero-limit lemma produces a nonzero bounded selected-window limit, so the branch exits the retained local Type II row through the bounded-window alternative. The two cases exhaust all spatially constant limits, so the branch cannot occur. □

Remark 2.19 (Minimal local inputs). The proof uses two local analytic inputs from the Type II argument: the localized representation for a nondegenerate selected core, the compact-ball pressure reconstruction with stability on smaller balls. The compact-cylinder A, E, C, D bounds and vanishing localized dissipation are part of the branch hypothesis, not separate analytic inputs. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are outside this single-core theorem; they are treated by the local multibubble and cascade exclusion theorem.

3. Repaired-Gauge Construction

3.1. **Setup and notation.** Let $u : \mathbb{R}^3 \times (T-\delta, T) \to \mathbb{R}^3$, $p : \mathbb{R}^3 \times (T-\delta, T) \to \mathbb{R}$ be a suitable weak Navier–Stokes solution with viscosity $\nu > 0$. For $z = (x_0, t_0)$ and $R > 0$,

$$Q_R(z) := B_R(x_0) \times (t_0 - R^2, t_0), \qquad Q_R^* := B_R \times I_*.$$

Fix a reference physical center-time $(x_\bullet, t_\star)$ around which $B_{R_*}(x_\bullet)$ is taken. For a fixed spatial center x_0 we also use

$$Q_R(t) := Q_R(x_0, t).$$

Fix a compact window $I_* := [\tau_0 - \ell, \tau_0 + \ell]$, $\ell > 0$, and write $I_{\mathrm{ph}} := [t_\star - \Lambda_{\max}^2 \ell,\ t_\star + \Lambda_{\max}^2 \ell]$.

For a cylinder $Q_R(x_0, t_0) = B_R(x_0) \times (t_0 - R^2, t_0)$, set

$$\begin{aligned}
A(u; Q_R(x_0,t_0)) &:= R^{-1} \sup_{s \in (t_0 - R^2, t_0)} \int_{B_R(x_0)} |u(x,s)|^2\, dx, \\
E(u; Q_R(x_0,t_0)) &:= R^{-1} \int_{Q_R(x_0,t_0)} |\nabla u|^2\, dx\, ds, \\
C(u; Q_R(x_0,t_0)) &:= R^{-2} \int_{Q_R(x_0,t_0)} |u|^3\, dx\, ds, \\
D(p; Q_R(x_0,t_0)) &:= R^{-2} \int_{t_0 - R^2}^{t_0} \int_{B_R(x_0)} \left| p(x,s) - \langle p(s) \rangle_{B_R(x_0)} \right|^{3/2} dx\, ds.
\end{aligned}$$

Here $\langle p(s) \rangle_{B_R(x_0)} := |B_R|^{-1} \int_{B_R(x_0)} p(x,s)\, dx$; pressure is always understood modulo time-dependent constants.

Definition 3.1 (Single-core hypotheses)**.** Let $(R_*, r_*, M, \eta, \kappa, \Lambda_{\min}, \Lambda_{\max})$ satisfy $\Lambda_{\min} \le \Lambda_{\max} < \infty$. We say (u,p) satisfies $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$ on $Q_{R_*}(x_\bullet, t_\star)$ if the following hold:

(SC1) $A(u; Q_{R_*}(x_\bullet, t)) + E(u; Q_{R_*}(x_\bullet, t)) + C(u; Q_{R_*}(x_\bullet, t)) + D(p; Q_{R_*}(x_\bullet, t)) \le M$ for every $t \in [t_\star - R_*^2, t_\star)$.

(SC2) There exists a physically selected core map $(x_{\mathrm{c}}(t), \rho_{\mathrm{c}}(t))$ on I_{ph} such that $\rho_{\mathrm{c}}(t) \in (0, R_*]$, $x_{\mathrm{c}}(t) \in B_{R_*/2}(x_\bullet)$, and for some $\Theta_0 > 0$,

$$\int_{B_{r_*}(0)} |\rho_{\mathrm{c}}(t) u(x_{\mathrm{c}}(t) + \rho_{\mathrm{c}}(t) y, t)|^3 \mathrm{d}y \ge \eta \quad \text{for a.e. } t \in I_{\mathrm{ph}}.$$

(SC3) There exists $0 < r_{\mathrm{sep}} \le r_*/2$ such that for every $0 < r \le r_{\mathrm{sep}}$, every competitor pair

$$\mathcal{C}_{\mathrm{core}}(t) := \left\{ (\bar{x}, \bar{\rho}) : |\bar{x} - x_\bullet| \le \frac{R_*}{2},\ 0 < \bar{\rho} \le R_* \right\},$$

$(\bar{x}, \bar{\rho}) \in \mathcal{C}_{\mathrm{core}}(t) \setminus \{(x_{\mathrm{c}}(t), \rho_{\mathrm{c}}(t))\}$ with

$$\rho_{\mathrm{c}}(t)^{-1} |\bar{x} - x_{\mathrm{c}}(t)| + \left| \frac{\bar{\rho}}{\rho_{\mathrm{c}}(t)} - 1 \right| \ge r_{\mathrm{sep}}$$

has separated local weighted mass in the same core-scale variables:

$$\left| \int_{B_r} |\rho_{\mathrm{c}}(t) u(x_{\mathrm{c}}(t) + \rho_{\mathrm{c}}(t) y, t)|^3\, \mathrm{d}y - \int_{B_r} |\bar{\rho}\, u(\bar{x} + \bar{\rho} y, t)|^3\, \mathrm{d}y \right| \ge \eta.$$

This quantitative non-competing core condition is imposed on the selected physical core directly; it is not a consequence of the repaired-gauge construction.

(SC4) For the selected core,
$$\sup_{t\in I_{\mathrm{ph}}}\int_{B_{2R_*}} |y|^{-p}|\rho_{\mathrm{c}}(t)u(x_{\mathrm{c}}(t)+\rho_{\mathrm{c}}(t)y,t)|^3\,dy \le M,$$
for some $0<p<3$.

(SC5) The preliminary frame associated with the selected core is the absolutely continuous representative
$$0<\Lambda_{\min}\le\Lambda_{\mathrm{pre}}(\tau)\le\Lambda_{\max},\quad \Lambda_{\mathrm{pre}}\in AC(I_*),\quad X_{\mathrm{pre}}\in AC(I_*;\mathbb{R}^3).$$
It is tied to the selected physical core by the time change $dt_{\mathrm{c}}/d\tau=\Lambda_{\mathrm{pre}}(\tau)^2$, $t_{\mathrm{c}}(\tau_0)=t_\star$: one has
$$X_{\mathrm{pre}}(\tau)=x_{\mathrm{c}}(t_{\mathrm{c}}(\tau)),\qquad \Lambda_{\mathrm{pre}}(\tau)=\rho_{\mathrm{c}}(t_{\mathrm{c}}(\tau))$$
for a.e. $\tau\in I_*$.

The parameter κ denotes the quantitative constants obtained in the local gauge estimates and is not used in items (SC1)–(SC5).

Remark 3.2. The selected core $(x_{\mathrm{c}},\rho_{\mathrm{c}})$ and its separation/nondegeneracy conditions are specified in physical variables; no repaired-gauge constraint is used in their definition. This avoids the circularity of defining the core via the chart to be constructed.

3.2. **Preliminary frame map from the selected core.** The transport map is constructed from the selected core; the core selection itself is part of the hypotheses.

Define
$$\frac{dt_{\mathrm{c}}}{d\tau}=\Lambda_{\mathrm{pre}}(\tau)^2,\qquad t_{\mathrm{c}}(\tau_0)=t_\star,$$
and use the compatibility in (SC5) to identify $\Lambda_{\mathrm{pre}}(\tau)=\rho_{\mathrm{c}}(t_{\mathrm{c}}(\tau))$ and $X_{\mathrm{pre}}(\tau)=x_{\mathrm{c}}(t_{\mathrm{c}}(\tau))$ a.e.

Proposition 3.3 (Preliminary frame map regularity)**.** *Under* $\mathrm{SC}(R_*,r_*,M,\eta,\kappa)$*, the map*
$$\Phi_{\mathrm{pre}}(\tau,Y):=(x,t)=(X_{\mathrm{pre}}(\tau)+\Lambda_{\mathrm{pre}}(\tau)Y,\ t_{\mathrm{c}}(\tau))$$
is well-defined and absolutely continuous in τ. Moreover, $1/\Lambda_{\mathrm{pre}}$, $\Lambda_{\mathrm{pre}}\in L^\infty(I_*)$, $X_{\mathrm{pre}}\in W^{1,1}_{\mathrm{loc}}(I_*;\mathbb{R}^3)$, and for every $R\le R_*$,
$$\Phi_{\mathrm{pre}}:Q^*_R\to\Phi_{\mathrm{pre}}(Q^*_R)\subset Q_{2R_*}(x_\bullet,t_\star)$$
is bi-Lipschitz with $\det\nabla_Y\Phi_{\mathrm{pre}}=\Lambda^3_{\mathrm{pre}}$*, hence Jacobian and inverse-Jacobian bounds depend only on $\Lambda_{\min},\Lambda_{\max}$.*

Proof. From (SC5), $\Lambda_{\mathrm{pre}}\in AC(I_*)$, $0<\Lambda_{\min}\le\Lambda_{\mathrm{pre}}\le\Lambda_{\max}<\infty$, and $X_{\mathrm{pre}}\in AC(I_*;\mathbb{R}^3)$. The scalar ODE
$$\frac{dt_{\mathrm{c}}}{d\tau}=\Lambda_{\mathrm{pre}}(\tau)^2,\qquad t_{\mathrm{c}}(\tau_0)=t_\star$$
therefore has an absolutely continuous strictly increasing solution on I_*, and
$$\frac{dt_{\mathrm{c}}}{d\tau}=\Lambda_{\mathrm{pre}}(\tau)^2\quad\text{a.e. on } I_*.$$
The compatibility condition (SC5) identifies this absolutely continuous frame with the selected physical core a.e.; no differentiability of x_{c} or ρ_{c} beyond the frame regularity in (SC5) is used. For fixed τ, $\nabla_Y\Phi_{\mathrm{pre}}(\tau,\cdot)=\Lambda_{\mathrm{pre}}(\tau)I$, so
$$\det\nabla_Y\Phi_{\mathrm{pre}}(\tau,\cdot)=\Lambda_{\mathrm{pre}}(\tau)^3,\qquad \Lambda^3_{\min}\le\det\nabla_Y\Phi_{\mathrm{pre}}\le\Lambda^3_{\max}.$$
Hence, if $Y_1,Y_2\in B_R$,
$$\Lambda_{\min}|Y_1-Y_2|\le|\Phi_{\mathrm{pre}}(\tau,Y_1)-\Phi_{\mathrm{pre}}(\tau,Y_2)|\le\Lambda_{\max}|Y_1-Y_2|,$$

and therefore $\Phi_{\rm pre}(\tau,\cdot)$ is bi-Lipschitz on B_R. Its inverse has Lipschitz constant $\Lambda_{\min}^{-1}$, so Jacobian and inverse-Jacobian bounds are controlled by $\Lambda_{\min}, \Lambda_{\max}$.

For image inclusions, if $x = X_{\rm pre}(\tau) + \Lambda_{\rm pre}(\tau)Y$ with $|Y| \le R$, then

$$|x - X_{\rm pre}(\tau)| \le \Lambda_{\max} R, \qquad X_{\rm pre}(\tau) \in B_{R_*/2}(x_\bullet),$$

so $x \in B_{R_*/2+\Lambda_{\max}R}(x_\bullet)$. As $\tau \in I_*$, we also have $t = t_c(\tau) \in I_{\rm ph}$. Therefore

$$\Phi_{\rm pre}(Q_R^*) \subset B_{R_*/2+\Lambda_{\max}R}(x_\bullet) \times I_{\rm ph}.$$

□

Definition 3.4 (Preliminary profile)**.** For $(x,t) = \Phi_{\rm pre}(\tau, Y)$, define

$$V_{\rm pre}(Y,\tau) := \Lambda_{\rm pre}(\tau)\, u(x,t), \qquad P_{\rm pre}(Y,\tau) := \Lambda_{\rm pre}(\tau)^2\, p(x,t).$$

Hence $u(x,t) = \Lambda_{\rm pre}(\tau)^{-1} V_{\rm pre}(Y,\tau)$ and $p(x,t) = \Lambda_{\rm pre}(\tau)^{-2} P_{\rm pre}(Y,\tau)$ under $\Phi_{\rm pre}$.

3.3. **Exact renormalized change of variables.** Define renormalized fields on Q_{2R}^* by

$$Y = \frac{x - X(\tau)}{\Lambda(\tau)}, \qquad u(x,t) = \Lambda(\tau)^{-1} V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2} P(Y,\tau),$$

where $dt/d\tau = \Lambda(\tau)^2$, $X : I_* \to \mathbb{R}^3$, $\Lambda : I_* \to (0,\infty)$, and $\Lambda, X \in AC(I_*)$. Write

$$a(\tau) := \frac{\Lambda'(\tau)}{\Lambda(\tau)}, \qquad b(\tau) := \frac{X'(\tau)}{\Lambda(\tau)}.$$

Lemma 3.5 (Exact differential identities)**.** *For smooth data and fixed (x,t), the chain rule gives*

$$\partial_t u = \Lambda^{-3} \left(\partial_\tau V - (b + aY) \cdot \nabla_Y V - aV \right),$$

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \qquad \Delta_x u = \Lambda^{-3} \Delta_Y V, \qquad \nabla_x p = \Lambda^{-3} \nabla_Y P.$$

In particular

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

Proof. For fixed (x,t), $\tau = \tau(t)$ and $Y = (x - X(\tau))/\Lambda(\tau)$. Differentiate

$$u(x,t) = \Lambda(\tau)^{-1} V\big(Y,\tau\big)$$

in t:

$$\partial_t u = -\frac{\Lambda'}{\Lambda^2} V + \frac{1}{\Lambda} \partial_\tau V \,\frac{d\tau}{dt} + \frac{1}{\Lambda} \nabla_Y V \cdot \frac{dY}{dt}.$$

Since $d\tau/dt = \Lambda^{-2}$ and

$$Y_\tau = -\Lambda^{-1} X' - \frac{\Lambda'}{\Lambda} Y, \qquad \frac{dY}{dt} = Y_\tau \,\frac{d\tau}{dt},$$

therefore

$$\partial_t u = \Lambda^{-3} \left(\partial_\tau V - \left(\frac{X'}{\Lambda} + \frac{\Lambda'}{\Lambda} Y \right) \cdot \nabla_Y V - \frac{\Lambda'}{\Lambda} V \right) = \Lambda^{-3} \left(\partial_\tau V - (b + aY) \cdot \nabla_Y V - aV \right).$$

For spatial derivatives, $Y = (x - X)/\Lambda$ gives $\nabla_x Y = \Lambda^{-1} I$, hence

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \qquad \Delta_x u = \Lambda^{-3} \Delta_Y V.$$

Similarly $p(x,t) = \Lambda^{-2} P(Y,\tau)$ yields

$$\nabla_x p = \Lambda^{-3} \nabla_Y P.$$

Taking divergence of $u = \Lambda^{-1} V$ gives

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

□

Theorem 3.6 (Renormalized Navier–Stokes equation)**.** *Assume u, p solve*

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu \Delta_x u = 0, \qquad \nabla_x \cdot u = 0$$

in distributions. Under the transformation above, V, P satisfy, on the image renormalized cylinder, the exact equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)\big(V + Y \cdot \nabla V\big) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0,$$

with the convention fixed by

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda}.$$

Conversely, any V, P satisfying this equation pull back to a distributional solution u, p of Navier–Stokes on the physical image.

Proof. To avoid hidden regularity loss, first write the physical equation tested against $\Phi \in C_c^\infty$ in the physical cylinder and then use the smooth pull-back formula (an approximation argument gives distributions). By Lemma 3.5,

$$\partial_t u + \big(u \cdot \nabla_x\big)u + \nabla_x p - \nu \Delta_x u = 0$$

becomes

$$\Lambda^{-3}(\partial_\tau V - (b + aY) \cdot \nabla V - aV) + \Lambda^{-3}(V \cdot \nabla)V + \Lambda^{-3}\nabla P - \Lambda^{-3}\nu \Delta V = 0.$$

Multiplying by Λ^3 gives

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a\big(V + Y \cdot \nabla V\big) + b \cdot \nabla V.$$

The divergence condition is

$$\nabla_x \cdot u = \Lambda^{-2} \nabla \cdot V = 0 \quad \Longrightarrow \quad \nabla \cdot V = 0.$$

Conversely, assume (V, P) solve this equation on the renormalized cylinder. Set u, p by the same change-of-variables map. Then the preceding identities run in reverse and give

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu \Delta_x u = 0, \qquad \nabla_x \cdot u = 0$$

in $\mathcal{D}'$ on the physical image. □

3.4. **Domain localization and support.** Let $R > 0$ be fixed and $0 < \ell \le \ell_0$. Set $Q_R^* = B_R \times I_*$. With Φ as above define physical support maps

$$\mathcal{Q}_R^{\mathrm{ph}} := \Phi(Q_R^*), \qquad \mathcal{Q}_{2R}^{\mathrm{ph}} := \Phi(Q_{2R}^*).$$

Proposition 3.7 (Compact support transport)**.** *Assume $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$ on I_*. Then*

$$X(\tau) \in B_{R_*/2}(x_\bullet), \qquad |x - X(\tau)| \le \Lambda_{\max} R,$$
$$\mathcal{Q}_R^{\mathrm{ph}} \subset B_{R_*/2 + \Lambda_{\max} R}(x_\bullet) \times I_{\mathrm{ph}},$$
$$\mathcal{Q}_{2R}^{\mathrm{ph}} \subset B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet) \times I_{\mathrm{ph}},$$

for every $\tau \in I_*$ and $x = X(\tau) + \Lambda(\tau)Y$ with $Y \in B_{2R}$. In particular, if $|c(\tau)| \le \delta$ from Proposition 3.18, then $X(\tau) \in B_{R_*/2 + \Lambda_{\max}\delta}(x_\bullet)$ on I_*. Hence, $\chi_R \in C_c^\infty(B_{2R})$ pulls back to a compactly supported physical cutoff that is smooth in the spatial variables and absolutely continuous in time under the AC chart, with support depending only on $R, \Lambda_{\min}, \Lambda_{\max}$.

Proof. Fix $\tau \in I_*$ and $Y \in B_R$. Since $x = X(\tau) + \Lambda(\tau)Y$,

$$|x - X(\tau)| \le |Y|\,\Lambda_{\max} \le \Lambda_{\max} R, \qquad \Lambda_{\max}^{-1}|x - X(\tau)| \le |Y|.$$

Because $X(\tau) \in B_{R_*/2}(x_\bullet)$, this gives

$$x \in B_{R_*/2+\Lambda_{\max}R}(x_\bullet).$$

Similarly, if $Y \in B_{2R}$, then $x \in B_{R_*/2+2\Lambda_{\max}R}(x_\bullet)$. Together with $t = t_c(\tau) \in I_{\mathrm{ph}}$, these give the claimed inclusions for $\mathcal{Q}_R^{\mathrm{ph}}$ and $\mathcal{Q}_{2R}^{\mathrm{ph}}$.

For pull-backs of cutoffs, Φ is bi-Lipschitz at every τ, so composition

$$\chi_R^{\mathrm{ph}}(x,t) := \chi_R\Big(\frac{x - X(\tau(t))}{\Lambda(\tau(t))}\Big)$$

is smooth in x for each time and is AC in time when the chart is AC. It is supported where $Y \in B_{2R}$, hence supported where $(x,t) \in \mathcal{Q}_{2R}^{\mathrm{ph}}$. The support size depends only on $R, \Lambda_{\min}, \Lambda_{\max}$ and the fixed C_c^∞ norm of χ_R, not on the particular branch. □

3.5. **Transport of local quantities under the chart.** Let $t = t(\tau) = t_c(\tau)$ and, for $R > 0$, define

$$J_\tau^-(R) := \big(t(\tau) - \Lambda_{\max}^2 R^2,\ t(\tau)\big].$$

Define renormalized local quantities on $Q_R^*(\tau)$ by

$$A_V(\tau,R) := \sup_{s\in(\tau-R^2,\tau)} \int_{B_R} |V(\cdot,s)|^2\,dY, \qquad E_V(\tau,R) := \int_{\tau-R^2}^{\tau}\int_{B_R} |\nabla V|^2\,dY\,ds,$$

$$C_V(\tau,R) := \int_{\tau-R^2}^{\tau}\int_{B_R} |V|^3\,dY\,ds, \qquad D_V(\tau,R) := \int_{\tau-R^2}^{\tau}\int_{B_R} \Big|P - \langle P\rangle_{B_R}\Big|^{3/2}\,dY\,ds.$$

For interval notation and a center X, define

$$A_u(J,R;X) := \sup_{s\in J} \int_{B_R(X)} |u(\cdot,s)|^2\,dx, \quad E_u(J,R;X) := \int_J \int_{B_R(X)} |\nabla u|^2\,dx\,ds,$$

$$C_u(J,R;X) := \int_J \int_{B_R(X)} |u|^3\,dx\,ds, \quad D_u(J,R;X) := \int_J \int_{B_R(X)} |p - \langle p\rangle_{B_R(X)}|^{3/2}\,dx\,ds.$$

In the inequalities below we abbreviate

$$A_u(J,R) := A_u(J,R;X(\tau)), \quad E_u(J,R) := E_u(J,R;X(\tau)),$$
$$C_u(J,R) := C_u(J,R;X(\tau)), \quad D_u(J,R) := D_u(J,R;X(\tau)).$$

and set

$$\Delta_X(\tau,R) := \sup_{s\in(\tau-R^2,\tau)} |X(s) - X(\tau)|, \qquad R_{\mathrm{ph}}(\tau,R) := \Lambda_{\max}R + \Delta_X(\tau,R).$$

Proposition 3.8 (Cylinder comparison on compact windows)**.** *Assume $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$ and formulas of Lemma 3.5. Then for every $\tau \in I_*$, $R > 0$,*

$$A_V(\tau,R) \le \Lambda_{\min}^{-1} A_u(J_\tau^-(R), R_{\mathrm{ph}}(\tau,R)),$$
$$E_V(\tau,R) \le \Lambda_{\min}^{-1} E_u(J_\tau^-(R), R_{\mathrm{ph}}(\tau,R)),$$
$$C_V(\tau,R) \le \Lambda_{\min}^{-2} C_u(J_\tau^-(R), R_{\mathrm{ph}}(\tau,R)),$$
$$D_V(\tau,R) \le \Lambda_{\min}^{-2} D_u(J_\tau^-(R), R_{\mathrm{ph}}(\tau,R)).$$

In particular, all four renormalized quantities are bounded whenever the physical quantities are bounded on the corresponding compact physical cylinders.

Proof. These are direct changes of variables under Φ_{pre}:

$$V(\cdot,\tau)=\Lambda(\tau)u(\cdot,t(\tau)),\qquad P(\cdot,\tau)=\Lambda(\tau)^2p(\cdot,t(\tau)),$$

$$\nabla_Y V=\Lambda^2\nabla_x u,\qquad dY\,d\tau=\Lambda^{-5}\,dx\,dt,\qquad d\tau=\Lambda^{-2}\,dt.$$

Insert these formulas into each definition and use $\Lambda_{\min}\le\Lambda\le\Lambda_{\max}$, plus

$$\tau-s\in(0,R^2)\implies t(\tau)-\Lambda_{\max}^2R^2\le t(s)\le t(\tau).$$

Moreover, if $s\in(\tau-R^2,\tau)$ and $x=X(s)+\Lambda(s)Y$ with $|Y|<R$, then

$$|x-X(\tau)|\le|X(s)-X(\tau)|+\Lambda(s)|Y|\le\Delta_X(\tau,R)+\Lambda_{\max}R=R_{\text{ph}}(\tau,R).$$

So the physical image of $B_R\times(\tau-R^2,\tau)$ is contained in $B_{R_{\text{ph}}(\tau,R)}(X(\tau))\times J_\tau^-(R)$, which yields the stated comparisons. □

3.6. **Local repaired gauge map.** Fix $R>0$, $0<p<3$, and choose a radial cutoff $\chi_R\in C_c^\infty(B_{2R})$ with $0\le\chi_R\le1$, $\chi_R\equiv1$ on B_R, and $Y\cdot\nabla\chi_R\le0$. Let the local energy profile space be

$$\mathcal{V}_R:=\left\{V\in L^3(B_{2R})\cap W^{1,3/2}(B_{2R};\mathbb{R}^3):|Y|^{-p/3}V\in L^3(B_{2R})\right\},$$

with norm

$$\|V\|_{\mathcal{V}_R}:=\|V\|_{L^3(B_{2R})}+\||Y|^{-p/3}V\|_{L^3(B_{2R})}+\|\nabla V\|_{L^{3/2}(B_{2R})}.$$

For differentiating the repaired scale and centering gauges we use the admissible gauge class

$$\mathcal{A}_{p,R}:=\left\{V:\begin{array}{l}V\in L^3(B_{4R})\cap W^{1,3}_{\text{loc}}(B_{4R};\mathbb{R}^3),\\ \displaystyle\int_{B_{4R}}|Y|^{-p}|V|^3\,dY<\infty,\quad\int_{B_{4R}}|Y|^{-p}|V|\,|Y\cdot\nabla V|\,dY<\infty,\\ \displaystyle\int_{B_{4R}}|Y|^{-p-1}|V|^3\,dY<\infty\end{array}\right\}.$$

The larger ball is only a domain buffer: all gauge integrals are supported in B_{2R}, while (λ,c) is restricted near $(1,0)$ so that $c+\lambda B_{2R}\subset B_{4R}$.

Definition 3.9 (Gauge actions)**.** For $(\lambda,c)\in(0,\infty)\times\mathbb{R}^3$, define

$$V_{\lambda,c}(Y):=\lambda V(c+\lambda Y).$$

For pressure profiles use

$$P_{\lambda,c}(Y):=\lambda^2P(c+\lambda Y).$$

We also write $V^{\lambda,c}:=V_{\lambda,c}$ and $P^{\lambda,c}:=P_{\lambda,c}$.

Definition 3.10 (Finite-dimensional gauge map)**.** For $V\in\mathcal{V}_R$ and $(\lambda,c)\in(0,\infty)\times\mathbb{R}^3$, set

$$\mathcal{F}(\lambda,c;V):=\big(\mathcal{F}_0,\mathcal{F}_1,\mathcal{F}_2,\mathcal{F}_3\big),$$

with

$$\mathcal{F}_0(\lambda,c;V):=\int_{\mathbb{R}^3}\chi_R(Y)\,|Y|^{-p}|V_{\lambda,c}(Y)|^3\mathrm{d}Y-\Theta_0,\qquad\mathcal{F}_j(\lambda,c;V):=\int_{\mathbb{R}^3}Y_j\,\chi_R(Y)\,|V_{\lambda,c}(Y)|^2\mathrm{d}Y$$

for $j=1,2,3$. The gauge conditions are $\mathcal{F}(\lambda,c;V)=0$. Codomain is $\mathbb{R}^4$, domain $(0,\infty)\times\mathbb{R}^3\times\mathcal{V}_R$.

Definition 3.11 (Localized modulation matrix)**.** For $J\subset I_*$ and a profile path $W:J\to\mathcal{V}_R$, define

$$J_W(\tau):=D_{(\lambda,c)}\mathcal{F}\big(1,0;W(\tau)\big)\in\mathbb{R}^{4\times4}.$$

The rows are ordered as the scale constraint followed by the three centering constraints.

Lemma 3.12 (Weighted integrability and differentiation under integrals)**.** *Assume $V, W \in \mathcal{A}_{p,R}$. Then $G_0(V) := \int \chi_R |Y|^{-p}|V|^3$ is finite and*

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big|_{\epsilon=0} \int \chi_R |Y|^{-p}|V+\epsilon W|^3 \,\mathrm{d}Y = \int 3\chi_R |Y|^{-p}|V|\, V\cdot W \,\mathrm{d}Y.$$

Proof. For $W \in \mathcal{A}_{p,R}$, define

$$F_\varepsilon(Y) := \chi_R(Y)|Y|^{-p}|V(Y)+\varepsilon W(Y)|^3.$$

For each Y, $\varepsilon \mapsto F_\varepsilon(Y)$ is C^1 and

$$\partial_\varepsilon F_\varepsilon = 3\chi_R|Y|^{-p}|V+\varepsilon W|(V+\varepsilon W)\cdot W.$$

For $|\varepsilon| \le 1$,

$$|F_\varepsilon(Y)| + |\partial_\varepsilon F_\varepsilon| \le C\chi_R|Y|^{-p}\big(|V|^3+|W|^3\big) \in L^1(B_{2R})$$

by the defining weighted L^3 condition of $\mathcal{A}_{p,R}$. Hence dominated convergence allows differentiation under the integral, and

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0} \int_{B_{2R}} \chi_R |Y|^{-p}|V+\epsilon W|^3 \,\mathrm{d}Y = \int_{B_{2R}} 3\chi_R |Y|^{-p}|V|\, V\cdot W \,\mathrm{d}Y.$$

□

Lemma 3.13 (Gauge covariance)**.** *For $V \in \mathcal{V}_R$, $(\lambda, c) \in (0,\infty)\times\mathbb{R}^3$, and $\alpha = 0,\dots,3$,*

$$\mathcal{F}_\alpha(1,0;V_{\lambda,c}) = \mathcal{F}_\alpha(\lambda,c;V).$$

Proof. For each component this is a direct substitution:

$$\mathcal{F}_0(1,0;V_{\lambda,c}) = \int \chi_R(Y)|Y|^{-p}|V_{\lambda,c}(Y)|^3 - \Theta_0 \,\mathrm{d}Y = \mathcal{F}_0(\lambda,c;V),$$

$$\mathcal{F}_j(1,0;V_{\lambda,c}) = \int Y_j\chi_R(Y)|V_{\lambda,c}(Y)|^2 \,\mathrm{d}Y = \mathcal{F}_j(\lambda,c;V), \quad j=1,2,3.$$

□

Proposition 3.14 (Differentiability of $\mathcal{F}$)**.** *For each fixed $V \in \mathcal{A}_{p,R}$, the map $(\lambda,c)\mapsto\mathcal{F}(\lambda,c;V)$ is C^1 in a neighborhood of $(1,0)$. In each component $\mathcal{F}_\alpha$ all derivatives under the integral sign are legitimate.*

Proof. Fix (λ,c) near $(1,0)$, and abbreviate

$$V_{\lambda,c}(Y) := \lambda V(c+\lambda Y).$$

By direct differentiation,

$$\partial_\lambda V_{\lambda,c} = \lambda^{-1}V_{\lambda,c} + \lambda^{-1}Y\cdot\nabla V_{\lambda,c}, \qquad \partial_{c_k}V_{\lambda,c} = \lambda\,\partial_k V(c+\lambda Y) = \lambda^{-1}\partial_k V_{\lambda,c}.$$

The neighborhood of $(1,0)$ is chosen so that the arguments of V remain inside B_{4R} for $Y\in\operatorname{supp}\chi_R$. The scale derivative is controlled by the two weighted integrability conditions

$$\int |Y|^{-p}|V|^3 < \infty, \qquad \int |Y|^{-p}|V|\,|Y\cdot\nabla V| < \infty,$$

transported by the smooth change of variables. The translation derivative in the scale component is treated by the source identity

$$3\int \chi_R|Y|^{-p}|V|\,V\cdot\partial_k V = p\int \chi_R Y_k|Y|^{-p-2}|V|^3 - \int(\partial_k\chi_R)|Y|^{-p}|V|^3,$$

obtained first for smooth compactly supported profiles and then by approximation in $\mathcal{A}_{p,R}$; the right-hand side is finite because $\int |Y|^{-p-1}|V|^3 < \infty$, while the cutoff term is supported on a compact annulus away from the origin and is controlled by local L^3. The translation components are supported in B_{2R} and are controlled by $V \in L^2_{\mathrm{loc}}$ and $\nabla V \in L^3_{\mathrm{loc}}$. These bounds give an integrable majorant for the difference quotients, so dominated convergence applies.

$$\mathcal{F}_0(\lambda, c; V) = \int_{\mathbb{R}^3} \chi_R(Y)\, |Y|^{-p} |V_{\lambda,c}|^3 \,\mathrm{d}Y - \Theta_0,$$

for which dominated convergence gives

$$\partial_\lambda \mathcal{F}_0 = \int_{\mathbb{R}^3} \chi_R |Y|^{-p}\, 3|V_{\lambda,c}|\, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c}\, dY,$$

$$\partial_{c_k} \mathcal{F}_0 = \int_{\mathbb{R}^3} \chi_R |Y|^{-p}\, 3|V_{\lambda,c}|\, V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c}\, dY.$$

Similarly

$$\mathcal{F}_j(\lambda, c; V) = \int_{\mathbb{R}^3} Y_j \chi_R(Y)\, |V_{\lambda,c}|^2 \,\mathrm{d}Y, \quad j = 1, 2, 3,$$

hence

$$\partial_\lambda \mathcal{F}_j \ = \int_{\mathbb{R}^3} 2 Y_j \chi_R\, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c}\, dY,$$

$$\partial_{c_k} \mathcal{F}_j = \int_{\mathbb{R}^3} 2 Y_j \chi_R\, V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c}\, dY.$$

Thus each component admits pointwise C^1 formulas and these are continuous in (λ, c) near $(1, 0)$. Therefore $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V) \in C^1$ near $(1, 0)$. □

Proposition 3.15 (Full local Jacobian at $(1,0)$)**.** *Let $V \in \mathcal{A}_{p,R}$. Set $V_0(Y) = V(Y)$. At $(\lambda, c) = (1, 0)$, the matrix*

$$J(V) := D_{(\lambda,c)} \mathcal{F}(1, 0; V) \in \mathbb{R}^{4\times 4}$$

has rows

$$J_\alpha = \big(\partial_\lambda \mathcal{F}_\alpha,\ \partial_{c_1} \mathcal{F}_\alpha,\ \partial_{c_2} \mathcal{F}_\alpha,\ \partial_{c_3} \mathcal{F}_\alpha\big), \quad \alpha = 0, \dots, 3.$$

Writing $W_{\mathrm{scl}} = V_0 + Y \cdot \nabla V_0$, one has

$$\partial_\lambda \mathcal{F}_0(1, 0; V) = 3 \int \chi_R |Y|^{-p} |V_0|\, V_0 \cdot W_{\mathrm{scl}} \,\mathrm{d}Y, \tag{2}$$

$$\partial_{c_k} \mathcal{F}_0(1, 0; V) = 3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0|\, V_0 \cdot \partial_k V_0 \,\mathrm{d}Y, \tag{3}$$

$$\partial_\lambda \mathcal{F}_j(1, 0; V) = 2 \int_{\mathbb{R}^3} Y_j \chi_R\, V_0 \cdot W_{\mathrm{scl}} \,\mathrm{d}Y, \tag{4}$$

$$\partial_{c_k} \mathcal{F}_j(1, 0; V) = 2 \int_{\mathbb{R}^3} \chi_R Y_j V_0 \cdot \partial_k V_0 \,\mathrm{d}Y, \tag{5}$$

for $j, k = 1, 2, 3$.

Proof. Differentiate the formulas in Definition 3.10 at $(\lambda, c) = (1, 0)$.

Since $V_{\lambda,c}(Y) = \lambda V(c + \lambda Y)$,

$$\partial_\lambda V_{\lambda,c} = \lambda^{-1} V_{\lambda,c} + \lambda^{-1} Y \cdot \nabla V_{\lambda,c}, \qquad \partial_{c_k} V_{\lambda,c} = \lambda\, \partial_k V(c + \lambda Y).$$

At $(1, 0)$, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = V + Y \cdot \nabla V$, $\partial_{c_k} V_{\lambda,c}|_{(1,0)} = \partial_k V$.

For $\mathcal{F}_0$, write

$$\mathcal{F}_0(\lambda, c; V) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 \, dY - \Theta_0.$$

Derivative in λ:

$$\partial_\lambda \mathcal{F}_0 \ = \int \chi_R |Y|^{-p} 3 |V_{\lambda,c}|\, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c}\, dY$$

Evaluating at $(1,0)$ yields (2).

For $k=1,2,3$, c_k-derivative of $\mathcal{F}_0$ has no cutoff term:

$$\partial_{c_k}\mathcal{F}_0(1,0;V)=\int \chi_R|Y|^{-p}\,3|V|\,V\cdot\partial_k V\,dY,$$

which is (3).

For $\mathcal{F}_j$,

$$\mathcal{F}_j(\lambda,c;V)=\int Y_j\chi_R(Y)\,|V_{\lambda,c}|^2\,dY.$$

Hence

$$\partial_\lambda\mathcal{F}_j=\int 2Y_j\chi_R\,V_{\lambda,c}\cdot\partial_\lambda V_{\lambda,c}\,dY,$$

and at $(1,0)$:

$$\partial_\lambda\mathcal{F}_j(1,0;V)=2\int Y_j\chi_R\,V\cdot W_{\mathrm{scl}}\,dY,$$

which is (4). Similarly,

$$\partial_{c_k}\mathcal{F}_j(1,0;V)=\int 2Y_j\chi_R\,V\cdot\partial_{c_k}V_{\lambda,c}\big|_{(1,0)}\,dY=2\int\chi_R Y_j V\cdot\partial_k V\,dY,$$

giving (5). $\square$

Proposition 3.16 (Translation block and Schur-complement invertibility)**.** *Let $V:J\to\mathcal{A}_{p,R}$ satisfy the repaired gauge constraints*

$$\mathcal{F}(1,0;V(\tau))=0\qquad\textit{for a.e. }\tau\in J.$$

Write the modulation matrix in block form

$$J(V(\tau))=\begin{pmatrix}\alpha(\tau) & u(\tau)^T\\ v(\tau) & A(\tau)\end{pmatrix},$$

where $\alpha=\partial_\lambda\mathcal{F}_0$, $u_\ell=\partial_{c_\ell}\mathcal{F}_0$, $v_j=\partial_\lambda\mathcal{F}_j$, and $A_{j\ell}=\partial_{c_\ell}\mathcal{F}_j$. Assume that for a.e. $\tau\in J$

$$m_\chi(\tau):=\int|V(Y,\tau)|^2\chi_R(Y)\,dY\ge m_0>0,$$

$$\int_{A_R}|V(Y,\tau)|^2\,dY\le\varepsilon_0,\qquad 3C_\chi\varepsilon_0\le\frac{m_0}{2},$$

where $A_R=\{R\le|Y|\le 2R\}$ and

$$C_\chi:=\max_{j,\ell}\|Y_j\partial_\ell\chi_R\|_{L^\infty}.$$

Assume also

$$C_u:=\operatorname*{ess\,sup}_{\tau\in J}\int|Y|^{-p-1}|V(Y,\tau)|^3\,dY<\infty$$

and, with $C_\nabla:=\||Y|\nabla\chi_R\|_{L^\infty}$,

$$6(p+C_\nabla)\left(\frac{2}{m_0}\right)C_u\,RC_\nabla\varepsilon_0\le\frac{p\Theta_0}{2}.$$

Then $J(V(\tau))$ is invertible for a.e. $\tau\in J$, and

$$\operatorname*{ess\,sup}_{\tau\in J}\|J(V(\tau))^{-1}\|_{\infty\to\infty}\le K(R,p,\Theta_0,m_0,C_u,\varepsilon_0,\chi_R).$$

Proof. The entries are finite by $V(\tau) \in \mathcal{A}_{p,R}$ and the compact support of χ_R. With the present parameter convention, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = V + Y \cdot \nabla V$. Hence the scale entry satisfies the signed transversality bound

$$\alpha(\tau) = p \int \chi_R |Y|^{-p} |V(Y,\tau)|^3 \, dY - \int (Y \cdot \nabla \chi_R) |Y|^{-p} |V(Y,\tau)|^3 \, dY \geq p\Theta_0$$

on the gauge surface, because $Y \cdot \nabla \chi_R \leq 0$. Thus $|\alpha| \geq p\Theta_0$.

For the centering block, integration by parts gives

$$A_{j\ell} = -\delta_{j\ell} \int |V|^2 \chi_R - \int |V|^2 Y_j \partial_\ell \chi_R.$$

Writing $A = -m_\chi I_3 + E$, the annular support of $\nabla \chi_R$ gives

$$|E_{j\ell}| \leq C_\chi \int_{A_R} |V|^2 \leq C_\chi \varepsilon_0, \qquad \|E\|_{\infty\to\infty} \leq 3C_\chi \varepsilon_0 \leq m_0/2.$$

Thus $A = -m_\chi(I_3 - m_\chi^{-1}E)$ and $\|m_\chi^{-1}E\|_{\infty\to\infty} \leq 1/2$, so A^{-1} exists and $\|A^{-1}\|_{\infty\to\infty} \leq 2/m_0$.

For the scale-to-translation block, the scale-constraint formula gives

$$|u_\ell| \leq p \int |Y|^{-p-1} |V|^3 \, dY + \int |\partial_\ell \chi_R| \, |Y|^{-p} |V|^3 \, dY \leq (p + C_\nabla) C_u,$$

so $\|u\|_\infty \leq (p + C_\nabla)C_u$. For the translation-to-scale block,

$$v_j = -\int |V|^2 Y_j (Y \cdot \nabla \chi_R) \, dY$$

because the centering constraint cancels the interior term. Therefore

$$\|v\|_\infty \leq 2R \, C_\nabla \varepsilon_0.$$

The Schur complement $S = \alpha - u^T A^{-1} v$ satisfies

$$|u^T A^{-1} v| \leq 3\|u\|_\infty \|A^{-1}\|_{\infty\to\infty} \|v\|_\infty \leq 6(p + C_\nabla) \left(\frac{2}{m_0}\right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Since $|\alpha| \geq p\Theta_0$, this gives $|S| \geq p\Theta_0/2$. The block inverse formula

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}u^T A^{-1} \\ -A^{-1}vS^{-1} & A^{-1} + A^{-1}vS^{-1}u^T A^{-1} \end{pmatrix}$$

then gives the displayed bound. □

3.7. **Retained repaired-gauge selection in time.**

Definition 3.17 (Retained nondegenerate repaired-gauge regime)**.** Let $J \subset I_*$ be compact. The retained nondegenerate repaired-gauge regime over the preliminary profile V_{pre} consists of branches for which there are

$$\lambda \in AC(J; (0,\infty)), \qquad c \in AC(J; \mathbb{R}^3),$$

such that, for a.e. $\tau \in J$,

$$V(\tau) := V_{\mathrm{pre}}^{\lambda(\tau), c(\tau)} \in \mathcal{A}_{p,R}, \qquad \mathcal{F}(1, 0; V(\tau)) = 0.$$

The path is quantitatively nondegenerate if the translation-block, annular, weighted first-moment, and Schur-complement bounds in Proposition 3.16 hold on J. If no such local AC selection exists, or if the quantitative nondegeneracy hypotheses fail persistently, the branch lies in the gauge-degenerate, multibubble, or cascade alternatives rather than in the retained regime.

Proposition 3.18 (Gauge regularity on the retained regime)**.** *Let (λ, c) be the repaired-gauge selection of the retained nondegenerate regime on J. Then*

$$\mathcal{F}\big(\lambda(\tau), c(\tau); V_{\mathrm{pre}}(\tau)\big) = 0 \quad \textit{for a.e. } \tau \in J,$$

and $(\lambda, c) \in AC(J; \mathbb{R}^4)$ by definition. If the path is quantitatively nondegenerate, then

$$\operatorname*{ess\,sup}_{\tau \in J} \|J(V(\tau))^{-1}\| < \infty.$$

Proof. The constraint follows from Lemma 3.13 applied to $V(\tau) = V_{\mathrm{pre}}^{\lambda(\tau), c(\tau)}$. The absolute continuity is part of Definition 3.17. The uniform inverse bound is Proposition 3.16. $\square$

Definition 3.19 (Repaired chart)**.** Define

$$\Lambda(\tau) := \Lambda_{\mathrm{pre}}(\tau)\lambda(\tau), \qquad X(\tau) := X_{\mathrm{pre}}(\tau) + \Lambda_{\mathrm{pre}}(\tau)c(\tau),$$

$$V(\tau) := V_{\mathrm{pre}}^{\lambda(\tau), c(\tau)}, \qquad P(\tau) := P_{\mathrm{pre}}^{\lambda(\tau), c(\tau)} \text{ (same physical pressure pull-back under } \Phi).$$

Theorem 3.20 (AC repaired parameters)**.** *Assume* $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$*, and suppose the branch lies in the retained nondegenerate repaired-gauge regime on J, with selection (λ, c). Then the corrected chart (Λ, X) is absolutely continuous on J, satisfies*

$$a(\tau) = \frac{\Lambda'(\tau)}{\Lambda(\tau)} \in L^1_{\mathrm{loc}}(J), \qquad b(\tau) = \frac{X'(\tau)}{\Lambda(\tau)} \in L^1_{\mathrm{loc}}(J),$$

and the gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$ hold a.e. Equivalently,

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\mathrm{pre}}(\tau)) = 0$$

for the repaired-gauge selection (λ, c).

Proof. By Definition 3.17, $(\lambda, c) \in AC(J)$ and the repaired gauge constraints hold for $V(\tau)$ a.e. By definition $V(\tau) = V_{\mathrm{pre}}^{\lambda(\tau), c(\tau)}$, so Lemma 3.13 applied at $(\lambda, c) = (\lambda(\tau), c(\tau))$ gives

$$\mathcal{F}(1, 0; V_{\mathrm{pre}}^{\lambda, c}) = \mathcal{F}(\lambda, c; V_{\mathrm{pre}})$$

implies $\mathcal{F}(1, 0; V(\tau)) = 0$ a.e.

Since $\Lambda = \Lambda_{\mathrm{pre}}\lambda$ and $X = X_{\mathrm{pre}} + \Lambda_{\mathrm{pre}}c$, and $\Lambda_{\mathrm{pre}}, X_{\mathrm{pre}}$ are AC by (SC5), the products and sums preserve AC, hence $(\Lambda, X) \in AC(I_*)$. Differentiating,

$$a = \frac{\Lambda'}{\Lambda} = \frac{\Lambda'_{\mathrm{pre}}}{\Lambda_{\mathrm{pre}}} + \frac{\lambda'}{\lambda} \in L^1_{\mathrm{loc}}, \qquad b = \frac{X'}{\Lambda} = \frac{X'_{\mathrm{pre}} + \Lambda'_{\mathrm{pre}}c + \Lambda_{\mathrm{pre}}c'}{\Lambda_{\mathrm{pre}}\lambda} \in L^1_{\mathrm{loc}}.$$

This gives the claimed regularity. $\square$

Remark 3.21 (Gauge degeneracy). Failure of invertibility of J is the gauge-degenerate alternative to the retained single-core regime for this fixed renormalization mechanism. In the Type II decomposition of alternatives this case belongs to the multibubble/cascade and scale-rigidity alternatives, rather than to the representation theorem.

3.8. **Pressure transport and decomposition.**

Lemma 3.22 (Renormalized pressure equation)**.** *If V, P satisfy the renormalized Navier–Stokes equation in Theorem 3.6, then*

$$-\Delta P = \partial_i \partial_j (V_i V_j) \qquad \textit{in } \mathcal{D}'(Q^*_{2R}).$$

Proof. Take test $\varphi \in C_c^\infty(Q_{2R}^*)$. Since $V, P \in L^2_{\mathrm{loc}} \times L^{3/2}_{\mathrm{loc}}$ on Q^*_{2R}, all pairings below are legitimate. Expand

$$\nabla \cdot \big((V \cdot \nabla)V\big) = \sum_{i,j} \partial_i \partial_j (V_i V_j) - \sum_{i,j} \partial_i V_i \, \partial_j V_j.$$

Because $\nabla \cdot V = 0$, this reduces to $\partial_i \partial_j (V_i V_j)$. Taking divergence of

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V$$

and using $\nabla \cdot V = 0$, $\nabla \cdot (V + Y \cdot \nabla V) = 0$, $\nabla \cdot (b \cdot \nabla V) = b \cdot \nabla(\nabla \cdot V) = 0$, and $\nabla \cdot \Delta V = \Delta(\nabla \cdot V) = 0$, the following identity holds in $\mathcal{D}'$:

$$\Delta P + \partial_i \partial_j (V_i V_j) = 0.$$

Hence $-\Delta P = \partial_i \partial_j (V_i V_j)$ on Q^*_{2R}. □

Theorem 3.23 (Localized pressure decomposition)**.** *Let $\zeta \in C_c^\infty(B_{2R})$, $\zeta \equiv 1$ on B_R, and assume $V \in L^1_{\mathrm{loc}}(I_*; L^3(B_{2R}))$, $P \in L^1_{\mathrm{loc}}(I_*; L^{3/2}(B_{2R}))$. For a.e. $\tau \in I_*$ there exists*

$$P(\cdot, \tau) = P_{\mathrm{loc}}(\cdot, \tau) + H(\cdot, \tau)$$

on B_R such that

$$\begin{cases} -\Delta P_{\mathrm{loc}}(\cdot, \tau) = \partial_i \partial_j \big(\zeta V_i V_j\big), \\ -\Delta H(\cdot, \tau) = 0 \text{ on } B_R, \quad ⨍_{B_R} H(\cdot, \tau)\, dY = 0. \end{cases}$$

Moreover

$$\|P_{\mathrm{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|^2_{L^3(B_{2R})}, \qquad \|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|P(\tau)\|_{L^{3/2}(B_{2R})} + \|V(\tau)\|^2_{L^3(B_{2R})}$$

for each $1 \le q < \infty$ and each $0 < r < R$, with constants depending only on R, r, and q.

Proof. For fixed τ, set

$$P_{\mathrm{loc}} = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j), \qquad H := P - P_{\mathrm{loc}}.$$

$\zeta V_i V_j \in L^1_{\mathrm{loc}}(I_*; L^{3/2}(B_{2R}))$, so the Calderón–Zygmund operator $f \mapsto \mathcal{R}_i \mathcal{R}_j f$ gives

$$P_{\mathrm{loc}}(\cdot, \tau) = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j)(\cdot, \tau), \qquad \|P_{\mathrm{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|^2_{L^3(B_{2R})}.$$

Because $\mathcal{R}_i \mathcal{R}_j$ is linear bounded on $L^{3/2}$,

$$P_{\mathrm{loc}} \in L^1_{\mathrm{loc}}(I_*; L^{3/2}(B_R)).$$

Since $-\Delta P_{\mathrm{loc}} = \partial_i \partial_j (\zeta V_i V_j)$ and $\zeta \equiv 1$ on B_R,

$$-\Delta H(\cdot, \tau) = \partial_i \partial_j \big((1 - \zeta) V_i V_j\big) = 0 \quad \text{in } B_R,$$

so $H(\cdot, \tau)$ is harmonic on B_R. Imposing $⨍_{B_R} H = 0$ fixes the time-dependent additive constant uniquely in each time slice. For $q \in [1, \infty)$ and $0 < r < R$, harmonicity and local L^p-to-L^q interior estimates imply

$$\|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|H(\tau)\|_{L^{3/2}(B_R)}.$$

Finally, $\|H(\tau)\|_{L^{3/2}(B_R)} \le \|P(\tau)\|_{L^{3/2}(B_R)} + \|P_{\mathrm{loc}}(\tau)\|_{L^{3/2}(B_R)}$, giving the announced bound in terms of P and V. □

Remark 3.24 (Measurability and normalization). If V, P are measurable in τ, then $\tau \mapsto P_{\mathrm{loc}}(\cdot, \tau)$ is measurable in $L^{3/2}(B_R)$ by boundedness of $\mathcal{R}_i \mathcal{R}_j$, hence $\tau \mapsto H(\cdot, \tau)$ is measurable in each $L^q(B_r)$, $0 < r < R$. The normalization $⨍_{B_R} H(\cdot, \tau)\, dY = 0$ fixes the time-dependent additive constant in H uniquely for each τ, and therefore the decomposition is a legitimate renormalized pressure splitting.

3.9. Modulation from differentiated constraints.

Definition 3.25 (Admissible modulation-differentiability row)**.** On a compact window $J \subset I_*$, a retained repaired-gauge branch satisfies the modulation-differentiability row if, for each $\alpha = 0, 1, 2, 3$,

$$G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$$

belongs to $AC(J)$, and there are functions $\mathcal{R}_\alpha \in L^1(J)$ such that

$$G'_\alpha(\tau) = -\mathcal{R}_\alpha(\tau) + a(\tau)\, \partial_\lambda \mathcal{F}_\alpha(1, 0; V(\tau)) + \sum_{k=1}^{3} b_k(\tau)\, \partial_{c_k} \mathcal{F}_\alpha(1, 0; V(\tau)) \tag{6}$$

for a.e. $\tau \in J$. The terms $\mathcal{R}_\alpha$ are defined as the limits obtained by testing the represented renormalized equation against smooth compactly supported truncations of the Gateaux gradients of $\mathcal{F}_\alpha$, then passing to the limit in the local energy and pressure topologies. Thus $\mathcal{R}_\alpha$ is a distributional truncated-test limit, not the unsupported pairing of an L^1_x coefficient with a raw $\partial_\tau V$.

If this row fails on a compact window, the branch is routed to the modulation-failure exit rather than to the retained modulation theorem.

Lemma 3.26 (Differentiated repaired constraints under the modulation row)**.** *If the repaired branch satisfies definition 3.25 on J, then the repaired constraints have the a.e. derivative identity (6) on J, with $\mathcal{R} = (\mathcal{R}_0, \dots, \mathcal{R}_3) \in L^1(J)^4$.*

Proof. This is exactly the retained-row assertion in definition 3.25. The definition records the admissible limiting procedure: one first pairs the equation with smooth compactly supported truncations of the Gateaux gradients and only then passes to the limit. No step uses an L^1_x-against-L^2_x pairing with $\partial_\tau V$. □

Lemma 3.27 (Directional derivatives in the profile variable)**.** *For $\mathcal{A}_\tau = V + Y \cdot \nabla V$ and $\alpha = 0, 1, 2, 3$,*

$$D_V \mathcal{F}_\alpha(1, 0; V)[\mathcal{A}_\tau] = \partial_\lambda \mathcal{F}_\alpha(1, 0; V), \qquad D_V \mathcal{F}_\alpha(1, 0; V)[\partial_k V] = \partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Indeed, from $V_{\lambda,c}(Y) = \lambda V(c + \lambda Y)$,

$$\partial_\lambda V_{\lambda,c}\big|_{(1,0)} = V + Y \cdot \nabla V = \mathcal{A}_\tau, \qquad \partial_{c_k} V_{\lambda,c}\big|_{(1,0)} = \partial_k V.$$

Substituting these into the Gateaux formulas from Proposition 3.14 gives the two identities.

Theorem 3.28 (Modulation system)**.** *Let (V, P) be the repaired pair on the compact window under consideration, and assume the branch satisfies the modulation-differentiability row definition 3.25. Writing $\theta := (a, b)$, let $\mathcal{R}_\alpha$ be the truncated-test limit functions from definition 3.25. Then for a.e. $\tau \in I_*$,*

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V)\, a(\tau) + \sum_{k=1}^{3} \partial_{c_k} \mathcal{F}_\alpha(1, 0; V)\, b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

Equivalently,

$$J(\tau)\theta(\tau) = \mathcal{R}(\tau).$$

Hence

$$\theta(\tau) = J(\tau)^{-1}\mathcal{R}(\tau) \in L^1_{\text{loc}}(I_*)^4, \qquad |\theta(\tau)| \le K\, |\mathcal{R}(\tau)|.$$

Proof. Set $G_\alpha(\tau) := \mathcal{F}_\alpha(1,0;V(\tau))$. By Theorem 3.20, $G_\alpha(\tau) = 0$ a.e.; by lemma 3.26, $G_\alpha \in AC$ and (6) holds. Hence $G'_\alpha = 0$ a.e., and (6) gives

$$\partial_\lambda \mathcal{F}_\alpha(1,0;V)\, a(\tau) + \sum_{k=1}^{3} \partial_{c_k} \mathcal{F}_\alpha(1,0;V)\, b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

which is equivalent to $J(\tau)\theta(\tau) = \mathcal{R}(\tau)$. The row gives $\mathcal{R} \in L^1_{\text{loc}}(I_*)^4$, and Proposition 3.16 gives $|J^{-1}(\tau)| \le K$ a.e.; hence

$$\theta(\tau) = J^{-1}(\tau)\mathcal{R}(\tau), \quad \theta \in L^1_{\text{loc}}(I_*), \quad |\theta| \le K|\mathcal{R}| \text{ a.e.}$$

as claimed. □

Corollary 3.29 (Regularity of a, b)**.** *For the repaired pair of Theorem 3.28, $a, b \in L^1_{\text{loc}}(I_*)$. If in addition $\mathcal{R} \in L^\infty_{\text{loc}}(I_*)$, then $a, b \in L^\infty_{\text{loc}}(I_*)$.*

Proof. By Theorem 3.28, $\theta = (a,b) = J^{-1}\mathcal{R}$ a.e., and $J^{-1} \in L^\infty_{\text{loc}}(I_*)$ by Theorem 3.16. The first claim is therefore immediate from $\mathcal{R} \in L^1_{\text{loc}}$, while the second follows if $\mathcal{R} \in L^\infty_{\text{loc}}$. □

3.10. Repaired suitable structure in renormalized variables.

Lemma 3.30 (AC chart-generated tests are admissible)**.** *Let (u,p) be suitable on a compact physical cylinder, and let $\Lambda, X \in AC(J)$ on a compact interval J, with $0 < \lambda_- \le \Lambda \le \lambda_+ < \infty$ and*

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda} \quad \textit{in } L^1(J).$$

Let $t = t(\tau)$ be defined by $dt/d\tau = \Lambda(\tau)^2$. If $\phi \in C_c^\infty(B_R \times J)$ is nonnegative and its chart pullback has a positive support buffer inside the physical cylinder, then

$$\psi(x,t) := \Lambda(\tau(t))^{-1} \phi\left(\frac{x - X(\tau(t))}{\Lambda(\tau(t))}, \tau(t)\right)^2$$

is an admissible nonnegative test for the physical local energy inequality. The chain-rule identities for $\nabla_x\psi$, $\Delta_x\psi$, and $\partial_t\psi$ hold for a.e. t.

Proof. For each fixed time, $x \mapsto \psi(x,t)$ is smooth and compactly supported; the AC chart gives

$$\psi,\ \nabla_x\psi,\ \Delta_x\psi \in L^\infty, \qquad \partial_t\psi \in L^1_t L^\infty_x$$

on the compact cylinder, with the time derivative computed by the a.e. chain rule. Extend ψ by zero across a small time buffer and set $\psi^\varepsilon := \rho_\varepsilon *_t \psi$, where ρ_ε is a nonnegative time mollifier and ε is smaller than the support buffer. Then $\psi^\varepsilon \ge 0$, $\psi^\varepsilon \in C_c^\infty$ in spacetime with compact support in the same physical cylinder, and

$$\psi^\varepsilon \to \psi, \quad \nabla_x\psi^\varepsilon \to \nabla_x\psi, \quad \Delta_x\psi^\varepsilon \to \Delta_x\psi \quad \text{in } L^\infty,$$

while

$$\partial_t\psi^\varepsilon \to \partial_t\psi \quad \text{in } L^1_t L^\infty_x.$$

Apply the physical local energy inequality to ψ^ε. The suitable bounds $|u|^2 \in L^\infty_t L^1_x$, $|\nabla u|^2 \in L^1$, and $|u|^3 + |p||u| \in L^1$ allow passage to the limit in the right-hand side by dominated convergence and in the nonnegative dissipation term by lower semicontinuity. This gives the physical local energy inequality with ψ. □

Proposition 3.31 (Transport of suitable local energy inequality)**.** *Suppose (u,p) is suitable on $\mathcal{Q}^{\text{ph}}_{2R}$, and (V,P) is obtained by the repaired map Φ. Set*

$$\tilde{a}(\tau) := \Lambda(\tau)^{-1}\Lambda'(\tau), \qquad \tilde{b}(\tau) := \Lambda(\tau)^{-1}X'(\tau).$$

Then for every nonnegative $\phi \in C_c^\infty(Q_R^*)$ and $\tau_1 < \tau_2$,

$$\begin{aligned}\mathscr{E}_\phi(V,P) := \frac{1}{2}|V|^2\partial_\tau(\phi^2) + \frac{\nu}{2}|V|^2\Delta_Y\phi^2 + \frac{1}{2}|V|^2\, V\cdot\nabla(\phi^2)\\ + (P_{\rm loc}+H)\, V\cdot\nabla(\phi^2) - \frac{\tilde a}{2}|V|^2\phi^2 - \frac{\tilde a}{2}|V|^2\, Y\cdot\nabla(\phi^2) - \frac{1}{2}|V|^2\,\tilde b\cdot\nabla(\phi^2).\end{aligned}$$

$$\begin{aligned}\frac{1}{2}\int_{B_R}|V(\tau_2)|^2\phi(\cdot,\tau_2)^2 + \nu\int_{\tau_1}^{\tau_2}\int_{B_R}|\nabla V|^2\phi^2\,dY\,d\tau \le \frac{1}{2}\int_{B_R}|V(\tau_1)|^2\phi(\cdot,\tau_1)^2\\ + \int_{\tau_1}^{\tau_2}\int_{B_R}\mathscr{E}_\phi(V,P)\,dY\,d\tau.\end{aligned}$$

The right-hand side is controlled on compact windows by $\|\tilde a\|_{L^1}, \|\tilde b\|_{L^1}, M$, and the local bounds from Theorem 3.23, and support bounds from Section 3.4. Hence (V,P) is locally suitable on Q_R^*.

Proof. Set

$$\psi(x,t) := \Lambda(\tau(t))^{-1}\phi\bigg(\frac{x-X(\tau(t))}{\Lambda(\tau(t))},\tau(t)\bigg)^2, \qquad \tau=\tau(t), \qquad d\tau/dt = \Lambda^{-2},$$

and let $t_i = t(\tau_i)$, $i=1,2$. By Proposition 3.3, Section 3.4, and lemma 3.30, the physical local energy inequality is valid with this nonnegative AC chart-generated test ψ.

$$\nabla_x\psi = \Lambda^{-2}\nabla_Y\phi^2, \quad \Delta_x\psi = \Lambda^{-3}\Delta_Y\phi^2, \quad \partial_t\psi = -\Lambda^{-3}\tilde a\phi^2 + \Lambda^{-3}\partial_\tau(\phi^2) - \Lambda^{-3}(\tilde b + \tilde a Y)\cdot\nabla_Y\phi^2.$$

Using $u = \Lambda^{-1}V$, $p = \Lambda^{-2}P$, $\nabla_x u = \Lambda^{-2}\nabla_Y V$, $dx\,dt = \Lambda^5\,dY\,d\tau$, each term in the physical inequality becomes

$$\frac{1}{2}\int |u|^2\psi\Big|_{t_1}^{t_2} = \frac{1}{2}\int |V|^2\phi^2\Big|_{\tau_1}^{\tau_2},$$

$$\nu\int|\nabla_x u|^2\psi\,dxdt = \nu\int|\nabla V|^2\phi^2\,dY\,d\tau,$$

$$\frac{1}{2}\int|u|^2\partial_t\psi\,dxdt = \frac{1}{2}\int|V|^2\left(\partial_\tau(\phi^2) - \tilde a\phi^2 - \tilde a\,Y\cdot\nabla_Y\phi^2 - \tilde b\cdot\nabla_Y\phi^2\right)dY\,d\tau,$$

$$\frac{\nu}{2}\int|u|^2\Delta_x\psi\,dxdt = \frac{\nu}{2}\int|V|^2\Delta_Y\phi^2\,dY\,d\tau,$$

$$\int\left(\frac{|u|^2}{2}u + p\,u\right)\cdot\nabla_x\psi\,dxdt = \int\left(\frac{1}{2}|V|^2V + (P_{\rm loc}+H)V\right)\cdot\nabla_Y\phi^2\,dY\,d\tau.$$

Substituting these identities and grouping terms gives the proposition's right-hand side and coefficients $\tilde a, \tilde b$. ☐

3.11. **Stability under subsequences.**

Proposition 3.32. *Let $(u^{(n)},p^{(n)})$ satisfy $\mathrm{SC}(R_*,r_*,M,\eta,\kappa)$ with the same data parameters and let $(\Lambda^{(n)},X^{(n)},V^{(n)},P^{(n)},a^{(n)},b^{(n)})$ be obtained from retained quantitatively nondegenerate repaired-gauge selections on the same I_* satisfying the modulation-differentiability row, with the constants in Proposition 3.16 uniform in n. Assume also that $(a^{(n)},b^{(n)})$ is uniformly integrable on every compact time subinterval of I_* (in particular this holds under the L^∞ modulation bounds used in the scale-rigid applications). After extraction $n_j\to\infty$, there exist subsequences and limits (Λ,X,V,P,a,b) such that*

$$\Lambda^{(n_j)}\to\Lambda, \quad X^{(n_j)}\to X \quad \textit{uniformly on compact time subintervals},$$

$$V^{(n_j)}\to V \textit{ in } L^2_{\rm loc}(I_*;L^2_{\rm loc}), \quad V^{(n_j)}\rightharpoonup V \textit{ in } L^2_{\rm loc}(I_*;H^1_{\rm loc}),$$

$$P^{(n_j)}\rightharpoonup P \textit{ in } L^{3/2}_{\rm loc}(Q_R^*), \quad (a^{(n_j)},b^{(n_j)})\rightharpoonup(a,b) \textit{ in } L^1_{\rm loc}(I_*)^4.$$

Moreover the limit chart satisfies

$$\Lambda' = a\Lambda, \qquad X' = b\Lambda$$

in distributions, equivalently

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda}$$

a.e. on I_*, and the limit obeys all conclusions below.

Proof. From SC and the retained repaired-gauge selections, $(V^{(n)}, P^{(n)})$ are uniformly bounded in $L^\infty_\tau L^2(B_{2R}) \cap L^2_\tau H^1(B_{2R})$ and $L^{3/2}_{\mathrm{loc}}(Q^*_R)$. Theorem 3.16 gives uniform invertibility bounds for $J^{(n)}$, and Theorem 3.28 gives $(a^{(n)}, b^{(n)})$ uniformly in L^1_{loc} (since $\mathcal{R}^{(n)} \in L^1_{\mathrm{loc}}$). Since on compact windows

$$(\Lambda^{(n)})' = a^{(n)}\Lambda^{(n)}, \qquad (X^{(n)})' = b^{(n)}\Lambda^{(n)},$$

and the charts have uniform upper and lower scale bounds, $\Lambda^{(n)}$ and $X^{(n)}$ are uniformly bounded in $W^{1,1}_{\mathrm{loc}}$. The assumed local uniform integrability and Dunford–Pettis provide weak subsequential limits for $(a^{(n)}, b^{(n)})$. Thus, by BV compactness and one-dimensional subsequences,

$$\Lambda^{(n_j)} \to \Lambda, \qquad X^{(n_j)} \to X \quad \text{uniformly on compact time subintervals,}$$

and, after extraction,

$$a^{(n_j)} \rightharpoonup a, \qquad b^{(n_j)} \rightharpoonup b, \qquad a^{(n_j)}\Lambda^{(n_j)} \rightharpoonup a\Lambda, \qquad b^{(n_j)}\Lambda^{(n_j)} \rightharpoonup b\Lambda$$

in L^1_{loc}. Since $\Lambda^{(n_j)} \to \Lambda$ uniformly on compact time intervals after extraction, the distributional derivative identities pass to the limit and give

$$\Lambda' = a\Lambda, \qquad X' = b\Lambda.$$

The chart bounds keep Λ locally bounded away from zero, so equivalently $a = \Lambda'/\Lambda$ and $b = X'/\Lambda$ a.e. From the renormalized equation,

$$\partial_\tau V^{(n)} \in L^1_\tau(H^{-1}_{\mathrm{loc}}) + L^{4/3}_\tau(L^{12/11}_{\mathrm{loc}}),$$

uniformly on compact windows. Aubin–Lions compactness on bounded balls gives

$$V^{(n_j)} \to V \text{ in } L^2_{\mathrm{loc}}(I_*; L^2_{\mathrm{loc}}), \quad V^{(n_j)} \rightharpoonup V \text{ in } L^2_{\mathrm{loc}}(I_*; H^1_{\mathrm{loc}}).$$

By Calderón–Zygmund and Theorem 3.23, $P^{(n_j)} \rightharpoonup P$ in $L^{3/2}_{\mathrm{loc}}(Q^*_R)$, and with normalized pressure split $P = P_{\mathrm{loc}} + H$, the same for P_{loc} and H. Passing to limits in $\mathcal{F}(1,0;V^{(n)}(\tau)) = 0$, Theorem 3.6, and the linear modulation system yields the corresponding limit constraints and equations. $\square$

3.12. **Final representation theorem.**

Theorem 3.33 (Chart and repaired-gauge representation)**.** *Assume* $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$*, and suppose the branch lies in the retained repaired-gauge regime with selection (λ, c) on $J = [\tau_0 - \ell, \tau_0 + \ell] \subset I_*$. Then there are fields (V, P) on Q^J_{2R}, with $(\Lambda, X) \in AC(J)$,*

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda}, \qquad a, b \in L^1_{\mathrm{loc}}(J),$$

such that

$$u(x,t) = \Lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right), \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

and $\mathcal{F}(1,0;V(\tau)) = 0$ for $\tau \in J$.

Proof. From (SC5) and Proposition 3.3, $(\Lambda_{\text{pre}}, X_{\text{pre}})$ are AC on I_*, so V_{pre} is defined on Q^*_{2R}. The retained repaired-gauge regime supplies $(\lambda, c) \in AC(J;(0,\infty)\times\mathbb{R}^3)$ and $\mathcal{F}(1,0;V^{\lambda,c}_{\text{pre}}) = 0$ a.e. By Definition 3.19 and Theorem 3.20, setting

$$\Lambda(\tau) = \Lambda_{\text{pre}}(\tau)\lambda(\tau), \qquad X(\tau) = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau), \qquad V(\tau) = V^{\lambda(\tau),c(\tau)}_{\text{pre}},$$

and $P(\tau) = P^{\lambda(\tau),c(\tau)}_{\text{pre}}$, gives $\Lambda, X \in AC(J)$ and

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda}, \qquad a, b \in L^1_{\text{loc}}(J),$$

Moreover,

$$\mathcal{F}(1,0;V(\tau)) = 0 \text{ for a.e. } \tau \in J.$$

Finally $u = \Lambda^{-1}V(\cdot,\tau)\circ\Phi^{-1}$ by definition of the repaired map, hence the physical identity with $\frac{dt}{d\tau} = \Lambda^2$ follows. □

Theorem 3.34 (Pressure and modulation)**.** *Assume, in addition to Theorem 3.33, that the retained repaired-gauge selection is quantitatively nondegenerate in the sense of Proposition 3.16 and satisfies the modulation-differentiability row of definition 3.25. Then (V,P) satisfies the renormalized Navier–Stokes equation with the logarithmic coefficient convention of Theorem 3.6; there is a decomposition $P = P_{\text{loc}} + H$ on Q^*_R as in Theorem 3.23; and (a,b) is the unique solution of*

$$J(\tau)\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = \mathcal{R}(\tau), \qquad \begin{pmatrix} a \\ b \end{pmatrix} \in L^1_{\text{loc}}(J)^4,$$

with bounds $|(a,b)| \le K|\mathcal{R}|$, where K depends on $R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R$.

Proof. Equation and coefficient convention are inherited directly from Theorem 3.6 applied to (Λ, X) from Theorem 3.33 and the definition of V. By Theorem 3.23 applied at each time slice, the decomposition on Q^J_R is defined and unique once $⨍_{B_R} H = 0$ is imposed. The modulation identity is Theorem 3.28; uniqueness follows from invertibility of J (Theorem 3.16) and the pointwise identity $(a,b) = J^{-1}\mathcal{R}$. Uniform bounds on (a,b) are obtained by the bound on J^{-1} there and the stated bound on $\mathcal{R}$. □

Theorem 3.35 (Compact-window representation theorem)**.** *Under the assumptions of theorems 3.33 and 3.34, for every compact window $J \subset I_*$ and each fixed $R > 0$, setting $Q^J_R := B_R \times J$, one has:*

1. *the exact renormalized equation on Q^J_{2R},*
2. *the four repaired-gauge constraints $\mathcal{F}(1,0;V(\tau)) = 0$,*
3. *bounds on $a, b \in L^1_{\text{loc}}(J)$ (and L^∞_{loc} under $\mathcal{R} \in L^\infty_{\text{loc}}$),*
4. *local pressure decomposition $P = P_{\text{loc}} + H$ with $L^{3/2}$ local bounds and harmonic remainder controls,*
5. *renormalized local suitable energy inequality on Q^J_R,*
6. *stability under subsequences in the topologies above for families whose modulation coefficients are locally uniformly integrable, in particular in the L^∞-modulated retained rows used below.*

The later reductions use only these conclusions.

Proof. From Theorem 3.33 and the choice of $J \subset I_*$, all quantities are defined on Q^J_{2R}. Items (1) and (2) are Theorems 3.6 and 3.20; item (3) is Theorem 3.28 together with the inverse bound from Proposition 3.16; item (4) is Theorem 3.23; item (5) is Proposition 3.31; item (6) is Proposition 3.32 under the stated local uniform integrability condition. Combining these results gives the stated compact-window representation. □

Remark 3.36. The preceding results separate two alternatives:

(1) the nondegenerate single-core repaired-gauge regime, where the representation theorems apply;
(2) gauge degeneracy ($\det J = 0$) or compact-structure failure, which belongs to the multi-bubble, cascade, or scale-rigidity alternatives.

4. Multibubble and Cascade Exclusion

Analytic inputs.

The local reduction theorem uses the following analytic inputs from the preceding sections, with profile decompositions and decoupling estimates in their standard critical-space sense [Gal01, BL83]. The remaining work is the multibubble, cascade, and terminal-frame decoupling.

(i) From Section 2, the compact single-core criterion theorem 2.18: a compact single-core finite-cost Type II branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded.

(ii) From Section 3, the renormalized equation theorem 3.6, pressure decomposition theorem 3.23, modulation system theorem 3.28, and repaired-gauge representation theorems 3.33 to 3.35: the renormalized equation, pressure decomposition modulo functions of time, bounded modulation coefficients, renormalized suitable structure, and stability of these structures under subsequences.

(iii) From Section 5, the compact-cylinder Caccioppoli estimate theorem 5.4 and rough-core exclusion corollary 5.7.

(iv) From Section 6, the collapse cost exclusion theorems 6.6 and 6.7 and autonomous reduced-limit theorem theorem 6.19 for scale-rigid or scale-collapsing branches.

4.1. **Local setting and active frames.** *Normalized local Navier–Stokes setting.* All quantities are written in rescaled local variables $(x,t) \in Q_R = B_R \times (-R^2, 0)$ with viscosity $\nu = 1$ (critical scaling has absorbed ν). A local pair (u_n, p_n) on such a cylinder satisfies

$$\partial_t u_n - \Delta u_n + (u_n \cdot \nabla) u_n + \nabla p_n = 0, \qquad \nabla \cdot u_n = 0,$$

and pressure is normalised through spatial averaging

$$(p_n)_{B_\rho}(t) := |B_\rho|^{-1} \int_{B_\rho} p_n(x,t)\, dx$$

to factor out time-dependent constants.

For $r > 0$, $x_0 \in \mathbb{R}^3$, and $t_0 \in \mathbb{R}$, let

$$Q_r(x_0, t_0) := B_r(x_0) \times (t_0 - r^2, t_0), \qquad Q_r := Q_r(0,0) = B_r \times (-r^2, 0).$$

Let (u_n, p_n) be parabolic rescalings around a potential singular point with positive local Caffarelli–Kohn–Nirenberg concentration on compact cylinders. Concretely, for a base triple (x_n, t_n, λ_n),

$$u_n(y,s) = \lambda_n u(x_n + \lambda_n y,\ t_n + \lambda_n^2 s), \qquad p_n(y,s) = \lambda_n^2 p(x_n + \lambda_n y,\ t_n + \lambda_n^2 s).$$

For each n and $R > 0$, define the scaling-invariant local quantities

$$A_n(R) := R^{-1} \operatorname*{ess\,sup}_{-R^2<t<0} \int_{B_R} |u_n(x,t)|^2\, dx,$$

$$E_n(R) := R^{-1} \int_{Q_R} |\nabla u_n(x,t)|^2\, dx\, dt,$$

$$C_n(R) := R^{-2} \int_{Q_R} |u_n(x,t)|^3\, dx\, dt, \qquad D_n(R) := R^{-2} \int_{-R^2}^{0} \int_{B_R} |p_n - (p_n)_{B_R}(t)|^{3/2}\, dx\, dt,$$

where

$$(p_n)_{B_R}(t) := |B_R|^{-1} \int_{B_R} p_n(x,t)\, dx.$$

For an active frame (x_n, λ_n), $\lambda_n > 0$, define the critical rescaling

$$\mathcal{T}_{x_n,\lambda_n} f(y) := \lambda_n f(x_n + \lambda_n y).$$

If $\phi \in L^3_\sigma(\mathbb{R}^3)$ is observed in this frame, its contribution is

$$\phi_n(y) = \mathcal{T}_{x_n,\lambda_n}\phi(y) = \lambda_n \phi(x_n + \lambda_n y).$$

The critical Jacobian identity is exact:

$$\|\mathcal{T}_{x_n,\lambda_n} f\|^3_{L^3(E)} = \int_{x_n+\lambda_n E} |f(x)|^3\,dx$$

for every measurable $E \subset \mathbb{R}^3$. In particular

$$\|\mathcal{T}_{x_n,\lambda_n} f\|^3_{L^3(\mathbb{R}^3)} = \|f\|^3_{L^3(\mathbb{R}^3)},$$

so $\mathcal{T}_{x_n,\lambda_n}$ is an isometry on $L^3(\mathbb{R}^3)$.

Definition 4.1 (Bounded selected-window limit)**.** A selected-window subsequence has a bounded selected-window limit on a compact cylinder Q_R if there exists $u_\infty \in L^\infty(Q_R)$ such that, after extracting a subsequence,

$$\forall \varepsilon > 0\ \exists N :\ \forall n \geq N,\ \|u_n - u_\infty\|_{L^3(Q_R)} < \varepsilon.$$

The limit is nonzero if $u_\infty \not\equiv 0$ on Q_R.

Definition 4.2 (Local Type II sequence)**.** A positive local concentration sequence is called a local Type II sequence if it satisfies definition 2.2; equivalently, no selected-window subsequence converges strongly in L^3 on any compact cylinder Q_R to a nonzero bounded selected-window limit.

Definition 4.3 (Active frame)**.** Fix a concentrating time sequence $t_n \uparrow T$ and a compact time interval $I \Subset (-\infty, 0]$. An active frame is a tuple

$$\mathfrak{F}^j = \big(x^j_n, \lambda^j_n, I, U^j_n, P^j_n\big), \qquad \lambda^j_n \downarrow 0,$$

where $x^j_n \in \mathbb{R}^3$ and

$$U^j_n(y,s) := \lambda^j_n u(x^j_n + \lambda^j_n y, t_n + (\lambda^j_n)^2 s), \qquad P^j_n(y,s) := (\lambda^j_n)^2 p(x^j_n + \lambda^j_n y, t_n + (\lambda^j_n)^2 s)$$

on $B_{2R} \times I$ for every fixed compact cylinder under consideration. The frame is active at threshold $\eta_* > 0$ if for some fixed $R_0 < \infty$

$$\liminf_{n\to\infty} \int_{Q_{R_0}} |U^j_n|^3\,dy\,ds \geq \eta_*^3.$$

Definition 4.4 (Relations between active frames)**.** Let $\mathfrak{F}^i$ and $\mathfrak{F}^j$ be two active frames. After passing to a subsequence exactly one of the following parameter regimes holds.

(i) They are *same-point comparable-scale* if

$$0 < c \leq \frac{\lambda^i_n}{\lambda^j_n} \leq C < \infty, \qquad \frac{|x^i_n - x^j_n|}{\max(\lambda^i_n, \lambda^j_n)} \leq C$$

for all large n.

(ii) They form a *same-point cascade* if, after relabeling,

$$\frac{\lambda^i_n}{\lambda^j_n} \to 0, \qquad \frac{|x^i_n - x^j_n|}{\lambda^j_n} = O(1).$$

The frame i is then the inner frame and j is an outer frame.

(iii) They are *separated-point* if

$$\frac{|x_n^i - x_n^j|}{\lambda_n^i} \to \infty \quad \text{and} \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} \to \infty.$$

Definition 4.5 (Terminal frame and perturbative remainder)**.** Given a finite active family, a terminal frame is chosen by selecting an active scale with minimal order:

$$\lambda_n^{j_*} \le C_j \lambda_n^j \qquad \text{for every active } j$$

after passing to a subsequence. Comparable frames at the same physical point are grouped before this choice; if several grouped frames remain tied, one is chosen by a fixed lexicographic ordering of their labels. A remainder r_n in a terminal frame is perturbative on $B_{2R} \times I$ if

$$\|r_n\|_{L_I^\infty L^3(B_{2R})} \to 0$$

and its equation residual and pressure satisfy

$$\|f_n\|_{L_I^1 H^{-1}(B_R)} + \|\nabla \pi_n^r\|_{L_I^1 H^{-1}(B_R)} \to 0$$

for every compact $B_R \times I$.

Definition 4.6 (Local multibubble/cascade branch)**.** A local multibubble/cascade branch is a positive-concentration local Type II branch for which at least one of the following alternatives occurs:

(i) the compact-cylinder suitable estimates $A_n + E_n + C_n + D_n$ are unbounded on some fixed rescaled cylinder;
(ii) positive CKN mass splits into two or more comparable active cores;
(iii) a strict same-point scale cascade appears inside bounded rescaled cylinders;
(iv) separated physical points remain active in different local rescaled frames;
(v) the localized scale or centering selection degenerates (no unique core selected).

Definition 4.7 (Active profile family)**.** An active profile family is a family

$$\{(\phi^j, x_n^j, \lambda_n^j)\}_{j \in J}, \qquad \phi^j \in L^3_\sigma(\mathbb{R}^3),$$

with finite or countable index set J, extracted along a positive local concentration sequence. In physical variables write

$$\Lambda_{\lambda, x_0} \phi(x) := \lambda^{-1} \phi\left(\frac{x - x_0}{\lambda}\right).$$

For every finite $J_0 \subset J$ the family gives an expansion

$$u(\cdot, t_n) = \sum_{j \in J_0} \Lambda_{\lambda_n^j, x_n^j} \phi^j + r_n^{J_0}$$

on the compact terminal windows under consideration. After the finite active subfamily has been selected, the corresponding remainder satisfies the perturbative bounds of definition 4.5. The parameters are considered only after grouping same-point comparable-scale profiles into compound profiles as in definition 4.8. A profile is active if

$$\|\phi^j\|_{L^3(\mathbb{R}^3)} \ge \eta_* > 0.$$

The family has finite critical mass when

$$\sum_{j \in J} \|\phi^j\|^3_{L^3(\mathbb{R}^3)} \le M_* < \infty.$$

It captures positive local concentration if every compact terminal-window sequence with

$$\liminf_{n\to\infty}\int_{Q_R}|u_n|^3>0$$

either contains an active profile observed in that window or is absorbed into the selected terminal frame; equivalently, no positive terminal-window concentration remains in the perturbative remainder.

Definition 4.8 (Compound profile)**.** Let $J_k\subset J$ be a same-point comparable-scale class. Choose one representative $(x_n^{j_k},\lambda_n^{j_k})$. After passing to a subsequence,

$$\rho_j:=\lim_{n\to\infty}\frac{\lambda_n^j}{\lambda_n^{j_k}}\in(0,\infty),\qquad z_j:=\lim_{n\to\infty}\frac{x_n^j-x_n^{j_k}}{\lambda_n^{j_k}}\in\mathbb{R}^3.$$

The compound profile in the representative frame is the L^3_σ-sum

$$\Phi^k(y):=\sum_{j\in J_k}\rho_j^{-1}\phi^j\left(\frac{y-z_j}{\rho_j}\right),$$

whenever J_k is finite; finiteness follows for active classes from lemma 4.9. If $\|\Phi^k\|_{L^3}<\varepsilon_\star$, where $\varepsilon_\star$ is the critical small-data threshold, the class is perturbative. Otherwise Φ^k is treated as one active object in its representative frame.

Lemma 4.9 (Finite active profile count)**.** *Every active profile family with finite critical mass contains only finitely many active profiles. More precisely,*

$$\#J\le M_*\eta_*^{-3}.$$

Proof. For each active profile $j\in J$,

$$\|\phi^j\|^3_{L^3(\mathbb{R}^3)}\ge\eta_*^3.$$

Let $J_0\subset J$ be finite. Summing over J_0 gives

$$\#J_0\,\eta_*^3\le\sum_{j\in J_0}\|\phi^j\|^3_{L^3(\mathbb{R}^3)}\le\sum_{j\in J}\|\phi^j\|^3_{L^3(\mathbb{R}^3)}\le M_*.$$

Hence $\#J_0\le M_*\eta_*^{-3}$. Taking the supremum over all finite subsets J_0 yields the same bound for the cardinality of J, so J is finite and

$$\#J\le M_*\eta_*^{-3}.$$

□

Lemma 4.10 (Exhaustive parameter partition)**.** *Let $\mathfrak{F}^i,\mathfrak{F}^j$ be two active frames. After passing to a subsequence, either the pair is same-point comparable-scale, same-point cascade, or separated-point in the sense of definition 4.4; if none of these relations holds in a selected compact frame, then one of the two contributions is locally invisible there in L^3.*

Proof. Pass to a subsequence so that

$$\frac{\lambda_n^i}{\lambda_n^j}\to\ell\in[0,\infty]$$

in the extended half-line. If $0<\ell<\infty$, then either

$$\frac{|x_n^i-x_n^j|}{\max(\lambda_n^i,\lambda_n^j)}$$

is bounded along a further subsequence, giving the same-point comparable-scale case, or it tends to infinity, giving separated points because the two scales are comparable.

If $\ell = 0$, then either

$$\frac{|x_n^i - x_n^j|}{\lambda_n^j}$$

is bounded, giving a same-point cascade with i inner and j outer, or it tends to infinity. In the latter case the j-profile is locally invisible in the i-frame by lemma 4.13. The case $\ell = \infty$ is the same after interchanging i and j. These alternatives exhaust all subsequential limits of the scale ratio and normalized center distance. □

Theorem 4.11 (Minimal bad-configuration extraction)**.** *Let a positive local Type II branch admit an active profile family with finite critical mass, perturbative remainder in the sense of definition 4.5, and concentration capture in the sense of definition 4.7. If the branch is not already a single active terminal frame satisfying the single-core hypotheses, then after passing to a subsequence one of the following occurs:*

(i) *compact-cylinder suitable control fails, and lemma 4.15 produces a bounded-cylinder concentration core;*
(ii) *there are at least two active frames in a same-point comparable-scale class, hence a compound profile in the sense of definition 4.8;*
(iii) *there is a same-point cascade;*
(iv) *there are separated active physical points;*
(v) *the scale-center selection of the selected compound or terminal frame is degenerate.*

In every case the positive local concentration is carried by an active frame or compound profile, not by the perturbative remainder.

Proof. If compact-cylinder suitable control fails, item (i) follows from lemma 4.15. Assume therefore that the compact-cylinder family is bounded on the windows under consideration.

By concentration capture, positive local concentration cannot remain solely in the perturbative remainder. Hence either one selected active frame carries the terminal-window concentration, or another active frame remains non-negligible. If there is one nondegenerate selected active frame and no competing active object, the branch is in the single-core regime. Since this case is excluded by hypothesis, either the selected scale-center map is degenerate, giving item (v), or at least two active frames remain.

For two active frames, apply lemma 4.10. Same-point comparable-scale frames give item (ii), same-point scale separation gives item (iii), and separated points give item (iv). The remaining alternative in lemma 4.10 is local invisibility of one contribution in the selected compact frame; such a contribution is absorbed into the perturbative remainder and cannot carry the retained positive local concentration by the capture property. This gives the extraction statement. □

4.2. **Elementary local decoupling estimates.**

Lemma 4.12 (Critical rescalings that escape are locally invisible)**.** *Let $\phi \in L^3(\mathbb{R}^3)$, let $R < \infty$, and define*

$$W_n(y) = \rho_n \phi(z_n + \rho_n y),$$

where $\rho_n > 0$. Assume that the sets

$$z_n + \rho_n B_R$$

either have measure tending to zero, or escape to spatial infinity in the sense that for every compact set $K \subset \mathbb{R}^3$,

$$(z_n + \rho_n B_R) \cap K = \varnothing$$

for all sufficiently large n. Then

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. By the critical change of variables,

$$\|W_n\|^3_{L^3(B_R)} = \int_{B_R} \rho_n^3 |\phi(z_n + \rho_n y)|^3 \, dy = \int_{z_n+\rho_n B_R} |\phi(x)|^3 \, dx.$$

Set

$$m_n := \int_{z_n+\rho_n B_R} |\phi(x)|^3 \, dx.$$

Since $\|W_n\|^3_{L^3(B_R)} = m_n$, it is enough to show $m_n \to 0$.

Fix $\varepsilon > 0$. We split into the two structural alternatives.

Case 1: shrinking measure. If $|z_n + \rho_n B_R| \to 0$, then by absolute continuity of the integral $|\phi|^3 \in L^1(\mathbb{R}^3)$ there exists $\delta > 0$ such that

$$|E| < \delta \Longrightarrow \int_E |\phi|^3 \, dx < \varepsilon.$$

Choose N with $|z_n + \rho_n B_R| < \delta$ for all $n \geq N$. Then

$$m_n = \int_{z_n+\rho_n B_R} |\phi|^3 \, dx < \varepsilon, \qquad n \geq N.$$

Case 2: escape to infinity. Assume now that for every compact set $K \subset \mathbb{R}^3$, $(z_n + \rho_n B_R) \cap K = \varnothing$ for all sufficiently large n. By absolute continuity at infinity, choose compact $K \subset \mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

For every such compact K, there is $N(K)$ with $(z_n + \rho_n B_R) \cap K = \varnothing$ for all $n \geq N(K)$. For these indices,

$$z_n + \rho_n B_R \subset \mathbb{R}^3 \setminus K,$$

and therefore, for $n \geq N(K)$,

$$m_n = \int_{z_n+\rho_n B_R} |\phi|^3 \, dx \leq \int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

In either case, for every $\varepsilon > 0$, $m_n < \varepsilon$ for n sufficiently large. Hence $m_n \to 0$, and therefore

$$\|W_n\|_{L^3(B_R)} \to 0.$$

□

Lemma 4.13 (Separated active profiles vanish in the selected frame)**.** *Let (x_n^p, λ_n^p) and (x_n^q, λ_n^q) be two active frames with separated physical centers in the sense that*

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then the q-profile observed in the p-frame converges to zero locally in L^3: for every $R < \infty$,

$$\left\| \frac{\lambda_n^p}{\lambda_n^q} \phi^q \left(\frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} y \right) \right\|_{L^3(B_R)} \to 0.$$

Proof. The displayed function has the form in lemma 4.12 with

$$z_n = \frac{x_n^p - x_n^q}{\lambda_n^q}, \qquad \rho_n = \frac{\lambda_n^p}{\lambda_n^q}.$$

The corresponding physical set in the q-profile variables is

$$z_n + \rho_n B_R = \frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} B_R.$$

For every $y \in B_R$,

$$|z_n + \rho_n y| \geq |z_n| - \rho_n R.$$

Set $E_n := z_n + \rho_n B_R$. Then

$$\operatorname{dist}(E_n, 0) = |z_n| - \rho_n R = \frac{\lambda_n^p}{\lambda_n^q} \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right).$$

Since $\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty$, there exists $N_0(R)$ such that

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \geq 1, \quad n \geq N_0(R),$$

and hence for such n, $\operatorname{dist}(E_n, 0) \geq \rho_n$.

There are two cases.

(a) If $\rho_n \to 0$, then

$$|z_n + \rho_n B_R| = \rho_n^3 |B_R| \to 0.$$

Then the first alternative in lemma 4.12 applies.

(b) Otherwise, after passing to a subsequence, $\rho_n \geq \rho_* > 0$. Then

$$\operatorname{dist}(z_n + \rho_n B_R, 0) \geq \rho_* \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right) \to \infty,$$

so the sets escape every fixed compact subset of $\mathbb{R}^3$. The second alternative in lemma 4.12 applies.

In both subcases the $L^3(B_R)$-norm of the displayed expression converges to zero. □

Lemma 4.14 (Strict outer scales vanish in the innermost frame)**.** *Let several active profiles have the same physical center and scales $\lambda_n^1, \dots, \lambda_n^J$. Suppose λ_n^1 is an innermost scale and*

$$\frac{\lambda_n^1}{\lambda_n^j} \to 0 \qquad (j \neq 1).$$

Then every outer profile $\phi^j \in L^3(\mathbb{R}^3)$, viewed in the λ_n^1-frame, converges to zero in $L^3(B_R)$ for each fixed $R < \infty$. Any constant ambient velocity arising from the regular part of the outer flow is absorbed into the translation modulation.

Proof. In the innermost variables, the j-th outer contribution has the form

$$W_n^j(y) = \frac{\lambda_n^1}{\lambda_n^j} \phi^j \left(\frac{x_n^1 - x_n^j}{\lambda_n^j} + \frac{\lambda_n^1}{\lambda_n^j} y \right).$$

Set

$$\rho_n := \frac{\lambda_n^1}{\lambda_n^j} \to 0, \qquad z_n^j := \frac{x_n^1 - x_n^j}{\lambda_n^j}.$$

Then

$$W_n^j(y) = \rho_n \phi^j \left(z_n^j + \rho_n y \right),$$

and by the critical change of variables,

$$\|W_n^j\|_{L^3(B_R)}^3 = \int_{z_n^j+\rho_n B_R} |\phi^j(x)|^3\,dx.$$

Since $\rho_n \to 0$, the translated domains have

$$|z_n^j + \rho_n B_R| = \rho_n^3 |B_R| \to 0.$$

By absolute continuity of the integral for $\phi^j \in L^3(\mathbb{R}^3)$,

$$\|W_n^j\|_{L^3(B_R)} \to 0.$$

To treat a general ϕ^j, let $\phi_\varepsilon^j \in C_c^\infty(\mathbb{R}^3)$ with $\|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon$. By critical isometry,

$$\|W_n^j - (W_n^j)_\varepsilon\|_{L^3(B_R)} \le \|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon,$$

where $(W_n^j)_\varepsilon$ is the same scaling of ϕ_ε^j. The compactly supported term has vanishing local norm by the shrinking-domain argument above, and then letting first $n \to \infty$, then $\varepsilon \downarrow 0$ implies the claimed convergence. A constant vector field from the regular part of an outer flow contributes only a Galilean drift in the selected frame; it is therefore incorporated into the translation modulation and does not remain as an exterior perturbation. □

Lemma 4.15 (Compactness failure produces another core)**.** *If the compact-cylinder suitable estimates are unbounded on a fixed rescaled cylinder, then after passing to a subsequence the local $C_n + D_n$ quantity is unbounded on a comparable cylinder. Consequently the branch contains an additional local CKN concentration core.*

Proof. Fix $r > 0$. Apply the local Caccioppoli inequality with outer radius $2r$. For each index n, suitable weak solutions satisfy, with C_r independent of n,

$$A_n(r) + E_n(r) \le C_r\big(1 + C_n(2r) + C_n(2r)^{2/3} + D_n(2r)\big),$$

where the pressure is normalized by subtracting spatial means $(p_n)_{B_{2r}}(t)$.

Suppose that $A_n(r) + E_n(r) + C_n(r) + D_n(r)$ is unbounded along a subsequence. If $C_n(2r) + D_n(2r)$ were bounded on this subsequence, the preceding estimate would give uniform boundedness of $A_n(r) + E_n(r)$. Monotonicity of the local integrals also gives $C_n(r) + D_n(r) \le C_n(2r) + D_n(2r)$, contradicting unboundedness of the full family on Q_r. Therefore, after extracting a subsequence,

$$C_n(2r) + D_n(2r) \to \infty.$$

Define the localized CKN concentration at variable center and radius

$$G_n(x,t,\rho) := \rho^{-2}\int_{Q_\rho(x,t)} |u_n|^3\,dx\,ds + \rho^{-2}\int_{Q_\rho(x,t)} |p_n - (p_n)_{B_\rho}(s)|^{3/2}\,dx\,ds.$$

Set

$$\mathbf{M}_n := \sup\{G_n(x,t,\rho) : 0 < \rho \le 2r,\ Q_\rho(x,t) \subset Q_{4r}\}.$$

The choice $(x,t,\rho) = (0,0,2r)$ is admissible because $Q_{2r}(0,0) \subset Q_{4r}$, so

$$\mathbf{M}_n \ge C_n(2r) + D_n(2r) \to \infty.$$

After extracting a further subsequence, we may assume $\mathbf{M}_n \ge 1$ for all n. Choose (x_n, t_n, ρ_n) with $0 < \rho_n \le 2r$, $Q_{\rho_n}(x_n, t_n) \subset Q_{4r}$, such that

$$G_n(x_n, t_n, \rho_n) \ge \tfrac{1}{2}\mathbf{M}_n \ge \tfrac{1}{2}.$$

Rescale around (x_n, t_n, ρ_n):

$$v_n(y,s) := \rho_n u_n(x_n + \rho_n y, t_n + \rho_n^2 s), \qquad q_n(y,s) := \rho_n^2 p_n(x_n + \rho_n y, t_n + \rho_n^2 s).$$

Using the critical Jacobian identity,

$$\int_{Q_1} |v_n|^3 \, dy \, ds + \int_{Q_1} |q_n - (q_n)_{B_1}(s)|^{3/2} \, dy \, ds = G_n(x_n, t_n, \rho_n) \geq \tfrac{1}{2}.$$

Hence this new frame carries positive local critical mass, i.e. a new local CKN concentration core, as required. ☐

4.3. Terminal active-frame decoupling.

Definition 4.16 (Admissible terminal active family)**.** Fix a compact cylinder $B_{2R} \times I$ and a selected active frame. A terminal active family is admissible on this cylinder if the following hold.

(i) The active profiles satisfy the finite-mass and active-mass conditions of definition 4.7.

(ii) Every nonselected active profile is either separated from the selected physical point in the sense of lemma 4.13, or has the same physical point and a strictly outer scale in the sense of lemma 4.14, after comparable same-point scales have been grouped.

(iii) The rescaled remainder r_n satisfies

$$\|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0,$$

and there exist fields (π_n^r, f_n) such that on $B_{2R} \times I$,

$$\partial_t r_n - \Delta r_n + a_n\big(r_n + y \cdot \nabla r_n\big) + b_n \cdot \nabla r_n + \nabla \pi_n^r = f_n, \qquad \nabla \cdot r_n = 0,$$

with

$$\|f_n\|_{L^1_I H^{-1}(B_R)} \to 0,$$
$$\|\nabla \pi_n^r\|_{L^1_I H^{-1}(B_R)} \to 0.$$

(iv) The selected velocity satisfies

$$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

(v) The selected modulation coefficients satisfy

$$\sup_n \big(\|a_n\|_{L^\infty(I)} + \|b_n\|_{L^\infty(I)}\big) < \infty.$$

(vi) Local pressure decomposition is stable under the splitting: if tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$, then the corresponding pressure gradients tend to zero in $L^1_I H^{-1}(B_R)$, after subtracting spatial means.

(vii) Cutoff commutators generated by localizing to B_R are controlled by the $L^\infty_I L^3$, $L^1_I L^{3/2}$, and pressure-gradient norms appearing above. In particular, for any compact $\chi_R \in C_c^\infty(B_{2R})$ with $\chi_R \equiv 1$ on B_R, the commutators satisfy

$$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla \chi_R, \nabla \cdot](u \otimes u)\|_{H^{-1}(B_R)} \leq C_R \|u\|^2_{L^3(B_{2R})}.$$

uniformly for $u \in L^3(B_{2R})$ and the same type of estimate for S_n-quadratic commutator terms.

Theorem 4.17 (Terminal profile-evolution decoupling)**.** *Consider an admissible terminal active family on $B_{2R} \times I$. Let U_n be the selected-frame velocity and let S_n be the sum, in that frame, of all nonselected active profiles together with the rescaled remainder. Then on $B_R \times I$:*

(i) $\nabla \cdot S_n = 0$ *distributionally;*

(ii) $S_n \to 0$ *in* $L^\infty_I L^3(B_{2R})$*;*

(iii) $U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0$ *in* $L^1_I L^{3/2}(B_R)$*;*

(iv)

$$\begin{aligned}&\exists\, \pi_n,\ F_n: \quad \partial_t S_n - \Delta S_n + a_n\big(y\cdot\nabla S_n + S_n\big) + b_n\cdot\nabla S_n + \nabla\pi_n = F_n,\\ &\|F_n\|_{L^1_I H^{-1}(B_R)} \to 0, \qquad \nabla\cdot S_n = 0 \quad \textit{in } \mathcal{D}'(B_R\times I).\\ &\qquad\qquad\qquad \|\nabla\pi_n\|_{L^1_I H^{-1}(B_R)} \to 0.\end{aligned}$$

(v) *every pressure source containing at least one factor of S_n has vanishing pressure gradient in $L^1_I H^{-1}(B_R)$;*
(vi) *localization errors introduced by compact cutoffs vanish in $L^1_I H^{-1}(B_R)$;*
(vii) *the effective modulation parameters of the selected frame remain bounded;*
(viii) *after subtracting S_n, the selected frame remains a positive finite-mass local Type II branch with pressure reconstruction and the compact-cylinder bounds required by the single-core analysis.*

Proof. Each profile in the family is divergence-free, and the rescaled remainder is divergence-free. Since divergence commutes with the Navier–Stokes critical rescalings, finite sums remain divergence-free in distributions. This establishes (i).

We prove (ii). By admissibility, every nonselected active profile is either separated from the selected physical point or is a same-point strictly outer scale. The separated profiles vanish locally by lemma 4.13, and the same-point outer profiles vanish locally by lemma 4.14. The remainder tends to zero in $L^\infty_I L^3(B_{2R})$ by definition 4.16. The number of active profiles is finite by lemma 4.9. Since every summand in S_n tends to zero in $L^\infty_I L^3(B_{2R})$, the triangle inequality gives

$$\|S_n\|_{L^\infty_I L^3(B_{2R})} \le \sum_{j\ne j_*} \|S_n^j\|_{L^\infty_I L^3(B_{2R})} + \|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0.$$

This establishes (ii).

For (iii), admissibility gives

$$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

Hence, by Hölder's inequality,

$$\|U_n\otimes S_n\|_{L^1_I L^{3/2}(B_R)} \le |I|\|U_n\|_{L^\infty_I L^3(B_R)}\|S_n\|_{L^\infty_I L^3(B_R)} \to 0,$$

and the same estimate applies to $S_n\otimes U_n$. Finally,

$$\|S_n\otimes S_n\|_{L^1_I L^{3/2}(B_R)} \le |I|\|S_n\|^2_{L^\infty_I L^3(B_R)} \to 0.$$

For (iv), write the represented Navier–Stokes equation for the full profile sum $U_n + S_n$ and subtract the represented equation for the selected component U_n. The remaining equation for S_n has the displayed linear left-hand side

$$\partial_t S_n - \Delta S_n + a_n\big(y\cdot\nabla S_n + S_n\big) + b_n\cdot\nabla S_n.$$

Collect all right-hand terms into

$$F_n := -\nabla\cdot(U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n) + f_n + E_n^{\mathrm{cut}} + E_n^{\mathrm{mod}},$$

where f_n is the remainder forcing from item (iii) of admissibility, and $E_n^{\mathrm{cut}}, E_n^{\mathrm{mod}}$ are localization and modulation errors. The nonlinear stresses tend to zero in $L^1_I L^{3/2}(B_R)$, hence in $L^1_I H^{-1}(B_R)$, because $L^{3/2}(B_R) \hookrightarrow H^{-1}(B_R)$ on a bounded three-dimensional ball. The residual term vanishes by definition 4.16. The cutoff terms vanish by the localization part of admissibility, recorded explicitly in (vi). This gives the displayed convergence in $L^1_I H^{-1}(B_R)$.

For (v), decompose the pressure source of $U_n + S_n$ into the pure selected source, the pure exterior source, and the two mixed sources. The mixed sources are

$$\partial_i\partial_j(U_{n,i}S_{n,j} + S_{n,i}U_{n,j}) \quad \text{and} \quad \partial_i\partial_j(S_{n,i}S_{n,j}).$$

By (iii), their tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$. The pressure-stability part of admissibility, equivalently local Calderón–Zygmund estimates plus harmonic pressure control modulo spatial means, gives convergence of the corresponding pressure gradients to zero in $L^1_I H^{-1}(B_R)$. Pure exterior sources are generated by profiles that are locally invisible in the selected frame; applying the same pressure stability after lemmas 4.13 and 4.14 gives their vanishing pressure gradients.

For (vi), let χ_R be a cutoff equal to one on B_R and supported in B_{2R}. The commutators are finite sums of terms of the following types:

$$(\nabla\chi_R)\cdot\nabla S_n, \qquad (\Delta\chi_R)S_n, \qquad (\nabla\chi_R)\cdot(U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n),$$

plus analogous pressure cutoff terms. Their supports lie in the fixed annulus $B_{2R}\setminus B_R$. By the localization part of admissibility, these commutators are controlled by the norms already shown to vanish in (ii), (iii), and (v). Hence all localization errors vanish in $L^1_I H^{-1}(B_R)$.

For (vii), the only exterior contribution that can remain nonzero at first order in a compact selected frame is a constant ambient velocity. This has already been absorbed into the translation modulation. The remaining exterior profiles vanish locally, so their contribution to the modulation equations is small in the same local L^3 topology used above. The selected modulation coefficients are bounded by admissibility; hence the effective parameters remain bounded.

For $(viii)$, the positive CKN mass of the selected profile is unchanged by subtracting a term that tends to zero in local L^3 and whose mixed pressure sources vanish by (v). The pressure reconstruction is stable under the same decomposition by the pressure-stability condition in definition 4.16. The compact-cylinder bounds pass to the selected component because the full family is bounded and the nonretained terms vanish on compact cylinders. Thus the selected frame remains a positive finite-mass local Type II branch with the analytic inputs needed for the single-core analysis. ☐

4.4. **Same-point and separated-point reductions.**

Theorem 4.18 (Same-point cascade reduction)**.** *Every same-point multibubble or strict same-point cascade reduces, in the selected innermost or compound frame, to one of the following: the single-core criterion, scale-rigidity, or a perturbative terminal branch covered by theorem 4.17, provided the terminal family generated in the selected innermost frame is admissible in the sense of definition 4.16.*

Proof. Step 1: scale-comparability classes. Split active same-point profiles into equivalence classes by

$$i\sim j \iff \exists c, C\in(0,\infty): c\le \frac{\lambda_n^i}{\lambda_n^j}\le C \quad \text{for all } n$$

after passing to a subsequence. Thus within each class there are constants c_{ij}, C_{ij} with $c_{ij}\le \lambda_n^i/\lambda_n^j \le C_{ij}$ for all n, so one may define a single common rescaling sequence (choose $\lambda_n^{(k)} = \lambda_n^{j_k}$) and write all class-$k$ profiles in the same coordinates. Within each class, finite sums of divergence-free profiles remain divergence-free.

Step 2: compound profile reduction inside each class. For each class set

$$\Phi_n^k := \sum_{j\in J_k} \lambda_n^j \phi^j(x_n^j + \lambda_n^j y).$$

If the L^3-size of this rescaled sum satisfies

$$\left\| \sum_{j\in J_k} \phi^j \right\|_{L^3(\mathbb{R}^3)} < \varepsilon_\star,$$

with $\varepsilon_\star$ the perturbative small-data threshold, then standard local small-data theory for Navier–Stokes in the critical space eliminates this class from the active alternatives.

If the class is above threshold, it is retained as a single compound active core; at most one active core from that class is needed for local selection.

Step 3: local alternatives. If the retained compound core is unique and its scale-centering Jacobian is nondegenerate, we obtain the single selected core alternative. If this Jacobian is degenerate, the localized choice of active center or scale is not unique, which is the gauge-degenerate alternative in definition 4.6.

Step 4: strict same-point scale separation. Assume now the branch is not reduced to comparable scales and choose the innermost active scale λ_n^1 in that physical point. By lemma 4.14, every profile with larger scale vanishes in the λ_n^1-frame in local L^3, up to constant drift terms. These constant drifts are absorbed into translation modulation by admissibility. Let S_n be the sum of all such outer profiles plus the rescaled remainder r_n. Then, by Step 2 and lemma 4.14,

$$S_n \to 0 \quad \text{in } L_I^\infty L^3(B_{2R}).$$

Subtracting the equation for the innermost selected profile from the full-frame equation gives a perturbation equation for the selected profile whose forcing belongs to $L_I^1 H^{-1}(B_R)$ and vanishes. Thus theorem 4.17 applies, and one of three outcomes holds:

(a) **Single-core branch:** after subtracting S_n, the branch is governed by one active component, so it reduces to the compact single-core branch of proposition 4.20;
(b) **Scale-rigid branch:** the relative Jacobian between comparable scales becomes rigid in the sense of proposition 4.21, so proposition 4.21 excludes it;
(c) **Purely perturbative branch:** $S_n \to 0$ and the selected profile satisfies a vanishing forcing perturbation in $L_I^1 H^{-1}(B_R)$. No independent same-point cascade mechanism remains; after subtracting S_n, the selected component is reduced to the single-core or scale-rigidity analysis.

These are the strict same-point scale-separation alternatives, so the same-point reduction is exhausted. □

Theorem 4.19 (Separated-point decoupling)**.** *Separated active physical points are locally invisible in each other's active rescaled frames. Consequently every separated-point multibubble reduces in each selected active frame to a single-core, scale-rigid, or perturbative terminal branch, provided the terminal family in that selected active frame is admissible in the sense of definition 4.16.*

Proof. Fix an active point p with frame (x_n^p, λ_n^p). Let $q \neq p$ be another active point. By separation,

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

Thus lemma 4.13 gives local L^3 convergence to zero of the q-profile in the p-frame on every compact ball. The same conclusion holds for each profile centered at any physical point different from p. Since only finitely many active profiles remain after the critical mass threshold is imposed, the sum of all profiles outside the p-frame tends to zero locally in L^3.

The pure pressure generated by a separated profile is locally invisible by the same change of variables whenever its profile pressure belongs to $L^{3/2}_{\mathrm{loc}}$ with the standard Calderón–Zygmund control. Mixed pressure terms contain at least one exterior velocity factor, hence vanish in $L^1 H^{-1}_{\mathrm{loc}}$ by the pressure part of theorem 4.17. Localization and modulation commutators are localized to $B_{2R} \setminus B_R$, so their H^{-1}-norm is controlled by the L^3-smallness already proved in lemmas 4.13 and 4.14 and by admissibility item (vii).

Therefore, in the selected p-frame, all other physical points contribute only a perturbative forcing tending to zero in L^1H^{-1} on compact cylinders. The selected frame remains a positive finite-mass branch with the same compact-cylinder and pressure reconstruction properties. Hence the branch reduces to the single-core case, the scale-rigid case, or the perturbative terminal branch. Since p was arbitrary, the same reduction holds in every active frame. □

4.5. **Multibubble and cascade exclusion.** The following statements isolate the analytic inputs needed for the multibubble reduction: the (already proved) single-core exclusion, the scale-rigidity reduction (proved below), and terminal active-frame admissibility (deduced here from the terminal nonlinear decoupling theorem theorem 8.27). They are stated explicitly so that the following theorem uses no unstated compactness, pressure, or profile completeness conclusion.

Proposition 4.20 (Local single-core exclusion)**.** *Every compact single-core branch with retained positive local velocity concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded from the local Type II regime.*

Proof. This is precisely the conclusion of the local single-core Type II criterion, theorem 2.18, applied to the represented compact single-core branch in the renormalized chart. The hypotheses listed here are the input data of that theorem: retained positive local velocity concentration provides the nondegenerate selected core, the nondegenerate scale-center selection together with the pressure reconstruction and bounded compact-cylinder suitable estimates are the inputs of the compactness statement proposition 2.10, bounded modulation is remark 2.11, and vanishing localized dissipation is definition 2.4. The conclusion of theorem 2.18 is exactly the stated exclusion. □

Proposition 4.21 (Local scale-rigidity exclusion)**.** *Every scale-rigid local branch produced by the same-point or separated-point reductions, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit or reduces to the single-core branch in proposition 4.20. Consequently no such branch is compatible with the local Type II condition.*

Proof. This is proved at the end of the subsection as proposition 4.40. The present proposition is a forward reference used by the multibubble reduction. □

Remark 4.22 (Status of proposition 4.21). The remainder of this subsection proves proposition 4.21 by an explicit local compactness stratification. The proof consists of the local lemmas lemmas 4.25 to 4.35 and 4.37 and theorem 4.36 and the main decomposition theorem 4.38. The final proposition 4.40 then proves the conclusion of proposition 4.21 from these results. All proofs use only the local Type II definition definition 2.2, the represented variables of theorem 3.35, the local pressure decomposition of theorem 3.23, the modulation L^∞ control of theorem 3.34, the same-point and separated-point reductions theorems 4.18 and 4.19, the minimal bad-configuration extraction theorem 4.11, the single-core exclusion theorem 2.18, the terminal CKN regularity criterion theorem 8.34, and Banach–Alaoglu plus Calderón–Zygmund on fixed cylinders [CZ52, Ste70]. No global PDE bound, no Liouville theorem, and no profile-decomposition classification are invoked.

4.5.1. *Local stratification of scale-rigid branches.* We now identify the named scale-rigid local stratum and prove that every branch placed there by theorems 4.18 and 4.19 either exits the local Type II class through definition 2.2 or relaxes back to one of the already excluded local strata.

Definition 4.23 (Scale-rigid local branch)**.** A scale-rigid local branch is a represented branch (V_n, P_n, a_n, b_n) on a fixed compact selected cylinder Q_{R_*}, produced by Step 3 or Step 4 of theorem 4.18 or by theorem 4.19, such that:

(R1) the compact-cylinder suitable estimates are uniformly bounded,

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M;$$

(R2) the branch carries retained positive local velocity concentration on a fixed inner cylinder $Q_{r_*} \Subset Q_{R_*}$,

$$C_{V_n}(r_*) \ge \eta_0;$$

If the retained concentration is pressure-only, the branch is assigned instead to the harmonic pressure row or to the exterior-pressure row.

(R3) the modulation coefficients are bounded on the selected cylinder,

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M;$$

(R4) the relative scale-center selection between the retained core and any previously selected core is rigid in one of the senses produced by theorem 4.18: either the relative scale-center Jacobian degenerates along the subsequence (the gauge-degenerate alternative of definition 4.6 (v)), or the inner core is the innermost member of a same-point cascade in which all outer profiles have been absorbed into the perturbative remainder by lemma 4.14.

Remark 4.24 (Inputs (R1)–(R3) are automatic on the selected cylinder). The bounds (R1)–(R3) are not extra hypotheses but are inherited from the preceding reductions. Bound (R1) is the compact-cylinder hypothesis carried through theorem 4.11; if it failed, lemma 4.15 would assign the branch to the multicore alternative, not to the scale-rigid stratum. Bound (R2) is the velocity-carrier part of the positive concentration capture in definition 4.7: after pressure-only carriers have been routed to the harmonic/exterior-pressure row, the retained scale-rigid row is entered only by a fixed positive L^3 velocity carrier on Q_{r_*}. Bound (R3) is the selected-cylinder modulation control of theorem 3.34, which is part of the represented-variable output of theorem 3.35. No additional analytic input is needed to enter the scale-rigid stratum.

Lemma 4.25 (Bounded selected-window limits exit Type II). *Let (V_n, P_n, a_n, b_n) be a represented sequence on a compact selected cylinder Q_r, and assume that, after passing to a selected-window subsequence, there is $V_* \in L^\infty(Q_r)$ with $V_* \not\equiv 0$ such that $V_n \to V_*$ strongly in $L^3(Q_r)$. Then the original sequence is not a local Type II sequence in the sense of definition 2.2.*

Proof. The hypothesis is exactly the statement that the subsequence has a nonzero bounded selected-window limit on Q_r in the sense of definition 2.1. By definition 2.2, a local Type II sequence has no selected-window subsequence with a nonzero bounded selected-window limit on any compact cylinder. Hence the sequence fails to be a local Type II sequence. □

Lemma 4.26 (Pressure and modulation stability under rigid-frame limits). *Let (V_n, P_n, a_n, b_n) be a scale-rigid local branch on Q_{R_*} in the sense of definition 4.23. Fix $\rho \in (r_*, R_*)$. After passing to a subsequence:*

(i) *$V_n \rightharpoonup V_*$ weakly in $L^3(Q_\rho) \cap L^2_t H^1_x(Q_\rho)$, and $V_n \to V_*$ strongly in $L^3(Q'_\rho)$ for every $\rho' \in (r_*, \rho)$;*
(ii) *the mean-normalized pressure $\widetilde{P}_n := P_n - \overline{P_n}_{B_\rho}$ converges weakly in $L^{3/2}(Q_\rho)$ to a limit P_* such that $-\Delta P_* = \partial_i\partial_j(V_{*,i}V_{*,j})$ on Q_ρ in the sense of distributions modulo a function of time;*
(iii) *$a_n \rightharpoonup^* a_*$ and $b_n \rightharpoonup^* b_*$ weakly-$*$ in $L^\infty(-R_*^2, 0)$, with $\|a_*\|_\infty + \|b_*\|_\infty \le M$;*
(iv) *the limit (V_*, P_*, a_*, b_*) is a suitable weak solution of the represented Navier–Stokes equation (1) on Q_ρ, with $V_* \in L^3(Q_\rho) \cap L^2_t H^1_x(Q_\rho)$.*

This lemma is only a compactness and stability statement. It does not assert boundedness of V_*. Bounded selected-window limits are obtained below from the separate no-subconcentration/CKN argument.

Proof. (i) Compactness of V_n. By (R1), the family (V_n) is bounded in $L^3(Q_\rho)$ (from $C_{V_n}(R_*)$), in $L^2_t H^1_x(Q_\rho)$ (from $A_{V_n}(R_*)+E_{V_n}(R_*)$), and in $L^\infty_t L^2_x(Q_\rho)$ (from $A_{V_n}(R_*)$). Banach–Alaoglu yields the weak limits in $L^3(Q_\rho)$ and $L^2_t H^1_x(Q_\rho)$ along a subsequence. The represented Navier–Stokes equation (1) bounds $\partial_s V_n$ in $L^{4/3}_t H^{-1}_x(Q_\rho)$ by Sobolev product estimates on the bounded cylinder, since each term on the right-hand side is controlled by $A_{V_n}(R_*)$, $E_{V_n}(R_*)$, $C_{V_n}(R_*)$, $D_{P_n}(R_*)$, and $\|a_n\|_\infty+\|b_n\|_\infty \le M$; the modulation correction $a_n(s)(V_n+y\cdot\nabla V_n)+b_n(s)\cdot\nabla V_n$ is bounded in $L^2_t L^{6/5}_x(Q_\rho) \subset L^{4/3}_t H^{-1}_x(Q_\rho)$ on the bounded cylinder. Hence Aubin–Lions [Aub63, Sim87] applied on the fixed cylinder Q_ρ yields strong convergence $V_n \to V_*$ in $L^3(Q_{\rho'})$ for every $\rho' \in (r_*, \rho)$.

(ii) Pressure stability. By (R1) the local pressure quantity $D_{P_n}(R_*)$ is bounded, equivalently $\widetilde{P}_n$ is bounded in $L^{3/2}(Q_\rho)$ by the local Calderón–Zygmund mean-normalization estimate [CZ52, Ste70] applied to the local pressure identity in theorem 3.23. Banach–Alaoglu yields $\widetilde{P}_n \rightharpoonup P_*$ weakly in $L^{3/2}(Q_\rho)$. Pass to the limit in the local pressure identity

$$-\Delta \widetilde{P}_n = \partial_i\partial_j(V_{n,i}V_{n,j}) - \partial_i\partial_j(\overline{V_{n,i}V_{n,j}}_{B_\rho})$$

on Q_ρ. The right-hand side converges in $\mathcal{D}'(Q_\rho)$ by the strong L^3 convergence of (i) (which implies weak $L^{3/2}$ convergence of $V_{n,i}V_{n,j}$ by Hölder). This proves (ii).

(iii) Modulation limits. By (R3), a_n, b_n are bounded in $L^\infty(-R_*^2, 0)$; Banach–Alaoglu applied to the dual of L^1 yields the weak-$*$ limits with the same bound M.

(iv) Suitable weak solution status. Each (V_n, P_n) is suitable on Q_ρ (by (R1) and the represented-equation output of theorem 3.35). The local energy inequality and the distributional momentum equation pass to the limit using (i)–(iii) and Hölder on the fixed cylinder Q_ρ: the dissipation term is lower semicontinuous under weak $L^2_t H^1_x$ convergence, the convection term converges by strong $L^3(Q_{\rho'})$ convergence, and the pressure-velocity pairing $\int \widetilde{P}_n V_n \cdot \nabla\chi$ converges by weak $L^{3/2}$ times strong L^3 on the fixed cylinder. The modulation correction converges by weak-$*$ L^∞ for (a_n, b_n) against the strong L^1_t convergence of $(V_n, y\cdot\nabla V_n, \nabla V_n)$ tested against fixed test functions on Q_ρ. This proves the represented suitable limit in the displayed local energy class. No L^∞ bound is inferred from the compact-cylinder estimates. ∎

Lemma 4.27 (Compact chart comparability in scale-rigid gauges)**.** *Let (V, P, a, b) be a represented branch on $Q_{2\rho}(z_0)$ obtained from the repaired-gauge chart of theorem 3.35. Assume that the associated chart (Λ, X) satisfies*

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda}, \qquad \|a\|_{L^\infty(I_{2\rho})} + \|b\|_{L^\infty(I_{2\rho})} \le M$$

on the time interval $I_{2\rho}$ of $Q_{2\rho}(z_0)$, and set $\Lambda_0 := \Lambda(s_0) > 0$ for $z_0 = (y_0, s_0)$. There exists $\rho_c = \rho_c(M, \Lambda_0, \rho) > 0$ such that, for every $0 < r \le \rho_c$, the chart on $Q_{2r}(z_0)$ obeys

$$\frac{\Lambda_0}{2} \le \Lambda(s) \le \frac{3\Lambda_0}{2}, \qquad |X(s) - X(s_0)| \le 6\Lambda_0 M r^2, \qquad s \in (s_0 - 4r^2, s_0].$$

After a harmless concentric shrink, if needed, the physical image remains inside the support buffer of proposition 3.7. Consequently propositions 3.7and 3.8 apply on the retained subcylinder with

$$\Lambda_{\min} = \Lambda_0/2, \qquad \Lambda_{\max} = 3\Lambda_0/2,$$

and represented and physical CKN quantities on comparable cylinders are equivalent with constants depending only on M, Λ_0, r, ρ and the fixed buffer.

Proof. The identities

$$a = \frac{\Lambda'}{\Lambda}, \qquad b = \frac{X'}{\Lambda}$$

are part of the exact repaired change of variables in lemma 3.5 and theorems 3.33 and 3.35. Hence, for $|s-s_0|\le 4r^2$,

$$\left|\log\frac{\Lambda(s)}{\Lambda_0}\right|\le\int_s^{s_0}|a(\sigma)|\,d\sigma\le 4Mr^2,\qquad |X(s)-X(s_0)|\le\int_s^{s_0}\Lambda(\sigma)|b(\sigma)|\,d\sigma.$$

Choose

$$\rho_c\le\min\{\rho/4,(\log(3/2)/(4M))^{1/2}\}$$

when $M>0$, and choose $\rho_c=\rho/4$ when $M=0$. Then $e^{-4Mr^2}\Lambda_0\le\Lambda(s)\le e^{4Mr^2}\Lambda_0$, which implies the displayed lower-upper scale bounds. Substituting the upper scale bound into the center integral gives

$$|X(s)-X(s_0)|\le(3\Lambda_0/2)\,M\,4r^2=6\Lambda_0Mr^2.$$

A final concentric shrink, depending only on the fixed distance to the boundary of the selected compact window, keeps the image inside the domain buffer. Propositions 3.7 and 3.8 then give the stated comparison of the represented and physical quantities. No solution boundedness, and no vanishing dissipation, is used. □

Lemma 4.28 (Two-sided local CKN transfer in comparable charts)**.** *Let (V,P) be related to a physical suitable pair (u,p) by the repaired scale-translation chart on $Q_{2r}(z_0)$, and assume the chart comparability conclusion of lemma 4.27 there:*

$$0<\Lambda_{\min}\le\Lambda(s)\le\Lambda_{\max}<\infty,\qquad |X(s)-X(s_0)|\le Mr^2.$$

Set

$$\mathcal{C}_V(z_0,r):=r^{-2}\int_{Q_r(z_0)}|V|^3+r^{-2}\int_{Q_r(z_0)}|P-(P)_{B_r}|^{3/2},$$

and define the physical CKN quantity $\mathcal{C}_u(x,t,\rho)$ analogously, with pressure normalized by spatial averages. There are constants

$$N_{\mathrm{tr}}<\infty,\qquad c_{\mathrm{tr}}\in(0,1),\qquad C_{\mathrm{tr}}<\infty,$$

depending only on $M,\Lambda_{\min},\Lambda_{\max}$ and the fixed support buffer, and physical cylinders

$$Q_\ell^{\mathrm{ph}}:=B_{\rho_\ell}(x_\ell)\times(t_\ell-\rho_\ell^2,t_\ell),\qquad c_{\mathrm{tr}}\Lambda_{\min}r\le\rho_\ell\le C_{\mathrm{tr}}\Lambda_{\max}r,\qquad 1\le\ell\le N_{\mathrm{tr}},$$

such that the physical image of $Q_{c_{\mathrm{tr}}r}(z_0)$ is contained in $\cup_\ell Q_\ell^{\mathrm{ph}}$, each Q_ℓ^{ph} is contained in the physical image of $Q_r(z_0)$, and

$$\mathcal{C}_u(x_\ell,t_\ell,\rho_\ell)\le C_{\mathrm{tr}}\mathcal{C}_V(z_0,r)\qquad(1\le\ell\le N_{\mathrm{tr}}).$$

Conversely, if a finite physical cylinder cover of the image of $Q_r(z_0)$ with comparable radii satisfies $\mathcal{C}_u(x_\ell,t_\ell,\rho_\ell)\le\varepsilon$ for every member of the cover, then

$$\mathcal{C}_V(z_0,c_{\mathrm{tr}}r)\le C_{\mathrm{tr}}N_{\mathrm{tr}}\varepsilon.$$

Proof. The chart is

$$x=X(s)+\Lambda(s)y,\qquad dt=\Lambda(s)^2\,ds,\qquad V(y,s)=\Lambda(s)u(x,t),\qquad P(y,s)=\Lambda(s)^2p(x,t).$$

The bounds on Λ and X imply that the image of a represented parabolic cylinder is trapped between parabolic cylinders whose radii are comparable to $\Lambda(s_0)r$. A standard Besicovitch covering with the bounded eccentricity supplied by $\Lambda_{\min},\Lambda_{\max}$ gives the finite cover and the bounded overlap constant N_{tr}.

For the velocity term the change of variables gives, on every comparison cylinder,

$$\rho_\ell^{-2}\int_{Q_\ell^{\mathrm{ph}}}|u|^3\,dx\,dt\le C_{\mathrm{tr}}r^{-2}\int_{Q_r(z_0)}|V|^3\,dy\,ds.$$

For the pressure term we use the invariant seminorm $\inf_{c(s)} \int |P - c(s)|^{3/2}$. Spatial averages are equivalent to this seminorm up to the universal factor $2^{3/2}$, because for every ball B and $q \geq 1$,

$$\int_B |F - F_B|^q \leq 2^q \inf_c \int_B |F - c|^q.$$

The same Jacobian identity therefore gives the displayed physical estimate for the pressure oscillation after subtracting spatial means. Applying the same argument to the inverse chart, whose scale is bounded between $\Lambda_{\max}^{-1}$ and $\Lambda_{\min}^{-1}$ on the comparison cover, gives the converse estimate. No direction of proposition 3.8 is used as a black box here; both estimates come from the exact change of variables and the pressure oscillation seminorm. □

Lemma 4.29 (CKN regularity in comparable repaired gauges). *Let (V, P, a, b) be a represented suitable weak solution on $Q_{2r}(z_0)$ obtained from the repaired-gauge chart of theorem 3.35. Assume that the chart satisfies the comparability conclusion of lemma 4.27 on this cylinder, with constants $\Lambda_{\min}, \Lambda_{\max}$, and that*

$$\|a\|_{L^\infty} + \|b\|_{L^\infty} \leq M$$

there. There are constants

$$\varepsilon_{\mathrm{rg}} = \varepsilon_{\mathrm{rg}}(M, \Lambda_{\min}, \Lambda_{\max}) > 0, \qquad c_{\mathrm{rg}} = c_{\mathrm{rg}}(M, \Lambda_{\min}, \Lambda_{\max}) \in (0, 1),$$

such that if

$$r^{-2} \int_{Q_r(z_0)} |V|^3 + r^{-2} \int_{Q_r(z_0)} |P - (P)_{B_r}|^{3/2} \leq \varepsilon_{\mathrm{rg}},$$

then V is bounded on $Q_{c_{\mathrm{rg}}r}(z_0)$, with a bound depending only on $M, \Lambda_{\min}, \Lambda_{\max}, r$, and the preceding displayed quantity.

Proof. Apply the forward direction of lemma 4.28 to $Q_r(z_0)$. Choose $\varepsilon_{\mathrm{rg}}$ so small that every physical cylinder in the resulting finite comparison cover has CKN quantity below the threshold $\varepsilon_{\mathrm{CKN}}$ from theorem 8.34. That theorem gives boundedness of the physical velocity on smaller physical cylinders. Pulling those estimates back through the same comparable repaired chart gives the asserted L^∞ bound for V on $Q_{c_{\mathrm{rg}}r}(z_0)$. Thus boundedness is obtained only from epsilon regularity after two-sided chart transfer and small CKN quantity, not from compactness or gauge degeneracy. □

Lemma 4.30 (Scaled local pressure decay). *Let (V, P) be a represented suitable pair on $Q_{2r}(z_0)$, and assume the localized pressure decomposition of theorem 3.23 is available with a cutoff supported in $B_{2r}(y_0)$ and equal to 1 on $B_r(y_0)$. For*

$$C_V(\rho; z_0) := \rho^{-2} \int_{Q_\rho(z_0)} |V|^3, \qquad D_P(\rho; z_0) := \rho^{-2} \int_{Q_\rho(z_0)} |P - (P)_{B_\rho}|^{3/2},$$

there is a universal constant C_{dec} such that, whenever $0 < \theta \leq 1/4$,

$$D_P(\theta r; z_0) \leq C_{\mathrm{dec}} \theta^{5/2} D_P(r; z_0) + C_{\mathrm{dec}} \theta^{-2} C_V(2r; z_0).$$

Proof. Write $z_0 = (y_0, s_0)$. For a.e. time $s \in (s_0 - r^2, s_0]$, apply theorem 3.23 on $B_r(y_0)$ with cutoff supported in $B_{2r}(y_0)$:

$$P(\cdot, s) = P_{\mathrm{loc}}(\cdot, s) + H(\cdot, s), \qquad -\Delta H(\cdot, s) = 0 \quad \text{in } B_r(y_0).$$

The Calderón–Zygmund part satisfies

$$\int_{B_{\theta r}(y_0)} |P_{\mathrm{loc}}|^{3/2} \leq C \int_{B_{2r}(y_0)} |V|^3.$$

After integrating over the shorter time interval and multiplying by $(\theta r)^{-2}$, this contributes at most $C\theta^{-2} C_V(2r; z_0)$.

For the harmonic part, the interior oscillation estimate for harmonic functions gives, for $q = 3/2$,

$$\int_{B_{\theta r}(y_0)} |H - (H)_{B_{\theta r}}|^q \le C\theta^{3+q} \int_{B_r(y_0)} |H - (H)_{B_r}|^q.$$

Since $3 + q = 9/2$, scaling the cylinder by $(\theta r)^{-2}$ leaves the factor $\theta^{5/2}$. Moreover

$$\int_{B_r} |H - (H)_{B_r}|^{3/2} \le C \int_{B_r} |P - (P)_{B_r}|^{3/2} + C \int_{B_r} |P_{\mathrm{loc}}|^{3/2},$$

and the last term is again bounded by $C \int_{B_{2r}} |V|^3$. Absorbing the resulting $C\theta^{5/2}C_V(2r; z_0)$ into $C\theta^{-2}C_V(2r; z_0)$ gives the displayed estimate. Spatial averages may be replaced by the minimizing constants at the cost of a universal factor, as in lemma 4.28. □

Lemma 4.31 (Dyadic persistence of CKN concentration)**.** *Let (V, P) be suitable on a compact cylinder Q_ρ, and let*

$$\mathcal{C}_{V,P}(z, r) := r^{-2} \int_{Q_r(z)} |V|^3 + r^{-2} \int_{Q_r(z)} |P - (P)_{B_r}|^{3/2}.$$

If $z_* \in Q_\rho$ is a Caffarelli–Kohn–Nirenberg concentration point in the sense that

$$\limsup_{r \downarrow 0} \mathcal{C}_{V,P}(z_*, r) \ge \kappa_* > 0,$$

then for every $0 < \theta < 1$ and every sufficiently small outer radius r_0 with $Q_{2r_0}(z_*) \Subset Q_\rho$, there exist $z_m \to z_*$, integers $L_m \to \infty$, and dyadic radii $\rho_m = \theta^{L_m} r_0$ such that

$$Q_{\rho_m}(z_m) \Subset Q_{r_0}(z_*), \qquad \mathcal{C}_{V,P}(z_m, \rho_m) \ge c_\theta \kappa_*,$$

where $c_\theta > 0$ depends only on θ.

Proof. Choose radii $\sigma_m \downarrow 0$ and points $z_m \to z_*$ with $Q_{\sigma_m}(z_m) \Subset Q_{r_0}(z_*)$ and $\mathcal{C}_{V,P}(z_m, \sigma_m) \ge \kappa_*/2$. Pick L_m so that

$$\theta^{L_m+1} r_0 < \sigma_m \le \theta^{L_m} r_0 =: \rho_m.$$

Then $L_m \to \infty$ and $\rho_m \le \theta^{-1}\sigma_m$. The velocity term on $Q_{\rho_m}(z_m)$ controls the velocity term on $Q_{\sigma_m}(z_m)$ with loss at most θ^2, because the normalization changes by $(\sigma_m/\rho_m)^2$. For the pressure term, spatial averages are compared through the minimizing constant:

$$\int_{B_{\sigma_m}} |P - (P)_{B_{\sigma_m}}|^{3/2} \le 2^{3/2} \int_{B_{\sigma_m}} |P - c|^{3/2} \le 2^{3/2} \int_{B_{\rho_m}} |P - c|^{3/2}.$$

Taking $c = (P)_{B_{\rho_m}}$, integrating in time, and accounting for the same normalization loss gives

$$\mathcal{C}_{V,P}(z_m, \rho_m) \ge 2^{-5/2}\theta^2 \mathcal{C}_{V,P}(z_m, \sigma_m).$$

Thus the conclusion holds with $c_\theta = 2^{-7/2}\theta^2$. □

Lemma 4.32 (Exterior pressure persistence is a structural exit)**.** *Let a scale-rigid branch have a pressure lower bound on shrinking selected cylinders that is not accounted for by the local source $\partial_i\partial_j(V_iV_j)$ in any compact selected frame after the decomposition of theorem 3.23. Then the branch enters the separated/exterior structural alternative of definition 4.6 (iv), and it does not remain in the residual scale-rigid stratum.*

Proof. The unaccounted part of the pressure is harmonic in the selected compact frame and is generated by sources at positive distance in the corresponding physical variables. If the exterior velocity source has a terminal CKN concentration on one of the annuli separating it from the selected core, then theorem 8.38 gives an exterior concentration branch. In the local variables this is a separated active physical point, hence the separated-point alternative definition 4.6 (iv) and the reduction theorem 4.19 apply.

If no such exterior concentration exists on the separating annuli, then theorem 8.41 gives regular exterior control and lemma 8.44 gives convergence of the exterior pressure contribution modulo functions of time on the selected compact frame. This contradicts the assumed persistent pressure lower bound there. Thus the only non-vanishing exterior-pressure case is already a separated/exterior structural exit. □

Lemma 4.33 (Persistent CKN mass activates an L^3 core or exits structurally)**.** *Let (V_n, P_n) be a compact-cylinder scale-rigid sequence on Q_{2R} with*

$$\sup_n \big(C_{V_n}(2R) + D_{P_n}(2R)\big) \le M_0,$$

and assume the pressure decomposition of theorem 3.23 is available on Q_{2R}. Fix $\kappa_0 > 0$ and choose $\theta \in (0, 1/8)$ so that

$$C_{\mathrm{dec}}\theta^{5/2} \le \frac{1}{4}.$$

Then there are constants $\eta_{\mathrm{act}} > 0$ and $L_0 \in \mathbb{N}$, depending only on M_0, R, κ_0, θ, the fixed-scale pressure bound below, and the pressure-decomposition constants, with the following property. Suppose $0 < R_0 < R$, $z_n \to z$, $Q_{2R_0}(z_n) \Subset Q_{2R}$ for all large n, and suppose the fixed outer scale used for the iteration has a uniform local pressure bound

$$D_{P_n}(R_0; z_n) \le D_0 < \infty.$$

Let $L_n \to \infty$, $\rho_n := \theta^{L_n} R_0$, and assume

$$\mathcal{C}_n(\rho_n; z_n) := \rho_n^{-2} \int_{Q_{\rho_n}(z_n)} |V_n|^3 + \rho_n^{-2} \int_{Q_{\rho_n}(z_n)} |P_n - (P_n)_{B_{\rho_n}}|^{3/2} \ge \kappa_0$$

for all large n. Then, after passing to a subsequence, either:

(a) *for some integers $0 \le j_n < L_n$, with $s_n := 2\theta^{j_n} R_0$,*

$$s_n^{-2} \int_{Q_{s_n}(z_n)} |V_n|^3 \ge \eta_{\mathrm{act}}^3,$$

so the rescaled branch at scale s_n contains an active L^3 core;

(b) *the retained pressure lower bound is exterior pressure persistence, hence the branch is assigned to the separated/exterior structural alternative by lemma 4.32.*

Proof. Choose L_0 so large that

$$(1/4)^{L_0} D_0 \le \kappa_0/16,$$

and choose $\eta_{\mathrm{act}} > 0$ so small that

$$C_{\mathrm{dec}}\theta^{-2} \sum_{m=0}^{\infty} (1/4)^m \eta_{\mathrm{act}}^3 \le \kappa_0/16, \qquad 4\theta^{-2}\eta_{\mathrm{act}}^3 \le \kappa_0/4.$$

For large n, $L_n \ge L_0$. Suppose item (a) fails. Then for each $0 \le j < L_n$,

$$C_{V_n}(2\theta^j R_0; z_n) < \eta_{\mathrm{act}}^3.$$

Iterating lemma 4.30 from the fixed outer scale R_0 down to $\theta^{L_n} R_0 = \rho_n$ gives

$$D_{P_n}(\rho_n; z_n) \le (1/4)^{L_n} D_{P_n}(R_0; z_n) + C_{\mathrm{dec}}\theta^{-2} \sum_{j=0}^{L_n-1} (1/4)^{L_n-1-j} C_{V_n}(2\theta^j R_0; z_n) \le \kappa_0/8.$$

Here the initial pressure is bounded by D_0 because the iteration starts from this fixed compact scale, not from the shrinking target scale. Moreover $Q_{\rho_n}(z_n) \subset Q_{2\theta^{L_n-1}R_0}(z_n)$, so failure of item (a) at $j = L_n - 1$ gives

$$\rho_n^{-2} \int_{Q_{\rho_n}(z_n)} |V_n|^3 \leq 4\theta^{-2}\eta_{\mathrm{act}}^3 \leq \kappa_0/4.$$

Hence $\mathcal{C}_n(\rho_n; z_n) < \kappa_0$, contradicting the assumed lower bound, unless the pressure term used in the lower bound is not the local pressure governed by the compact-frame decomposition. The latter possibility is exactly exterior pressure persistence and is item (b) by lemma 4.32. ☐

Lemma 4.34 (Activated cores are captured by the finite active ledger)**.** *Let a scale-rigid branch inherit an active profile family in the sense of definition 4.7, with finite critical mass M_* and concentration capture. Suppose lemma 4.33 produces represented scales $s_n > 0$ and centers z_n such that*

$$s_n^{-2} \int_{Q_{s_n}(z_n)} |V_n|^3 \geq \eta_{\mathrm{act}}^3.$$

After lowering the active threshold once, if necessary, to $\eta_*^{\mathrm{new}} := \min\{\eta_*, \eta_{\mathrm{act}}/2\}$, exactly one of the following occurs after passing to a subsequence:

- (a) *the activated core is already represented by an existing active profile or compound profile in the inherited family;*
- (b) *it is absorbed into the selected terminal frame, so it does not create an independent residual obstruction;*
- (c) *it is assigned to the profile-completeness obligation rather than to the scale-rigid residual branch.*

In particular, under the stated concentration-capture hypothesis, pressure-core activation cannot create an unregistered active object inside the residual scale-rigid branch.

Proof. Rescale the branch around (z_n, s_n) inside the represented window; in physical variables the corresponding scale is the selected physical scale multiplied by s_n, hence still tends to zero along the Type II sequence. The displayed lower bound is exactly the active-frame condition in definition 4.3 on Q_1, with threshold η_{act}. The concentration-capture clause in definition 4.7 says that a compact terminal-window sequence with positive L^3 mass is either seen by the active family or is absorbed into the selected terminal frame; it cannot remain in the perturbative remainder. Hence alternatives (a) and (b) hold unless the frame is genuinely absent from the captured profile family. That last possibility is not a scale-rigid residual mechanism; it is precisely failure of the terminal active-family completeness/capture input and is assigned to that separate profile-completeness obligation. Therefore the scale-rigid refinement below only uses active objects that already belong to the finite captured family, together with objects absorbed into the selected terminal frame. ☐

Lemma 4.35 (No sub-concentration gives a bounded selected-window limit)**.** *Let (V_n, P_n, a_n, b_n) be a scale-rigid local branch and let (V_*, P_*) be a compact limit supplied by lemma 4.26 on Q_ρ. Then, after shrinking once to a compact subcylinder $Q_{\rho'} \Subset Q_\rho$, exactly one of the following alternatives holds:*

- (a) *(V_*, P_*) has no Caffarelli–Kohn–Nirenberg concentration point in $Q_{\rho'}$. Then $V_* \in L^\infty(Q_{\rho''})$ for every $Q_{\rho''} \Subset Q_{\rho'}$, and the selected subsequence has a nonzero bounded selected-window limit on some such $Q_{\rho''}$.*
- (b) *(V_*, P_*) has a Caffarelli–Kohn–Nirenberg concentration point in $Q_{\rho'}$. Then the original branch either contains a further captured active L^3 concentration core after rescaling*

around that point, the pressure mass is exterior and the branch is already in the separated/exterior structural alternative, or the core is assigned to the separate profile-completeness obligation. According to its relative scale-center position, every captured core enters one of the already named multicore, same-point cascade, or separated/exterior structural alternatives, rather than remaining a residual scale-rigid branch.

Proof. Let $\varepsilon_{\mathrm{rg}}$ be the represented-gauge regularity threshold from lemma 4.29. Before using that threshold, apply lemma 4.27 on each retained compact subcylinder. The chart stability in proposition 3.32, together with the L^∞ modulation bound (R3), gives a limiting chart (Λ_*, X_*) with $a_* = \Lambda_*'/\Lambda_*$ and $b_* = X_*'/\Lambda_*$ on the retained subwindow. Thus the compactness of the subcylinder and (R3) give a uniform lower-upper scale comparison after one shrink, so the constants in lemma 4.29 depend only on the retained chart geometry and not on any unproved boundedness of V_n or V_*.

Assume first that there is no CKN concentration point of (V_*, P_*) in $Q_{\rho'}$. For every point $z \in Q_{\rho'}$ there is a parabolic cylinder $Q_{r_z}(z) \Subset Q_\rho$ such that the scale-invariant velocity-pressure quantity of (V_*, P_*) on $Q_{r_z}(z)$ is below $\varepsilon_{\mathrm{rg}}$. A finite subcover of each $Q_{\rho''} \Subset Q_{\rho'}$, followed by lemma 4.29 on the cover, gives $V_* \in L^\infty(Q_{\rho''})$.

It remains to check that the bounded limit is nonzero. If $V_* \equiv 0$ on the retained inner cylinder, then the strong L^3 convergence from lemma 4.26 and the zero-velocity argument of lemma 2.16 imply

$$C_{V_n}(r_*) \to 0,$$

contradicting the retained lower bound (R2) in definition 4.23. Hence $V_* \not\equiv 0$ on some smaller compact cylinder. Since $V_n \to V_*$ strongly in L^3 there, the selected subsequence has a nonzero bounded selected-window limit.

Now suppose a CKN concentration point z_* remains in $Q_{\rho'}$. Then there is $\kappa_* > 0$ such that

$$\limsup_{r \downarrow 0} \mathcal{C}_{V_*,P_*}(z_*, r) \geq \kappa_*.$$

Choose $R_0 > 0$ with $Q_{2R_0}(z_*) \Subset Q_\rho$, and choose $\theta \in (0, 1/8)$ satisfying the pressure-decay smallness condition in lemma 4.33. By lemma 4.31, there are $z_m \to z_*$, $L_m \to \infty$, and $\rho_m = \theta^{L_m} R_0$ such that

$$\mathcal{C}_{V_*,P_*}(z_m, \rho_m) \geq c_\theta \kappa_*.$$

For each fixed m, the strong L^3 convergence of V_n and weak lower semicontinuity of the pressure oscillation seminorm in $L^{3/2}/\{\text{spatial constants}\}$ imply, along a diagonal subsequence,

$$\mathcal{C}_{V_n,P_n}(z_m, \rho_m) \geq \frac{1}{2} c_\theta \kappa_*.$$

Relabeling the diagonal sequence, apply lemma 4.33 with $\kappa_0 = \frac{1}{2} c_\theta \kappa_*$ on the compact cylinder where (R1) holds. The additional fixed-scale pressure bound required by that lemma holds because R_0 is fixed, $Q_{2R_0}(z_m) \Subset Q_\rho$ for all large m, and the mean-normalized pressures are uniformly bounded in $L^{3/2}(Q_\rho)$ by lemma 4.26; comparison of spatial averages on the fixed nested balls gives $D_{P_n}(R_0; z_m) \leq D_0(R_0, \rho, M)$. If the persistent CKN mass is produced by an interior velocity core, then lemma 4.34 either identifies that core with the finite captured active ledger or assigns it to the separate profile-completeness obligation, after the one-time lowering of the active threshold. If the mass is carried by harmonic pressure generated outside the selected compact frame, lemma 4.32 assigns the branch to the separated/exterior structural alternative, so it exits the residual scale-rigid stratum.

The relative position of this new core with respect to the retained scale is exhausted by lemma 4.10: it is comparable at the same point, strictly separated in scale at the same point, separated in physical center, or locally invisible in the retained frame. The invisible case cannot carry the retained positive concentration by the capture clause in definition 4.7. The remaining cases are precisely the multicore, same-point cascade, and separated/exterior structural alternatives already named in the manuscript. Thus the branch does not remain in the residual scale-rigid stratum in this case. ☐

Theorem 4.36 (Single-core reduction compatibility). *Let (V_n, P_n, a_n, b_n) be a scale-rigid local branch in the sense of definition 4.23, and assume the branch reduces, after the rigid-frame limit of lemma 4.26, to a single retained nondegenerate core on a fixed inner cylinder $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. Then, after passing to a subsequence, one of the following alternatives holds:*

(i) *the branch has a nonzero bounded selected-window limit;*
(ii) *the original selected sequence has vanishing localized dissipation on a retained compact cylinder, hence is a compact single-core vanishing-dissipation branch in the sense of definition 2.3 and is excluded by theorem 2.18;*
(iii) *a further active concentration core is produced, so the branch leaves the single-core reduction and enters a multicore, cascade, or separated/exterior structural alternative.*

Proof. If the original selected sequence has vanishing localized dissipation on some $Q_{\rho'}$, $Q_{r_*} \Subset Q_{\rho'} \Subset Q_{R_*}$, then we first verify definition 2.3. Suitability and the represented equation are supplied by theorem 3.35. Compact-cylinder bounds, positive selected-core CKN concentration, and bounded modulation are exactly (R1), (R2), and (R3) of definition 4.23. The nondegenerate scale-center selection is the hypothesis of this single-core case. The vanishing localized dissipation is the present hypothesis on the original selected sequence, not on a residual after subtracting a limit. Thus the branch satisfies every clause of definition 2.3, and theorem 2.18 excludes it. This is item (ii).

The compact limit on a slightly smaller cylinder is covered by lemma 4.35. If the no-subconcentration alternative holds, item (i) follows. If a further active concentration core is produced, item (iii) follows. ☐

Lemma 4.37 (Degenerate Jacobian resolution). *Let (V_n, P_n, a_n, b_n) be a represented sequence reaching the gauge-degenerate alternative definition 4.6 (v), i.e. along the subsequence the relative scale-center Jacobian of the selected compound or terminal frame degenerates. Then, after passing to a further subsequence and reselecting the active center in the sense of definition 4.7 on the same physical point, exactly one of the following four resolutions occurs:*

(D1) *the reselection produces an additional active core or active compound carrying positive L^3 mass at the fixed active threshold; hence the branch enters the multicore, same-point cascade, or separated/exterior alternative definition 4.6 (ii)–(iv) and is treated by theorem 4.11;*
(D2) *the reselection collapses the compound to a single nondegenerate core and the branch enters the single-core compatibility dichotomy governed by theorem 4.36;*
(D3) *the reselection produces a perturbative compound that is below the critical small-data threshold of Step 2 of theorem 4.18, hence is absorbed into the perturbative remainder and treated by theorem 4.17;*
(D4) *the reselection still has degenerate scale-center Jacobian on every further subsequence and has no further active subconcentration; in this case the no-subconcentration alternative of lemma 4.35 gives a nontrivial bounded selected-window limit on a compact cylinder.*

Proof. By the gauge-degeneracy alternative definition 4.6 (v), the localized scale or centering selection of the selected compound or terminal frame is non-unique along the subsequence. Apply the parameter partition lemma lemma 4.10 to the selected compound together with any candidate alternate selection produced by the gauge degeneracy. Three mutually exclusive structural outcomes are possible.

Case A: alternate selection is comparable to the original at the same physical point. Then both selections lie in the same scale comparability class of Step 1 of theorem 4.18. Step 2 of that theorem groups them into a single compound profile. If the compound is below the small-data threshold, we are in case (D3); otherwise the compound is above threshold and Step 3 of the theorem applies. If the compound is then nondegenerate, we are in case (D2); if it is still degenerate, the gauge degeneracy persists on the reselected compound, and we proceed to Case D below.

Case B: alternate selection is at a strictly different scale at the same physical point. By lemma 4.14, the strictly outer profile vanishes locally in the innermost frame; absorbing it into the perturbative remainder lands us in case (D3). If instead the reselection produces a strictly inner core not previously detected, then either compact-cylinder suitable control fails, in which case lemma 4.15 produces the new CKN core, or compact control persists and lemma 4.33 converts the retained CKN mass into an active L^3 core unless the branch has already exited to the separated/exterior pressure alternative. These outcomes are case (D1).

Case C: alternate selection is at a separated physical point. By theorem 4.19, the new selection is locally invisible in the original frame, hence absorbed into the perturbative remainder; this is case (D3). If the new selection retains positive CKN concentration in its own frame, then lemma 4.33 makes it an additional separated active core, or else assigns it to the separated/exterior pressure alternative. Both outcomes are case (D1) via definition 4.6 (iv).

Case D: persistent degeneracy. Suppose every reselection on every further subsequence still has degenerate scale-center Jacobian. Apply lemma 4.26 and then lemma 4.35. If the compact limit has a CKN subconcentration point, that lemma produces an additional active core; this is case (D1). If no such subconcentration remains, the same lemma gives a nonzero bounded selected-window limit; this is case (D4). No L^∞ bound is inferred from the Jacobian degeneracy itself. □

Theorem 4.38 (Scale-rigid stratum decomposition)**.** *Every scale-rigid local branch in the sense of definition 4.23 enters one of the following named local strata after passing to a subsequence:*

- (i) *the multicore alternative definition 4.6 (ii)–(iv), governed by theorem 4.11;*
- (ii) *the compact single-core stratum, governed by theorem 4.36;*
- (iii) *the perturbative remainder of definition 4.7, governed by theorem 4.17;*
- (iv) *the bounded selected-window limit alternative definition 2.1, which by lemma 4.25 is incompatible with definition 2.2.*

Proof. By definition 4.23 (R4), the branch enters the scale-rigid stratum either via a degenerate relative scale-center Jacobian or as an innermost cascade core after outer profiles have been absorbed perturbatively.

Degenerate Jacobian case. Apply lemma 4.37. Cases (D1)–(D4) of that lemma give items (i)–(iv) respectively.

Innermost cascade case. By Step 4 of theorem 4.18, after subtracting the outer-profile sum $S_n \to 0$ in $L^\infty_I L^3(B_{2R})$ and absorbing constant drifts into the translation modulation, the residual is governed by theorem 4.17. That theorem produces three outcomes already enumerated as (a) single-core, (b) scale-rigid (degenerate Jacobian), or (c) purely perturbative. Case (a) is item (ii) here via theorem 4.36; case (c) is item (iii); case (b) loops back into the degenerate Jacobian case treated above and is therefore reduced by lemma 4.37 to one of (i)–(iv). □

Lemma 4.39 (Finite refinement of scale-rigid structural exits)**.** *Let ω be a scale-rigid local branch produced by theorem 4.18 or theorem 4.19, with the inherited active family of finite critical mass and with concentration capture in the sense of definition 4.7. Resolve every structural exit of theorem 4.38 by applying only theorem 4.11, theorem 4.18, theorem 4.19, and theorem 4.17, never* *theorem 4.42. Then the refinement terminates after finitely many ledger-decreasing refinements. At the terminal residual branch one has either a nonzero bounded selected-window limit or a compact single-core vanishing-dissipation branch excluded by theorem 2.18.*

Proof. After the active threshold has been lowered once, every core produced by lemma 4.33 is handled by lemma 4.34: it is already in the captured inherited family, is absorbed into the terminal frame, or exits to the separate profile-completeness obligation. The residual scale-rigid ledger therefore uses only the captured active family. Write η_*^{new} for the lowered active threshold. By lemma 4.9, the resulting family contains at most

$$N_* := \lfloor M_*(\eta_*^{\mathrm{new}})^{-3} \rfloor$$

active profiles. Group same-point comparable-scale profiles into compound profiles as in definition 4.8. There are therefore only finitely many active objects and only finitely many unordered pair relations among them.

Define the refinement ledger to consist of:

(L1) every active object not yet selected, grouped, absorbed, or declared terminal;
(L2) every pair relation not yet classified as comparable same-point, strict same-point cascade, separated point, or locally invisible;
(L3) every degenerate scale-center selection not yet resolved by lemma 4.37;
(L4) every perturbative remainder not yet removed by theorem 4.17.

The ledger is finite because it is built from a finite active family and finitely many compounds. Order ledgers lexicographically by the number of unresolved entries in (L1), (L2), (L3), and (L4).

At each nonterminal step, no multibubble exclusion theorem is used. If theorem 4.38 gives a multicore, same-point cascade, or separated/exterior structural exit, apply theorem 4.11 to select the corresponding active object. Comparable same-point objects are grouped into a compound profile; below-threshold compounds are removed by the small-data step in theorem 4.18, while above-threshold compounds count as active objects. Strict same-point cascades are reduced by theorem 4.18; separated physical centers are reduced by theorem 4.19. Perturbative exterior and remainder terms are discarded only through theorem 4.17, whose output preserves the selected positive concentration and produces no new active object.

Each move strictly decreases the ledger. Adding a previously unseen active core consumes one entry of (L1), and this can happen at most N_* times. Grouping comparable same-point objects, ordering a same-point cascade, separating two physical centers, or declaring one object locally invisible resolves a previously unresolved pair and consumes one entry of (L2). Applying lemma 4.37 resolves the selected degenerate frame or produces an active structural exit already counted in (L1)–(L2), so it consumes one entry of (L3). Applying theorem 4.17 to a perturbative remainder consumes one entry of (L4). Once an entry is resolved it is not reintroduced: later passes use the grouped compound, the selected innermost representative, the separated-frame assignment, or the discarded perturbative remainder. Hence no cycle is possible.

After the ledger is exhausted, item (i) of theorem 4.38 cannot occur again, because it would create either a new active object or an unresolved active relation, contradicting exhaustion of (L1)–(L2). Item (iii) cannot carry the retained positive concentration by the capture clause in definition 4.7; a purely perturbative remainder has already been absorbed through (L4). Item (iv) is exactly the nonzero bounded selected-window-limit alternative.

It remains only to handle item (ii). Apply theorem 4.36. Its bounded-limit alternative gives the first terminal outcome. Its further-active-core alternative would add a new ledger entry, impossible after ledger exhaustion. Its vanishing-localized-dissipation alternative verifies definition 2.3 for the original selected sequence and is excluded by theorem 2.18. These are precisely the two terminal residual outcomes claimed. □

Proposition 4.40 (Scale-rigidity exclusion)**.** *Every scale-rigid local branch produced by theorem 4.18 or theorem 4.19, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit on a compact cylinder (and so fails definition 2.2 by lemma 4.25) or reduces to the compact single-core branch governed by theorem 2.18 (equivalently by the local proposition proposition 4.20). Consequently no such branch is compatible with the local Type II condition. This proves the earlier scale-rigidity exclusion statement recorded as proposition 4.21 in the stated class.*

Proof. Let ω be a scale-rigid local branch. Theorem 4.38 places ω into one of items (i)–(iv).

If item (i) occurs, the branch enters a named multicore, same-point cascade, or separated/exterior structural exit. Resolve that exit by the finite refinement procedure of lemma 4.39. This procedure uses only theorem 4.11, theorem 4.18, theorem 4.19, and theorem 4.17; it does not invoke theorem 4.42, so there is no circular use of proposition 4.21. The refinement terminates at a bounded selected-window limit or at the compact single-core vanishing-dissipation branch.

In item (ii), apply theorem 4.36. Its bounded-limit case is the first alternative in the proposition. Its vanishing-localized-dissipation case verifies definition 2.3 for the original selected sequence and is excluded by theorem 2.18; this is the "reduces to the single-core branch" alternative. Its further-active-core case is resolved by the same finite refinement lemma used for item (i).

In item (iii), the branch is the perturbative remainder; by theorem 4.17 it produces no independent multibubble obstruction. The retained positive concentration is carried by the selected component, not by the perturbative remainder, so the residual selected component is reduced to item (ii) or item (iv) by lemma 4.39.

In item (iv), by lemma 4.25 the branch is not a local Type II sequence in the sense of definition 2.2. This is the "nonzero bounded selected-window limit" alternative.

In every case the branch either has a nonzero bounded selected-window limit or reduces to the single-core branch. This is the conclusion of proposition 4.21. ☐

Proposition 4.41 (Terminal admissibility in active frames)**.** *Assume finite local critical norm, terminal windowed state-space completeness in the sense of theorem 8.16, local perturbative residual stability, the exterior annulus alternative corollary 8.47, the repaired-gauge representation of theorem 3.35, and local Caccioppoli regularity. Then every terminal family produced by the same-point and separated-point reductions is admissible on each compact selected cylinder in the sense of definition 4.16.*

Proof. The terminal nonlinear local-state decoupling theorem, theorem 8.27, applies under the stated hypotheses and produces, on every compact terminal cylinder, the conclusions of definition 8.13. We verify each clause of definition 4.16.

Items (i) and (ii) are the active-family enumeration of definition 4.7 together with the same-point and separated-point grouping of lemmas 4.13 and 4.14; they are properties of the active family that survive the terminal-frame construction by definition of the terminal active class.

Item (iii) is the velocity decoupling lemma 8.18, the linear residual decoupling lemma 8.24, and the pressure decoupling lemma 8.23; together they give the $L^\infty_I L^3$ vanishing of the rescaled remainder r_n and the existence of (π^r_n, f_n) with the stated $L^1_I H^{-1}(B_R)$ decay.

Item (iv) is the retained $L^\infty_I L^3(B_{2R})$ bound established in the proof of lemma 8.26, which uses the finite local critical norm and the retained local-state bounds.

Item (v) is the bounded-modulation hypothesis, supplied by the repaired-gauge representation theorem 3.35 and theorem 3.34.

Item (vi) is the local pressure-decomposition stability of lemma 8.23, whose proof relies only on the Calderón–Zygmund pressure-tail decoupling lemma 8.12 together with the kernel boundedness on $L^{3/2}$ provided by the classical Calderón–Zygmund theorem [CZ52, Ste70].

Item (vii) follows from the Leibniz formula

$$[\Delta, \chi_R]u = (\Delta\chi_R)u + 2\nabla\chi_R \cdot \nabla u$$

and the identity

$$[\nabla\chi_R, \nabla\cdot](u \otimes u) = (\nabla\chi_R)(u \otimes u).$$

These are combined with the dual estimate

$$\|fu\|_{H^{-1}(B_R)} \le C_R\|f\|_{C^1(\overline{B}_R)}\|u\|_{L^{6/5}(B_R)}$$

and the embedding $L^3(B_R) \hookrightarrow L^{6/5}(B_R)$ on bounded balls. Indeed, for $\varphi \in H^1_0(B_R)$,

$$\langle \nabla\chi_R \cdot \nabla u, \varphi\rangle = -\langle u, (\Delta\chi_R)\varphi + \nabla\chi_R \cdot \nabla\varphi\rangle,$$

giving $\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} \le C_R\|u\|_{L^3(B_{2R})}$. The pointwise bound $|(\nabla\chi_R)(u \otimes u)| \le \|\nabla\chi_R\|_\infty|u|^2$ gives $\|(\nabla\chi_R)(u \otimes u)\|_{L^{3/2}(B_R)} \le C_R\|u\|^2_{L^3(B_{2R})}$, and $L^{3/2} \hookrightarrow H^{-1}$ on B_R supplies the displayed bound. The combined estimate

$$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla\chi_R, \nabla\cdot](u\otimes u)\|_{H^{-1}(B_R)} \le C_R\big(\|u\|_{L^3(B_{2R})} + \|u\|^2_{L^3(B_{2R})}\big) \le C'_R\|u\|^2_{L^3(B_{2R})}$$

holds whenever $\|u\|_{L^3(B_{2R})}$ is bounded below by a fixed constant, which is the case in the active selected frame. The same estimate applies to the S_n-quadratic commutator terms thanks to the uniform $L^\infty_I L^3(B_{2R})$ control of U_n and the $L^1_I L^{3/2}(B_R)$ control of the cross stresses produced by lemma 8.20.

All items in definition 4.16 are therefore satisfied, which is the asserted admissibility. □

Theorem 4.42 (Local multibubble and cascade exclusion)**.** *Use the local single-core exclusion proposition 4.20, the scale-rigidity exclusion proposition 4.21, and the terminal admissibility in active frames proposition 4.41. Let a positive local Type II branch admit an active family satisfying the hypotheses of theorem 4.11. Then no local multibubble, cascade, or gauge-degenerate Type II case can occur in this class.*

Proof. Let a local Type II branch be in the multibubble/cascade class of definition 4.6. Apply theorem 4.11. The positive local concentration is carried by an active frame or compound local state, and one of the alternatives listed there occurs.

Case 1: compactness-failure reduction. Compactness failure yields, after changing radius by a fixed factor and rescaling as in lemma 4.15, a bounded-cylinder concentration branch. Relabel this branch as an active core. After passing to a subsequence, the new core and every previously retained active core have one of three relations: comparable scale at the same physical point, strictly separated scale at the same physical point, or separated physical center. Comparable same-point cores are grouped into compound local states. A compound state with no detecting compact cylinder is removed by local perturbative residual stability; an active compound state is treated as one selected core in its active frame. If the scale-center selection inside such a compound core is degenerate, the branch is gauge-degenerate and belongs to the non-uniqueness alternative in definition 4.6, rather than to a separate core dynamics.

Case 1a: gauge degeneracy. If the scale-center selection of the selected compound or terminal frame is degenerate, then the branch is in the gauge-degenerate alternative of definition 4.6. By the repaired-gauge representation from Section 3, nondegeneracy is the hypothesis needed to enter the single-core branch. Persistent degeneracy therefore belongs to the scale-rigidity alternative in proposition 4.21; if the degeneracy disappears after passing to a subsequence, the branch enters the single-core or terminal-frame alternatives below.

Case 1b: same-point comparable scales. Same-point comparable-scale active frames are grouped into a compound local state by the local state-space selection rule. If the compound state has no detecting compact cylinder, it is part of the perturbative residual. If it has the local detection threshold and its scale-center selection is nondegenerate, it is one selected active core and enters the single-core branch. If the corresponding scale-center selection is degenerate, it is covered by Case 1a. Thus comparable same-point states do not produce an additional multibubble mechanism.

Case 2: same-point cascading. Strict same-point scale cascades are treated by theorem 4.18. In the innermost frame, all outer scales are locally invisible after constant drifts are absorbed. The branch therefore reduces to one of the three terminal alternatives

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

Case 3: separated-point interactions. For separated physical points, theorem 4.19 gives the same three terminal alternatives as in Case 2:

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

The single-core alternative is excluded by proposition 4.20. The scale-rigid alternative is excluded by proposition 4.21, which includes the scale-collapse and autonomous-limit exclusions in Section 6 in this setting. The terminal perturbative alternative gives no independent multibubble obstruction: theorem 4.17 removes all exterior profiles and remainders from the selected active equation in $L^1 H^{-1}_{\mathrm{loc}}$, preserves the positive finite-mass selected branch, and leaves only the single-core or scale-rigid possibilities already excluded. Consequently every alternative in definition 4.6 is incompatible with the local Type II regime. □

Corollary 4.43 (Removal of the multibubble alternative)**.** *Under the hypotheses of theorem 4.42, every failure of the single nondegenerate compact core alternative is excluded.*

Proof. By definition 4.6, failure of the single nondegenerate compact core alternative means that at least one of the following occurs: the compact-cylinder suitable estimates are unbounded; positive local CKN mass splits between active cores; a strict same-point scale cascade occurs; separated physical points remain active in distinct local frames; or the localized scale or center selection is degenerate. The compactness-failure case produces an additional core by lemma 4.15. Same-point and separated cases are reduced by theorems 4.18 and 4.19. Theorem 4.42 excludes the remaining cases. Hence no failure of the single nondegenerate compact core alternative remains. □

5. Local Caccioppoli and Rough-Core Exclusion

5.1. **Analytic inputs and notation.** The renormalized local energy inequality from the preceding renormalized-chart analysis is used as an analytic input. The renormalized suitable structure and pressure reconstruction are not rederived in this section. The estimates below are local implications under the displayed compact-cylinder conditions; the terminal assembly verifies or routes these conditions by the ordered local state-space analysis.

Proposition 5.1 (Renormalized local energy inequality)**.** *Let (V, P, a, b) be defined on a compact renormalized cylinder $Q_{2R,J} := B_{2R} \times J$, where $J \subset \mathbb{R}$ is a bounded interval. Assume*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu\Delta V = a(\tau)V + a(\tau)(y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0, \tag{7}$$

with $\nu > 0$, $\nabla \cdot V = 0$, and $a, b \in L^\infty(J)$. For every nonnegative $\phi \in C_c^\infty(Q_{2R,J})$ and every $\tau_1 < \tau_2$ in J,

$$\begin{aligned}
&\int_{B_{2R}} \frac{|V(\cdot,\tau_2)|^2}{2}\phi(\cdot,\tau_2) - \int_{B_{2R}} \frac{|V(\cdot,\tau_1)|^2}{2}\phi(\cdot,\tau_1) \\
&\quad + \nu\int_{\tau_1}^{\tau_2}\int_{B_{2R}} \phi\,|\nabla V|^2 \le \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2}(\partial_\tau\phi + \nu\Delta\phi) \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2}V\cdot\nabla\phi + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} PV\cdot\nabla\phi \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} a\frac{|V|^2}{2}(-\phi - y\cdot\nabla\phi) \\
&\quad - \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2}b\cdot\nabla\phi.
\end{aligned} \tag{8}$$

The pressure is understood modulo functions of time, and the mean-normalized pressure on compact balls belongs to the local class used below.

Proof. Proposition 3.31 transports the physical local energy inequality of Caffarelli–Kohn–Nirenberg [CKN82] into the renormalized variables (y, τ) and yields the inequality stated above with test function of the form ϕ^2, where $\phi \in C_c^\infty(Q_R^*)$ is nonnegative. Setting $\Phi := \phi^2$ and noting that every nonnegative $\Phi \in C_c^\infty(Q_{2R,J})$ is the limit, in $C_c^k(Q_{2R,J})$ for every k, of squares of nonnegative smooth compactly supported functions $\phi_\varepsilon := (\Phi + \varepsilon^2)^{1/2} - \varepsilon$, the inequality extends by approximation to all such Φ; this is the standard reformulation used in the theory of suitable weak solutions [CKN82, Lin98]. The compact-window representation theorem theorem 3.35 provides the cylinder $Q_{2R,J} = B_{2R} \times J$ and the divergence-free renormalized equation (7), while theorem 3.23 supplies the localized pressure decomposition with the mean-normalized pressure on compact balls. Identifying the modulation coefficients $a = \tilde{a}$ and $b = \tilde{b}$ of proposition 3.31 with those of (7) and grouping the renormalized error terms exactly as in (8) yields the stated inequality. □

Throughout, for a space–time set $E \subset \mathbb{R}^3 \times \mathbb{R}$, write

$$\int_E f := \int_E f(y,\tau)\,dy\,d\tau.$$

For $r > 0$ and a bounded interval $J \subset \mathbb{R}$, define

$$Q_{r,J} := B_r \times J, \qquad |Q_{r,J}| := |B_r|\,|J|.$$

For a ball $B \subset \mathbb{R}^3$, denote the spatial mean of pressure by

$$(P)_B(\tau) := \frac{1}{|B|} \int_B P(y,\tau)\, dy.$$

The local quantities are

$$\mathcal{A}_J(r) := \operatorname*{ess\,sup}_{\tau \in J} \int_{B_r} |V(y,\tau)|^2\, dy,$$
$$\mathcal{H}_J(r) := \int_J \int_{B_r} |\nabla V|^2\, dy\, d\tau,$$
$$\mathcal{C}_J(r) := \int_J \int_{B_r} |V|^3\, dy\, d\tau, \qquad \mathcal{D}_J(r) := \int_J \int_{B_r} |P - (P)_{B_r}(\tau)|^{3/2}\, dy\, d\tau.$$

Lemma 5.2 (Finite-measure conversion)**.** *For every bounded interval J and every $r > 0$,*

$$\int_J \int_{B_r} |V|^2 \le |Q_{r,J}|^{1/3} \mathcal{C}_J(r)^{2/3} \le |Q_{r,J}|^{1/3} \big(1 + \mathcal{C}_J(r)\big).$$

Proof. Hölder's inequality on $Q_{r,J}$ gives

$$\int_J \int_{B_r} |V|^2 \le |Q_{r,J}|^{1/3} \left(\int_J \int_{B_r} |V|^3 \right)^{2/3}.$$

The second inequality follows from $s^{2/3} \le 1 + s$ for $s \ge 0$. □

Lemma 5.3 (Comparison of pressure normalizations)**.** *Let $1 \le p < \infty$ and $0 < r \le R$. For almost every τ such that $P(\tau) \in L^p(B_R)$,*

$$\|P(\tau) - (P)_{B_r}(\tau)\|_{L^p(B_r)} \le 2 \|P(\tau) - (P)_{B_R}(\tau)\|_{L^p(B_R)}.$$

Consequently,

$$\mathcal{D}_J(r) \le 2^{3/2} \mathcal{D}_J(R)$$

whenever the right-hand side is finite.

Proof. Set $c = (P)_{B_R}(\tau)$. Then

$$\|P - (P)_{B_r}\|_{L^p(B_r)} \le \|P - c\|_{L^p(B_r)} + |B_r|^{1/p} |(P)_{B_r} - c|.$$

Since

$$|(P)_{B_r} - c| \le |B_r|^{-1} \int_{B_r} |P - c| \le |B_r|^{-1/p} \|P - c\|_{L^p(B_r)},$$

the displayed pointwise estimate follows. Raising to the power $3/2$ and integrating in time gives the last assertion. □

5.2. **Compact-cylinder Caccioppoli estimate.**

Theorem 5.4 (Compact-cylinder Caccioppoli estimate)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume proposition 5.1 on $Q_{2R,J}$ and assume*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty.$$

Then there is a constant

$$C = C\big(R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \operatorname{dist}(J', \partial J)\big)$$

such that

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) \le C \Big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\Big).$$

The estimate is outer-to-inner: all right-hand quantities are measured on $B_{2R} \times J$, while the conclusion is on $B_R \times J'$.

Proof. **Step 1: localized cut-off and the two energy inequalities.** Set

$$\delta := \operatorname{dist}(J', \partial J) > 0.$$

Choose $\eta \in C_c^\infty(J)$ and $\zeta \in C_c^\infty(B_{2R})$ with

$$0 \le \eta \le 1, \qquad 0 \le \zeta \le 1, \qquad \eta \equiv 1 \text{ on } J', \qquad \zeta \equiv 1 \text{ on } B_R,$$

and

$$\|\eta'\|_{L^\infty(J)} \le C_\eta\, \delta^{-1}, \qquad \|\nabla\zeta\|_{L^\infty(B_{2R})} \le C_\zeta R^{-1}, \qquad \|\Delta\zeta\|_{L^\infty(B_{2R})} \le C_\zeta R^{-2}.$$

Set

$$\phi(y,\tau) := \eta(\tau)\, \zeta(y)^2.$$

Then $\phi \in C_c^\infty(Q_{2R,J})$, $0 \le \phi \le 1$, and

$$\phi \equiv 1 \text{ on } B_R \times J'.$$

Choose $\tau_- < \inf J'$ and $\tau_+ > \sup J'$ in J with $\eta(\tau_-) = \eta(\tau_+) = 0$. Applying (8) on (τ_-, τ_+) gives

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi\, |\nabla V|^2 \le |R_\tau| + |R_\Delta| + |R_{\mathrm{conv}}| + |R_{\mathrm{pres}}| + |R_{\mathrm{mod}}|,$$

where

$$R_\tau := \int_J \int_{B_{2R}} \frac{|V|^2}{2}\, \partial_\tau \phi, \qquad R_\Delta := \nu \int_J \int_{B_{2R}} \frac{|V|^2}{2}\, \Delta\phi,$$

$$R_{\mathrm{conv}} := \int_J \int_{B_{2R}} \big(\tfrac{|V|^2}{2} V\big) \cdot \nabla\phi, \qquad R_{\mathrm{pres}} := \int_J \int_{B_{2R}} \big(P - (P)_{B_{2R}}(\tau)\big) V \cdot \nabla\phi,$$

$$R_{\mathrm{mod}} := \int_J \int_{B_{2R}} a \frac{|V|^2}{2} (-\phi - y \cdot \nabla\phi) - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla\phi.$$

For almost every $\sigma \in J'$, applying (8) on (τ_-, σ) gives the same right-hand estimates and leaves the terminal term $\frac{1}{2}\int_{B_R} |V(\sigma)|^2$ on the left. Therefore the bounds below control both $\mathcal{H}_{J'}(R)$ and $\mathcal{A}_{J'}(R)$. Define

$$L := \int_J \int_{B_{2R}} |V|^2, \qquad C_3 := \mathcal{C}_J(2R), \qquad D_{3/2} := \mathcal{D}_J(2R).$$

By lemma 5.2,

$$L \le |Q_{2R,J}|^{1/3} C_3^{2/3} \le C(R, |J|)(1 + C_3).$$

Since $\phi(y,\tau) = \eta(\tau)\zeta(y)^2$,

$$\partial_\tau \phi = \eta'(\tau)\zeta(y)^2, \qquad \nabla\phi = 2\eta(\tau)\zeta(y)\nabla\zeta(y), \qquad \Delta\phi = \eta(\tau)\Delta(\zeta(y)^2),$$

hence

$$\|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} \le C_\eta \delta^{-1}, \qquad \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \le 2C_\zeta R^{-1}, \qquad \|\Delta\phi\|_{L^\infty(Q_{2R,J})} \le C_\zeta R^{-2}.$$

Step 2: derivative terms from the cut-off.

$$|R_\tau| \le \frac{1}{2} \|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} L \le C\delta^{-1}(1 + C_3),$$

$$|R_\Delta| \le \frac{\nu}{2} \|\Delta\phi\|_{L^\infty(Q_{2R,J})} L \le C\nu R^{-2}(1 + C_3).$$

Step 3: convection term. Using $|\nabla\phi| \le CR^{-1}$,

$$|R_{\mathrm{conv}}| \le \frac{1}{2} \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \int_J \int_{B_{2R}} |V|^3 \le CR^{-1} C_3.$$

Step 4: pressure term. For each time τ,

$$\int_{B_{2R}} (P - (P)_{B_{2R}}(\tau))V \cdot \nabla\phi = \int_{B_{2R}} PV \cdot \nabla\phi,$$

with

$$\int_{B_{2R}} (P)_{B_{2R}}(\tau)V \cdot \nabla\phi = (P)_{B_{2R}}(\tau)\int_{B_{2R}} V \cdot \nabla\phi = -(P)_{B_{2R}}(\tau)\int_{B_{2R}} \phi\, \nabla \cdot V = 0$$

(using $\phi \in C_c^\infty(B_{2R})$ and $\nabla \cdot V = 0$). Thus subtracting any time-dependent pressure constant is legitimate in the pressure flux term. Hence

$$|R_{\text{pres}}| \le \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \int_J \|P - (P)_{B_{2R}}(\tau)\|_{L_x^{3/2}(B_{2R})} \|V(\tau,\cdot)\|_{L_x^3(B_{2R})}\, d\tau \le CR^{-1} D_{3/2}^{2/3} C_3^{1/3}.$$

Young's inequality gives

$$|R_{\text{pres}}| \le C(R)(C_3 + D_{3/2}).$$

Step 5: modulation terms. The local energy inequality already contains the modulation terms after the spatial integrations by parts from the repaired-gauge equation:

$$\int_{B_{2R}} (y \cdot \nabla V) \cdot V\, \phi = \frac{1}{2}\int_{B_{2R}} y \cdot \nabla(|V|^2)\phi = -\frac{1}{2}\int_{B_{2R}} |V|^2 \nabla \cdot (y\phi),$$

and

$$\int_{B_{2R}} (b(\tau) \cdot \nabla V) \cdot V\, \phi = \frac{1}{2}\int_{B_{2R}} b(\tau) \cdot \nabla(|V|^2)\phi = -\frac{1}{2}\int_{B_{2R}} |V|^2 b(\tau) \cdot \nabla\phi.$$

Thus we estimate the additional terms directly. Split

$$R_{\text{mod}} = R_a + R_{ay} + R_b,$$

where

$$R_a := -\int_J\int_{B_{2R}} a\, \frac{|V|^2}{2}\phi, \quad R_{ay} := -\int_J\int_{B_{2R}} a\, \frac{|V|^2}{2}\, y \cdot \nabla\phi, \quad R_b := -\int_J\int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla\phi.$$

$$M_a := \|a\|_{L^\infty(J)}, \qquad M_b := \|b\|_{L^\infty(J)}.$$

For R_a,

$$|R_a| \le \frac{1}{2} M_a L \le CM_a(1 + C_3).$$

For the scale-gradient part, $|y| \le 2R$ on B_{2R} and $\|\nabla\phi\|_{L^\infty(Q_{2R,J})} \le CR^{-1}$, hence

$$|y \cdot \nabla\phi| \le C.$$

Therefore

$$|R_{ay}| \le CM_a L \le CM_a(1 + C_3).$$

For R_b,

$$|R_b| \le CM_b R^{-1} L \le CM_b R^{-1}(1 + C_3).$$

Step 6: absorption and conclusion. Collecting estimates from Steps 2–5 yields

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi |\nabla V|^2 \le C\big(1 + C_3 + D_{3/2}\big),$$

where C depends only on

$$R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \delta.$$

Since $\phi \equiv 1$ on $B_R \times J'$,

$$\mathcal{H}_{J'}(R) \le C\big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\big).$$

Repeating the same estimates with terminal time $\sigma \in J'$ gives

$$\frac{1}{2}\int_{B_R} |V(y,\sigma)|^2\,dy \le C\big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\big)$$

for almost every $\sigma \in J'$. Taking the essential supremum over $\sigma \in J'$ and renaming C proves the theorem. ☐

Definition 5.5 (Local compact-cylinder control)**.** Local Caffarelli–Kohn–Nirenberg control on $Q_{r,J}$ means

$$\mathcal{C}_J(r) + \mathcal{D}_J(r) < \infty.$$

Local windowed H^1 control on $Q_{r,J}$ means

$$\mathcal{H}_J(r) < \infty.$$

Corollary 5.6 (Outer control implies inner suitable control)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume proposition 5.1 on $Q_{2R,J}$ and $\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty$. Then*

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) < \infty.$$

More quantitatively, the bound is the one in theorem 5.4. If an argument uses pressure means on a smaller ball, lemma 5.3 converts those normalizations to the outer-ball normalization above.

Proof. The estimate follows from theorem 5.4. The comparison of pressure normalizations follows from lemma 5.3. ☐

Corollary 5.7 (Rough-core exclusion on compact windows)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume proposition 5.1 on $Q_{2R,J}$. If*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty,$$

then

$$\mathcal{H}_{J'}(R) < \infty.$$

Equivalently, if $\mathcal{H}_{J'}(R) = +\infty$ for an inner cylinder $Q_{R,J'}$, then for every enclosing cylinder $Q_{2R,J}$ with $J' \Subset J$ on which the local energy inequality and bounded modulation hypotheses hold,

$$\mathcal{C}_J(2R) = +\infty \qquad \text{or} \qquad \mathcal{D}_J(2R) = +\infty.$$

Proof. The direct implication is the gradient part of theorem 5.4. The last assertion is its contrapositive with the same outer-to-inner geometry: finite outer $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ forces finite inner $\mathcal{H}$ on $Q_{R,J'}$. ☐

Remark 5.8 (Meaning of rough core)*.* Here a rough core means failure of local windowed H^1 control on compact inner renormalized cylinders while the modulation coefficients are bounded. Corollary 5.7 shows that such a failure is not independent of compact-cylinder Caffarelli–Kohn–Nirenberg control: it can occur only if the velocity-pressure bounds $\mathcal{C} + \mathcal{D}$ fail on every relevant larger enclosing compact cylinder.

5.3. **Consequences for the assembly.** We shall use the following consequences in the final assembly.

(1) On every compact renormalized window satisfying the local energy inequality and bounded modulation, finite $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ implies finite $\mathcal{A} + \mathcal{H}$ on $Q_{R,J'}$ whenever $J' \Subset J$.
(2) The pressure normalization may be taken on any comparable ball, after using lemma 5.3.
(3) Failure of local windowed H^1 control on an inner compact cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on each larger enclosing cylinder to which theorem 5.4 applies.

6. Scale-Collapse Cost and Autonomous Limits

Notation and functional conventions. For a bounded spatial region $K \subset \mathbb{R}^3$, we use

$$(f,g)_{L^2(K)} := \int_K f \cdot g \, dx, \qquad \langle F, \phi \rangle_{H^{-1}(K),H_0^1(K)} := \int_0^T \langle F(\cdot,s), \phi(\cdot,s) \rangle_{H^{-1}(K),H_0^1(K)} \, ds$$

whenever $F \in L^1(0,T;H^{-1}(K))$ and $\phi \in L^\infty(0,T;H_0^1(K))$. Weak limits and weak convergences are taken in the distributional sense on C_c^∞-test functions.

$$\|f\|_{L_t^p L_x^q} := \left(\int_0^T \|f(\cdot,s)\|_{L^q(\mathbb{R}^3)}^p \, ds \right)^{1/p},$$

for any fixed $T > 0$, with the usual modification $p = \infty$.

6.1. **Renormalized scale-collapse setting.** *Position of the result.* The scale-collapse analysis has two components:

(1) Sections 6.2–6.4 contain the cost exclusion arguments, which do not use compactness.
(2) Section 6.5 contains the autonomous compactness statement under explicitly stated compactness hypotheses.

Throughout the scale-collapse analysis, the renormalized representation established earlier is assumed: on the time interval under consideration, the fields (V,P) satisfy (9) and its distributional weak formulation. In particular, the solenoidal formulation is obtained by restricting the test fields below to divergence-free fields.

Let (u,p) be a divergence-free suitable weak solution of

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

in $\mathbb{R}^3$, with viscosity $\nu > 0$. Let $\lambda(\tau) > 0$, $X(\tau) \in \mathbb{R}^3$, and a measurable gauge function $c(t)$ be given, and define the renormalized fields by

$$u(x,t) = \lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau\right), \qquad p(x,t) = \lambda(\tau)^{-2} P\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau\right) + c(t).$$

Pressure is defined only up to an arbitrary function of time; $c(t)$ has no effect on ∇p, so it is irrelevant for all subsequent estimates. Assume that (V,P) satisfies on $(0,\infty) \times \mathbb{R}^3$

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0. \tag{9}$$

For every $\psi \in C_c^\infty((0,\infty) \times \mathbb{R}^3; \mathbb{R}^3)$ with $\nabla \cdot \psi = 0$, the pair (V,P) satisfies the weak formulation

$$\begin{aligned} 0 = &\int_0^\infty \int_{\mathbb{R}^3} \big(-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi + \nu \nabla V : \nabla \psi \big) \, dy \, d\tau \\ &+ \int_0^\infty \int_{\mathbb{R}^3} \big(-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi + P \, \nabla \cdot \psi \big) \, dy \, d\tau. \end{aligned}$$

For every $0 < T < \infty$, restricting the τ-integration to $(0,T)$ gives the same identity on $(0,T) \times \mathbb{R}^3$. Equivalently, for every $\psi \in C_c^\infty((0,\infty) \times \mathbb{R}^3; \mathbb{R}^3)$, not necessarily divergence free,

$$\begin{aligned} 0 = &\int_0^\infty \int_{\mathbb{R}^3} \big(-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi \nu \nabla V : \nabla \psi - P \, \nabla \cdot \psi \big) \, dy \, d\tau \\ &+ \int_0^\infty \int_{\mathbb{R}^3} \big(-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi \big) \, dy \, d\tau. \end{aligned}$$

Define positive and negative parts of the scalar field a by

$$a_+(\tau) := \max\{a(\tau), 0\}, \qquad a_-(\tau) := \max\{-a(\tau), 0\}.$$

Then for a.e. τ, $a = a_+ - a_-$, $a_+, a_- \geq 0$, and $a_+ a_- = 0$. Assume

$$a \in L^1_{\mathrm{loc}}([\tau_0, \infty)), \qquad b \in L^1_{\mathrm{loc}}([\tau_0, \infty); \mathbb{R}^3),$$

and the scale convention

$$\frac{d}{d\tau} \log \lambda(\tau) = a(\tau) \tag{10}$$

for a.e. $\tau \geq \tau_0$, with $\log \lambda$ absolutely continuous on compact intervals.

All integrals below are interpreted in $[0, \infty]$. When we write $\int_{\tau_0}^{\infty}$, partial integrals are taken over $[\tau_0, T]$ and monotone convergence is used as $T \to \infty$.

Fix $R > 0$. Choose $\Phi \in C_c^\infty(\mathbb{R}^3; [0,1])$ satisfying

$$\Phi(y) = 1 \text{ if } |y| \leq R, \quad \operatorname{supp}_y \Phi \subset B_{2R}. \tag{11}$$

Define the localized kinetic mass

$$M_R(\tau) := \int_{\mathbb{R}^3} |V(y,\tau)|^2 \Phi(y)\, dy.$$

Since Φ is fixed in τ, M_R is measurable in τ, and only this measurability is used below.

Definition 6.1 (Nonvanishing localized core)**.** A renormalized branch has nonvanishing localized core if there exist constants $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

In applications, this lower bound is supplied by the selected-core construction; here it is stated as an explicit hypothesis.

Definition 6.2 (Scale-collapse and absolute scale-drift densities)**.** Define

$$\mathcal{C}_{\mathrm{sc}}(\tau) := a_-(\tau) M_R(\tau), \qquad \mathcal{C}_{\mathrm{abs}}(\tau) := \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \Phi(y)\, dy + |a(\tau)| M_R(\tau).$$

$\mathcal{C}_{\mathrm{sc}}$ is the negative drift weighted by localized core mass, while $\mathcal{C}_{\mathrm{abs}}$ controls both collapse drift and local dissipation. Their total costs are extended real-valued by

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau, \qquad \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\, d\tau.$$

Definition 6.3 (Finite-cost branch)**.** A branch has finite scale-collapsing cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau < \infty.$$

It has finite absolute cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\, d\tau < \infty.$$

6.2. **Elementary alternative.**

Lemma 6.4 (Finite-or-infinite alternative)**.** *Let $f \geq 0$ be measurable on $[\tau_0, \infty)$. Then exactly one of the following holds:*

$$\int_{\tau_0}^{\infty} f(\tau)\, d\tau < \infty \quad \text{or} \quad \int_{\tau_0}^{\infty} f(\tau)\, d\tau = \infty.$$

Proof. (1) For $T > \tau_0$, define partial costs

$$F(T) := \int_{\tau_0}^{T} f(\tau)\, d\tau \in [0, \infty].$$

(2) If $T_2 > T_1$, then by nonnegativity of f,

$$F(T_2) - F(T_1) = \int_{T_1}^{T_2} f(\tau)\, d\tau \geq 0,$$

so F is monotone nondecreasing.

(3) A monotone function on $[\tau_0, \infty)$ has an extended limit $L \in [0, \infty]$,

$$L := \lim_{T \to \infty} F(T).$$

By definition of improper integral, this limit equals $\int_{\tau_0}^{\infty} f(\tau)\, d\tau$.

(4) Therefore L is either finite or equal to $+\infty$, and these cases are mutually exclusive.

Hence exactly one item in the lemma holds. □

Lemma 6.5 (Logarithmic scale identity)**.** *Under* (10)*, for all* τ_1, τ_2 *with* $\tau_0 \leq \tau_1 < \tau_2 < \infty$*,*

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Proof. (1) Since $a \in L^1_{\mathrm{loc}}$, the map $\log \lambda$ is absolutely continuous on $[\tau_1, \tau_2]$, hence has a representative with a.e. derivative equal to a.

(2) By the fundamental theorem of calculus for absolutely continuous functions, for every $\tau \in [\tau_1, \tau_2]$,

$$\log \lambda(\tau) = \log \lambda(\tau_1) + \int_{\tau_1}^{\tau} a(s)\, ds.$$

(3) Setting $\tau = \tau_2$ yields the identity.

□

6.3. **Scale collapse versus localized core lower bounds.**

Theorem 6.6 (Collapse with nonvanishing localized core forces infinite scale-collapsing cost)**.** *Assume:*

(1) $\lambda(\tau) \to 0$ *as* $\tau \to \infty$*;*

(2) the branch has nonvanishing localized core.

Then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. (1) Choose $m_0 > 0$ and $\tau_1 \geq \tau_0$ from the nonvanishing core assumption so that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

(2) By Lemma 6.5, for any $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Taking $\tau_2 \to \infty$ and using $\lambda(\tau_2) \to 0$,

$$\lim_{\tau_2 \to \infty} \int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = -\infty.$$

(3) Decompose $a = a_+ - a_-$. Then for each $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\, d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau \geq - \int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau.$$

Hence

$$\int_{\tau_1}^{\tau_2} a_-(\tau)\,d\tau \geq -\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau.$$

(4) Taking limits and using Step 2 yields

$$\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty.$$

Indeed, by Step 2,

$$-\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau \to +\infty.$$

Since the partial integrals of a_- dominate this quantity, they are unbounded. Because $a_- \geq 0$, monotone convergence gives $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$.

(5) Split the cost into early and late times:

$$\int_{\tau_0}^{\infty} a_- M_R\,d\tau = \int_{\tau_0}^{\tau_1} a_- M_R\,d\tau + \int_{\tau_1}^{\infty} a_- M_R\,d\tau.$$

The first term is finite or infinite independently of the argument; the second term satisfies

$$\int_{\tau_1}^{\infty} a_-(\tau) M_R(\tau)\,d\tau \geq m_0 \int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$$

by the bound in (1) and Step 4. Hence the total integral is infinite.

□

6.4. **Finite-cost exclusions.**

Theorem 6.7 (Scale-collapse cost exclusion)**.** *No branch with finite scale-collapsing cost (Definition 6.3) can satisfy simultaneously $\lambda(\tau) \to 0$ and nonvanishing localized core.*

Proof. Assume, for contradiction, that such a branch exists. The two hypotheses are exactly those of Theorem 6.6; therefore

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau = \infty.$$

This contradicts finite scale-collapsing cost. Therefore no such branch exists. □

Lemma 6.8 (Absolute-cost domination)**.** *For a.e. $\tau \in [\tau_0, \infty)$,*

$$\mathcal{C}_{\mathrm{abs}}(\tau) \geq \mathcal{C}_{\mathrm{sc}}(\tau).$$

Hence, pointwise and after integrating in τ,

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau \leq \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\,d\tau,$$

as extended real numbers. In particular, finite absolute cost implies finite scale-collapsing cost, and equivalently

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau = \infty \quad\Longrightarrow\quad \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\,d\tau = \infty.$$

Proof. At each time τ,

$$\mathcal{C}_{\mathrm{abs}}(\tau) = \nu \int |\nabla V|^2 \Phi + (a_+ + a_-) M_R \geq (a_+ + a_-) M_R \geq a_- M_R = \mathcal{C}_{\mathrm{sc}}(\tau),$$

since $a_+, a_- \geq 0$ and $\nu \int |\nabla V|^2 \Phi \geq 0$. Integrating over $[\tau_0, \infty)$ yields the claimed implication. □

Corollary 6.9 (Absolute scale-cost exclusion)**.** *No branch with finite absolute cost in the sense of Definition 6.3 can satisfy simultaneously $\lambda(\tau) \to 0$ and nonvanishing localized core.*

Proof. Assume instead such a branch exists. By Theorem 6.6, $\int \mathcal{C}_{\mathrm{sc}} = \infty$. By Lemma 6.8, the absolute cost is pointwise above $\mathcal{C}_{\mathrm{sc}}$, so its integral must also be infinite, contradicting finite absolute cost. □

Remark 6.10. The exclusions above are purely integral and use neither compactness nor any autonomous-limit argument.

6.5. **Autonomous reduced-limit theorem from the selected-window estimates.** The preceding exclusions are integral statements and do not rely on compactness. On a fixed finite autonomous window, compactness requires not only the cost bound but also the selected-window estimates supplied by the preceding renormalized representation, pressure reconstruction, and local Caccioppoli estimates.

Analytic inputs. The preceding sections supply the following ingredients on selected compact renormalized windows:

(1) the repaired-gauge renormalized equation and weak formulation for (V, P);
(2) pressure reconstruction modulo spatial means, with local $L^{3/2}$ bounds on compact cylinders;
(3) local windowed bounds $V \in L^\infty_t L^2_x \cap L^2_t H^1_x$ on compact cylinders;
(4) a retained compact core lower bound preventing the selected profile from vanishing.

Together with the thick-window convergence of the modulation coefficients, these estimates imply the compactness and limit passage below.

Definition 6.11 (Thick autonomous windows)**.** A branch has *thick autonomous windows* if there exist constants $T > 0$, $c > 0$, $M_1 \in (0, \infty)$, a sequence (τ_n) with $\tau_n \to \infty$, and limits $a_\infty \in \mathbb{R}$, $b_\infty \in \mathbb{R}^3$, such that for every n:

(i)
$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) M_R(\tau)\, d\tau \geq c;$$

(ii)
$$0 \leq M_R(\tau) \leq M_1 \quad \text{for a.e. } \tau \in [\tau_n, \tau_n + T];$$

(iii)
$$a(\tau_n + \cdot) \to a_\infty \text{ in } L^1(0, T), \qquad b(\tau_n + \cdot) \to b_\infty \text{ in } L^1(0, T; \mathbb{R}^3).$$

Remark 6.12. This notion isolates a limiting drift coefficient $a_\infty < 0$. The compactness to extract a profile limit is obtained from the selected-window compactness estimates, not from the scalar cost inequalities of Sections 6.2–6.4.

Lemma 6.13 (Thick drift implies negative autonomous limit parameter)**.** *On thick autonomous windows, the limiting autonomous coefficient is negative:*

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. (1) From (ii), for a.e. $\tau \in [\tau_n, \tau_n + T]$, $0 \leq M_R(\tau) \leq M_1$. Therefore

$$\frac{a_-(\tau) M_R(\tau)}{M_1} \leq a_-(\tau).$$

Therefore

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau)\, d\tau \geq \frac{1}{M_1 T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) M_R(\tau)\, d\tau \geq \frac{c}{M_1}.$$

(2) Set $a_n(s) := a(\tau_n + s)$. By (iii), $a_n \to a_\infty$ in $L^1(0,T)$. The map $r \mapsto r^-$ is 1-Lipschitz, so $a_n^- \to a_\infty^-$ in $L^1(0,T)$:

$$\|a_n^- - a_\infty^-\|_{L^1(0,T)} \le \|a_n - a_\infty\|_{L^1(0,T)} \to 0.$$

Thus

$$\lim_{n\to\infty} \frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)\,d\tau = \frac{1}{T}\int_0^T (a_\infty)^-\,ds = (a_\infty)^-.$$

(3) Combine Step 1 and Step 2: $(a_\infty)^- \ge c/M_1$, i.e. $a_\infty \le -c/M_1 < 0$.

□

Proposition 6.14 (Selected-window suitable estimates)**.** *Let (τ_n) be a thick autonomous-window sequence and set*

$$V_n(y,s) := V(y,\tau_n+s), \qquad P_n(y,s) := P(y,\tau_n+s),$$
$$\alpha_n(s) := a(\tau_n+s), \qquad \beta_n(s) := b(\tau_n+s), s \in (0,T).$$

Assume that the loss-of-retained-active-core row does not occur on these selected windows. Then, after passing to a subsequence, for every bounded Lipschitz domain $K \Subset \mathbb{R}^3$, there is a constant $C_K < \infty$, independent of n, such that

$$\|V_n\|_{L^\infty(0,T;L^2(K))} + \|V_n\|_{L^2(0,T;H^1(K))} + \|P_n - (P_n)_K\|_{L^{3/2}((0,T)\times K)} \le C_K,$$

where

$$(P_n)_K(s) := \frac{1}{|K|}\int_K P_n(y,s)\,dy.$$

Each V_n is divergence free and there exist $\chi \in C_c^\infty(B_R)$, $\chi \ge 0$, $\eta > 0$, and $N \in \mathbb{N}$ such that for all $n \ge N$,

$$\int_{\mathbb{R}^3} |V_n(y,s)|^2 \chi(y)\,dy \ge \eta \quad \text{for a.e. } s \in (0,T).$$

Proof. The localized lower bound is given by the retained-window branch of theorem 6.21 below, after excluding the loss-of-retained-active-core row on the chosen windows. With that lower bound in hand, proposition 6.15 below derives the local $L_t^\infty L_x^2$, $L_t^2 H_x^1$, and pressure bounds from the repaired-gauge renormalized representation theorem 3.35, the localized pressure decomposition theorem 3.23, and the compact-cylinder Caccioppoli estimate theorem 5.4. These two inputs together yield all items in the statement. □

Proposition 6.15 (Derivation of the selected-window estimates from the preceding sections)**.** *Let a repaired-gauge branch satisfy the renormalized representation and pressure reconstruction of the compact-window representation theorem, theorem 3.35, on the windows $[\tau_n, \tau_n + T]$. Assume that on each compact cylinder $(0,T)\times K$ the local compact-cylinder Caffarelli–Kohn–Nirenberg quantities are finite uniformly in n, and that the selected core has the localized lower bound stated in Proposition 6.14. Then Proposition 6.14 holds.*

Proof. The renormalized representation theorem supplies the shifted equation, divergence-free condition, and pressure reconstruction modulo functions of time. The pressure reconstruction gives, after subtracting spatial means,

$$\sup_n \|P_n - (P_n)_K\|_{L^{3/2}((0,T)\times K)} < \infty$$

on every compact K. The local Caccioppoli estimate applied on a slightly larger compact cylinder gives the inner bounds

$$\sup_n \|V_n\|_{L^\infty(0,T;L^2(K))} + \sup_n \|V_n\|_{L^2(0,T;H^1(K))} < \infty.$$

These are precisely the local velocity and pressure estimates in Proposition 6.14. The retained selected-core lower bound is the last condition in that proposition. Hence all items in Proposition 6.14 follow from the representation, pressure, and local Caccioppoli results. ☐

Lemma 6.16 (Local time regularity on autonomous windows)**.** *Assume thick autonomous windows and Proposition 6.14. Let $K_0 \Subset K_1 \Subset \mathbb{R}^3$ be bounded Lipschitz domains, and choose an integer m so large that $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$. Then*

$$\{\partial_s V_n\}_{n\geq 1} \quad \textit{is bounded in } L^1(0,T;H^{-m}(K_1)).$$

Proof. Fix $\zeta \in C_c^\infty(K_1)$ with $\zeta \equiv 1$ on a neighborhood of K_0. Let $\phi \in H_0^m(K_1;\mathbb{R}^3)$. Since $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$, there is C_m such that

$$\|\phi\|_{L^\infty(K_1)} + \|\nabla\phi\|_{L^\infty(K_1)} \leq C_m\|\phi\|_{H^m(K_1)}.$$

The shifted equation is

$$\partial_s V_n = -(V_n\cdot\nabla)V_n - \nabla P_n + \nu\Delta V_n + \alpha_n(V_n + y\cdot\nabla V_n) + \beta_n\cdot\nabla V_n$$

in distributions. We estimate each term against ϕ.

For the convection term, integration by parts gives

$$|\langle (V_n\cdot\nabla)V_n,\phi\rangle| = \left|\int_{K_1} V_{n,i}V_{n,j}\partial_j\phi_i\,dy\right| \leq C_m\|V_n(s)\|^2_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the viscous term,

$$|\langle \nu\Delta V_n,\phi\rangle| = \nu\left|\int_{K_1}\nabla V_n : \nabla\phi\,dy\right| \leq \nu C_m\|\nabla V_n(s)\|_{L^2(K_1)}|K_1|^{1/2}\|\phi\|_{H^m(K_1)}.$$

For the pressure term, subtracting the spatial mean does not change the gradient:

$$|\langle\nabla P_n,\phi\rangle| = \left|\int_{K_1}(P_n - (P_n)_{K_1})\,\nabla\cdot\phi\,dy\right| \leq C_m|K_1|^{1/3}\|P_n - (P_n)_{K_1}\|_{L^{3/2}(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the V_n-part of the scaling drift,

$$|\alpha_n(s)|\left|\int_{K_1} V_n\cdot\phi\,dy\right| \leq C_m|\alpha_n(s)|\,|K_1|^{1/2}\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the $y\cdot\nabla V_n$-part, integrate by parts:

$$\int_{K_1}(y\cdot\nabla V_n)\cdot\phi\,dy = -\int_{K_1} V_n\cdot(y\cdot\nabla\phi + 3\phi)\,dy,$$

because ϕ has zero trace. Hence

$$|\alpha_n(s)|\left|\int_{K_1}(y\cdot\nabla V_n)\cdot\phi\,dy\right| \leq C_{K_1,m}|\alpha_n(s)|\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

Similarly,

$$\left|\int_{K_1}(\beta_n\cdot\nabla V_n)\cdot\phi\,dy\right| = \left|\int_{K_1} V_n\cdot(\beta_n\cdot\nabla\phi)\,dy\right| \leq C_m|\beta_n(s)|\,|K_1|^{1/2}\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

Combining the estimates and taking the supremum over $\|\phi\|_{H^m(K_1)} \leq 1$, we obtain

$$\begin{aligned}\|\partial_s V_n(s)\|_{H^{-m}(K_1)} \leq C_{K_1,m}\big(&\|V_n(s)\|^2_{L^2(K_1)} + \|\nabla V_n(s)\|_{L^2(K_1)}\\ &+ \|P_n(s) - (P_n)_{K_1}(s)\|_{L^{3/2}(K_1)} + (|\alpha_n(s)| + |\beta_n(s)|)\|V_n(s)\|_{L^2(K_1)}\big).\end{aligned}$$

Integrating in $s\in(0,T)$, using Proposition 6.14, and using $\alpha_n\to a_\infty$, $\beta_n\to b_\infty$ in L^1, hence uniform L^1-bounds for α_n,β_n, proves the claimed $L^1(0,T;H^{-m})$ bound. ☐

Lemma 6.17 (Local compactness on autonomous windows)**.** *Assume thick autonomous windows and Proposition 6.14. Then after passing to a subsequence there exist*

$$U \in L^\infty(0,T;L^2_{\mathrm{loc}}(\mathbb{R}^3)) \cap L^2(0,T;H^1_{\mathrm{loc}}(\mathbb{R}^3)), \qquad \Pi \in L^{3/2}_{\mathrm{loc}}((0,T)\times\mathbb{R}^3),$$

such that, for every compact $K \Subset \mathbb{R}^3$,

(a)
$$V_n \to U \quad \textit{strongly in } L^2(0,T;L^2(K));$$

(b)
$$V_n \rightharpoonup U \quad \textit{weakly in } L^2(0,T;H^1(K));$$

(c)
$$V_n \rightharpoonup^* U \quad \textit{weakly-* in } L^\infty(0,T;L^2(K));$$

(d)
$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \textit{weakly in } L^{3/2}((0,T)\times K),$$

where the local distributions $\nabla\Pi_K$ agree on overlaps and define a pressure gradient $\nabla\Pi$ modulo functions of time.

Proof. Fix $K_0 \Subset K_1 \Subset \mathbb{R}^3$. Proposition 6.14 gives boundedness of V_n in $L^2(0,T;H^1(K_1))$, and Lemma 6.16 gives boundedness of $\partial_s V_n$ in $L^1(0,T;H^{-m}(K_1))$ for m large. Since

$$H^1(K_1) \Subset L^2(K_0) \hookrightarrow H^{-m}(K_1),$$

the Aubin–Lions–Simon compactness theorem implies, after extracting a subsequence,

$$V_n \to U \quad \text{strongly in } L^2(0,T;L^2(K_0)).$$

The local $L^\infty L^2$ and $L^2 H^1$ bounds also give, after extraction,

$$V_n \rightharpoonup^* U \quad \text{in } L^\infty(0,T;L^2(K_0)), \qquad V_n \rightharpoonup U \quad \text{in } L^2(0,T;H^1(K_0)).$$

The pressure bound in Proposition 6.14 gives weak compactness of $P_n-(P_n)_{K_0}$ in $L^{3/2}((0,T)\times K_0)$. A diagonal extraction over balls B_j gives a single subsequence for every compact K. On overlaps, two pressure limits can differ only by a function of s, because both represent the same distributional pressure gradient in the limit equation. Thus the gradients assemble into $\nabla\Pi$, with Π defined modulo functions of time. □

Lemma 6.18 (Compatibility of local pressure limits)**.** *Let $K_1, K_2 \Subset \mathbb{R}^3$ be bounded Lipschitz domains with nonempty connected overlap $K_{12} := K_1 \cap K_2$. Suppose that, after passing to a common subsequence,*

$$P_n - (P_n)_{K_i} \rightharpoonup \Pi_i \quad \textit{in } L^{3/2}((0,T)\times K_i), \qquad i = 1,2.$$

Then there exists $c_{12} \in L^{3/2}(0,T)$ such that

$$\Pi_1(y,s) - \Pi_2(y,s) = c_{12}(s) \quad \textit{for a.e. } (y,s) \in K_{12}\times(0,T).$$

Consequently the pressure gradient $\nabla\Pi$ obtained from the local limits is globally well-defined on $(0,T)\times\mathbb{R}^3$, with the pressure itself defined modulo functions of time.

Proof. For each n,

$$\big(P_n - (P_n)_{K_1}\big) - \big(P_n - (P_n)_{K_2}\big) = (P_n)_{K_2}(s) - (P_n)_{K_1}(s)$$

on K_{12}. Hence its spatial gradient is zero in $\mathcal{D}'((0,T)\times K_{12})$. Passing to the weak limit gives

$$\nabla_y(\Pi_1 - \Pi_2) = 0 \quad \text{in } \mathcal{D}'((0,T)\times K_{12}).$$

Since K_{12} is connected, the standard distributional characterization of functions with zero spatial gradient implies that $\Pi_1 - \Pi_2 = c_{12}(s)$ for some $c_{12} \in L^{3/2}(0,T)$. The last assertion follows because adding a time-dependent scalar to pressure does not change $\nabla\Pi$. □

Theorem 6.19 (Autonomous reduced-limit theorem)**.** *Assume that a branch has thick autonomous windows and satisfies Proposition 6.14 on the same window sequence. Let (U,Π) be the limit extracted in Lemma 6.17. Then*

$$\nabla\cdot U=0,$$
$$U\not\equiv 0$$

and (U,Π) satisfies

$$\partial_s U+(U\cdot\nabla)U+\nabla\Pi=\nu\Delta U+a_\infty(U+y\cdot\nabla U)+b_\infty\cdot\nabla U,\qquad \nabla\cdot U=0, \tag{12}$$

in $\mathcal{D}'((0,T)\times\mathbb{R}^3)$. Equivalently, for every $\varphi\in C_c^\infty((0,T)\times\mathbb{R}^3;\mathbb{R}^3)$,

$$\int_0^T\int\big(-U\cdot\partial_s\varphi-(U\otimes U):\nabla\varphi+\nu\nabla U:\nabla\varphi-\Pi\,\nabla\cdot\varphi-a_\infty(U+y\cdot\nabla U)\cdot\varphi-(b_\infty\cdot\nabla U)\cdot\varphi\big)\,dy\,ds=0.$$

Remark 6.20. The theorem asserts only the compact finite-window autonomous limit from the selected-window estimates. A rigidity contradiction for an arbitrary autonomous limit would require a separate Liouville theorem for the class containing (U,Π). The retained compact scale-collapse branch used in the terminal proof is handled differently: failures of retained inputs are named local exits, and the remaining retained autonomous case is routed by theorem 6.34 into the scale-rigid discharge.

Proof. Step 1: divergence-free property of the limit.

Each V_n is divergence free by Proposition 6.14, so for any compactly supported scalar $\psi\in C_c^\infty((0,T)\times\mathbb{R}^3)$,

$$\int_{(0,T)\times\mathbb{R}^3}V_n\cdot\nabla\psi\,dy\,ds=0.$$

Hence, for every such ψ, since $\nabla\psi\in C_c^\infty((0,T)\times\mathbb{R}^3)$ and $V_n\to U$ in $L^2(0,T;L^2_{\mathrm{loc}})$,

$$\int_{(0,T)\times\mathbb{R}^3}(V_n-U)\cdot\nabla\psi\,dy\,ds\to 0,$$

we deduce

$$\int_{(0,T)\times\mathbb{R}^3}U\cdot\nabla\psi\,dy\,ds=0.$$

Hence $\nabla\cdot U=0$ in $\mathcal{D}'((0,T)\times\mathbb{R}^3)$.

Step 2: nontriviality of the limit profile. Define for $s\in(0,T)$

$$f_n(s):=\int_{\mathbb{R}^3}|V_n(y,s)|^2\chi(y)\,dy,\qquad f(s):=\int_{\mathbb{R}^3}|U(y,s)|^2\chi(y)\,dy.$$

Here $K_\chi:=\operatorname{supp}\chi$, so $\chi^{1/2}V_n,\chi^{1/2}U\in L^2(0,T;L^2(K_\chi))$. Then

$$f_n-f=\int_{\mathbb{R}^3}(V_n-U)\cdot(V_n+U)\chi\,dy.$$

Applying Cauchy–Schwarz in space,

$$|f_n(s)-f(s)|\le\|\chi^{1/2}(V_n(s)-U(s))\|_{L^2(\mathbb{R}^3)}\|\chi^{1/2}(V_n(s)+U(s))\|_{L^2(\mathbb{R}^3)}.$$

Integrating in s and applying Cauchy–Schwarz in time gives

$$\|f_n-f\|_{L^1(0,T)}\le\|\chi^{1/2}(V_n-U)\|_{L^2(0,T;L^2)}\|\chi^{1/2}(V_n+U)\|_{L^2(0,T;L^2)}.$$

Since K_χ is compact, Proposition 6.14 and the membership $U\in L^2(0,T;L^2(K_\chi))$ give

$$\sup_n\|\chi^{1/2}(V_n+U)\|_{L^2(0,T;L^2)}\le C_\chi\left(\sup_n\|V_n\|_{L^2(0,T;L^2(K_\chi))}+\|U\|_{L^2(0,T;L^2(K_\chi))}\right)<\infty.$$

Moreover

$$\|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)} \le \|\chi\|_{L^\infty}^{1/2} \|V_n - U\|_{L^2(0,T;L^2(K_\chi))} \to 0.$$

Therefore

$$\|f_n - f\|_{L^1(0,T)} \to 0.$$

Hence $f_n \to f$ in $L^1(0,T)$.

Proposition 6.14 gives for large n, $f_n(s) \ge \eta$ for a.e.s. Therefore

$$\int_0^T f(s)\,ds = \lim_{n\to\infty} \int_0^T f_n(s)\,ds \ge \eta T > 0.$$

Hence $f \not\equiv 0$, so $U \not\equiv 0$.

Step 3: shifted weak formulation for V_n. Take any test field $\varphi \in C_c^\infty((0,T)\times\mathbb{R}^3;\mathbb{R}^3)$, and set

$$K_\varphi := \overline{\{x \in \mathbb{R}^3 : \exists s \in (0,T),\ \varphi(x,s) \neq 0\}} \subset \mathbb{R}^3, \qquad K_\varphi \text{ bounded.}$$

Since (V,P) satisfies (9) in distributions, the change of variables $\tau = \tau_n + s$ gives the shifted weak formulation

$$\begin{aligned} 0 = &\int_0^T \int_{\mathbb{R}^3} \Big(-V_n \cdot \partial_s \varphi - (V_n \otimes V_n) : \nabla\varphi + \nu \nabla V_n : \nabla \varphi - P_n \nabla \cdot \varphi \Big)\,dy\,ds \\ &+ \int_0^T \int_{\mathbb{R}^3} \Big(-a(\tau_n + s)(V_n + y \cdot \nabla V_n)\cdot\varphi - (b(\tau_n+s)\cdot\nabla V_n)\cdot\varphi \Big)\,dy\,ds. \end{aligned} \tag{13}$$

Step 4: term-by-term passage to the limit. Write $\alpha_n(s) := a(\tau_n + s)$, $\beta_n(s) := b(\tau_n+s)$.

(i) Time derivative.

By strong convergence in $L^2(0,T;L^2(K_\varphi))$,

$$\left| \int_0^T \int (V_n - U)\cdot \partial_s\varphi\,dy\,ds \right| \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \|\partial_s \varphi\|_{L^2(0,T;L^2(K_\varphi))} \to 0.$$

Hence

$$\int V_n \cdot \partial_s \varphi \to \int U \cdot \partial_s \varphi.$$

(ii) Convection term. Decompose

$$V_n \otimes V_n - U \otimes U = (V_n - U)\otimes V_n + U \otimes (V_n - U).$$

Then

$$\begin{aligned} &\left| \int_0^T \int ((V_n \otimes V_n - U\otimes U) : \nabla\varphi)\,dy\,ds \right| \\ &\quad \le \|\nabla\varphi\|_{L^\infty_{x,t}} \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \big(\|V_n\|_{L^2(0,T;L^2(K_\varphi))} + \|U\|_{L^2(0,T;L^2(K_\varphi))} \big) \to 0, \end{aligned}$$

using boundedness of V_n in $L^2(0,T;L^2(K_\varphi))$. Therefore $\int (V_n \otimes V_n) : \nabla\varphi \to \int (U\otimes U):\nabla\varphi$.

(iii) Viscous term. From weak convergence of V_n in $L^2(0,T;H^1_{\mathrm{loc}})$,

$$\int_0^T \int \nabla V_n : \nabla \varphi\,dy\,ds \to \int_0^T \int \nabla U : \nabla\varphi\,dy\,ds.$$

(iv) Pressure term. Let K be a bounded Lipschitz domain with $K_\varphi \Subset K$. Since φ is compactly supported in K,

$$\int_K (P_n)_K(s)\,\nabla\cdot\varphi(y,s)\,dy = (P_n)_K(s) \int_K \nabla \cdot \varphi(y,s)\,dy = 0.$$

Therefore

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi \, dy \, ds = \int_0^T \int_K (P_n - (P_n)_K) \nabla \cdot \varphi \, dy \, ds.$$

By Lemma 6.17,

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{in } L^{3/2}((0,T) \times K),$$

and $\nabla \cdot \varphi \in L^3((0,T) \times K)$. The compatibility Lemma 6.18 permits us to write the resulting pressure distribution as Π, modulo functions of time. Hence

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi \, dy \, ds \to \int_0^T \int_K \Pi_K \nabla \cdot \varphi \, dy \, ds.$$

The value of the last integral is independent of the chosen K, because replacing Π_K by $\Pi_K + c(s)$ changes the integral by $\int_0^T c(s) \int_K \nabla \cdot \varphi \, dy \, ds = 0$. The limit is therefore the pressure term $\int \Pi \nabla \cdot \varphi$.

(v) Drift term with a: part involving V_n. Define

$$Y_n^0(s) := \int_{\mathbb{R}^3} V_n(y,s) \cdot \varphi(y,s) \, dy, \qquad Y^0(s) := \int_{\mathbb{R}^3} U(y,s) \cdot \varphi(y,s) \, dy.$$

For each s, Cauchy–Schwarz gives

$$|Y_n^0(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \|\varphi(s)\|_{L^2(K_\varphi)}.$$

By Proposition 6.14, there is $C_0 < \infty$ such that $\sup_n \|Y_n^0\|_{L^\infty(0,T)} \le C_0$. Moreover,

$$Y_n^0 \to Y^0 \quad \text{in } L^2(0,T),$$

because $V_n \to U$ in $L^2(0,T; L^2(K_\varphi))$. Now

$$\int_0^T \alpha_n(s) Y_n^0(s) \, ds = \int_0^T (\alpha_n(s) - a_\infty) Y_n^0(s) \, ds + a_\infty \int_0^T Y_n^0(s) \, ds.$$

For the first term,

$$\left| \int_0^T (\alpha_n - a_\infty) Y_n^0 \, ds \right| \le \sup_{s \in (0,T)} |Y_n^0(s)| \, \|\alpha_n - a_\infty\|_{L^1(0,T)} \to 0.$$

For the second term, strong convergence in $L^2(0,T)$ implies $\int_0^T Y_n^0 \to \int_0^T Y^0$, so

$$\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi \, dy \, ds \to a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi \, dy \, ds.$$

Thus the signed contribution in (13) satisfies

$$-\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi \, dy \, ds \to -a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi \, dy \, ds.$$

(vi) Drift term with a: part involving $y \cdot \nabla V_n$. Since $\varphi \in C_c^\infty$, integrating by parts in y gives

$$\int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi = -\int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy.$$

Define

$$Y_n^1(s) := \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy, \qquad Y^1(s) := \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy.$$

For each s, Cauchy–Schwarz gives

$$|Y_n^1(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \|y \cdot \nabla \varphi(s) + 3\varphi(s)\|_{L^2(K_\varphi)}.$$

The $L^\infty(0,T;L^2(K_\varphi))$ bound in Proposition 6.14 gives

$$\sup_n \|Y_n^1\|_{L^\infty(0,T)} \le C_1 < \infty.$$

Also,

$$\|Y_n^1 - Y^1\|_{L^2(0,T)} \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \|y \cdot \nabla\varphi + 3\varphi\|_{L^\infty(0,T;L^2(K_\varphi))} \to 0.$$

Therefore

$$-\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi \, dy \, ds \to a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla\varphi + 3\varphi) \, dy \, ds.$$

Integrating by parts once more in the limit gives

$$a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla\varphi + 3\varphi) \, dy \, ds = -a_\infty \int_0^T \int_{\mathbb{R}^3} (y \cdot \nabla U) \cdot \varphi \, dy \, ds.$$

(vii) Term with b. Since $\beta_n(s)$ is independent of y,

$$\int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi = -\int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla\varphi) \, dy,$$

so

$$-\int_0^T \int (\beta_n \cdot \nabla V_n) \cdot \varphi \, dy \, ds = \int_0^T \int V_n \cdot (\beta_n \cdot \nabla\varphi) \, dy \, ds.$$

Let

$$Z_n(s) := \left(\int_{\mathbb{R}^3} V_n \cdot \partial_1\varphi \, dy, \int_{\mathbb{R}^3} V_n \cdot \partial_2\varphi \, dy, \int_{\mathbb{R}^3} V_n \cdot \partial_3\varphi \, dy \right),$$
$$Z(s) := \left(\int_{\mathbb{R}^3} U \cdot \partial_1\varphi \, dy, \int_{\mathbb{R}^3} U \cdot \partial_2\varphi \, dy, \int_{\mathbb{R}^3} U \cdot \partial_3\varphi \, dy \right).$$

Then

$$\int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla\varphi) \, dy = \beta_n(s) \cdot Z_n(s), \qquad \int_{\mathbb{R}^3} U \cdot (b_\infty \cdot \nabla\varphi) \, dy = b_\infty \cdot Z(s).$$

Consequently

$$\left| \int_0^T (\beta_n - b_\infty) \cdot Z_n \, ds \right| \le \|\beta_n - b_\infty\|_{L^1(0,T)} \sup_{s \in (0,T)} |Z_n(s)| \to 0,$$

because

$$|Z_n(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \left(\sum_{j=1}^3 \|\partial_j\varphi(s)\|^2_{L^2(K_\varphi)} \right)^{1/2}$$

and Proposition 6.14 bounds the right-hand side uniformly in n and s. Furthermore,

$$\|Z_n - Z\|_{L^2(0,T;\mathbb{R}^3)} \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \left(\sum_{j=1}^3 \|\partial_j\varphi\|^2_{L^\infty(0,T;L^2(K_\varphi))} \right)^{1/2} \to 0.$$

Therefore

$$\int_0^T b_\infty \cdot Z_n(s) \, ds \to \int_0^T b_\infty \cdot Z(s) \, ds.$$

Hence

$$-\int_0^T \int (\beta_n \cdot \nabla V_n) \cdot \varphi \, dy \, ds \to \int_0^T \int U \cdot (b_\infty \cdot \nabla\varphi) \, dy \, ds.$$

Since b_∞ is constant in y, integration by parts yields

$$\int_0^T \int U \cdot (b_\infty \cdot \nabla\varphi)\, dy\, ds = -\int_0^T \int (b_\infty \cdot \nabla U) \cdot \varphi\, dy\, ds.$$

Step 5: assemble the limit. Passing to the limit in (13) in each term (Steps 4(i)–4(vii)) gives

$$\int_0^T \int \big(-U\cdot\partial_s\varphi - (U\otimes U) : \nabla\varphi + \nu\nabla U : \nabla\varphi - \Pi\nabla\cdot\varphi - a_\infty(U + y\cdot\nabla U)\cdot\varphi - (b_\infty\cdot\nabla U)\cdot\varphi\big)\, dy\, ds = 0.$$

Since this holds for every compactly supported vector field φ, equation (12) holds in distributions on $(0,T)\times\mathbb{R}^3$. $\square$

6.6. **Decomposition of alternatives of the scale-collapse branch.** The cost argument in Sections 6.2 and 6.4 excludes the finite-cost scale-collapse branch. The remaining scale-collapse obstruction is treated by a decomposition of alternatives, following the local estimate and autonomous-reduction argument of the preceding sections. The conclusion separates the excluded branches from the residual alternatives; it does not assert a Liouville theorem for every autonomous negative-drift equation.

Theorem 6.21 (Windowwise retained-core alternative)**.** *Let (τ_n) be a selected fixed-length terminal window sequence of length $T>0$ for a represented Type II branch produced by the selected-core construction of the preceding sections, and write*

$$V_n(y,s) := V(y,\tau_n + s), \qquad s\in(0,T).$$

Then, after passing to a subsequence, exactly one of the following alternatives holds:

(i) *the loss-of-retained-active-core row occurs on these selected windows;*

(ii) *there exist $R>0$, a cutoff $\chi\in C_c^\infty(B_R)$, $\chi\geq 0$, and $\eta>0$ such that*

$$\int_{\mathbb{R}^3} |V_n(y,s)|^2\chi(y)\, dy \geq \eta \quad \textit{for a.e. } s\in(0,T)$$

for all sufficiently large n.

In particular, whenever the loss-of-retained-active-core row is excluded on the selected windows, the translated sequence satisfies the localized nontriviality condition in Proposition 6.14.

Proof. The selected-core construction marks, on each admissible selected window, an inner compact renormalized core carrying positive localized kinetic mass on the time set used in the window analysis. If no subsequence admitted a uniform lower floor of the displayed form with one fixed inner cutoff, then after shrinking once inside the candidate core every further selected-window subsequence would lose that retained active mass; this is exactly the loss-of-retained-active-core row in the local first-failure ledger. Therefore, once that row is excluded, a subsequence exists on which one compact inner cutoff and one positive threshold work uniformly on all remaining windows, yielding alternative (ii). The final assertion is just a restatement of (ii). $\square$

Theorem 6.22 (Autonomous modulation selection)**.** *Let a represented scale-collapse branch satisfy the modulation bounds obtained in the repaired gauge construction. Suppose that a sequence of windows $[\tau_n,\tau_n+T]$ is selected in the autonomous regime. Then, after passing to a subsequence, there exist constants $a_\infty\in\mathbb{R}$ and $b_\infty\in\mathbb{R}^3$ such that*

$$a(\tau_n+\cdot)\to a_\infty \quad \textit{in } L^1(0,T), \qquad b(\tau_n+\cdot)\to b_\infty \quad \textit{in } L^1(0,T;\mathbb{R}^3).$$

Proof. The autonomous-window selection in the repaired-gauge modulation analysis gives convergence of the modulation functions to constant autonomous parameters on the fixed window length T. Passing to the corresponding subsequence gives the displayed L^1-convergences. $\square$

Theorem 6.23 (Scale-collapse decomposition of alternatives)**.** *Let a represented Type II branch have genuine scale collapse and nonvanishing selected core. Then exactly one of the following alternatives occurs after passing to selected terminal windows:*

(i) the branch has finite scale-collapsing cost, which is excluded by Theorem 6.7;
(ii) the branch has finite absolute cost, which is excluded by Corollary 6.9;
(iii) the branch admits thick autonomous windows in the sense of Definition 6.11, satisfies the selected-window estimates of Proposition 6.14, and has autonomous modulation convergence;
(iv) the branch has a thin-drift alternative: no selected fixed-length window carries a uniform positive lower bound for the averaged weighted negative drift;
(v) the branch has an autonomous-modulation alternative: thick windows exist, but no subsequence has a constant-coefficient L^1 modulation limit in the sense needed for Theorem 6.19;
(vi) the branch has a compactness alternative: the local estimate package or the retained-core input needed for Proposition 6.14 fails on the selected windows.

Proof. The scale-collapse selection theorem decomposes terminal scale-collapse windows according to the averaged negative-drift mass

$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)M_R(\tau)\,d\tau.$$

If the accumulated cost is finite, alternatives (i) or (ii) apply according to the cost functional under consideration. If the averaged negative-drift mass stays bounded below on a subsequence, one obtains thick windows; otherwise the branch lies in the thin-drift alternative. On thick windows, either the local estimate package holds and the retained-window branch of the selected-core alternative is available, in which case Proposition 6.15 together with the retained-window branch of Theorem 6.21 give Proposition 6.14; if the local estimate package or the retained-core alternative fails on the selected windows, the branch lies in the compactness alternative. If the estimates hold but no subsequence has the required constant-coefficient modulation limit, the branch lies in the autonomous-modulation alternative. The remaining case is the thick autonomous alternative. ☐

Definition 6.24 (Retained compact scale-collapse state)**.** A represented scale-collapse branch is in the retained compact scale-collapse state if, on the selected terminal windows, it has:

(i) the repaired-gauge representation and local pressure reconstruction of theorems 3.23 and 3.35;
(ii) the compact-cylinder suitable estimates used in proposition 6.14;
(iii) thick autonomous selected windows in the sense of definition 6.11, together with a subsequential constant-coefficient L^1 modulation limit in the sense used by theorem 6.19;
(iv) bounded modulation on the selected compact cylinders;
(v) a retained active Caffarelli–Kohn–Nirenberg mass floor on some fixed inner selected cylinder;
(vi) the finite scale-collapsing-cost and finite absolute-cost alternatives have already been removed by theorem 6.7 and corollary 6.9;
(vii) no unresolved multicore, same-point cascade, separated/exterior, rough-core, autonomous-modulation-failure, or compactness-package first-failure label after the ordered local state-space tests have been applied.

Lemma 6.25 (Failure of retained compact scale-collapse is a named exit)**.** *Let a represented genuine scale-collapse branch with nonvanishing selected core be tested against definition 6.24 after the ordered scale-collapse stratification has been applied. If the branch is not in the retained* *compact scale-collapse state, then its first failed retained-state clause is one of the named local exits: representation/pressure failure, thin-drift, finite scale-cost or finite absolute-cost exclusion, compactness package first failure routed by theorem 6.31, autonomous-modulation failure, no retained active core, multicore/same-point cascade, separated/exterior concentration, or rough-core loss.*

Proof. The clauses of definition 6.24 are ordered in the same way as the local tests. Failure of clause (i) is exactly the representation or pressure-reconstruction exit. Failure of (ii) is the compactness alternative in theorem 6.23, which is routed by theorem 6.31. Failure of (iii) is either the thin-drift alternative or the autonomous-modulation alternative in that theorem. Failure of (iv) is a modulation-control exit in the repaired-gauge state-space tests. Failure of (v) means the selected scale-collapse branch carries no retained active Caffarelli–Kohn–Nirenberg core and is therefore not the retained terminal branch. Failure of (vi) is precisely one of the two cost alternatives already excluded by theorem 6.7 and corollary 6.9. Failure of (vii) is, by definition, one of the listed ordered local state-space exits. Thus no failed retained-state input remains unlabelled. □

Definition 6.26 (Selected scale-collapse compactness package)**.** Let (V_n, P_n, a_n, b_n) be a represented selected scale-collapse window sequence on $B_{2R} \times I$, where

$$V_n(y,s) = V(y, \tau_n + s), \qquad P_n(y,s) = P(y, \tau_n + s), \qquad s \in I,$$

and $I \Subset \mathbb{R}$ is a fixed model interval. The sequence satisfies the selected scale-collapse compactness package on $B_{2R} \times I$ if the following rows all hold with constants independent of n:

(i) the repaired-gauge equation, divergence-free condition, local energy inequality, and local pressure reconstruction modulo functions of time hold on the window;

(ii) there is $C_{R,I} < \infty$ such that

$$\sup_n \Big(\|V_n\|_{L^\infty(I;L^2(B_{2R}))} + \|V_n\|_{L^2(I;H^1(B_{2R}))} + \|P_n - c_{n,2R}\|_{L^{3/2}(B_{2R}\times I)} \Big) \le C_{R,I};$$

(iii) the modulation has a local selected-window bound

$$\sup_n \int_I \big(|a_n(s)| + |b_n(s)|\big)\, ds \le C_{R,I};$$

(iv) the retained selected core is active on an inner cylinder: for some $0 < r < R$, a nonnegative $\chi \in C_c^\infty(B_r)$, and $\eta_* > 0$,

$$\int_{B_r} |V_n(y,s)|^2 \chi(y)\, dy \ge \eta_* \quad \text{for a.e. } s \in I$$

after passing to a tail of the sequence;

(v) the critical local upper rows have passed uniformly: for every compact subcylinder $B_\rho \times J \Subset B_{2R} \times I$ used by the terminal local state-space test, there is $C_{\rho,J} < \infty$, independent of n, such that

$$\sup_n \|V_n\|_{L^\infty(J;L^3(B_\rho))} \le C_{\rho,J}.$$

Failure of this uniform row is not part of the retained package; it is routed by lemma 9.73.

This package is a local compact-cylinder statement. It contains no whole-space L^3 bound, no global critical profile decomposition, and no assumption that localized dissipation vanishes.

Lemma 6.27 (Selected compactness package gives local strong compactness)**.** *Assume the selected scale-collapse compactness package of definition 6.26 on $B_{2R} \times I$. Then, after passing to a subsequence,*

$$V_n \to V_\infty \quad \textit{strongly in } L^3(B_R \times I).$$

For the pressure, for every $B_\rho \Subset B_{R_1} \Subset B_R$ and every cutoff $\zeta \in C_c^\infty(B_R)$ with $\zeta \equiv 1$ on B_{R_1}, define

$$P_\infty^{\mathrm{loc}} := \mathcal{R}_i\mathcal{R}_j\big(\zeta(V_\infty \otimes V_\infty)_{ij}\big).$$

Then either

$$P_n - c_{n,R_1}(s) - P_\infty^{\mathrm{loc}} \to 0 \quad \textit{strongly in } L^{3/2}(B_\rho \times I),$$

and hence the pressure gradients converge locally modulo functions of time on the retained branch, or the branch has already entered the pressure/exterior row routed by lemma 8.22.

Proof. Choose m so large that

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-m}(B_{2R}).$$

The local $L_s^\infty L_y^2$, $L_s^2 H_y^1$, pressure, and modulation bounds in the compactness package allow the repaired-gauge equation to be tested against $H_0^m(B_{2R})$. The same estimates as in lemma 6.16 give

$$\sup_n \|\partial_s V_n\|_{L^1(I;H^{-m}(B_R))} < \infty.$$

Aubin–Lions–Simon applied to

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-m}(B_R)$$

therefore gives strong convergence in $L^2(B_R \times I)$, after passing to a subsequence. The uniform $L_s^\infty L_y^2 \cap L_s^2 H_y^1$ bound gives the standard parabolic interpolation estimate

$$\sup_n \|V_n\|_{L^{10/3}(B_R\times I)} < \infty.$$

Interpolating the strong L^2 convergence with this uniform $L^{10/3}$ bound gives strong convergence in $L^p(B_R \times I)$ for every $2 \le p < 10/3$, in particular for $p = 3$.

For the pressure, fix $B_\rho \Subset B_{R_1} \Subset B_R$ and a cutoff $\zeta \equiv 1$ on B_{R_1}, $\operatorname{supp}\zeta \Subset B_R$. With the local pressure P_∞^{loc} defined in the statement, the pressure atlas gives, modulo functions of time on B_{R_1},

$$P_n - c_{n,R_1}(s) - P_\infty^{\mathrm{loc}} = \mathcal{R}_i\mathcal{R}_j\big(\zeta(V_n \otimes V_n - V_\infty \otimes V_\infty)_{ij}\big) + H_{n,R_1}.$$

The strong $L^3(B_R \times I)$ convergence gives

$$V_n \otimes V_n \to V_\infty \otimes V_\infty \quad \text{strongly in } L^{3/2}(B_R \times I),$$

so Calderón–Zygmund continuity gives strong convergence of the local part in $L^{3/2}(B_{R_1}\times I)$. If the mean-zero harmonic remainder H_{n,R_1} does not converge to zero in $L^{3/2}(B_\rho \times I)$, then lemma 8.22 with $(q,p) = (3/2, 3/2)$ assigns the branch to the pressure/exterior row. Otherwise the pressure convergence is strong locally modulo functions of time. In all cases pressure compactness has been obtained by a local decomposition or routed to a named row; no weak lower-semicontinuity of pressure mass is used. ☐

Lemma 6.28 (Pressure-only compactness failure is a pressure/exterior row)**.** *Let a selected scale-collapse window sequence satisfy the repaired-gauge, local pressure-reconstruction, and velocity rows of definition 6.26 on $B_{2R} \times I$, and suppose that, after passing to a subsequence,*

$$V_n \to V_\infty \quad \textit{strongly in } L^3(B_R \times I).$$

Fix $B_\rho \Subset B_{R_1} \Subset B_R$ and $\zeta \in C_c^\infty(B_R)$ with $\zeta \equiv 1$ on B_{R_1}, and set

$$P_\infty^{\mathrm{loc}} := \mathcal{R}_i\mathcal{R}_j\big(\zeta(V_\infty \otimes V_\infty)_{ij}\big).$$

Write the normalized pressure defect on B_{R_1}, modulo functions of time, as

$$Q_n := P_n - c_{n,R_1}(s) - P_\infty^{\mathrm{loc}} = Q_n^{\mathrm{loc}} + H_{n,R_1},$$

where

$$Q_n^{\mathrm{loc}} = \mathcal{R}_i\mathcal{R}_j\big(\zeta(V_n \otimes V_n - V_\infty \otimes V_\infty)_{ij}\big)$$

and H_{n,R_1} is mean-zero harmonic on B_{R_1}. If the normalized pressure defect has a positive lower bound on the smaller ball,

$$\limsup_{n\to\infty} \|Q_n\|_{L^{3/2}(B_\rho\times I)} > 0,$$

then the branch is in the ordered pressure/exterior alternative.

Proof. The strong $L^3(B_R \times I)$ convergence gives

$$V_n \otimes V_n - V_\infty \otimes V_\infty \to 0 \quad \text{strongly in } L^{3/2}(B_R \times I).$$

Calderón–Zygmund continuity gives

$$Q_n^{\mathrm{loc}} \to 0 \quad \text{strongly in } L^{3/2}(B_{R_1} \times I).$$

Thus the positive lower bound for Q_n on $B_\rho \times I$ forces

$$\limsup_{n\to\infty} \|H_{n,R_1}\|_{L^{3/2}(B_\rho\times I)} > 0.$$

By lemma 8.22, together with the quantitative annular-tail converse lemma 8.21, such a non-vanishing harmonic part can only come from pressure-reconstruction failure or from a far-field pressure carrier. These are exactly the pressure/exterior rows of the ordered local state space. □

Lemma 6.29 (Failure of selected compactness produces a local defect label)**.** *Let a represented genuine scale-collapse branch have a nonvanishing selected core and be tested on canonical selected windows $B_{2R} \times I$. If the selected scale-collapse compactness package of definition 6.26 fails on those windows, then after passing to a subsequence the first failed row is one of the following local labels:*

(i) *representation or pressure-reconstruction failure;*
(ii) *autonomous-modulation failure or modulation-driven recentering;*
(iii) *loss of retained active core, including pressure-only critical-mass detection;*
(iv) *rough-core loss of local windowed H^1 control;*
(v) *exterior or separated-core concentration;*
(vi) *multicore, comparable-core, or same-point cascade;*
(vii) *a critical local L^3 upper-mass failure routed by lemma 9.73.*

If none of these labels occurs, the compactness package holds and lemma 6.27 gives strong local compactness on the selected windows.

Proof. The rows in definition 6.26 are tested in the displayed order. Failure of the repaired-gauge equation, local energy inequality, or pressure reconstruction is item (i). Failure of the local selected-window modulation bound is exactly the autonomous-modulation or recentering row, giving item (ii).

If the retained active lower floor fails, the selected candidate is not the retained terminal component. The terminal local critical-mass routing lemma lemma 9.73 records this as lower active-core loss or, when the CKN signal is carried only by pressure, as the pressure-only critical-mass row. This gives item (iii).

Assume the preceding rows hold and the local velocity/pressure estimate row fails. If the failure is the $L^2_sH^1_y$ part on a retained compact cylinder, then it is the rough-core row by theorem 10.7, giving item (iv). If the failure is that the compact-cylinder $L^\infty_sL^2_y$ or pressure mass cannot be bounded on any fixed enclosing ball, apply a finite cover of the selected retained core and the local pressure atlas. Either a positive velocity concentration remains in one member of the finite cover, or the mass leaves every fixed member. In the first case the ordered local state-space decomposition reads a compact active label: a comparable retained core, multicore label, or same-point cascade. In the second case the label is exterior or separated-core concentration. If the positive diagnostic is pressure-only, lemma 6.28 routes it to the pressure/exterior row. This gives items (iii), (v), and (vi).

It remains to consider failure of the critical local upper row. By lemma 9.73, such a failure produces an additional compact active label at the same retained frame, a comparable finite-mass retained class, a multibubble/multicore label, a cascade label, a rough-core label, or an exterior/separated label. These are exactly items (iv)–(vii). Comparable same-point classes are not left as a separate compactness defect: they are grouped by lemma 9.72 and lemma 8.3.

If no listed first failure occurs, all rows in the selected scale-collapse compactness package hold. The local strong compactness conclusion then follows from lemma 6.27. Thus selected-window compactness failure has no unlabelled terminal form. □

Lemma 6.30 (Comparable compact defects are grouped before retention)**.** *Suppose a selected compactness defect produced by lemma 6.29 is based at the same physical point as the retained selected scale-collapse core and has scale comparable to the retained frame. Then either it is absorbed into the retained compound core before the retained scale-collapse state is declared, or the branch has already entered the multicore/cascade/exterior local alternative.*

Proof. If the relative scale-center parameters stay in a compact subset of the same-point comparable chart, lemma 8.3 assembles the corresponding positive local states into one compound state, and lemma 9.72 says that any surviving comparable retained class is already absorbed into the selected retained component. If the relative parameters leave every compact subset, the ordered state-space compactification records separated scale, separated center, exterior leakage, or same-point cascade before the retained compact scale-collapse state is reached. Hence a comparable compactness defect cannot remain as a separate unresolved retained compactness failure. □

Theorem 6.31 (Selected compactness failure is routed)**.** *Let a represented genuine scale-collapse branch have nonvanishing selected core and be tested on the canonical selected windows supplied by the local state-space construction. Then the compactness alternative in theorem 6.23 is not an independent terminal exit. More precisely, either the selected scale-collapse compactness package holds on every retained compact subcylinder after passing to a subsequence, or the first failed row is routed to one of the named local alternatives: representation/pressure failure, autonomous-modulation failure, loss of retained active core or pressure-only critical mass, rough-core loss, exterior/separated-core concentration, multicore/same-point cascade, or a critical local upper-mass label routed by lemma 9.73.*

Proof. Choose a countable nested exhaustion of the retained compact selected core by compact cylinders

$$B_{R_j} \times I_j \Subset B_{2R_j} \times I_j, \qquad j = 1, 2, \ldots.$$

Apply lemma 6.29 to each member of this exhaustion in the ordered local test. If a row fails on some member, take the least j for which a failure occurs and then the first failed row in the ordered list on that cylinder. The lemma assigns it to one of the named local alternatives. If comparable compact defects occur at the retained physical point and comparable scale, then lemma 6.30 absorbs them into the retained compound core or routes them to the multicore/cascade/exterior rows.

If no row fails on the exhaustion, the package holds on each compact subcylinder. Apply lemma 6.27 on the first cylinder, then on the second along the extracted subsequence, and continue diagonally. The diagonal subsequence gives strong local L^3 compactness and pressure compactness modulo functions of time, with any nonvanishing harmonic pressure remainder routed by lemma 8.22. Therefore compactness failure is represented by a local first-failure classification. It either returns a compact retained sequence or enters a more specific named local row; it is not a separate terminal obstruction. □

Lemma 6.32 (Retained autonomous scale collapse enters the scale-rigid stratum)**.** *Let a represented Type II branch be in the retained compact scale-collapse state of definition 6.24. Suppose it has thick autonomous windows and satisfies proposition 6.14 on the same window sequence. Then the shifted selected-window sequence*

$$V_n(y, s) = V(y, \tau_n + s), \qquad P_n(y, s) = P(y, \tau_n + s),$$

restricted to a fixed compact cylinder carrying the retained active mass, is a scale-rigid local branch in the sense of definition 4.23, or it is the compact single-core branch excluded by theorem 2.18, unless the branch has already entered one of the named local alternatives excluded from the retained compact state.

Proof. The compact-cylinder suitable bounds in definition 6.24(ii) give definition 4.23(R1) on a fixed compact selected cylinder after shrinking once. The retained active CKN mass floor in (v) gives (R2) on an inner cylinder. Bounded modulation in (iv) gives (R3).

It remains to verify the scale-rigid structural clause (R4). Apply the minimal active-family extraction theorem 4.11 to the selected retained core and to the nearest previously selected active scale-center, if such a scale-center is present. If no such active relation is present, the branch is the compact single-core state rather than a genuine scale-collapse residual; this is the single-core branch excluded by theorem 2.18, not a remaining case in this lemma.

Otherwise the pair is an input to the same-point or separated-point reduction, theorems 4.18 and 4.19. Those reductions are an ordered partition. Their non-rigid outputs are exactly the named exits: another active core at comparable scale, a strict same-point cascade not yet absorbed, a separated physical point, an exterior pressure/concentration label, a rough-core label, or failure of the compact estimates/modulation convergence needed for the autonomous scale-collapse passage. The comparable, same-point, separated, and exterior outcomes are the multicore, same-point cascade, or separated/exterior alternatives of definition 4.6; the rough-core outcome is the local windowed-H^1 failure alternative; and the last two outcomes are the compactness or autonomous-modulation alternatives of theorem 6.23. All are excluded from the retained compact state by definition 6.24(vii).

After those exits are removed, the same ordered reductions leave only the structural outputs named in definition 4.23(R4): either the relative scale-center Jacobian degenerates, or the selected core is the innermost member after all strict outer same-point states have been absorbed into the perturbative remainder by lemma 4.14. Thus the shifted sequence is produced by the scale-rigid reduction and satisfies (R4) on the selected compact cylinder. □

Theorem 6.33 (Scale-collapse first failures are routed local alternatives)**.** *Let a represented branch have genuine scale collapse and a nonvanishing selected core. If one of the retained compact scale-collapse inputs fails before the branch reaches definition 6.24, then the first failure is one of the named local alternatives: finite scale cost, finite absolute scale cost, autonomous-modulation failure, compactness-package first failure routed to a named local row, exterior/separated-core loss, rough-core loss, multicore/same-point cascade, or scale-rigid residual behavior; the thin-drift first failure is routed by theorem 9.99 to one of these rows or to a closed retained row. None of these first failures remains inside the retained compact scale-collapse branch.*

Proof. Apply the ordered scale-collapse stratification theorem 6.23 and the first-failure ledger of lemma 6.25. Finite scale cost and finite absolute scale cost are the cost alternatives excluded by theorem 6.7 and corollary 6.9. Thin drift is first recorded as the scale-drift boundary face of the scale-collapse stratification, but theorem 9.99 proves that it is not terminal: persistent windows return to the thick-window ledger, while diffuse occupation blocks route to another named row or to a closed retained row. Failure of autonomous modulation is a repaired-gauge/modulation boundary face. Failure of compactness on the selected windows is routed by theorem 6.31: either the local compactness package holds after passing to a subsequence, or the first failed row is a representation/pressure row, active-core or pressure-only row, rough-core row, exterior/separated-core row, multicore/same-point cascade row, or critical local upper-mass row. Comparable same-point defects are grouped by lemma 6.30. Thus compactness failure is not a separate retained terminal state. The remaining rigid residual is the scale-rigid local branch discharged by proposition 4.40. These are exactly the named local alternatives used in the state-space decomposition, so none is a retained compact scale-collapse state. □

Theorem 6.34 (Compact scale-collapse rigidity from local routing)**.** *Let a represented Type II branch have genuine scale collapse, nonvanishing selected core, and belong to the retained compact scale-collapse state of definition 6.24. Then no compact scale-collapse branch remains. In particular, compact scale-collapse rigidity is supplied by the retained local state-space branch itself, not by an independent stationary input.*

Proof. Apply the scale-collapse stratification theorem 6.23. The finite scale-collapsing cost and finite absolute cost cases are excluded by theorem 6.7 and corollary 6.9.

The thin-drift, autonomous-modulation, and compactness alternatives are not discarded by definition. A branch satisfying the retained compact scale-collapse-state hypotheses has thick autonomous selected windows, selected-window estimates, and autonomous modulation convergence by definition 6.24(ii),(iii). Thus the thin-drift, autonomous-modulation, and compactness alternatives of theorem 6.23 are incompatible with the present retained state.

It remains to consider the thick autonomous case. By lemma 6.32, the shifted selected-window sequence is either a scale-rigid local branch or the compact single-core branch, unless the branch has already left the retained compact state. Since the present branch is retained, the exit case is unavailable. If the compact single-core branch occurs, it is excluded by theorem 2.18. If the scale-rigid branch occurs, proposition 4.40 applies: the branch either has a nonzero bounded selected-window limit, which exits the retained local Type II row by lemma 4.25, or it reduces to the compact single-core branch, again excluded by theorem 2.18. Hence the thick autonomous retained case is impossible.

□

Theorem 6.35 (Scale-collapse alternatives)**.** *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. Then one of the following mutually exclusive outcomes occurs:*

(i) the finite scale-collapsing cost exclusion applies;
(ii) the finite absolute cost exclusion applies;
(iii) the branch has a thin-drift alternative, which is routed by theorem 9.99;
(iv) the branch has an autonomous-modulation alternative;
(v) the branch has a compactness-package first failure, which is routed by theorem 6.31;
(vi) the branch enters the retained compact scale-collapse state, in which case theorem 6.34 excludes it.

Proof. Apply theorem 6.23. The finite-cost alternatives are items (i) and (ii). The thin-drift, modulation, and compactness alternatives are items (iv)–(vi) of that theorem; the thin-drift item is routed by theorem 9.99, and the compactness item is routed by theorem 6.31. It remains only to treat item (iii), the thick autonomous branch with selected-window estimates and autonomous modulation convergence. If one of the retained compact-state inputs in definition 6.24 fails, then theorem 6.33 assigns the branch to a named local alternative, hence to one of the preceding nonretained outcomes. Otherwise the branch is in the retained compact scale-collapse state and theorem 6.34 excludes it. ☐

7. Critical Tightness and Windowed-Control Estimates

This section records the local estimates used to exclude failure of uniform critical tightness and failure of local windowed H^1 control. The statements are phrased as analytic alternatives for a renormalized Navier–Stokes solution $V(\tau)$.

7.1. **Uniform critical tightness.**

Definition 7.1 (Uniform critical tightness)**.** A renormalized orbit $V(\tau)$, $\tau \geq \tau_0$, is uniformly critically tight if

$$\forall \varepsilon > 0 \ \exists R_\varepsilon < \infty : \sup_{\tau \geq \tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3 \, dy < \varepsilon.$$

The non-tight alternative is the failure of this property.

Lemma 7.2 (Critical precompactness implies uniform tightness)**.** *If*

$$\mathcal{O} := \{V(\tau) : \tau \geq \tau_0\}$$

is contained in a compact subset of $L^3(\mathbb{R}^3)$, then V is uniformly critically tight.

Proof. Compact subsets of $L^3(\mathbb{R}^3)$ are uniformly tight. Let $K \subset L^3$ be compact and fix $\varepsilon > 0$. Set $\delta = \varepsilon^{1/3}/2$. Choose a finite δ-net $f_1, \dots, f_N$ in L^3. For each i, choose R_i so that

$$\|f_i\|_{L^3(|y|>R_i)} < \delta.$$

Let $R = \max_i R_i$. For any $f \in K$, choose i with $\|f - f_i\|_3 < \delta$. Then

$$\|f\|_{L^3(|y|>R)} \leq \|f - f_i\|_3 + \|f_i\|_{L^3(|y|>R)} < \varepsilon^{1/3}.$$

Taking the cube gives

$$\int_{|y|>R} |f(y)|^3 \, dy < \varepsilon.$$

Applying this to the compact set containing $\mathcal{O}$ gives uniform critical tightness. □

Definition 7.3 (Compact core with vanishing critical remainder)**.** The orbit has a compact critical core with vanishing remainder if there exist a compact set $\mathcal{K}^{\mathrm{c}} \subset L^3(\mathbb{R}^3)$, a map $Q : [\tau_0, \infty) \to \mathcal{K}^{\mathrm{c}}$, and a remainder $r(\tau) \in L^3(\mathbb{R}^3)$, such that

$$V(\tau) = Q(\tau) + r(\tau), \qquad \lim_{\tau \to \infty} \|r(\tau)\|_{L^3} = 0,$$

and, for every $T > \tau_0$, the set

$$\{V(\tau) : \tau \in [\tau_0, T]\}$$

is uniformly L^3-tight.

Lemma 7.4 (Compact core plus vanishing remainder implies tightness)**.** *If definition 7.3 holds, then V is uniformly critically tight.*

Proof. Fix $\varepsilon > 0$ and set $\delta = \varepsilon^{1/3}$. By compactness of $\mathcal{K}^{\mathrm{c}}$, apply lemma 7.2 to $\mathcal{K}^{\mathrm{c}}$ and choose R_1 such that

$$\sup_{q \in \mathcal{K}^{\mathrm{c}}} \|q\|_{L^3(|y|>R_1)} < \delta/3.$$

Since $\|r(\tau)\|_3 \to 0$, choose T so that

$$\|r(\tau)\|_3 < \delta/3 \qquad \text{for all } \tau \geq T.$$

Then for $\tau \geq T$,

$$\|V(\tau)\|_{L^3(|y|>R_1)} \leq \|Q(\tau)\|_{L^3(|y|>R_1)} + \|r(\tau)\|_3 < 2\delta/3.$$

By finite-initial-window tightness on $[\tau_0, T]$, choose R_2 such that

$$\sup_{\tau\in[\tau_0,T]} \int_{|y|>R_2} |V(y,\tau)|^3\, dy < \varepsilon.$$

With $R = \max(R_1, R_2)$, the L^3-norm of the tail of $V(\tau)$ is $< \delta$ for all $\tau \geq T$, while the tail integral is $< \varepsilon$ on $[\tau_0, T]$. Cubing on the tail $\tau \geq T$ gives the required L^3-tail integral $< \varepsilon$ for all $\tau \geq \tau_0$. □

Definition 7.5 (No-splitting profile condition)**.** A critical profile decomposition has no critical-mass splitting if it has:

(i) a primary compact core family $\mathcal{K}^{\mathrm{c}} \subset L^3$;
(ii) asymptotic L^3-decoupling of core, secondary profiles, and remainder;
(iii) single-profile saturation, so every non-primary profile has zero critical L^3-mass;
(iv) a remainder whose L^3-norm vanishes on the renormalized tail;
(v) finite-initial-window tightness.

Lemma 7.6 (No splitting gives compact core with vanishing remainder)**.** *If definition 7.5 holds, then definition 7.3 holds.*

Proof. By single-profile saturation and nonnegative L^3-mass decoupling, every secondary profile has zero critical L^3-mass and therefore vanishes in L^3. The profile decomposition then reduces to a primary compact core plus a remainder whose L^3-norm tends to zero on the renormalized tail. These provide the core $Q(\tau)$, compact set $\mathcal{K}^{\mathrm{c}}$, and remainder $r(\tau)$ required by definition 7.3, with finite-initial-window tightness supplied as part of definition 7.5. □

Corollary 7.7 (No splitting gives uniform critical tightness)**.** *If definition 7.5 holds, then V is uniformly critically tight.*

Proof. Lemma 7.6 gives a compact core with vanishing critical remainder, and lemma 7.4 gives uniform critical tightness. □

Definition 7.8 (Finite critical profile approximation)**.** The orbit admits a finite critical profile approximation if there are finitely many profiles

$$\mathcal{B}^{II}_{NS} := \{B_1, \ldots, B_N\} \subset L^3(\mathbb{R}^3)$$

such that, for every $\delta > 0$, there is τ_δ with

$$\forall \tau \geq \tau_\delta\ \exists i(\tau) \in \{1, \ldots, N\} : \|V(\tau) - B_{i(\tau)}\|_{L^3} < \delta,$$

and the finite initial window

$$\{V(\tau) : \tau \in [\tau_0, \tau_\delta]\}$$

is uniformly L^3-tight for each $\delta > 0$.

Lemma 7.9 (Finite critical profile approximation implies tightness)**.** *If definition 7.8 holds, then V is uniformly critically tight.*

Proof. Fix $\varepsilon > 0$ and set $\delta = \varepsilon^{1/3}$. Since the profile family is finite and each $B_i \in L^3(\mathbb{R}^3)$, there is R_1 such that

$$\max_{1\leq i\leq N} \|B_i\|_{L^3(|y|>R_1)} < \delta/3.$$

Choose the approximation tolerance $\delta/3$. For $\tau \geq \tau_{\delta/3}$, pick $i(\tau)$ with $\|V(\tau) - B_{i(\tau)}\|_3 < \delta/3$. Then

$$\|V(\tau)\|_{L^3(|y|>R_1)} \leq \|V(\tau) - B_{i(\tau)}\|_3 + \|B_{i(\tau)}\|_{L^3(|y|>R_1)} < 2\delta/3.$$

By finite-initial-window tightness, choose R_2 so that the L^3 tail is $< \varepsilon$ on $[\tau_0, \tau_{\delta/3}]$. With $R = \max(R_1, R_2)$, the tail integral is $< \varepsilon$ for every $\tau \geq \tau_0$, so the orbit is uniformly L^3-tight. □

Definition 7.10 (Compact parameter realization)**.** The orbit has a compact parameter realization if there exist a compact parameter set Θ_{II}, a continuous map

$$\Theta_{II} \ni \theta \mapsto V_\theta \in L^3(\mathbb{R}^3),$$

a curve $\theta(\tau) \in \Theta_{II}$, and a remainder $r(\tau) \to 0$ in L^3, such that

$$V(\tau) = V_{\theta(\tau)} + r(\tau),$$

and, for every $T > \tau_0$, the set

$$\{V(\tau) : \tau \in [\tau_0, T]\}$$

is uniformly L^3-tight.

Lemma 7.11 (Compact realization implies tightness)**.** *If definition 7.10 holds, then V is uniformly critically tight.*

Proof. The continuous image of compact Θ_{II} in L^3 is compact. Thus

$$\mathcal{K}^c := \{V_\theta : \theta \in \Theta_{II}\}$$

is compact in L^3. The representation $V(\tau) = V_{\theta(\tau)} + r(\tau)$ with $r(\tau) \to 0$ in L^3 gives the compact-core vanishing-remainder situation, because finite-initial-window tightness is included in definition 7.10. Lemma 7.4 gives uniform critical tightness. □

Theorem 7.12 (Exclusion of the non-tight alternative)**.** *For a renormalized Type II Navier–Stokes solution, any one of the following conditions gives uniform critical tightness:*

(i) $\mathcal{O} \Subset L^3(\mathbb{R}^3)$*;*
(ii) *compact core with vanishing critical remainder;*
(iii) *no-splitting profile condition;*
(iv) *finite critical profile approximation;*
(v) *compact parameter realization.*

Proof. The five alternatives are handled respectively by lemmas 7.2, 7.4, 7.9 and 7.11 and corollary 7.7. In every case V is uniformly critically tight, so the failure of uniform critical tightness is unavailable. □

Corollary 7.13 (After exclusion of the non-tight alternative)**.** *Assume a finite-cost renormalized Type II solution not excluded by the compact cost-divergence criterion has a two-alternative classification whose only remaining alternatives are failure of uniform critical tightness and failure of tail windowed H^1 control. If definition 7.1 holds, then the only remaining alternative is failure of tail windowed H^1 control.*

Proof. The two-alternative classification gives exactly one of the two alternatives. Uniform critical tightness rules out failure of tightness, so only the windowed H^1-failure alternative remains. □

7.2. Tail windowed H^1 control.

Lemma 7.14 (Uniform local pressure control)**.** *Assume*

$$M_{3,R} := \sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(B_{4R})} < \infty$$

and

$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathbb{R}^3$$

for each $\tau \geq \tau_0$. Assume also the local tail estimate

$$\mathcal{T}_R(\tau) := \sup_{x \in B_R} \int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^5}\, dz, \qquad \sup_{\tau \geq \tau_0} \mathcal{T}_R(\tau) \leq C R^{-4} M_{3,R}^2.$$

Then there is a measurable choice of constants $c_R(\tau) \in \mathbb{R}$ such that

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_{3,R}^2.$$

Proof. The local pressure decomposition gives

$$\|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C \left(\|V(\tau)\|_{L^3(B_{4R})}^2 + R^2 \mathcal{T}_R(\tau) \right).$$

The tail hypothesis gives

$$\sup_{\tau \geq \tau_0} \mathcal{T}_R(\tau) \leq C R^{-4} M_{3,R}^2,$$

and the local critical bound gives

$$\|V(\tau)\|_{L^3(B_{4R})} \leq M_{3,R}.$$

Combining the two estimates yields

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_{3,R}^2.$$

The constants $c_R(\tau)$ can be chosen by fixing the displayed pressure decomposition, hence measurably. □

Proposition 7.15 (Renormalized Caccioppoli gives uniform windowed gradients)**.** *Assume V, P are smooth on compact renormalized cylinders and solve*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Assume local compact-cylinder critical control on the outer cylinders used in the estimate, bounded modulation

$$M_{ab} := \sup_{\tau \geq \tau_0} (|a(\tau)| + |b(\tau)|) < \infty,$$

and pressure normalized as in lemma 7.14. Then for every $R > 0$ there exists

$$C_R = C(R, \nu, M_{3,R}, M_{ab})$$

such that

$$\sup_{T \geq \tau_0 + 1} \int_T^{T+1} \int_{B_R} |\nabla V(y, \tau)|^2 \, dy \, d\tau \leq C_R.$$

Proof. Fix $R > 0$ and $T \geq \tau_0 + 1$. Use the renormalized Caccioppoli estimate with outer radius $2R$, inner radius R, and time intervals

$$I_{2R} := (T-1, T+2), \qquad I_R := [T, T+1].$$

Thus $R_{\text{out}} - R_{\text{in}} = R$ and the inner time length is 1. Keeping the pressure term before taking absolute values gives

$$\begin{aligned} \nu \int_T^{T+1} \int_{B_R} |\nabla V|^2 \leq C_R \iint_{(T-1,T+2) \times B_{2R}} |V|^2 + C_R \iint_{(T-1,T+2) \times B_{2R}} |V|^3 \\ + C_R \mathcal{P}_{T,R} + C_R M_{ab} \iint_{(T-1,T+2) \times B_{2R}} |V|^2, \end{aligned}$$

where C_R absorbs the powers of R^{-1}, R^{-2}, and ν, and

$$\mathcal{P}_{T,R} := \left| \iint_{(T-1,T+2) \times B_{2R}} P \, V \cdot \nabla \zeta \right|.$$

The L^2-term is controlled by the local critical bound and boundedness of B_{2R}:

$$\int_{B_{2R}} |V(\tau)|^2 \, dy \le |B_{2R}|^{1/3} \|V(\tau)\|^2_{L^3(B_{2R})} \le C_R M^2_{3,R}.$$

Since the outer time interval has length 3,

$$\iint_{(T-1,T+2)\times B_{2R}} |V|^2 \le C_R M^2_{3,R},$$

and similarly

$$\iint_{(T-1,T+2)\times B_{2R}} |V|^3 \le 3M^3_{3,R}.$$

For any time-dependent constant $c(\tau)$,

$$\int_{B_{2R}} c(\tau) V \cdot \nabla\zeta \, dy = -\int_{B_{2R}} c(\tau)\zeta \, \nabla \cdot V \, dy = 0.$$

Thus P may be replaced by $P - c_{2R}(\tau)$. Hölder's inequality gives

$$\mathcal{P}_{T,R} \le C_R \iint_{(T-1,T+2)\times B_{2R}} |P - c_{2R}(\tau)| \, |V| \, dy \, d\tau.$$

For almost every τ,

$$\int_{B_{2R}} |P - c_{2R}(\tau)| \, |V(\tau)| \, dy \le \|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})} \|V(\tau)\|_{L^3(B_{2R})}.$$

The pressure estimate with radius $2R$ gives

$$\|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})} \le C_R M^2_{3,R},$$

and the local critical bound gives

$$\|V(\tau)\|_{L^3(B_{2R})} \le M_{3,R}.$$

Hence $\mathcal{P}_{T,R} \le C_R M^3_{3,R}$. Substituting these estimates into Caccioppoli yields

$$\nu \int_T^{T+1} \int_{B_R} |\nabla V|^2 \le C_R \left(M^2_{3,R} + M^3_{3,R} + M_{ab} M^2_{3,R} \right),$$

uniformly in $T \ge \tau_0 + 1$. Dividing by ν and renaming the constant proves the claim. □

Corollary 7.16 (Tail windowed local H^1 bound)**.** *Under the hypotheses of proposition 7.15, for every $R > 0$,*

$$\sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_R)} \, d\tau < \infty.$$

Proof. The gradient part is proposition 7.15. The L^2-part follows from the same bounded-set Hölder estimate:

$$\int_T^{T+1} \|V(\tau)\|^2_{L^2(B_R)} \, d\tau \le C_R \int_T^{T+1} \|V(\tau)\|^2_{L^3(B_{2R})} \, d\tau \le C_R M^2_{3,R}.$$

Adding the two bounds gives the asserted $H^1(B_R)$ estimate. □

Corollary 7.17 (Exclusion of failure of local windowed H^1 control)**.** *Assume the renormalized Type II solution satisfies the repaired-gauge renormalized Navier–Stokes equation on compact cylinders, local compact-cylinder critical control*

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(B_{2R})}<\infty$$

for each relevant R, bounded modulation

$$\sup_{\tau\geq\tau_0}(|a(\tau)|+|b(\tau)|)<\infty,$$

and pressure reconstruction

$$-\Delta P=\partial_i\partial_j(V_iV_j).$$

Then failure of local windowed H^1 control,

$$\exists m\geq 1:\sup_{T\geq\tau_0+1}\int_T^{T+1}\|V(\tau)\|_{H^1(B_m)}^2\,d\tau=\infty$$

is impossible.

Proof. Apply corollary 7.16 with $R=m$. The resulting finite bound contradicts the displayed failure of local windowed H^1 control. □

7.3. Modulation and weighted forcing inputs.

Definition 7.18 (Modulation matrix Schur data)**.** Write the repaired modulation matrix in block form

$$M(V)=\begin{pmatrix}\alpha(V) & u(V)^T\\ v(V) & A(V)\end{pmatrix}.$$

Assume

$$\alpha(V(\tau))\geq\alpha_0>0,\qquad \|A(V(\tau))^{-1}\|_{\infty\to\infty}\leq C_A,$$
$$\|u(V(\tau))\|_\infty\leq C'_u,\qquad \|v(V(\tau))\|_\infty\leq C'_v,$$

and

$$S(V(\tau)):=\alpha(V(\tau))-u(V(\tau))^TA(V(\tau))^{-1}v(V(\tau))\geq\sigma_0>0.$$

Lemma 7.19 (Uniform inverse from Schur data)**.** *Under definition 7.18,*

$$\sup_{\tau\geq\tau_0}\|M(V(\tau))^{-1}\|_{\infty\to\infty}\leq\sigma_0^{-1}+3\sigma_0^{-1}C'_uC_A+\sigma_0^{-1}C_AC'_v+C_A+3C_A^2C'_vC'_u\sigma_0^{-1}.$$

Proof. For each τ, write

$$M(V(\tau))=\begin{pmatrix}\alpha & u^T\\ v & A\end{pmatrix},\qquad S=\alpha-u^TA^{-1}v.$$

By the translation-block inverse bound, A^{-1} exists and is uniformly bounded. By the Schur gap, S^{-1} exists and $|S^{-1}|\leq\sigma_0^{-1}$. The block inverse formula gives

$$M^{-1}=\begin{pmatrix}S^{-1} & -S^{-1}u^TA^{-1}\\ -A^{-1}vS^{-1} & A^{-1}+A^{-1}vS^{-1}u^TA^{-1}\end{pmatrix}.$$

Using the ℓ^∞ matrix norm and the finite dimension of the mixed blocks,

$$\|S^{-1}u^TA^{-1}\|_{\infty\to\infty}\leq 3\sigma_0^{-1}C'_uC_A,$$
$$\|A^{-1}vS^{-1}\|_{\infty\to\infty}\leq\sigma_0^{-1}C_AC'_v,$$

and

$$\|A^{-1}vS^{-1}u^TA^{-1}\|_{\infty\to\infty}\leq 3C_A^2C'_vC'_u\sigma_0^{-1}.$$

Adding the four block estimates gives the displayed uniform bound. □

Lemma 7.20 (Scale-normalization nondegeneracy gives Schur data)**.** *Assume the scale normalization constraint is satisfied, the scale derivative obeys*

$$\alpha(V) = DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0 > 0,$$

the translation block has a uniform inverse bound, and the mixed blocks obey the weighted first-moment bound and the Schur-compatible estimate

$$\alpha(V(\tau)) - u(V(\tau))^T A(V(\tau))^{-1} v(V(\tau)) \geq \frac{\alpha_0}{2}.$$

Then the hypotheses of definition 7.18 hold with $\sigma_0 = \alpha_0/2$.

Proof. The scale-normalization identity gives $\alpha(V) = DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0$, hence the scale transversality bound. The translation-block inverse hypothesis gives the uniform bound on A^{-1}. The mixed-block hypotheses give the bounds on u and v, and the displayed estimate is the Schur gap with $\sigma_0 = \alpha_0/2$. □

Definition 7.21 (Weighted scale-force components)**.** Let $0 < p < 3$ and set

$$H(V, P) = \nu\Delta V - (V \cdot \nabla)V - \nabla P.$$

Define

$$I_\Delta(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V \cdot \Delta V \, dy,$$

$$I_{\mathrm{nl}}(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V \cdot ((V \cdot \nabla)V) \, dy,$$

and

$$I_P(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V \cdot \nabla P \, dy,$$

whenever these integrals are absolutely convergent or are defined by the distributional pairing used for the renormalized solution.

Lemma 7.22 (Scale-force decomposition)**.** *If*

$$\sup_\tau |I_\Delta(\tau)| \leq C_\Delta, \qquad \sup_\tau |I_{\mathrm{nl}}(\tau)| \leq C_{\mathrm{nl}}, \qquad \sup_\tau |I_P(\tau)| \leq C_P,$$

then

$$\sup_{\tau \geq \tau_0} \left| \int |y|^{-p}|V|V \cdot H(V, P) \, dy \right| \leq \nu C_\Delta + C_{\mathrm{nl}} + C_P.$$

Proof. By the definition of H,

$$\int |y|^{-p}|V|V \cdot H(V, P) \, dy = \nu I_\Delta - I_{\mathrm{nl}} - I_P.$$

The triangle inequality gives the displayed uniform bound. □

Lemma 7.23 (Weighted fourth moment controls the nonlinear scale component)**.** *Assume $V(\tau)$ is divergence-free and belongs to the approximation class needed to justify the weighted integration by parts. If*

$$\sup_{\tau \geq \tau_0} \int_{\mathbb{R}^3} |y|^{-p-1}|V(y, \tau)|^4 \, dy < \infty,$$

then

$$|I_{\mathrm{nl}}(\tau)| \leq \frac{p}{3} \int_{\mathbb{R}^3} |y|^{-p-1}|V(y, \tau)|^4 \, dy.$$

Proof. For smooth compactly supported approximants,

$$|V|V\cdot((V\cdot\nabla)V)=\frac{1}{3}V\cdot\nabla(|V|^3).$$

Since $\nabla\cdot V=0$, integration by parts gives

$$I_{\rm nl}=\frac{1}{3}\int|y|^{-p}V\cdot\nabla(|V|^3)\,dy=-\frac{1}{3}\int|V|^3V\cdot\nabla(|y|^{-p})\,dy.$$

Because

$$\nabla(|y|^{-p})=-p|y|^{-p-2}y,$$

we obtain

$$I_{\rm nl}=\frac{p}{3}\int|y|^{-p-2}(V\cdot y)|V|^3\,dy.$$

Therefore

$$|I_{\rm nl}|\le\frac{p}{3}\int|y|^{-p-1}|V|^4\,dy.$$

Taking the supremum in τ proves the assertion. The general admissible case follows by the same approximation convention used for the weighted scale normalization. □

Corollary 7.24 (Pointwise scale-force bound)**.** *If*

$$\sup_\tau|I_\Delta(\tau)|<\infty,\qquad\sup_{\tau\ge\tau_0}\int|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty,\qquad\sup_\tau|I_P(\tau)|<\infty,$$

then

$$\sup_{\tau\ge\tau_0}\left|\int|y|^{-p}|V|V\cdot H(V,P)\,dy\right|<\infty.$$

Proof. Lemma 7.23 bounds $I_{\rm nl}$. Together with the Laplacian and pressure bounds, this gives the hypotheses of lemma 7.22. □

Lemma 7.25 (Bounded modulation from the modulation system)**.** *Assume*

$$M(V(\tau))\begin{pmatrix}a(\tau)\\b(\tau)\end{pmatrix}=-F(V(\tau)),$$

$$\sup_{\tau\ge\tau_0}\|M(V(\tau))^{-1}\|_{\infty\to\infty}<\infty,\qquad\sup_{\tau\ge\tau_0}\|F(V(\tau))\|_\infty<\infty.$$

Then

$$\sup_{\tau\ge\tau_0}(|a(\tau)|+\|b(\tau)\|_\infty)<\infty.$$

Proof. The system gives

$$\begin{pmatrix}a(\tau)\\b(\tau)\end{pmatrix}=-M(V(\tau))^{-1}F(V(\tau)),$$

so

$$\max\{|a(\tau)|,\|b(\tau)\|_\infty\}\le\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty.$$

Taking the supremum in τ gives bounded modulation. □

Proposition 7.26 (Windowed H^1 exclusion from modulation bounds)**.** *Assume a renormalized Type II solution has the repaired-gauge orbit and pressure reconstruction, bounded critical L^3 norm, uniformly invertible modulation matrix, uniformly bounded modulation forcing vector, and the compact-cylinder regularity needed for Caccioppoli. Then*

$$\forall R>0:\qquad\sup_{T\ge\tau_0+1}\int_T^{T+1}\|V(\tau)\|^2_{H^1(B_R)}\,d\tau<\infty,$$

and failure of local windowed H^1 control is excluded for this solution.

Proof. The inverse matrix and forcing bounds imply bounded modulation by lemma 7.25. The renormalized solution supplies the renormalized equation and pressure reconstruction. Bounded critical norm gives the local compact-cylinder critical bounds, and the regularity hypothesis allows the Caccioppoli estimate. Therefore corollary 7.16 applies. Failure of local windowed H^1 control is the negation of the displayed tail windowed H^1 estimate on some bounded ball. □

Corollary 7.27 (Two-alternative exclusion after windowed control)**.** *Assume a finite-cost renormalized Type II solution not excluded by the compact cost-divergence criterion has reached the two-alternative classification. If the hypotheses of proposition 7.26 are supplied for that solution, then failure of local windowed H^1 control is excluded. If, in addition, uniform critical tightness is available, then no finite-cost alternative remains outside the compact cost-divergence criterion.*

Proof. Proposition 7.26 gives the positive tail windowed H^1 bound, contradicting failure of local windowed H^1 control. If uniform critical tightness is also available, then failure of tightness is excluded as well, and the two-alternative classification has no remaining finite-cost alternative outside the compact cost-divergence criterion. □

Theorem 7.28 (Exclusion of local windowed H^1 failure)**.** *If every renormalized Type II solution in the class satisfies the hypotheses of proposition 7.26, then no such solution can satisfy*

$$\exists m \geq 1 : \sup_{T \geq \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_m)}\, d\tau = \infty.$$

Proof. Apply proposition 7.26 to each renormalized solution. The displayed condition is the negation of the resulting windowed H^1-bound for $R = m$. □

7.4. **Window-Integrated Modulation Forcing and Weighted Scale Components.** This subsection records sufficient analytic estimates for the modulation forcing. The estimates are stated directly for the differentiated modulation conditions and for the weighted scale component; no pointwise modulation bound is used unless it is explicitly assumed.

Let the modulation forcing operator be

$$H(V, P) := \nu \Delta V - (V \cdot \nabla)V - \nabla P.$$

For gauge functionals G_k, $0 \leq k \leq 3$, define

$$F_k(V) := DG_k(V)[H(V, P)].$$

Thus the modulation system has the form

$$M(V(\tau)) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -F(V(\tau)).$$

Lemma 7.29 (Forcing bound for the translation constraints)**.** *Let*

$$G_j(V) = \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y)\, dy, \qquad 1 \leq j \leq 3,$$

where $\psi_R \in C_c^\infty(B_{2R})$, and set

$$\Phi_j(y) := y_j \psi_R(y).$$

For

$$F_j(V) := DG_j(V)[H(V, P)] = 2 \int_{\mathbb{R}^3} \Phi_j\, V \cdot H(V, P)\, dy,$$

assume, uniformly for $\tau \geq \tau_0$,

$$\|V(\tau)\|_{L^3(B_{2R})} \leq M_3, \qquad \|\nabla V(\tau)\|_{L^2(B_{2R})} \leq M_\nabla,$$

and

$$\inf_{c\in\mathbb{R}} \|P(\tau) - c\|_{L^{3/2}(B_{2R})} \leq M_P.$$

Then

$$\sup_{\tau\geq\tau_0} \max_{1\leq j\leq 3} |F_j(V(\tau))| \leq C(R,\nu,\psi_R)\left(M_\nabla^2 + M_3^2 + M_3^3 + M_P M_3\right).$$

Proof. Fix j. Since Φ_j is supported in B_{2R},

$$F_j = 2\nu I_\Delta - 2I_{\mathrm{nl}} - 2I_P,$$

where

$$I_\Delta := \int \Phi_j V\cdot\Delta V, \qquad I_{\mathrm{nl}} := \int \Phi_j V\cdot(V\cdot\nabla V), \qquad I_P := \int \Phi_j V\cdot\nabla P.$$

For the Laplacian term, using compact support of Φ_j and summing over the component index α,

$$\begin{aligned} I_\Delta &= \sum_{\alpha=1}^{3}\int \Phi_j V_\alpha \Delta V_\alpha\, dy \\ &= -\sum_{\alpha,k}\int \partial_k(\Phi_j V_\alpha)\partial_k V_\alpha\, dy \\ &= -\int \Phi_j|\nabla V|^2\, dy - \sum_{\alpha,k}\int(\partial_k\Phi_j)V_\alpha\partial_k V_\alpha\, dy. \end{aligned}$$

Equivalently,

$$I_\Delta = -\int \Phi_j|\nabla V|^2 - \int(\nabla\Phi_j\cdot\nabla V)\cdot V.$$

Consequently, by the Cauchy–Schwarz inequality,

$$|I_\Delta| \leq \|\Phi_j\|_{L^\infty}\|\nabla V\|^2_{L^2(B_{2R})} + \|\nabla\Phi_j\|_{L^\infty}\|V\|_{L^2(B_{2R})}\|\nabla V\|_{L^2(B_{2R})}.$$

Hölder's inequality on the bounded set B_{2R} gives

$$\|V\|_{L^2(B_{2R})} \leq |B_{2R}|^{1/6}\|V\|_{L^3(B_{2R})} \leq C_R M_3.$$

Therefore

$$|I_\Delta| \leq C_R(M_\nabla^2 + M_3 M_\nabla) \leq C_R(M_\nabla^2 + M_3^2).$$

The last inequality is Young's inequality, $ab \leq (a^2+b^2)/2$.

For the nonlinear term,

$$V\cdot(V\cdot\nabla V) = V_i V_\alpha\partial_i V_\alpha = \frac{1}{2}V_i\partial_i(|V|^2) = \frac{1}{2}V\cdot\nabla(|V|^2).$$

Since $\nabla\cdot V = 0$ and Φ_j is compactly supported,

$$I_{\mathrm{nl}} = \frac{1}{2}\int \Phi_j V\cdot\nabla(|V|^2) = -\frac{1}{2}\int |V|^2 V\cdot\nabla\Phi_j.$$

Thus

$$|I_{\mathrm{nl}}| \leq \frac{1}{2}\|\nabla\Phi_j\|_{L^\infty}\|V\|^3_{L^3(B_{2R})} \leq C_R M_3^3.$$

For the pressure term, replace P by $P - c(\tau)$. Since $\nabla\cdot V = 0$, the identity

$$\nabla\cdot(\Phi_j V) = V\cdot\nabla\Phi_j$$

holds distributionally. Therefore

$$I_P = \int \Phi_j V\cdot\nabla(P-c) = -\int(P-c)V\cdot\nabla\Phi_j.$$

Consequently

$$|I_P| \le \|\nabla \Phi_j\|_{L^\infty} \|P - c\|_{L^{3/2}(B_{2R})} \|V\|_{L^3(B_{2R})} \le C_R M_P M_3.$$

Taking the infimum over c, summing the three estimates in F_j, and then taking the supremum in τ proves the bound. □

Remark 7.30 (Local source of the gradient bound). The pointwise local gradient bound in lemma 7.29 is not a free terminal assumption in the retained proof. Whenever the pointwise lemma is invoked on a retained compact window, the bound is supplied at admissible times by the local Caccioppoli/regularity package, or else the first failed row is the local H^1-loss or rough-core row before the translation estimate is used. Thus the translation forcing estimate never obtains windowed H^1-control by feeding that same conclusion back into its hypotheses.

Definition 7.31 (Weighted scale forcing)**.** Let

$$G_{\mathrm{sc}}(V) = \int_{\mathbb{R}^3} |y|^{-p} |V|^3 \, dy - \Theta_0, \qquad 0 < p < 3.$$

The weighted scale component is

$$F_0(V) := DG_{\mathrm{sc}}(V)[H(V,P)] = 3 \int_{\mathbb{R}^3} |y|^{-p} |V| V \cdot H(V,P) \, dy.$$

All weighted pairings in this subsection are assumed to be defined either as absolutely convergent integrals or as specified distributional pairings represented by $L^1_{\mathrm{loc}}(d\tau)$ functions.

Lemma 7.32 (Weighted scale forcing bound)**.** *Assume*

$$\sup_{\tau \ge \tau_0} \left| \int_{\mathbb{R}^3} |y|^{-p} |V| V \cdot H(V,P) \, dy \right| < \infty.$$

Then

$$\sup_{\tau \ge \tau_0} |F_0(V(\tau))| < \infty.$$

Proof. By the derivative formula in definition 7.31,

$$|F_0(V(\tau))| \le 3 \left| \int_{\mathbb{R}^3} |y|^{-p} |V| V \cdot H(V,P) \, dy \right|.$$

Taking the supremum in τ gives the result. □

Corollary 7.33 (Pointwise modulation forcing bound)**.** *Assume the hypotheses of lemma 7.29 for the three translation components and the weighted scale forcing hypothesis of lemma 7.32. Then*

$$\sup_{\tau \ge \tau_0} \|F(V(\tau))\|_\infty < \infty.$$

Proof. The vector $F(V) \in \mathbb{R}^4$ has the scale component F_0 and the three translation components F_j, $1 \le j \le 3$. The scale component is bounded by lemma 7.32, and the translation components are bounded by lemma 7.29. Taking the maximum over the four components proves the ℓ^∞-bound. □

Corollary 7.34 (Analytic inputs for pointwise modulation forcing)**.** *Pointwise boundedness of the modulation forcing follows from the following analytic estimates:*

(i) *local L^3, local H^1, and local pressure-oscillation control on the support of the compactly supported centering constraints;*

(ii) *weighted control of the Laplacian, pressure, and fourth-moment terms in the scale component, sufficient to bound the weighted scale forcing.*

The scale-component estimate is independent of the algebraic scale transversality identity

$$DG_{\rm sc}(V)[Z_{\rm sc}(V)] = p\Theta_0.$$

Proof. The first item is the set of hypotheses used in lemma 7.29. The second item bounds the weighted scale component through lemma 7.32, or through the explicit decomposition in corollary 7.51 below in the corresponding integrated setting. Corollary 7.33 then gives the modulation forcing bound. The displayed transversality identity controls the modulation matrix, not the forcing vector, so it is a separate algebraic input. □

Lemma 7.35 (Window-integrated forcing gives window-integrated modulation)**.** *Assume*

$$\sup_{\tau\ge\tau_0} \|M(V(\tau))^{-1}\|_{\infty\to\infty} \le C_M < \infty$$

and

$$\sup_{T\ge\tau_0} \int_T^{T+1} \|F(V(\tau))\|_\infty \, d\tau \le C_F < \infty.$$

Then

$$\sup_{T\ge\tau_0} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau \le 2C_M C_F.$$

Proof. For almost every τ,

$$\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -M(V(\tau))^{-1} F(V(\tau)).$$

Hence

$$\max\{|a(\tau)|, \|b(\tau)\|_\infty\} \le C_M \|F(V(\tau))\|_\infty.$$

Since

$$|a(\tau)| + \|b(\tau)\|_\infty \le 2\max\{|a(\tau)|, \|b(\tau)\|_\infty\},$$

we have, for every $T \ge \tau_0$,

$$\begin{aligned} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau &\le 2C_M \int_T^{T+1} \|F(V(\tau))\|_\infty \, d\tau \\ &\le 2C_M C_F. \end{aligned}$$

Taking the supremum over T proves the claim. □

Lemma 7.36 (Selection from an L^1 modulation bound)**.** *Let $I_n = [T_n, T_n + 1]$, where $T_n \to \infty$. Let*

$$d_n : I_n \to [0, \infty), \qquad q_n : I_n \to [0, \infty)$$

be measurable functions satisfying

$$\int_{I_n} d_n(\tau) \, d\tau \to 0, \qquad \sup_n \int_{I_n} q_n(\tau) \, d\tau \le Q < \infty.$$

Then, after choosing measurable representatives, there are times $\tau_n \in I_n$ such that

$$d_n(\tau_n) \to 0$$

and, if $Q > 0$,

$$q_n(\tau_n) \le 2Q \qquad \textit{for every } n.$$

If $Q = 0$, the times may be chosen so that

$$q_n(\tau_n) = 0, \qquad d_n(\tau_n) \to 0.$$

Proof. Assume first that $Q > 0$, and set

$$E_n := \{\tau \in I_n : q_n(\tau) \le 2Q\}.$$

By Chebyshev's inequality,

$$|I_n \setminus E_n| \le \frac{1}{2Q} \int_{I_n} q_n(\tau)\, d\tau \le \frac{1}{2}.$$

Thus $|E_n| \ge 1/2$. Since $d_n \ge 0$,

$$\inf_{\tau \in E_n} d_n(\tau) \le \frac{1}{|E_n|} \int_{E_n} d_n(\tau)\, d\tau \le 2 \int_{I_n} d_n(\tau)\, d\tau.$$

Choose $\tau_n \in E_n$, outside the null set on which the representatives are undefined, such that

$$d_n(\tau_n) \le 2 \int_{I_n} d_n(\tau)\, d\tau + \frac{1}{n}.$$

The right-hand side tends to zero, hence $d_n(\tau_n) \to 0$, and the definition of E_n gives $q_n(\tau_n) \le 2Q$.

If $Q = 0$, then $q_n = 0$ almost everywhere on I_n. Applying the same averaging argument to d_n on the full-measure set where $q_n = 0$ gives the asserted times. □

Corollary 7.37 (Selection on unit windows)**.** *Assume*

$$q(\tau) := |a(\tau)| + \|b(\tau)\|_\infty$$

satisfies

$$\sup_{T \ge \tau_0} \int_T^{T+1} q(\tau)\, d\tau < \infty.$$

For every sequence of unit windows $I_n = [T_n, T_n + 1]$ and every nonnegative measurable density d_n satisfying

$$\int_{I_n} d_n(\tau)\, d\tau \to 0,$$

there are times $\tau_n \in I_n$ such that

$$d_n(\tau_n) \to 0, \qquad \sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty.$$

Proof. Apply lemma 7.36 with $q_n = q|_{I_n}$ and

$$Q = \sup_n \int_{I_n} q(\tau)\, d\tau.$$

This Q is finite because of the uniform window L^1-bound. If $Q > 0$, the selected times satisfy

$$|a(\tau_n)| + \|b(\tau_n)\|_\infty = q(\tau_n) \le 2Q.$$

If $Q = 0$, the same lemma gives $q(\tau_n) = 0$. In both cases $d_n(\tau_n) \to 0$ and the modulation is uniformly bounded along the selected times. □

Lemma 7.38 (Product-form windowed modulation estimate)**.** *Assume*

$$\sup_{T \ge \tau_0} \int_T^{T+1} \|M(V(\tau))^{-1}\|_{\infty \to \infty} \|F(V(\tau))\|_\infty\, d\tau < \infty.$$

Then

$$\sup_{T \ge \tau_0} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big)\, d\tau < \infty.$$

Proof. The modulation system gives

$$\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -M(V(\tau))^{-1}F(V(\tau)).$$

Therefore

$$|a(\tau)| + \|b(\tau)\|_\infty \le 2\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty.$$

Integrating this pointwise estimate on $[T, T+1]$ and taking the supremum in T gives the result. □

Lemma 7.39 (Integrated translation forcing bound)**.** *Let G_j, Φ_j, $H(V,P)$, and $F_j(V)$ be as in lemma 7.29. Assume that, for some $R > 0$,*

$$\sup_{\tau\ge\tau_0} \|V(\tau)\|_{L^3(B_{2R})} \le M_3,$$

$$\sup_{T\ge\tau_0} \int_T^{T+1} \|\nabla V(\tau)\|^2_{L^2(B_{2R})}\, d\tau \le A_R,$$

and

$$\sup_{\tau\ge\tau_0} \inf_{c\in\mathbb{R}} \|P(\tau) - c\|_{L^{3/2}(B_{2R})} \le M_P.$$

Then

$$\sup_{T\ge\tau_0} \int_T^{T+1} \max_{1\le j\le 3} |F_j(V(\tau))|\, d\tau \le C(R,\nu,\psi_R)(A_R + M_3^2 + M_3^3 + M_P M_3).$$

Proof. Fix j. The proof of lemma 7.29 gives, for almost every τ,

$$|F_j(V(\tau))| \le C(R,\nu,\psi_R)\left(\|\nabla V(\tau)\|^2_{L^2(B_{2R})} + M_3\|\nabla V(\tau)\|_{L^2(B_{2R})} + M_3^3 + M_P M_3\right).$$

Integrate this estimate over $I_T = [T, T+1]$. The first term contributes at most A_R. For the mixed term, Cauchy's inequality on the unit interval gives

$$\int_{I_T} M_3\|\nabla V(\tau)\|_{L^2(B_{2R})}\, d\tau \le M_3\left(\int_{I_T} \|\nabla V(\tau)\|^2_{L^2(B_{2R})}\, d\tau\right)^{1/2} \le M_3 A_R^{1/2}.$$

Since

$$M_3 A_R^{1/2} \le \frac{1}{2}(M_3^2 + A_R),$$

the mixed term is bounded by $C(A_R + M_3^2)$. The terms M_3^3 and $M_P M_3$ are constant on the unit window. Hence

$$\int_T^{T+1} |F_j(V(\tau))|\, d\tau \le C(R,\nu,\psi_R)(A_R + M_3^2 + M_3^3 + M_P M_3).$$

Since $\max_{1\le j\le 3}|F_j| \le \sum_{j=1}^3 |F_j|$, taking the maximum over the three centering constraints changes only the constant. □

Remark 7.40 (Independence of the windowed gradient hypothesis)*.* The windowed gradient input in lemma 7.39 is an independent hypothesis whenever the integrated translation estimate is used to prove windowed H^1-control. It may come, for example, from a preliminary Caccioppoli estimate depending only on already available data, from a direct approximation or regularity estimate on the support of the centering functions, or from an independent verification obtained without using the conclusion of the windowed H^1 estimate under consideration.

Lemma 7.41 (Window-integrated scale forcing)**.** *Assume the window-integrated weighted scale-forcing bound*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|d\tau<\infty,$$

where the weighted pairing is interpreted as in definition 7.31. Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}|F_0(V(\tau))|\,d\tau<\infty.$$

Proof. By the identity

$$F_0(V)=3\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy$$

we have, for each $T\geq\tau_0$,

$$\int_T^{T+1}|F_0(V(\tau))|\,d\tau=3\int_T^{T+1}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|d\tau.$$

Taking the supremum over T gives the asserted L^1-bound. □

Lemma 7.42 (Full integrated modulation forcing)**.** *Assume the integrated translation forcing estimate*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau<\infty$$

and the window-integrated weighted scale-forcing bound in lemma 7.41. Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|F(V(\tau))\|_\infty\,d\tau<\infty.$$

Proof. The vector $F(V)\in\mathbb{R}^4$ consists of F_0 and the three translation components F_j, $1\leq j\leq 3$. Since

$$\|F(V(\tau))\|_\infty\leq|F_0(V(\tau))|+\max_{1\leq j\leq 3}|F_j(V(\tau))|,$$

for almost every τ. Integrating this inequality on $[T,T+1]$ gives

$$\int_T^{T+1}\|F(V(\tau))\|_\infty\,d\tau\leq\int_T^{T+1}|F_0(V(\tau))|\,d\tau+\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau.$$

The first term is uniformly bounded in T by lemma 7.41, and the second term is uniformly bounded by the assumed integrated translation forcing estimate. Taking the supremum over T proves the claim. □

Corollary 7.43 (Integrated modulation criterion on selected windows)**.** *Assume the uniform inverse bound for $M(V(\tau))$, the integrated translation forcing estimate, and the window-integrated weighted scale-forcing bound. Then*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\big(|a(\tau)|+\|b(\tau)\|_\infty\big)\,d\tau<\infty.$$

Consequently every sequence of unit windows with vanishing nonnegative measurable density contains selected times where this density vanishes asymptotically and the modulation remains uniformly bounded.

Proof. Lemma 7.42 gives the window-integrated bound on $\|F(V)\|_\infty$. Combining this with the uniform inverse bound in lemma 7.35 gives the window-integrated modulation bound. The selection conclusion is corollary 7.37. □

Definition 7.44 (Weighted scale-force components)**.** Let $w(y) = |y|^{-p}$, $0 < p < 3$. Define

$$F_0^{\Delta}(V) := 3\nu \int_{\mathbb{R}^3} w|V|V \cdot \Delta V \, dy,$$

$$F_0^{\mathrm{nl}}(V) := -3 \int_{\mathbb{R}^3} w|V|V \cdot (V \cdot \nabla V) \, dy,$$

and

$$F_0^{P}(V, P) := -3 \int_{\mathbb{R}^3} w|V|V \cdot \nabla P \, dy.$$

The three pairings are required to be compatible with the decomposition of $H(V, P)$ into diffusion, transport, and pressure.

Lemma 7.45 (Integrated scale-force decomposition)**.** *For every time at which the three pairings in definition 7.44 are defined,*

$$F_0(V(\tau)) = F_0^{\Delta}(V(\tau)) + F_0^{\mathrm{nl}}(V(\tau)) + F_0^{P}(V(\tau), P(\tau)).$$

Proof. Substitute

$$H(V, P) = \nu\Delta V - (V \cdot \nabla)V - \nabla P$$

into

$$F_0(V) = 3 \int w|V|V \cdot H(V, P) \, dy.$$

Then

$$F_0(V) = 3\nu \int w|V|V \cdot \Delta V \, dy - 3 \int w|V|V \cdot (V \cdot \nabla V) \, dy$$
$$- 3 \int w|V|V \cdot \nabla P \, dy.$$

The three terms coincide with $F_0^{\Delta}(V)$, $F_0^{\mathrm{nl}}(V)$, and $F_0^{P}(V, P)$, respectively. Thus linearity of the weighted pairing gives the stated decomposition. ☐

Lemma 7.46 (Component bounds imply integrated scale-force control)**.** *Assume*

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^{\Delta}(V(\tau))| \, d\tau < \infty,$$

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^{\mathrm{nl}}(V(\tau))| \, d\tau < \infty,$$

and

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^{P}(V(\tau), P(\tau))| \, d\tau < \infty.$$

Then the window-integrated weighted scale-forcing bound in lemma 7.41 holds.

Proof. By lemma 7.45,

$$|F_0(V(\tau))| \leq |F_0^{\Delta}(V(\tau))| + |F_0^{\mathrm{nl}}(V(\tau))| + |F_0^{P}(V(\tau), P(\tau))|$$

for almost every τ. Integrating on $[T, T + 1]$, taking the supremum in T, and applying the three component bounds gives

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0(V(\tau))| \, d\tau < \infty.$$

Since F_0 is three times the weighted pairing in definition 7.31, this is equivalent to the window-integrated weighted scale-forcing bound. ☐

Definition 7.47 (Singular-weight convective integration by parts)**.** The singular-weight convective integration-by-parts condition means that, for almost every τ,

$$-\int_{\mathbb{R}^3} w\, V \cdot \nabla(|V|^3)\, dy = \int_{\mathbb{R}^3} |V|^3 V \cdot \nabla w\, dy,$$

with no boundary contribution from $y = 0$ or from spatial infinity, either classically or by convergence of smooth compactly supported divergence-free approximants.

Lemma 7.48 (Convective scale-force bound)**.** *Assume the condition of definition 7.47 and*

$$\sup_{T \geq \tau_0} \int_T^{T+1} \int_{\mathbb{R}^3} |y|^{-p-1} |V(y,\tau)|^4\, dy\, d\tau < \infty.$$

Then

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^{\mathrm{nl}}(V(\tau))|\, d\tau < \infty.$$

More precisely,

$$|F_0^{\mathrm{nl}}(V(\tau))| \leq p \int_{\mathbb{R}^3} |y|^{-p-1} |V(y,\tau)|^4\, dy$$

for almost every τ.

Proof. Fix such a time τ and suppress it from the notation. Since

$$\partial_i(|V|^3) = 3|V| V_j \partial_i V_j$$

for each i, and therefore

$$V \cdot \nabla(|V|^3) = 3|V| V_i V_j \partial_i V_j = 3|V|\, V \cdot ((V \cdot \nabla)V),$$

we have

$$w|V| V \cdot (V \cdot \nabla V) = \frac{1}{3} w\, V \cdot \nabla(|V|^3).$$

Therefore

$$F_0^{\mathrm{nl}}(V) = -3 \int w|V| V \cdot (V \cdot \nabla V) = -\int w\, V \cdot \nabla(|V|^3).$$

Using $\nabla \cdot V = 0$ and definition 7.47,

$$-\int w\, V \cdot \nabla(|V|^3) = \int |V|^3 V \cdot \nabla w.$$

Since

$$\nabla w(y) = -p|y|^{-p-2} y, \qquad y \neq 0,$$

we have

$$|V \cdot \nabla w| = p|y|^{-p-2} |V \cdot y| \leq p|y|^{-p-1} |V|.$$

Thus

$$|F_0^{\mathrm{nl}}(V)| \leq p \int |y|^{-p-1} |V|^4\, dy.$$

Integrating over $[T, T+1]$ and taking the supremum in T proves the window-integrated convective bound. ∎

Definition 7.49 (Annular convective regularity)**.** Annular convective regularity means that, for almost every τ, $V(\tau)$ is divergence free and, on every compact annulus $0 < r < |y| < R < \infty$, the chain-rule identities

$$\partial_i(|V|^3) = 3|V| V_j \partial_i V_j, \qquad V \cdot \nabla(|V|^3) = 3|V| V_i V_j \partial_i V_j$$

hold in the sense of distributions, either classically or by passage to the limit from smooth divergence-free approximants on annuli.

Lemma 7.50 (Annular regularity gives singular-weight integration by parts)**.** *Assume annular convective regularity and the weighted L^4-window bound*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy\,d\tau<\infty.$$

Then the singular-weight convective integration-by-parts condition of definition 7.47 holds for almost every τ.

Proof. Fix a time τ for which

$$\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty$$

and annular convective regularity holds. Such times form a full-measure set on every unit window.

Let $\eta\in C^\infty([0,\infty))$ satisfy $0\leq\eta\leq 1$, $\eta(s)=0$ for $s\leq 1$, and $\eta(s)=1$ for $s\geq 2$. Define

$$\chi_{\varepsilon,R}(y):=\eta(|y|/\varepsilon)\eta(R/|y|).$$

Then $\chi_{\varepsilon,R}$ is supported in $\{\varepsilon<|y|<R\}$, equals one on $\{2\varepsilon<|y|<R/2\}$, and satisfies

$$|\nabla\chi_{\varepsilon,R}(y)|\leq C\varepsilon^{-1}\mathbf{1}_{\{\varepsilon<|y|<2\varepsilon\}}+CR^{-1}\mathbf{1}_{\{R/2<|y|<R\}}.$$

By annular convective regularity, the following integration by parts is valid on the annulus $\{\varepsilon<|y|<R\}$. Since $\nabla\cdot V=0$ and $w\chi_{\varepsilon,R}|V|^3V$ is compactly supported in this annulus,

$$0=\int\nabla\cdot\left(w\chi_{\varepsilon,R}|V|^3V\right)dy.$$

Expanding the divergence gives

$$0=\int|V|^3V\cdot\nabla(w\chi_{\varepsilon,R})\,dy+\int w\chi_{\varepsilon,R}V\cdot\nabla(|V|^3)\,dy,$$

and hence

$$-\int w\chi_{\varepsilon,R}V\cdot\nabla(|V|^3)=\int|V|^3V\cdot\nabla(w\chi_{\varepsilon,R}).$$

Expanding the derivative,

$$-\int w\chi_{\varepsilon,R}V\cdot\nabla(|V|^3)=\int\chi_{\varepsilon,R}|V|^3V\cdot\nabla w+\int w|V|^3V\cdot\nabla\chi_{\varepsilon,R}.$$

On the inner cutoff region $\{\varepsilon<|y|<2\varepsilon\}$, one has $|y|\simeq\varepsilon$, and on the outer cutoff region $\{R/2<|y|<R\}$, one has $|y|\simeq R$. Therefore the derivative bound for $\chi_{\varepsilon,R}$ implies

$$\begin{aligned}\left|\int w|V|^3V\cdot\nabla\chi_{\varepsilon,R}\right|&\leq C\int_{\{\varepsilon<|y|<2\varepsilon\}}|y|^{-p-1}|V|^4\,dy\\&\quad+C\int_{\{R/2<|y|<R\}}|y|^{-p-1}|V|^4\,dy.\end{aligned}$$

The integrand $|y|^{-p-1}|V|^4$ belongs to $L^1(\mathbb{R}^3)$ at the fixed time. Hence absolute continuity of the Lebesgue integral gives

$$\int_{\{\varepsilon<|y|<2\varepsilon\}}|y|^{-p-1}|V|^4\,dy\to 0\qquad\text{as }\varepsilon\downarrow 0,$$

and integrability at infinity gives

$$\int_{\{R/2<|y|<R\}}|y|^{-p-1}|V|^4\,dy\to 0\qquad\text{as }R\uparrow\infty.$$

Moreover

$$|\chi_{\varepsilon,R}|V|^3V\cdot\nabla w|\le p|y|^{-p-1}|V|^4\in L^1(\mathbb{R}^3),$$

so dominated convergence gives

$$\int\chi_{\varepsilon,R}|V|^3V\cdot\nabla w\to\int|V|^3V\cdot\nabla w.$$

Passing first $\varepsilon\downarrow 0$ and then $R\uparrow\infty$ yields

$$-\int_{\mathbb{R}^3}w\,V\cdot\nabla(|V|^3)\,dy=\int_{\mathbb{R}^3}|V|^3V\cdot\nabla w\,dy,$$

with no boundary contribution at $y=0$ or at infinity. □

Corollary 7.51 (Reduced integrated scale-force criterion)**.** *Assume annular convective regularity, the weighted L^4-window bound, and the two component estimates*

$$\sup_{T\ge\tau_0}\int_T^{T+1}|F_0^{\Delta}(V(\tau))|\,d\tau<\infty,$$

$$\sup_{T\ge\tau_0}\int_T^{T+1}|F_0^{P}(V(\tau),P(\tau))|\,d\tau<\infty.$$

Then the window-integrated weighted scale-forcing bound holds.

Proof. Lemma 7.50 gives the singular-weight integration-by-parts condition. Lemma 7.48 then gives the integrated convective scale-force bound. Combining this with the diffusive and pressure component estimates, and applying lemma 7.46, gives the window-integrated weighted scale-forcing bound. □

Corollary 7.52 (Remaining analytic alternatives for the scale component)**.** *Before the singular-weight integration by parts is justified, failure of the window-integrated weighted scale-forcing bound is reduced to failure of at least one of the following four analytic statements:*

(i) *the integrated diffusive weighted scale-force bound;*
(ii) *the singular-weight convective integration-by-parts identity;*
(iii) *the weighted L^4-window bound;*
(iv) *the integrated weighted pressure scale-force bound.*

If annular convective regularity is available, then the second item is replaced by failure of annular convective regularity itself, together with the same weighted L^4-window input.

Proof. Lemma 7.46 shows that the diffusive, convective, and pressure component bounds imply the integrated scale-force bound. Lemma 7.48 reduces the convective component to the singular-weight integration-by-parts identity and the weighted L^4-window bound. Finally, lemma 7.50 shows that annular convective regularity and the weighted L^4-window bound imply the singular-weight integration-by-parts identity. Therefore the listed failures exhaust the remaining ways in which this scale-component estimate can fail. □

7.5. **Navier–Stokes Cost-Divergence Exclusion.**

Definition 7.53 (Type II exclusion conclusion)**.** For a renormalized Type II solution, the exclusion conclusion means that after the active supercritical scaling condition and the compact cost-divergence criterion have both been established, the solution is excluded from the Type II class under consideration. This is stronger than cost divergence alone, which only states that the solution satisfies the compact Type II cost-divergence criterion.

Definition 7.54 (Local cost-to-exclusion data)**.** Let ω be a repaired-gauge Navier–Stokes Type II branch on a terminal renormalized tail $[\tau_0, \infty)$. Local cost-to-exclusion data for the validated compact cost consist of a nonnegative locally integrable cost $\mathcal{D}_\omega$, a locally absolutely continuous corrected localized energy $\mathcal{H}_\omega$, a constant $c_\omega > 0$, and a nonnegative locally integrable remainder $\mathcal{R}_\omega$ such that:

(i) the validated compact cost-divergence conclusion is
$$\int_{\tau_0}^{\infty} \mathcal{D}_\omega(\tau)\, d\tau = \infty;$$

(ii) $\mathcal{H}_\omega(\tau_0) < \infty$ and
$$\inf_{\tau \geq \tau_0} \mathcal{H}_\omega(\tau) > -\infty;$$

(iii) the corrected localized energy inequality
$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + c_\omega \mathcal{D}_\omega(\tau) + \mathcal{R}_\omega(\tau) \leq 0$$
holds in distributions on (τ_0, ∞).

For the identity cost one takes $\mathcal{D}_\omega = \tilde{\mathfrak{D}}_{R_0}$. For an explicitly comparable localized cost, $\mathcal{D}_\omega$ is the localized cost whose divergence is validated by the compact criterion, provided that the same cost is controlled by the corrected localized energy inequality on the terminal tail.

Definition 7.55 (Energy-controlled validated cost)**.** Let (V, P, a, b) be a repaired-gauge branch and let
$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla V|^2 \phi_{R_0} + a_+(\tau) \int |V|^2 \phi_{R_0}$$
be the identity localized Navier–Stokes cost. A validated compact cost $\mathfrak{C}_{\mathrm{II}}^{NS}$ is energy-controlled by the identity cost on a terminal tail if there are constants $C_1 < \infty$ and $\tau_1 \geq \tau_0$ such that
$$0 \leq \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq C_1 \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \text{for a.e. } \tau \geq \tau_1.$$
The identity cost is energy-controlled with $C_1 = 1$ and $\tau_1 = \tau_0$. For a non-identity cost, this is the upper comparison needed for a corrected localized energy inequality controlling $\tilde{\mathfrak{D}}_{R_0}$ to also control the validated cost.

Definition 7.56 (Local monotonicity carrier for a validated cost)**.** Let ω be a repaired-gauge Navier–Stokes Type II branch on a terminal renormalized tail $[\tau_0, \infty)$, and let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be the validated compact cost for that branch. A local monotonicity carrier for $\mathfrak{C}_{\mathrm{II}}^{NS}$ consists of a nonnegative locally integrable carrier cost $\mathcal{A}_\omega$, a locally absolutely continuous corrected localized energy $\mathcal{H}_\omega$, a nonnegative locally integrable remainder $\mathcal{R}_\omega$, and constants $c_A > 0$, $0 < C_A < \infty$ such that:

(i) $\mathcal{H}_\omega(\tau_0) < \infty$ and
$$\inf_{\tau \geq \tau_0} \mathcal{H}_\omega(\tau) > -\infty;$$

(ii) the carrier monotonicity inequality
$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + c_A \mathcal{A}_\omega(\tau) + \mathcal{R}_\omega(\tau) \leq 0$$
holds in distributions on (τ_0, ∞);

(iii) the validated compact cost is controlled by the carrier on the terminal tail:
$$0 \leq \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq C_A \mathcal{A}_\omega(\tau) \qquad \text{for a.e. } \tau \geq \tau_0.$$

The carrier is local: $\mathcal{A}_\omega$ may be the fixed cutoff identity cost, a moving-cutoff identity cost, or another localized cost with an explicit terminal comparison to the validated compact cost.

Lemma 7.57 (Monotonicity carrier gives local cost-to-exclusion data)**.** *Let ω be a repaired-gauge Navier–Stokes Type II branch. Assume that the validated compact cost $\mathfrak{C}_{\mathrm{II}}^{NS}$ is nonnegative, locally integrable on finite renormalized intervals, and satisfies*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\, d\tau = \infty.$$

If $\mathfrak{C}_{\mathrm{II}}^{NS}$ admits a local monotonicity carrier in the sense of definition 7.56, then ω carries the local cost-to-exclusion data of definition 7.54 with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$.

Proof. Let $(\mathcal{A}_\omega, \mathcal{H}_\omega, \mathcal{R}_\omega^{\mathrm{car}}, c_A, C_A)$ be the carrier data. The lower bound and finite initial value of $\mathcal{H}_\omega$ are exactly the energy bounds required in definition 7.54. The terminal comparison gives

$$\mathcal{A}_\omega(\tau) \geq C_A^{-1} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \qquad \text{for a.e. } \tau \geq \tau_0.$$

Substituting this into the carrier monotonicity inequality yields

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{c_A}{C_A}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) + \mathcal{R}_\omega^{\mathrm{car}}(\tau) \leq 0$$

in distributions. Thus definition 7.54 holds with

$$\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}, \qquad c_\omega = c_A/C_A, \qquad \mathcal{R}_\omega = \mathcal{R}_\omega^{\mathrm{car}}.$$

The divergent-cost item is the validated compact cost-divergence conclusion. □

Lemma 7.58 (Fixed-cutoff monotonicity gives local cost-to-exclusion data)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b). Let ϕ_{R_0} be the cutoff used in the validated compact cost, let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be a nonnegative locally integrable validated compact cost, and set*

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V(y,\tau)|^2 \phi_{R_0}(y)\, dy.$$

Assume, after increasing τ_0 if necessary, that:

(i) *$\mathcal{E}_\phi(\tau_0) < \infty$;*
(ii) *the fixed-cutoff localized energy identity (14) holds on the terminal tail;*
(iii) *the fixed-cutoff error density B_{R_0} of definition 9.9 satisfies*

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(iv) *the validated compact cost $\mathfrak{C}_{\mathrm{II}}^{NS}$ is energy-controlled by $\tilde{\mathfrak{D}}_{R_0}$ in the sense of definition 7.55;*
(v) *the validated compact cost-divergence conclusion is*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\, d\tau = \infty.$$

Then ω carries the local cost-to-exclusion data of definition 7.54 with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$.

Proof. Let C_1, τ_1 be the constants in the energy-control comparison. All statements are terminal. Since $\mathfrak{C}_{\mathrm{II}}^{NS}$ is locally integrable on finite intervals, removing any finite initial subinterval does not change divergence of its tail integral. After the allowed increase and relabeling of τ_0, we may therefore assume that $\tau_0 \geq \tau_1$, that $\mathcal{E}_\phi(\tau_0) < \infty$, and that the upper comparison holds on the whole tail.

Define the corrected localized energy

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\, ds.$$

The finite-error hypothesis makes the correction tail finite for every $\tau \geq \tau_0$ and locally absolutely continuous. Since $\mathcal{E}_\phi \geq 0$ and $B_{R_0} \geq 0$, one has

$$\inf_{\tau \geq \tau_0} \mathcal{H}_\omega(\tau) \geq 0, \qquad \mathcal{H}_\omega(\tau_0) < \infty.$$

Theorem 9.11 gives, in distributions,

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \leq 0.$$

By energy-control of the validated cost,

$$\tilde{\mathfrak{D}}_{R_0}(\tau) \geq C_1^{-1}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \quad \text{for a.e. } \tau \geq \tau_0.$$

Therefore

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2C_1}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq 0.$$

This is the corrected localized energy inequality in definition 7.54, with

$$\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}, \qquad c_\omega = (2C_1)^{-1}, \qquad \mathcal{R}_\omega = 0.$$

The divergent-cost item is exactly the validated compact cost-divergence conclusion. Hence all local cost-to-exclusion data hold. □

Lemma 7.59 (Moving-cutoff monotonicity gives local cost-to-exclusion data)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b), and let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be a nonnegative locally integrable validated compact cost. Let*

$$\Phi(\tau, y) = \phi(y/R(\tau))$$

be a moving cutoff as in definition 9.18, with $R(\tau) \geq R_0$ locally absolutely continuous and nondecreasing, and set

$$\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\Phi(\tau,y)\,dy, \qquad \tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu\int |\nabla V|^2\Phi + a_+(\tau)\int |V|^2\Phi.$$

Assume, after increasing τ_0 if necessary, that:

(i) *$\mathcal{E}_R(\tau_0) < \infty$;*
(ii) *the localized energy identity is valid with the time-dependent cutoff Φ;*
(iii) *the moving finite-error condition holds, for instance through summable moving-annulus errors:*

$$\int_{\tau_0}^\infty B_{\mathrm{mov}}(\tau)\,d\tau < \infty;$$

(iv) *the validated compact cost is controlled by the moving identity cost on the terminal tail: for some $C_m < \infty$,*

$$0 \leq \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq C_m\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \textit{for a.e. } \tau \geq \tau_0;$$

(v) *the validated compact cost-divergence conclusion is*

$$\int_{\tau_0}^\infty \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\,d\tau = \infty.$$

Then ω carries the local cost-to-exclusion data of definition 7.54 with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$.

Proof. The moving finite-error condition gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$; if it is verified through definition 9.20, this is precisely lemma 9.21. Define

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\mathrm{mov}}(s)\,ds.$$

The finite moving-error condition makes the correction tail finite and locally absolutely continuous. Since $\mathcal{E}_R \geq 0$ and $B_{\mathrm{mov}} \geq 0$,

$$\inf_{\tau \geq \tau_0} \mathcal{H}_\omega(\tau) \geq 0, \qquad \mathcal{H}_\omega(\tau_0) < \infty.$$

Theorem 9.19 gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \leq 0$$

in distributions. Thus $\tilde{\mathfrak{D}}_{R(\tau)}$ is a local monotonicity carrier for the validated compact cost, with $c_A = 1/2$, $C_A = C_m$, and $\mathcal{R}_\omega = 0$. Applying lemma 7.57 gives the claimed local cost-to-exclusion data. □

Lemma 7.60 (Corrected monotonicity bounds the validated cost)**.** *Let ω carry local cost-to-exclusion data in the sense of definition 7.54, except do not assume the divergence condition in item* (i)*. Then*

$$\int_{\tau_0}^{\infty} \mathcal{D}_\omega(\tau)\, d\tau < \infty.$$

Proof. Since $\mathcal{H}_\omega$ is locally absolutely continuous and the corrected localized energy inequality holds in distributions, integration over $[\tau_0, T]$ gives, for every $T > \tau_0$,

$$\mathcal{H}_\omega(T) - \mathcal{H}_\omega(\tau_0) + c_\omega \int_{\tau_0}^{T} \mathcal{D}_\omega(\tau)\, d\tau + \int_{\tau_0}^{T} \mathcal{R}_\omega(\tau)\, d\tau \leq 0.$$

The remainder is nonnegative, hence

$$c_\omega \int_{\tau_0}^{T} \mathcal{D}_\omega(\tau)\, d\tau \leq \mathcal{H}_\omega(\tau_0) - \mathcal{H}_\omega(T) \leq \mathcal{H}_\omega(\tau_0) - \inf_{\sigma \geq \tau_0} \mathcal{H}_\omega(\sigma).$$

The right-hand side is finite and independent of T. Letting $T \to \infty$ and using monotone convergence for the nonnegative cost proves the claim. □

Theorem 7.61 (Local Navier–Stokes cost-to-exclusion theorem)**.** *Let ω be a renormalized Navier–Stokes Type II branch on a retained terminal compact stratum. If ω has active supercritical scaling, the validated compact Type II cost-divergence conclusion, and local cost-to-exclusion data of definition 7.54, then ω is excluded from the Type II class under consideration. Equivalently, on this local stratum,*

$$\Pi_{\mathrm{NS,II}}(\omega) : \mathcal{S}_\omega \wedge \mathcal{B}_\omega \Longrightarrow \mathcal{E}_\omega$$

is a theorem.

Proof. Assume, for contradiction, that the branch remains in the Type II class on the retained compact stratum. The local cost-to-exclusion data give a corrected localized energy inequality for the same nonnegative cost $\mathcal{D}_\omega$ whose divergent tail integral is the validated compact cost-divergence conclusion. By lemma 7.60,

$$\int_{\tau_0}^{\infty} \mathcal{D}_\omega(\tau)\, d\tau < \infty.$$

This contradicts the validated compact cost-divergence conclusion. Hence the branch cannot remain in the Type II class under consideration. □

Definition 7.62 (Pointwise cost-divergence exclusion data)**.** For a renormalized Type II solution ω, the pointwise cost-divergence exclusion data are available when the following local analytic statements hold:

(i) class membership: ω lies in the renormalized Navier–Stokes Type II class and satisfies the Type II analytic data before the Type II cost criterion is evaluated;

(ii) cost compatibility: the active compact Type II cost criterion uses either the Navier–Stokes identity-cost criterion

$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0},$$

or a localized cost explicitly comparable to $\tilde{\mathfrak{D}}_{R_0}$ on a terminal tail in the sense of definition A.23; in either case its conclusion is the validated compact Type II cost-divergence conclusion for ω;

(iii) stability of the active scaling condition: the active supercritical scaling condition remains in force after the repaired-gauge representation, cost comparison, and criterion evaluation are attached;

(iv) stability of the cost-divergence conclusion: every renormalized continuation preserving the active scaling condition and the validated compact Type II cost-divergence conclusion remains in the same compact Type II cost-divergence regime, unless it first enters one of the local alternatives already isolated above;

(v) local monotonicity-carrier data: on the retained compact stratum, the validated compact cost admits a carrier in the sense of definition 7.56. This may be supplied by the fixed or moving carrier subidentity of definition 7.85, the corresponding finite-error condition, and the corresponding cost comparison. Exact fixed or moving localized energy identities are stronger sufficient inputs through lemmas 7.58 and 7.59.

Lemma 7.63 (Canonical cost compatibility)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b). Assume that the localized identity cost*

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy$$

is measurable, nonnegative, and locally integrable on finite τ-intervals. If the compact Type II criterion is evaluated with the canonical Navier–Stokes identity cost, then

$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0}$$

is a valid Type II scale-exclusion cost, the comparison holds with

$$c_0 = 1, \qquad \tau_1 = \tau_0,$$

and the validated compact cost-divergence conclusion for ω is the identity-cost divergence statement

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

If the compact criterion is not evaluated with this identity cost, the branch is not being treated by the canonical compatibility statement and must instead come with an explicit comparison as in lemma 7.64.

Proof. The first assertion is the identity-cost case of the localized cost definitions. Since $\tilde{\mathfrak{D}}_{R_0}$ is assumed measurable, nonnegative, and locally integrable on finite intervals, the canonical choice

$$\mathfrak{C}_{\mathrm{II}}^{NS} := \tilde{\mathfrak{D}}_{R_0}$$

satisfies the defining properties of a Type II scale-exclusion cost. The comparison inequality is the identity

$$\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) = \tilde{\mathfrak{D}}_{R_0}(\tau) \geq 1 \cdot \tilde{\mathfrak{D}}_{R_0}(\tau)$$

for a.e. $\tau \geq \tau_0$, so $c_0 = 1$ and $\tau_1 = \tau_0$. Therefore divergence of the active compact cost is precisely divergence of $\tilde{\mathfrak{D}}_{R_0}$ on the terminal tail. This is the canonical case of lemma A.24 and theorem A.25. ∎

Lemma 7.64 (Comparable cost compatibility)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V,P,a,b). Let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be a nonnegative, measurable, locally integrable localized cost on finite τ-intervals. Assume that, on a later terminal tail $[\tau_1,\infty)$, there is a constant $c_0>0$ such that*

$$\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \ge c_0\,\tilde{\mathfrak{D}}_{R_0}(\tau) \quad \text{for a.e. } \tau \ge \tau_1.$$

Then every identity-cost divergence conclusion

$$\int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau = \infty$$

implies the corresponding compact Type II cost-divergence conclusion

$$\int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\,d\tau = \infty.$$

Consequently the use of $\mathfrak{C}_{\mathrm{II}}^{NS}$ introduces no additional exit from the compact-cost stratum beyond the explicit failure of this comparison.

Proof. For every $T>\tau_1$, the assumed comparison gives

$$\int_{\tau_1}^{T} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\,d\tau \ge c_0 \int_{\tau_1}^{T} \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau.$$

Letting $T\to\infty$ proves the implication from identity-cost divergence to divergence of the selected compact Type II cost. This is precisely the one-sided comparison needed by definition A.23 and used in theorem A.25. If the inequality is unavailable, the selected diagnostic cost is not connected to the Navier–Stokes identity dissipation on this terminal tail; this is a cost-comparison failure, not a new unlabelled Type II branch. $\square$

Lemma 7.65 (Local cost well-posedness)**.** *Let (V,P,a,b) be a repaired-gauge branch on a finite renormalized interval $I=[\alpha,\beta]$, and let $\phi_{R_0}\in C_c^\infty(B_{R_0})$, $\phi_{R_0}\ge 0$. Assume:*

(i) *$V\in L^2(I;H^1(B_{R_0}))$;*
(ii) *$a\in L^1(I)$;*
(iii) *either $V\in L^\infty(I;L^3(\mathbb{R}^3))$, or more locally*

$$\operatorname*{ess\,sup}_{\tau\in I} \|V(\tau)\|_{L^2(B_{R_0})} < \infty.$$

Then the localized identity cost

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\,dy + a_+(\tau) \int |V(y,\tau)|^2 \phi_{R_0}(y)\,dy$$

is measurable, nonnegative, and integrable on I. Consequently, on every compact renormalized window satisfying these local bounds, $\tilde{\mathfrak{D}}_{R_0}$ is a well-defined finite-interval cost.

Proof. The two summands are measurable because V is strongly measurable in the local Sobolev spaces on the finite interval and a_+ is measurable. They are nonnegative since $\nu>0$, $\phi_{R_0}\ge 0$, and $a_+\ge 0$.

For the gradient term, boundedness of ϕ_{R_0} and the support condition give

$$\int_I \int |\nabla V|^2 \phi_{R_0} \le \|\phi_{R_0}\|_{L^\infty} \int_I \|\nabla V(\tau)\|_{L^2(B_{R_0})}^2\,d\tau < \infty.$$

For the drift term, first suppose

$$\operatorname*{ess\,sup}_{\tau\in I} \|V(\tau)\|_{L^2(B_{R_0})} < \infty.$$

Then

$$\int_I a_+(\tau) \int |V|^2 \phi_{R_0}\, dy\, d\tau \le \|\phi_{R_0}\|_{L^\infty} \left(\operatorname*{ess\,sup}_{\tau \in I} \|V(\tau)\|^2_{L^2(B_{R_0})} \right) \|a_+\|_{L^1(I)} < \infty.$$

If instead $V \in L^\infty(I; L^3(\mathbb{R}^3))$, then Hölder on B_{R_0} gives the local L^2 bound

$$\|V(\tau)\|^2_{L^2(B_{R_0})} \le |B_{R_0}|^{1/3} \|V(\tau)\|^2_{L^3(\mathbb{R}^3)}$$

for a.e. $\tau \in I$, reducing to the previous case. Thus both summands are integrable on I. □

Lemma 7.66 (Local modulation integrability from absolutely continuous repaired gauges)**.** *Let u, p be a Navier–Stokes solution on a terminal physical time interval, and let (V, P) be its repaired-gauge renormalization on a terminal renormalized tail. Assume that, on every compact renormalized subinterval, the physical concentration chart and the repaired-gauge path satisfy definitions 9.46 and 9.48. Let a, b be the final modulation coefficients in the repaired-gauge renormalized equation, as in lemma 9.50. Then*

$$a, b \in L^1_{\mathrm{loc}}([\tau_0, \infty)).$$

In particular, for every unit terminal window

$$I_n = [\tau_0 + n, \tau_0 + n + 1],$$

one has $a \in L^1(I_n)$. If the absolutely continuous chart, the absolutely continuous repaired gauge, or the resulting local integrability of the modulation coefficients is unavailable on such a compact window, then the branch does not lie in the retained compact-cost stratum and falls into the corresponding representation, repaired-gauge, or modulation-integrability alternative.

Proof. On each compact renormalized window, the assumed absolutely continuous chart and repaired-gauge path place the branch under the hypotheses of theorem 9.55. In particular, lemma 9.54 gives

$$a, b \in L^1_{\mathrm{loc}},$$

and lemma 9.59 identifies these locally integrable functions as the final coefficients in the repaired-gauge equation. Restricting to I_n gives $a \in L^1(I_n)$. If the absolutely continuous local chart, repaired-gauge path, or resulting coefficient integrability is missing, then the ordered stratification records the first missing input as a representation, repaired-gauge, or modulation-integrability failure. □

Lemma 7.67 (Local state data give local L^2 windows)**.** *Let (V, P, a, b) be a repaired-gauge Type II branch on a terminal tail. Fix $R_0 < \infty$ and a unit terminal window I_n. Suppose the retained local suitable-energy data on the compact repaired-gauge cylinder supply*

$$\operatorname*{ess\,sup}_{\tau \in I_n} \|V(\tau)\|_{L^2(B_{R_0})} < \infty,$$

Then

$$\operatorname*{ess\,sup}_{\tau \in I_n} \|V(\tau)\|_{L^2(B_{R_0})}$$

is finite. Thus the local L^2-input required for the finite-window identity cost is available on every retained compact cylinder from local suitable-energy data alone. Failure of local L^2 control on the selected compact cylinder is recorded by the ordered local validation as a local-energy or rough-core exit.

Proof. The displayed local $L^\infty_\tau L^2_y(B_{R_0})$ bound is precisely the compact-cylinder local-energy component of the retained state data. Hence the claimed finiteness is immediate. The final assertion is exactly the first-failure convention in lemma 7.70: a missing compact-cylinder local energy bound is not kept inside the retained compact-cost branch and is not repaired by a whole-space critical estimate. □

Corollary 7.68 (Compact-window data give cost well-posedness)**.** *Let (V, P, a, b) be a repaired-gauge branch on a terminal tail. Fix $R_0 < \infty$. Suppose that, for every unit terminal window*

$$I_n = [\tau_0 + n, \tau_0 + n + 1], \qquad n \in \mathbb{N},$$

the branch has local windowed $H^1(B_{R_0})$ control and satisfies one of the local L^2-window hypotheses of lemma 7.67. Suppose also that the local modulation-integrability conclusion of lemma 7.66 holds. Then $\widetilde{\mathfrak{D}}_{R_0}$ is a measurable nonnegative cost, integrable on each I_n. If the branch also has the identity cost or an explicitly comparable localized cost, then the compact Type II cost criterion can be evaluated on every retained terminal window.

Failure of the conclusion is exactly one of the local failures in the hypotheses: compact-cylinder L^2 control, windowed H^1 control, modulation-integrability, or cost-comparison failure.

Proof. Lemma 7.67 supplies the $L^\infty(I_n; L^2(B_{R_0}))$ input on every unit window, and lemma 7.66 supplies $a \in L^1(I_n)$. The assumed windowed $H^1(B_{R_0})$ control gives

$$V \in L^2(I_n; H^1(B_{R_0})).$$

Thus the hypotheses of lemma 7.65 hold on each I_n. The identity-cost case is then covered by lemma 7.63; the explicitly comparable case is covered by lemma 7.64. Hence the criterion is locally meaningful on every retained terminal window. Conversely, if the conclusion fails before criterion evaluation, at least one of the listed local inputs is absent; by the ordered stratification, that absence is recorded as the corresponding local alternative. □

Proposition 7.69 (Active scaling stability criterion)**.** *Let ω be an admissible local Type II branch with active supercritical scaling on a terminal tail. A passage from a raw Type II chart to a repaired-gauge chart is called scale-admissible on a later tail if its raw and repaired physical scales satisfy*

$$0 < m \leq \frac{\lambda_{\mathrm{rep}}(t)}{\lambda_{\mathrm{raw}}(t)} \leq M < \infty$$

on that tail. On the compact-cost stratum, the active supercritical scaling condition is unchanged by the following operations:

(i) *restriction to a later terminal tail;*
(ii) *passage from an admissible raw Type II chart to a scale-admissible repaired-gauge representation;*
(iii) *replacement of the canonical cost by an explicitly comparable localized cost;*
(iv) *evaluation of the compact Type II cost criterion.*

If the repaired chart is not scale-admissible on any later tail, the branch is not retained in the compact-cost stratum; it is assigned instead to the corresponding representation, repaired-gauge, or scale-degeneracy alternative.

Proof. The active supercritical scaling condition is a terminal property of the scale-center chart and is invariant under removal of a finite initial renormalized interval. Thus restriction from $[\tau_0, \infty)$ to $[\tau_1, \infty)$, $\tau_1 \geq \tau_0$, cannot change the terminal scale alternative.

Suppose next that a scale-admissible repaired-gauge representation is available. Two-sided comparability of λ_{rep} and λ_{raw} implies that any terminal statement depending only on the scale class, such as supercriticality relative to the Type I rate, holds for one scale if and only if it holds for the other after passing to the same tail. The repaired gauge changes coordinates and modulation variables inside this comparable scale class; it does not replace the branch by a different terminal scaling alternative. If the representation cannot be constructed, or if every repaired chart loses scale comparability, then the first failure is a representation, repaired-gauge, or scale-degeneracy alternative.

Cost comparison is scale-passive. It changes only the nonnegative diagnostic functional used to record cost divergence and does not alter the center, physical scale, renormalized time, or active scaling hypothesis. If the comparison is unavailable, the failure is cost-comparison failure. Finally, criterion evaluation is a logical operation applied to the already constructed branch and cost. It does not modify the branch or its scale. Hence active scaling is stable through the listed operations exactly on the local stratum on which those operations are admissible. ☐

Lemma 7.70 (No unrecorded exit from compact cost divergence)**.** *Let ω be an admissible local Type II branch with active supercritical scaling on a terminal tail. Before declaring that a continuation lies in the compact cost-divergence stratum, run the ordered local validation tests:*

(i) *Type II analytic data and exhaustion data;*
(ii) *raw-chart selection, repaired-gauge representation, and scale admissibility;*
(iii) *local compactness decomposition into the alternatives of theorem 10.11;*
(iv) *retained finite critical-mass window, read as the compact-window finite critical-mass input of lemma 9.73;*
(v) *the critical-tightness/exterior-leakage test at the retained scale;*
(vi) *local windowed H^1 control;*
(vii) *local modulation integrability, cost well-posedness, and compatibility of the chosen compact Type II cost;*
(viii) *local monotonicity-carrier data for the validated compact cost, supplied by the fixed-cutoff criterion, the moving-cutoff criterion, or another explicit localized carrier;*
(ix) *the compact Type II cost-divergence conclusion for the validated cost.*

Then exactly one of the following occurs:

(a) *all tests pass, and ω remains in the compact Type II cost-divergence regime on the retained terminal branch;*
(b) *the first failed test gives one of the already named local alternatives: ordered exhaustion failure, representation failure, repaired-gauge or scale-degeneracy failure, multibubble/multicore splitting, cascade, exterior tightness failure, loss of local windowed H^1 control, finite-cost scale-collapse alternative, scale-rigid residual alternative, critical-mass alternative, modulation-integrability failure, cost-compatibility failure, carrier-subidentity failure, finite fixed-cutoff error failure, summable moving-annulus error failure, carrier-comparison failure for the validated cost, or failure of compact cost divergence for the validated cost.*

Thus a renormalized continuation preserving active scaling and validated compact cost divergence cannot leave the compact cost-divergence regime without being recorded by the local compactness stratification.

Proof. The proof is the ordered-alternative argument. First test whether the Type II analytic and exhaustion data hold. If not, the first failure is one of the ordered exhaustion failures of definition A.3. If they hold, test for raw-chart selection, repaired-gauge representation, and scale admissibility. Failure at this stage is one of the ordered representation failures of definition A.11, possibly with the specific scale-degeneracy recorded by proposition 7.69.

Once a scale-admissible repaired branch is available, apply the local compactness decomposition theorem 10.11. If the branch falls into the multibubble, multicore, cascade, gauge-degenerate, finite-cost scale-collapse, local H^1-loss, or scale-rigid residual alternatives, it has left the compact single-core cost-divergence stratum through a named local case. In particular, an additional exterior concentration mechanism is not discarded; if it retains critical mass it appears as a multibubble or multicore branch, and if it escapes the retained compact region it appears as failure of critical tightness.

The first alternative in theorem 10.11 is retained only as a compact single-core branch; it is not yet identified with compact cost divergence. On that retained branch the remaining tests below decide whether the branch belongs to the compact cost-divergence stratum, has finite validated cost, or exits through one of the remaining compact tests.

On the retained compact single-core branch, test the finite critical-mass row using the compact-window version of lemma 9.73: lower active mass failure means non-retention, and failure of a finite local upper bound produces a comparable retained label or a named adjacent active label. Thus failure of the finite critical-mass row is recorded as a critical-mass or adjacent local alternative, not as a hidden global estimate. Once the row passes, the remaining compact tests are critical tightness and local windowed H^1 control. Failure of either is precisely one of the alternatives in definition A.26.

If all compact tests pass, local cost well-posedness and compatibility are evaluated. Local modulation integrability is supplied by lemma 7.66; the local cost well-posedness is supplied by corollary 7.68. The compatibility is the identity case of lemma 7.63 or the explicit comparison case of lemma 7.64; failure of both is a cost-compatibility failure. Next test whether the validated compact cost has a local monotonicity carrier. In the fixed-cutoff case this means the fixed carrier subidentity, finite fixed-cutoff error, and energy-control comparison. In the moving-cutoff case this means the moving carrier subidentity, summable moving-annulus errors, and moving-cost comparison. Exact localized energy identities are sufficient for these subidentities, but the one-sided suitable local energy inequality is the actual input needed. Failure of these inputs is recorded as a carrier-subidentity failure, a finite fixed-cutoff error failure, a summable moving-annulus error failure, or a carrier-comparison failure for the validated cost. Finally, evaluate the compact cost-divergence conclusion for the validated cost. If it fails, the branch is a finite-cost or non-divergent compact branch and is not counted as a compact cost-divergence branch. The tests are ordered, so the first failure is unique. If no failure occurs, the branch has the analytic data, scale-admissible repaired-gauge representation, compact single-core local compactness position, the retained finite critical-mass window, critical tightness, local windowed H^1 control, local modulation integrability, a compatible compact cost, local monotonicity-carrier data, and the validated compact cost-divergence conclusion. Together with the assumed active scaling, these hypotheses place the branch in the compact Type II cost-divergence regime. Therefore there is no unrecorded exit. □

Theorem 7.71 (Carrier validation alternative on the retained compact stratum)**.** *Let ω be an admissible local Type II branch with active supercritical scaling. Suppose that the ordered validation tests in lemma 7.70 have reached the retained compact single-core branch, that cost well-posedness and cost compatibility have passed, and that the validated compact cost-divergence conclusion holds. Then exactly one of the following alternatives occurs:*

(i) *the validated compact cost admits local monotonicity-carrier data in the sense of definition 7.56;*

(ii) *the branch leaves the retained compact cost-divergence stratum through the first failed carrier-validation test: carrier-subidentity failure, finite fixed-cutoff error failure, summable moving-annulus error failure, or carrier-comparison failure for the validated cost.*

Consequently, on the retained compact cost-divergence stratum, where the adjacent carrier-failure alternatives are not part of the stratum, the validated compact cost admits local monotonicity-carrier data.

Proof. After the branch has passed the Type II analytic, representation, compactness, retained finite critical mass, critical-tightness, local windowed H^1, modulation, cost well-posedness, and cost-compatibility tests, the next ordered test in lemma 7.70 is precisely the local monotonicity-carrier test for the validated compact cost. If this test passes, alternative (i) holds. If it fails, the first failed input is recorded as one of the named adjacent carrier failures listed in lemma 7.70: failure of the carrier subidentity, failure of the fixed finite-error condition, failure of the moving finite-error condition through the summable moving-annulus test, or failure of the carrier comparison for the validated cost. The ordered first-failure convention makes the two alternatives disjoint and exhaustive.

By definition, the retained compact cost-divergence stratum keeps only the branch for which all preceding compact tests and the carrier-validation test have passed before the validated compact cost-divergence conclusion is attached. Therefore a branch that remains in this stratum cannot realize one of the carrier-failure alternatives, and the local monotonicity carrier exists. □

Proposition 7.72 (Local verification of the cost-exclusion data)**.** *On a retained terminal branch in the compact Type II cost-divergence stratum, the pointwise data of definition 7.62 are local analytic statements, not additional global assumptions. More precisely, assume that the branch:*

(i) *has the local Type II analytic data and has not entered an exhaustion, representation, modulation-integrability, multibubble/multicore, cascade, exterior-tightness, scale-collapse, scale-rigid, critical-mass, or local H^1-loss alternative;*
(ii) *admits a scale-admissible repaired-gauge representation on retained windows from absolutely continuous local scale-center and repaired-gauge data;*
(iii) *has the retained finite critical-mass window, critical tightness at the retained scale, local windowed H^1 control, active supercritical scaling, and the compact cost-divergence conclusion for the validated cost;*
(iv) *uses either the canonical identity cost $\widetilde{\mathfrak{D}}_{R_0}$ or an explicitly comparable localized cost;*
(v) *remains in the retained compact cost-divergence stratum after the carrier-validation alternatives of theorem 7.71 have been evaluated.*

Then class membership, cost compatibility, stability of the active scaling condition, stability of the validated cost-divergence conclusion, and the cost-to-exclusion implication all hold for this branch. Consequently, on this local stratum, all clauses of definition 7.62 are verified; no estimate outside the retained local stratum and no exclusion of secondary exterior concentration are being assumed.

Proof. Class membership is the retained local Type II analytic-data alternative after the ordered exhaustion and representation tests have passed. The local inputs needed to evaluate the compact cost are available by lemmas 7.66 and 7.67 and corollary 7.68. Cost compatibility then follows from lemma 7.63 in the identity-cost case and from lemma 7.64 in the explicitly comparable-cost case. Stability of the active scaling condition is proposition 7.69. Stability of the validated cost-divergence conclusion, with all exits assigned to named local alternatives, is lemma 7.70. These prove clauses (i)–(iv) of definition 7.62. Since the branch remains in the retained compact cost-divergence stratum after carrier validation, theorem 7.71 gives local monotonicity-carrier data for the validated compact cost. These data provide the local cost-to-exclusion data through lemma 7.57. Then theorem 7.61 gives $\Pi_{\mathrm{NS,II}}(\omega)$, which is the analytic content of clause (v). □

Definition 7.73 (Classwise retained compact coverage)**.** The renormalized Type II class has classwise retained compact coverage for the validated compact cost if every branch with active supercritical scaling and validated compact cost divergence either remains in the retained compact cost-divergence stratum after the ordered validation tests of lemma 7.70, or has already entered one of the named adjacent alternatives in that ordered stratification.

Theorem 7.74 (Classwise retained compact coverage from ordered validation)**.** *Every admissible local Type II branch with active supercritical scaling and validated compact Type II cost divergence has classwise retained compact coverage in the sense of definition 7.73.*

Proof. Apply the ordered validation tests of lemma 7.70 to the branch. The lemma gives two possibilities. If all tests pass, the branch remains in the compact Type II cost-divergence regime on the retained terminal branch, which is precisely the retained compact cost-divergence stratum.

If a test fails, the first failed test is one of the named local alternatives in lemma 7.70. Since the present theorem assumes the validated compact cost-divergence conclusion, the last listed failure, failure of compact cost divergence for the validated cost, cannot occur. Every other first failure is an adjacent compactness stratum: exhaustion, representation, repaired-gauge or scale-degeneracy failure, multibubble/multicore splitting, cascade, exterior tightness failure, local H^1-loss, finite-cost scale-collapse, scale-rigid residual behavior, critical-mass failure, modulation-integrability failure, cost-compatibility failure, carrier-subidentity failure, fixed-cutoff finite-error failure, moving-annulus summability failure, or carrier-comparison failure. Therefore every branch satisfying the hypotheses either remains retained or has already entered a named adjacent alternative. This is exactly classwise retained compact coverage. ☐

Remark 7.75 (Scope of classwise retained compact coverage). Proposition 7.72 is deliberately local. It proves the interface hypotheses for a branch only after the branch has been placed in the validated compact cost-divergence stratum by the compactness stratification. Branches with multibubble, cascade, exterior leakage, scale-collapse, scale-rigid, or rough-core behavior are not forced into this stratum; they are assigned to their own local alternatives. Thus the cost-divergence exclusion step is unconditional on the retained compact stratum, while a classwise statement is obtained only after the adjacent strata have been treated by their own local arguments. The coverage part itself is no longer an assumption; it is supplied by theorem 7.74.

Definition 7.76 (Abstract cost-divergence exclusion theorem applicability)**.** An abstract Type II cost-divergence exclusion theorem is applicable to a renormalized Type II solution if the following analytic identifications are available:

(i) an abstract Type II exclusion theorem based on localized renormalization-cost divergence;
(ii) hypotheses supplying supercritical scaling, a Type II concentration scenario with bounded physical energy, and a localized energy monotonicity formula at the active scale $\lambda(t)$;
(iii) an identification of the theorem's renormalization-cost integral with a validated compact Navier–Stokes cost $\mathfrak{C}_{\mathrm{II}}^{NS}$, where either
$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0}$$
or $\mathfrak{C}_{\mathrm{II}}^{NS}$ is a localized cost satisfying the terminal one-sided comparison with $\tilde{\mathfrak{D}}_{R_0}$ in definition A.23;
(iv) a domain embedding placing the repaired-gauge Navier–Stokes orbit and its scale variables into the parabolic domain expected by the theorem without changing the active scaling condition, validated cost-divergence conclusion, or exclusion conclusion;
(v) identification of the theorem's conclusion with definition 7.53.

Lemma 7.77 (Applicability of the abstract cost-divergence theorem)**.** *Assume a renormalized Type II solution has the active supercritical scaling condition, the validated compact Type II cost-divergence conclusion, and the abstract applicability data of definition 7.76. Then the solution is excluded from the Type II class under consideration.*

Proof. The abstract Type II theorem is applied with supercritical scaling, bounded-energy Type II concentration, and the localized energy monotonicity formula at the active scale. The cost identification places the validated compact Navier–Stokes cost in the theorem's renormalization-cost integral. In the identity case this cost is $\tilde{\mathfrak{D}}_{R_0}$. In the comparable-cost case, the terminal one-sided comparison in definition A.23 and lemma 7.64 show that the validated Navier–Stokes cost-divergence conclusion implies divergence of the localized cost used by the abstract theorem. The domain embedding places the repaired-gauge Navier–Stokes solution in the theorem's parabolic domain, and the conclusion is identified with definition 7.53. Therefore the theorem gives exclusion from the Type II class. □

Theorem 7.78 (Pointwise Navier–Stokes cost-divergence exclusion)**.** *Let ω be a renormalized Type II solution. If ω has the active supercritical scaling condition, the validated compact Type II cost-divergence conclusion, and the pointwise data of definition 7.62, then ω is excluded from the Type II class under consideration.*

Proof. Class membership and stability of the active scaling condition ensure that ω is evaluated in the renormalized Type II class and that the active supercritical scaling condition is still the relevant one. Cost compatibility ensures that the validated compact cost-divergence conclusion is the Navier–Stokes compact cost conclusion for ω, not an external or incomparable cost statement. Stability of the cost-divergence conclusion rules out an exit from the compact Type II cost-divergence regime to a Type II case outside the compact cost-divergence criterion inside the same class.

It remains only to justify the transition from cost divergence to exclusion. The local monotonicity-carrier data in definition 7.62 give the local cost-to-exclusion data by lemma 7.57. Therefore theorem 7.61 applies to the active supercritical scaling condition and the validated compact cost-divergence conclusion, giving the claimed exclusion. In an abstract theorem setting, the same implication is also licensed by lemma 7.77. □

Corollary 7.79 (Cost-divergence exclusion on the retained compact stratum)**.** *Let ω be a renormalized Type II solution with active supercritical scaling and validated compact Type II cost divergence. If ω remains in the retained compact cost-divergence stratum after the ordered validation tests of lemma 7.70, then ω is excluded from the Type II class under consideration.*

Proof. By proposition 7.72, the pointwise cost-divergence exclusion data of definition 7.62 are verified on this retained stratum. Applying theorem 7.78 gives the exclusion conclusion. □

Definition 7.80 (Adjacent-stratum closure for the compact-cost regime)**.** The compact-cost regime has adjacent-stratum closure if every named adjacent alternative produced by lemma 7.70 for a branch with active supercritical scaling and validated compact Type II cost divergence has its own stratum-local argument giving the same Type II exclusion conclusion, either directly or by placing the branch in a named non-compact-cost alternative whose closure theorem already gives the Type II exclusion conclusion.

Corollary 7.81 (Uniform Navier–Stokes cost-divergence exclusion)**.** *Assume adjacent-stratum closure for the compact-cost regime in the sense of definition 7.80. Then every renormalized Type II solution with active supercritical scaling condition and validated compact Type II cost-divergence conclusion is either excluded in its adjacent stratum or excluded by corollary 7.79.*

Proof. By theorem 7.74, a branch with active supercritical scaling and validated compact cost divergence either lies in one of the named adjacent strata or remains in the retained compact cost-divergence stratum. The adjacent cases are handled by the adjacent-stratum closure hypothesis. In the retained compact case, corollary 7.79 applies. □

Definition 7.82 (Adjacent exits from cost-divergence exclusion)**.** If the active supercritical scaling condition and validated compact cost-divergence conclusion are present but the branch is not retained in the compact cost-divergence stratum, then the first exit from the cost-exclusion argument lies in the ordered list:

class membership, cost compatibility, stability of the active scaling condition,
stability of the validated cost-divergence conclusion, local monotonicity-carrier validation.

On the retained compact stratum none of these exits occurs. The final item is expanded by theorem 7.71 into the carrier-subidentity, fixed-finite-error, moving-annulus, and carrier-comparison adjacent strata.

Lemma 7.83 (Carrier exits are adjacent strata, not retained compact branches)**.** *Assume a renormalized Type II solution already has the active supercritical scaling condition and the validated compact Type II cost-divergence conclusion. If one of the exits in definition 7.82 occurs, then the branch is not in the retained compact cost-divergence stratum. If no such exit occurs, the local cost-to-exclusion data are available.*

Proof. The retained compact cost-divergence stratum is reached only after the ordered tests of lemma 7.70 have passed. Hence a failure of class membership, cost compatibility, active-scaling stability, cost-divergence stability, or carrier validation is by definition an exit to an adjacent stratum. Conversely, if no such exit occurs, the carrier-validation test has passed. By theorem 7.71, the validated compact cost admits local monotonicity-carrier data, and then lemma 7.57 gives the local cost-to-exclusion data. □

Definition 7.84 (Pre-carrier validated compact-cost regime)**.** A renormalized Type II branch lies in the pre-carrier validated compact-cost regime if, on a terminal tail, it has active supercritical scaling and the validated compact Type II cost-divergence conclusion, and the ordered validation tests of lemma 7.70 have passed through the following stages:

(i) the Type II analytic data and ordered exhaustion tests;
(ii) raw-chart selection, repaired-gauge representation, pressure reconstruction, modulation data, and scale-admissibility of the repaired chart;
(iii) the local compactness decomposition without entering a multibubble, multicore, cascade, exterior-tightness, local H^1-loss, finite-cost scale-collapse, or scale-rigid residual alternative;
(iv) the retained finite critical-mass window, namely the compact-window finite critical-mass input supplied by lemma 9.73;
(v) critical tightness at the retained scale and local windowed H^1 control;
(vi) local modulation integrability, local cost well-posedness, and compatibility of the chosen compact Type II cost with the validated Navier–Stokes cost.

This regime deliberately stops immediately before local monotonicity-carrier validation. Thus it contains all compact-cost interface data needed to ask for the fixed-cutoff or moving-cutoff carrier estimates, but it does not assert that those estimates have already been proved. The finite critical-mass row is read locally in the final no-Type-II assembly: failure of compact-window lower or upper critical mass is routed by the local terminal critical-mass and exterior/state-space alternatives.

Definition 7.85 (Fixed and moving carrier subidentities)**.** For the fixed cutoff ϕ_{R_0}, the fixed carrier subidentity is the distributional estimate

$$\frac{d}{d\tau}\mathcal{E}_\phi(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le B_{R_0}(\tau), \qquad \mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V|^2\phi_{R_0}.$$

For a moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$, the moving carrier subidentity is the distributional estimate

$$\frac{d}{d\tau}\mathcal{E}_R(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le B_{\mathrm{mov}}(\tau), \qquad \mathcal{E}_R(\tau) = \frac{1}{2}\int |V|^2\Phi.$$

Here B_{R_0} and B_{mov} are the absolute error densities of definitions 9.9 and 9.18. The subidentity is the one-sided estimate needed for corrected monotonicity; an exact localized energy identity is sufficient but not necessary.

Lemma 7.86 (Suitable local energy gives carrier subidentities)**.** *Assume the repaired-gauge representation of a pre-carrier branch includes the local suitable energy inequality on compact renormalized cylinders. Then the fixed carrier subidentity holds for every nonnegative compactly supported fixed cutoff. The moving carrier subidentity holds for every nonnegative finite-valued moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ with $R \in AC_{\mathrm{loc}}$, after the usual approximation of R by smooth positive radii on compact τ-intervals.*

Proof. Apply the local suitable energy inequality in the physical variables to the pullback of the nonnegative renormalized test function. The repaired-gauge change of variables is locally absolutely continuous in time and smooth in space on compact cylinders, so the distributional chain rule gives the renormalized local energy inequality with the drift terms $a(V + y \cdot \nabla V)$ and $b \cdot \nabla V$. For a fixed cutoff this is the fixed localized energy identity with equality replaced by $\le$. Adding $\frac{1}{2}\tilde{\mathfrak{D}}_{R_0}$ to both sides and bounding the cutoff, pressure, translation, and scale terms by their absolute values gives the fixed carrier subidentity.

For a moving cutoff, use smooth positive approximations $R_k \to R$ in AC_{loc} on each compact time interval. The local energy inequality applies to $\Phi_k(\tau, y) = \phi(y/R_k(\tau))$, and the preceding argument gives the moving subidentity with $B_{\mathrm{mov},k}$. Passing to the limit in distributions uses the local finite-energy and local H^1 bounds already present on the pre-carrier regime. The term $\partial_\tau\Phi$ is interpreted as the L^1_{loc} derivative coming from $R \in AC_{\mathrm{loc}}$. The limit is exactly the moving carrier subidentity. □

Definition 7.87 (Fixed-cutoff localized-energy closure criterion)**.** A branch in the pre-carrier validated compact-cost regime satisfies the fixed-cutoff localized-energy closure criterion if, for the cutoff ϕ_{R_0} used to evaluate the validated compact cost and for

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V(y, \tau)|^2 \phi_{R_0}(y)\, dy,$$

the following local analytic statements hold on a terminal tail:

(i) $\mathcal{E}_\phi(\tau_0) < \infty$;
(ii) the fixed carrier subidentity of definition 7.85 is valid for the repaired-gauge branch;
(iii) the fixed-cutoff error density B_{R_0} of definition 9.9 has finite tail integral,

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(iv) the validated compact cost is energy-controlled by $\tilde{\mathfrak{D}}_{R_0}$ in the sense of definition 7.55.

Definition 7.88 (Moving-cutoff localized-energy closure criterion)**.** A branch in the pre-carrier validated compact-cost regime satisfies the moving-cutoff localized-energy closure criterion if there is a finite-valued nondecreasing locally absolutely continuous radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$, such that, on a terminal tail:

(i)

$$\mathcal{E}_R(\tau_0) = \frac{1}{2}\int |V(y, \tau_0)|^2 \Phi(\tau_0, y)\, dy < \infty;$$

(ii) the moving carrier subidentity of definition 7.85 is valid with the time-dependent cutoff Φ;
(iii) the moving-annulus errors are summable in the sense of definition 9.20, which implies

$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\, d\tau < \infty$$

by lemma 9.21;
(iv) the validated compact cost is controlled by the moving identity cost on the terminal tail: for some $C_m < \infty$,

$$0 \le \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le C_m \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \text{for a.e. } \tau \ge \tau_0.$$

Theorem 7.89 (Non-carrier adjacent exits are identified before carrier validation)**.** *Let ω be an admissible local Type II branch with active supercritical scaling and validated compact Type II cost divergence. Apply the ordered tests of lemma 7.70. If a first failed test exists and is not one of the carrier-validation failures*

carrier subidentity, fixed finite error, moving-annulus summability, carrier comparison,

then the branch has already failed to enter the pre-carrier validated compact-cost regime of definition 7.84. More precisely, its first failure is one of the following named non-carrier strata:

ordered exhaustion failure, representation failure, repaired-gauge or scale-degeneracy failure,

multibubble or multicore splitting, cascade, exterior-tightness failure, local H^1-loss,

finite-cost scale-collapse, scale-rigid residual behavior, critical-mass failure,

modulation-integrability failure, cost-well-posedness failure, or cost-compatibility failure.

In any ambient class in which the corresponding local closure theorems for these named strata are imposed, such a branch is not a remaining compact-cost branch.

Proof. The validation procedure in lemma 7.70 is ordered. The pre-carrier regime is the set of branches for which all tests before local monotonicity-carrier validation have passed. Therefore any first failure before the carrier-validation row prevents membership in definition 7.84.

The same lemma lists every possible first failure before carrier validation: the exhaustion failures are those of definition A.3; representation and repaired-gauge failures are those of definitions A.11 and A.18 together with scale-degeneracy from proposition 7.69; the compactness failures are the alternatives in theorem 10.11; critical-mass failures are those routed by lemma 9.73; compact tightness and local H^1-loss are the tests of definition A.26; and the modulation, cost-well-posedness, and compatibility failures are exactly the failures recorded in lemmas 7.63, 7.64 and 7.66 and corollary 7.68. Thus the displayed list is exhaustive for non-carrier first failures.

If the surrounding Type II class includes the local closure theorem assigned to the named stratum, then the branch is handled there. It is therefore not a branch remaining to be treated by the retained compact-cost localized-energy argument. $\square$

Theorem 7.90 (Pre-carrier reduction to carrier validation)**.** *Let ω be in the pre-carrier validated compact-cost regime. Then exactly one of the following alternatives occurs:*

(i) *the validated compact cost admits local monotonicity-carrier data in the sense of definition 7.56;*
(ii) *the branch exits the retained compact cost-divergence stratum through the first failed carrier-validation test: carrier-subidentity failure, fixed-cutoff finite-error failure, moving-annulus summability failure, or carrier-comparison failure for the validated cost.*

If the first alternative occurs, ω is excluded by the retained compact-cost argument.

Proof. By definition of the pre-carrier regime, every ordered validation test before carrier validation has passed, and the validated compact cost-divergence conclusion is already present. Thus the next unresolved row in lemma 7.70 is precisely the local monotonicity-carrier row. Applying theorem 7.71 gives the two alternatives listed in the statement, and the first-failure convention makes them disjoint.

If the carrier exists, then lemma 7.57 gives the local cost-to-exclusion data, and theorem 7.61 excludes the branch. Equivalently, the same conclusion is corollary 7.79. □

Theorem 7.91 (Fixed-cutoff carrier closure)**.** *Let ω be in the pre-carrier validated compact-cost regime. If ω satisfies the fixed-cutoff localized-energy closure criterion of definition 7.87, then ω is excluded from the Type II class under consideration.*

Proof. The pre-carrier regime supplies active supercritical scaling, local well-posedness and compatibility of the validated compact cost, and the validated compact cost-divergence conclusion. The fixed-cutoff localized-energy criterion supplies finite initial localized energy, the fixed carrier subidentity, finite fixed-cutoff error, and energy-control comparison of the validated cost with $\tilde{\mathfrak{D}}_{R_0}$. Let C_1, τ_1 be the constants in that comparison and increase the tail initial time so that the comparison holds from τ_0 onward. Define

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\, ds.$$

The finite-error condition makes $\mathcal{H}_\omega$ locally absolutely continuous and bounded below. The fixed carrier subidentity gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le 0$$

in distributions. Since $\mathfrak{C}_{\mathrm{II}}^{NS} \le C_1 \tilde{\mathfrak{D}}_{R_0}$, this implies

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2C_1}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le 0.$$

Together with the validated compact cost-divergence conclusion, these are the local cost-to-exclusion data with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$. Equivalently, the fixed criterion is a successful carrier-validation row. Since all pre-carrier tests and the validated compact cost-divergence conclusion are already present, ω is on the retained compact cost-divergence stratum after carrier validation. Applying theorem 7.61 gives the Type II exclusion conclusion. □

Theorem 7.92 (Moving-cutoff carrier closure)**.** *Let ω be in the pre-carrier validated compact-cost regime. If ω satisfies the moving-cutoff localized-energy closure criterion of definition 7.88, then ω is excluded from the Type II class under consideration.*

Proof. The pre-carrier regime supplies active supercritical scaling, local well-posedness and compatibility of the validated compact cost, and the validated compact cost-divergence conclusion. The moving-cutoff carrier criterion supplies the finite initial moving localized energy, the moving carrier subidentity, summable moving-annulus errors, and the terminal comparison

$$0 \le \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le C_m \tilde{\mathfrak{D}}_{R(\tau)}(\tau).$$

By lemma 9.21, the summability condition gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$. Define

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\mathrm{mov}}(s)\, ds.$$

The finite moving-error condition makes $\mathcal{H}_\omega$ locally absolutely continuous and bounded below. The moving carrier subidentity gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le 0$$

in distributions. The moving comparison gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2C_m}\mathfrak{C}^{NS}_{\mathrm{II}}(\tau) \le 0.$$

Together with the validated compact cost-divergence conclusion, these are the local cost-to-exclusion data with $\mathcal{D}_\omega = \mathfrak{C}^{NS}_{\mathrm{II}}$. Equivalently, the moving criterion is a successful carrier-validation row. Since all pre-carrier tests and the validated compact cost-divergence conclusion are already present, ω is on the retained compact cost-divergence stratum after carrier validation. Applying theorem 7.61 excludes the branch. □

Lemma 7.93 (Pre-carrier local energy gives finite initial localized energy)**.** *Let ω be in the pre-carrier validated compact-cost regime, represented by V on a terminal tail. Fix any later time $T \ge \tau_0$. After possibly replacing the tail by one whose initial time $\tau_1 \in [T, T+1]$ is a compact-cylinder local-energy Lebesgue time, every finite $R < \infty$ satisfies*

$$\int_{B_R} |V(y,\tau_1)|^2\,dy < \infty.$$

Consequently the initial localized-energy clauses in the fixed and moving localized-energy closure criteria are available after a harmless terminal restriction for every compactly supported fixed cutoff and every moving cutoff with finite initial radius. This is a local compact-cylinder statement; it does not require $V(\tau) \in L^3(\mathbb{R}^3)$ or a whole-space critical bound.

Proof. The physical branch is suitable and the repaired-gauge chart is locally absolutely continuous on compact renormalized cylinders in the pre-carrier regime. Pulling back the local energy class gives

$$V \in L^\infty_{\mathrm{loc}}([\tau_0,\infty); L^2(B_R)) \quad \text{for every } R < \infty$$

after choosing the usual representative, or at minimum

$$V \in L^2_{\mathrm{loc}}([\tau_0,\infty); L^2(B_R))$$

with local-energy Lebesgue times on every compact interval. If the stronger representative is unavailable at a prescribed endpoint, choose $\tau_1 \in [T, T+1]$ which is a Lebesgue time simultaneously for the finitely many compact supports used by the fixed or moving cutoff under consideration. This is possible because the local L^2-in-time norms are finite on $[T, T+1]$.

At such a τ_1, every compactly supported fixed cutoff has finite localized energy. If $\Phi(\tau, y) = \phi(y/R(\tau))$ and $R(\tau_1) < \infty$, then $\Phi(\tau_1, \cdot)$ is compactly supported, so the same local L^2 finiteness gives $\mathcal{E}_R(\tau_1) < \infty$. Passing from τ_0 to τ_1 removes only a finite initial renormalized interval and therefore does not change any terminal divergence or tail-integrability alternative. □

Lemma 7.94 (Nested moving cutoffs give carrier comparison)**.** *Let ω be in the pre-carrier validated compact-cost regime, and assume the validated compact cost is energy-controlled by the fixed identity cost:*

$$0 \le \mathfrak{C}^{NS}_{\mathrm{II}}(\tau) \le C_1 \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \textit{for a.e. } \tau \ge \tau_1.$$

Let $\Phi(\tau, y)$ be a moving cutoff such that, after increasing the initial time if necessary,

$$\Phi(\tau, y) \ge \phi_{R_0}(y) \qquad \textit{for all } y \in \mathbb{R}^3 \textit{ and a.e. } \tau \ge \tau_1.$$

Then the validated compact cost is controlled by the moving identity cost:

$$0 \le \mathfrak{C}^{NS}_{\mathrm{II}}(\tau) \le C_1 \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \textit{for a.e. } \tau \ge \tau_1.$$

In particular, for the canonical identity cost the carrier comparison is automatic with $C_1 = 1$.

Proof. Because $\Phi(\tau,\cdot) \ge \phi_{R_0}$ and both cutoffs are nonnegative,

$$\nu \int |\nabla V|^2 \phi_{R_0} \le \nu \int |\nabla V|^2 \Phi(\tau,\cdot),$$

and, since $a_+(\tau) \ge 0$,

$$a_+(\tau) \int |V|^2 \phi_{R_0} \le a_+(\tau) \int |V|^2 \Phi(\tau,\cdot).$$

Adding these two inequalities gives

$$\tilde{\mathfrak{D}}_{R_0}(\tau) \le \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \text{for a.e. } \tau \ge \tau_1.$$

Combining this with the fixed energy-control comparison yields the displayed moving comparison. The canonical identity cost is energy-controlled by definition 7.55 with $C_1 = 1$. □

Definition 7.95 (Finite-error carrier selection)**.** The pre-carrier validated compact-cost regime has finite-error carrier selection if every branch in that regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies, after passing to a later terminal tail if necessary, at least one of the following two local alternatives:

(i) the fixed carrier subidentity of definition 7.85 is valid and

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(ii) there exists a finite-valued nondecreasing locally absolutely continuous radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$, such that

$$\Phi(\tau, y) \ge \phi_{R_0}(y) \qquad \text{for all } y \text{ and a.e. } \tau,$$

the moving carrier subidentity of definition 7.85 is valid with the time-dependent cutoff Φ, and the moving-annulus errors are summable in the sense of definition 9.20.

This condition contains only the local one-sided energy estimate and finite-error selection statements. It does not include finite initial localized energy or carrier comparison, because those are supplied by lemmas 7.93 and 7.94.

Definition 7.96 (Annular finite-error selection)**.** The pre-carrier validated compact-cost regime has annular finite-error selection if every branch in that regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies, after passing to a later terminal tail if necessary, at least one of the following two alternatives:

(i) the fixed error density has finite tail integral,

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(ii) there exists a finite-valued nondecreasing locally absolutely continuous nested radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and $\Phi(\tau, y) = \phi(y/R(\tau)) \ge \phi_{R_0}(y)$, such that the moving-annulus errors are summable in the sense of definition 9.20.

This is the purely finite-error part of the carrier problem. It does not include local energy subidentities, finite initial energy, or carrier comparison.

Definition 7.97 (Componentwise fixed-or-moving finite-error estimates)**.** For a branch in the pre-carrier validated compact-cost regime, the componentwise annular finite-error estimates hold if, after passing to a later terminal tail if necessary, either the six fixed-cutoff components

$$I_1 = \left| \int |V|^2 \Delta \phi_{R_0} \right|, \qquad I_2 = \left| \int |V|^2 V \cdot \nabla \phi_{R_0} \right|, \qquad I_3 = \left| \int P\, V \cdot \nabla \phi_{R_0} \right|,$$
$$I_4 = |b| \left| \int |V|^2 \nabla \phi_{R_0} \right|, \qquad I_5 = a_- M_\phi, \qquad I_6 = a_+ \int |V|^2 \big(-y \cdot \nabla \phi_{R_0}\big)$$

belong to $L^1(\tau_0, \infty)$, or there is a finite-valued nondecreasing locally absolutely continuous nested radius $R(\tau) \geq R_0$, with $R(\tau) \to \infty$ and $\Phi(\tau, y) = \phi(y/R(\tau)) \geq \phi_{R_0}(y)$, such that each of the seven moving annular components

$$J_1 = \left| \int |V|^2 \partial_\tau \Phi \right|, \qquad J_2 = \left| \int |V|^2 \Delta \Phi \right|, \qquad J_3 = \left| \int |V|^2 V \cdot \nabla \Phi \right|,$$
$$J_4 = \left| \int P\, V \cdot \nabla \Phi \right|, \qquad J_5 = |b| \left| \int |V|^2 \nabla \Phi \right|,$$
$$J_6 = a_- \int |V|^2 \Phi, \qquad J_7 = a_+ \int |V|^2 \big(-y \cdot \nabla \Phi\big)$$

belongs to $L^1(\tau_0, \infty)$.

Remark 7.98 (What the componentwise estimates isolate). The quantities in definition 7.97 are exactly the fixed and moving carrier-error components: in the fixed-cutoff case, viscous cutoff, convection, pressure-tail, center drift, negative scale drift, and the positive scale-shell drift; in the moving-cutoff case, cutoff transition, viscous cutoff, convection, pressure-tail, center drift, negative scale drift, and the positive scale-shell drift. These diagnostics need not be closed by a single global tail estimate. In the stratified use of Section 7, persistent failure of the cutoff, viscous, convection, or pressure diagnostics is read as shell mass that cannot be evacuated from selected windows and is therefore assigned to exterior leakage, exterior concentration, loss of critical tightness, or local windowed H^1-loss. Persistent failure of the translation diagnostic is read as a modulation-driven adjacent branch, while persistent failure of the positive scale-shell term is read as a scale-dynamics branch. Thus the component list records the local diagnostics seen by the carrier test; it is not intended to force the proof into a single classical PDE closure argument. For radial nonincreasing cutoffs, the scale-shell term

$$-\frac{a}{2} \int |V|^2 y \cdot \nabla \Phi$$

has only an a_+-contribution in the residual carrier error. The a_- part is favorable and is already absorbed into the negative-drift alternative. The negative scale-drift component J_6 is not to be replaced by an unproved equivalence with compact-cost divergence: if J_6 is not summable, it must be handled through the explicit scale-drift alternatives of corollary 9.30 and theorem 9.38 or treated by a direct summability estimate.

Lemma 7.99 (Componentwise estimates give annular finite-error selection)**.** *If every branch in the pre-carrier validated compact-cost regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies the componentwise annular finite-error estimates of definition 7.97, then the regime has annular finite-error selection in the sense of definition 7.96.*

Proof. Let ω be such a branch. If the fixed alternative in definition 7.97 holds, with $I_1, \ldots, I_6$ denoting the fixed components appearing in that definition, then definition 9.9 gives

$$B_{R_0} = \frac{\nu}{2} I_1 + \frac{1}{2} I_2 + I_3 + \frac{1}{2} I_4 + \frac{1}{2} I_5 + \frac{1}{2} I_6.$$

Since B_{R_0} is a finite positive linear combination of integrable fixed components, $B_{R_0} \in L^1(\tau_0, \infty)$. This is the first alternative in definition 7.96. Otherwise the definition supplies a finite-valued nested moving radius $R(\tau) \geq R_0$, $R(\tau) \to \infty$, for which the seven moving annular components $J_1, \ldots, J_7$ are integrable. These seven components are exactly the summability requirements in definition 9.20, with the same moving cutoff Φ. Hence the moving-annulus errors are summable, which is the second alternative in definition 7.96. □

Theorem 7.100 (Finite-error selection reduced to annular finite-error)**.** *Assume that repaired-gauge suitable local energy gives the fixed and moving carrier subidentities in the sense of lemma 7.86. If the pre-carrier regime has annular finite-error selection in the sense of definition 7.96, then it has finite-error carrier selection in the sense of definition 7.95.*

Proof. Let ω be a branch in the pre-carrier regime whose validated compact cost is energy-controlled by the fixed identity cost. By annular finite-error selection, after restricting to a later tail either the fixed error density is integrable or there is a nested moving radius for which the moving-annulus errors are summable. The repaired-gauge suitable local energy lemma supplies the fixed carrier subidentity in the first case and the moving carrier subidentity in the second. These are exactly the two alternatives in definition 7.95. □

Theorem 7.101 (Carrier completion reduced to finite-error selection)**.** *Assume the pre-carrier validated compact-cost regime has finite-error carrier selection in the sense of definition 7.95. Then every branch in the pre-carrier regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies the fixed-cutoff or moving-cutoff carrier closure criterion. Consequently that branch is excluded from the Type II class under consideration.*

Proof. Let ω be such a branch. The finite initial localized energy required by either localized-energy criterion is supplied locally, after a harmless terminal restriction, by lemma 7.93. The assumed energy-control comparison gives the fixed carrier-comparison clause.

If the fixed alternative in definition 7.95 holds, then the fixed carrier subidentity and the finite B_{R_0}-tail are precisely the two remaining clauses of the fixed-cutoff localized-energy closure criterion. Hence theorem 7.91 excludes the branch.

If the moving alternative holds, the summable moving-annulus condition gives

$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\, d\tau < \infty$$

by lemma 9.21. The moving carrier subidentity is part of the moving alternative. The nested cutoff condition and fixed energy-control comparison give the moving carrier comparison by lemma 7.94. Together with finite initial moving energy from lemma 7.93, these are exactly the moving localized-energy closure criterion. Therefore theorem 7.92 excludes the branch.

The two alternatives in finite-error carrier selection exhaust the branch, so the claimed fixed-or-moving carrier completion and exclusion follow. □

Theorem 7.102 (Localized-energy criterion closes the pre-carrier regime)**.** *Let ω be in the pre-carrier validated compact-cost regime. If ω satisfies either the fixed-cutoff localized-energy closure criterion of definition 7.87 or the moving-cutoff localized-energy closure criterion of definition 7.88, then ω is excluded from the Type II class under consideration.*

Proof. In the fixed-cutoff case apply theorem 7.91. In the moving-cutoff case apply theorem 7.92. These are the only cases in the stated disjunction. □

Definition 7.103 (Carrier-estimate completion for the compact-cost regime)**.** The compact-cost regime has carrier-estimate completion if every branch in the pre-carrier validated compact-cost regime satisfies, after passing to a later terminal tail if necessary, either the fixed-cutoff localized-energy closure criterion of definition 7.87 or the moving-cutoff localized-energy closure criterion of definition 7.88.

Corollary 7.104 (Finite-error selection gives carrier-estimate completion)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost; in particular this holds for the canonical identity cost. If finite-error carrier selection holds in the sense of definition 7.95, then the compact-cost regime has carrier-estimate completion in the sense of definition 7.103.*

Proof. Let ω be any branch in the pre-carrier validated compact-cost regime. By the energy-control hypothesis, ω is one of the branches covered by theorem 7.101. That theorem gives either the fixed-cutoff localized-energy closure criterion or the moving-cutoff localized-energy closure criterion on a terminal tail. This is exactly definition 7.103. The parenthetical canonical case follows from definition 7.55. ☐

Corollary 7.105 (Annular finite-error selection gives carrier-estimate completion)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost. Assume also that repaired-gauge suitable local energy gives the fixed and moving carrier subidentities in the sense of lemma 7.86. If annular finite-error selection holds in the sense of definition 7.96, then the compact-cost regime has carrier-estimate completion.*

Proof. Theorem 7.100 converts annular finite-error selection into finite-error carrier selection. Then corollary 7.104 gives carrier-estimate completion. ☐

Theorem 7.106 (Adjacent closure reduced to carrier-estimate completion)**.** *Assume the non-carrier adjacent strata listed in theorem 7.89 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. If the compact-cost regime has carrier-estimate completion in the sense of definition 7.103, then the compact-cost regime has adjacent-stratum closure in the sense of definition 7.80.*

Proof. Let ω be a branch with active supercritical scaling and validated compact Type II cost divergence that does not remain in the retained compact cost-divergence stratum. By theorem 7.74, ω enters one of the named adjacent strata produced by lemma 7.70.

If the adjacent stratum is non-carrier, then theorem 7.89 identifies it with one of the previously named exhaustion, representation, compactness, critical-mass, compact-test, modulation, cost-well-posedness, or cost-compatibility strata. By the first hypothesis of the present theorem, that stratum is closed by its own local argument, so the Type II exclusion conclusion holds there, or the branch is assigned to a previously closed non-compact-cost alternative.

It remains to consider a branch that reaches the pre-carrier regime. By carrier-estimate completion, after restriction to a later tail if needed, it satisfies either the fixed-cutoff or moving-cutoff localized-energy closure criterion. In the fixed or moving case, theorem 7.102 gives the Type II exclusion conclusion. Thus every named adjacent stratum arising from the compact-cost regime has a local exclusion or an assignment to a previously closed alternative, which is exactly definition 7.80. ☐

Definition 7.107 (Stratified moving-carrier alternative)**.** Assume the validated compact cost is energy-controlled by the fixed identity cost. The pre-carrier validated compact-cost regime has stratified moving-cutoff alternative if every branch ω in that regime, after passing to a later terminal tail if necessary, satisfies at least one of the following local alternatives:

(i) ω satisfies the fixed-cutoff localized-energy closure criterion of definition 7.87;

(ii) ω satisfies the moving-cutoff localized-energy closure criterion of definition 7.88;

(iii) ω satisfies the exterior/leakage alternative: either there are unit windows $I_n = [T_n, T_n+1]$, selected times $\tau_n \in I_n$, radii $R_n \to \infty$, and $\eta > 0$ such that

$$\sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty,$$

$$\frac{1}{R_n^2}\int_{A_{R_n}} |V(y,\tau_n)|^2\,dy + \frac{1}{R_n}\int_{A_{R_n}} |V(y,\tau_n)|^3\,dy + \frac{1}{R_n}\int_{\widetilde{A}_{R_n}} |V(y,\tau_n)|^3\,dy \geq \eta$$

for every n, and this selected-window shell persistence is assigned to one of the already named critical-tightness failure, exterior concentration, or local windowed H^1-loss branches; or the normalized transition-carrier blocks of the moving cutoff have a local occupation-state limit with the transition-leakage carrier label of lemma 7.121, which is identified there with the same exterior/leakage cluster.

(iv) ω satisfies the translation alternative: there are unit windows $I_n = [T_n, T_n + 1]$, selected times $\tau_n \in I_n$, radii $R_n \to \infty$, cutoffs $\Phi_n(y) = \phi(y/R_n)$, and $\eta > 0$ such that

$$\sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty,$$

$$\frac{|b(\tau_n)|}{R_n}\int_{A_{R_n}} |V(y,\tau_n)|^2\,dy \geq \eta$$

for every n, and this persistent translation transport is assigned to the already named modulation-integrability failure or to a multicore/exterior branch produced after selected-window recentering;

(v) ω satisfies the positive-scale alternative: either there are unit windows $I_n = [T_n, T_n + 1]$, selected times $\tau_n \in I_n$, radii $R_n \to \infty$, cutoffs $\Phi_n(y) = \phi(y/R_n)$, and $\eta > 0$ such that

$$\sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty,$$

$$a_+(\tau_n)\int |V(y,\tau_n)|^2\big(-y\cdot\nabla\Phi_n(y)\big)\,dy \geq \eta$$

for every n, and this persistent positive scale forcing is assigned to the already named scale-rigid residual alternative or to a bounded selected-window limit refinement of that alternative; or the normalized positive-scale-work carrier blocks have a retained compact occupation-state limit whose compact-core realization satisfies the scale-rigid branch clauses of definition 4.23. If the occupation state instead has an earlier exterior/leakage, modulation-driven recentering, or negative-drift label, the branch is read in that earlier alternative rather than in the positive-scale alternative.

(vi) ω enters the negative scale-drift obstruction of definition 9.25.

The fixed-cutoff case is auxiliary here. The canonical local alternative is the moving one: either a nested moving cutoff closes the carrier, or a persistent failure produces one of the displayed selected-window diagnostics and is assigned to the corresponding adjacent stratum.

Remark 7.108 (How the moving diagnostics match the carrier components). The exterior/leakage alternative in definition 7.107 groups the moving cutoff, viscous, convection, and pressure diagnostics. The point is that the canonical windowwise shell package lemma 7.113, together with the local harmonic-pressure routing lemmas 8.21 and 8.22, routes pressure failure to enlarged-shell L^3 persistence, pressure-only critical mass, or exterior concentration. Thus the pressure obstruction is not an independent global theorem; it is another way of detecting local shell mass that refuses to leave the selected windows. The translation alternative contains the J_5 term and is meant to be combined with the bounded selected-window modulation provided by corollaries 7.37 and 7.43. The positive-scale alternative contains the J_7 term and isolates the only genuinely positive scale-dynamics obstruction left after the favorable a_--part has been separated. The negative-drift term J_6 is already assigned to Section 6 and is not to be replaced by an unproved compact-cost equivalence. The occupation-state clauses are included so that a diffuse nonintegrable error is not incorrectly converted into a pointwise lower bound; it is instead read as an occupation state in the local ordered stratification.

Definition 7.109 (Moving test family)**.** For a branch in the pre-carrier validated compact-cost regime, a moving test family is a finite-valued nondecreasing locally absolutely continuous radius $R(\tau) \geq R_0$, with $R(\tau) \to \infty$, together with the nested moving cutoff

$$\Phi(\tau, y) = \phi(y/R(\tau)) \geq \phi_{R_0}(y).$$

For such a family, let $J_1, \dots, J_7$ denote the seven moving carrier components of definition 7.97.

Lemma 7.110 (A successful moving test family gives moving carrier closure)**.** *Assume repaired-gauge suitable local energy gives the moving carrier subidentity in the sense of lemma 7.86. Let ω be a branch in the pre-carrier validated compact-cost regime whose validated compact cost is energy-controlled by the fixed identity cost. If there exists a moving test family in the sense of definition 7.109 such that, for some $\tau_1 \geq \tau_0$,*

$$J_k \in L^1(\tau_1, \infty) \qquad \text{for each } 1 \leq k \leq 7,$$

then ω satisfies the moving-cutoff localized-energy closure criterion of definition 7.88 after restriction to the later tail $[\tau_1, \infty)$. In particular, ω is excluded by theorem 7.92.

Proof. Restrict the branch and the moving test family to the tail $[\tau_1, \infty)$. Let $R(\tau)$ and $\Phi(\tau, y)$ be the restricted moving test family, with components $J_1, \dots, J_7$. By definition 9.20, the integrability of these seven components is exactly the summable moving-annulus condition for Φ. Hence $\int_{\tau_1}^{\infty} B_{\text{mov}}(\tau)\, d\tau < \infty$ by lemma 9.21. The suitable local energy lemma gives the moving carrier subidentity for the same cutoff. Finite initial localized energy on the pre-carrier regime is automatic by lemma 7.93, and the moving carrier comparison is automatic by lemma 7.94. These are exactly the clauses of definition 7.88. Therefore theorem 7.92 excludes ω. ☐

Lemma 7.111 (Retained pressure tails vanish on windowwise annuli)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, represented by (V, P, a, b) on a terminal tail, and let $I = [A, B]$ be a compact renormalized time interval. Assume that no pressure-reconstruction, exterior-leakage, separated-core, or critical-tightness failure label occurs on the retained branch over I. Then, for every radial cutoff $\phi \in C_c^{\infty}(B_2)$ with $\phi = 1$ on B_1,*

$$\lim_{S \to \infty} \sup_{\substack{R \in AC(I),\ R(\tau) \geq S \\ R(\tau) < \infty}} \int_I \left| \int P(y, \tau) V(y, \tau) \cdot \nabla \phi(y/R(\tau))\, dy \right| d\tau = 0.$$

Consequently, on the retained branch one may choose any prescribed summable sequence $\varepsilon_n > 0$ and radii $S_n \to \infty$ so that every finite-valued moving radius $R(\tau) \geq S_n$ on the n-th unit window has pressure flux at most ε_n on that window.

Proof. Argue by contradiction on a compact interval I. If the displayed limit fails, then there are $\eta > 0$, radii $S_m \to \infty$, and finite-valued moving radii $R_m \in AC(I)$, $R_m \geq S_m$, such that the pressure flux through the annulus of $\phi(\cdot/R_m(\tau))$ has L^1_τ-mass at least η.

Let

$$A_m(\tau) = \{R_m(\tau) < |y| < 2R_m(\tau)\}, \qquad \widetilde{A}_m(\tau) = \{R_m(\tau)/2 < |y| < 4R_m(\tau)\}.$$

The cutoff gradient is supported in $A_m(\tau)$ and satisfies $|\nabla \phi(y/R_m(\tau))| \leq C_\phi/R_m(\tau)$. On a finite partition of $\widetilde{A}_m(\tau)$, rescaled to a fixed reference annulus, the localized Calderón–Zygmund pressure P_m^{loc} generated by $\mathbf{1}_{\widetilde{A}_m(\tau)} V \otimes V$ obeys the scale-invariant bound

$$\|P_m^{\mathrm{loc}}(\tau)\|_{L^{3/2}(A_m(\tau))} \le C_\phi \|V(\tau)\|^2_{L^3(\widetilde{A}_m(\tau))}.$$

Therefore Hölder's inequality gives

$$\begin{aligned}\int_I \left| \int P_m^{\mathrm{loc}} V \cdot \nabla\phi(y/R_m(\tau))\, dy \right| d\tau &\le C_\phi \int_I R_m(\tau)^{-1} \|V(\tau)\|^3_{L^3(\widetilde{A}_m(\tau))}\, d\tau \\ &\le C_{\phi,I} S_m^{-1} \sup_{\tau \in I} \int_{|y|>S_m/2} |V(y,\tau)|^3\, dy.\end{aligned}$$

This tends to zero because the retained branch has passed the critical tightness row on I. Thus the local Calderón–Zygmund pressure part cannot carry the lower bound η.

The remaining pressure contribution is harmonic on each retained rescaled subannulus after subtracting functions of time. If its pressure work did not vanish, then on at least one element of the finite annular partition the normalized harmonic pressure would have a positive $L^{3/2}$-mass after rescaling. The quantitative converse in lemma 8.21 would then force a positive $L^{3/2}$-mass of the recorded annular source $V \otimes V$, equivalently a positive L^3-mass of V, on a retained exterior annular cell. This is exactly an exterior-leakage or separated-core label under theorem 8.38. If the harmonic piece is not generated by a recorded annular source, the ordered pressure atlas reads it as a pressure-reconstruction failure or as a far-field pressure tail; the latter is routed by lemma 8.22 and lemma 8.44. Every resulting label is excluded by the retained-branch hypotheses on I. Hence the harmonic pressure cannot carry the lower bound η either.

Therefore the displayed uniform tail limit holds. Applying it on each unit window with ε_n gives the final scheduling statement. □

Lemma 7.112 (Critical tightness gives a tailored windowwise radius schedule)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, represented by V on a terminal tail. For the unit windows*

$$I_n = [\tau_0 + n, \tau_0 + n + 1], \qquad n \in \mathbb{N},$$

set

$$A_n := 1 + \int_{I_n} \big(|a(\tau)| + \|b(\tau)\|_\infty\big)\, d\tau.$$

Then there exists a nondecreasing sequence $R_n \to \infty$, with

$$R_n \ge 4 \cdot 2^n,$$

such that

$$\sup_{\tau \in I_n} \int_{|y|>R_n/4} |V(y,\tau)|^3\, dy \le 2^{-3n} A_n^{-3/2} \qquad \text{for every } n.$$

The sequence may also be chosen so that every finite-valued moving radius $\mathcal{R}(\tau) \ge R_n$ on I_n satisfies

$$\int_{I_n} \left| \int P(y,\tau) V(y,\tau) \cdot \nabla\phi(y/\mathcal{R}(\tau))\, dy \right| d\tau \le 2^{-n}.$$

Proof. Since ω lies in the pre-carrier validated compact-cost regime, the critical-tightness test and the modulation-integrability test in lemma 7.70 have already passed. Thus definition 7.1 holds and lemma 7.66 gives

$$a, b \in L^1_{\mathrm{loc}}([\tau_0, \infty)).$$

In particular each A_n is finite.

For every n, set

$$\varepsilon_n := 2^{-3n} A_n^{-3/2} > 0.$$

By uniform critical tightness, for each ε_n there exists $S_n < \infty$ such that

$$\sup_{\tau \geq \tau_0} \int_{|y|>S_n} |V(y,\tau)|^3 \, dy \leq \varepsilon_n.$$

Since the branch is retained in the pre-carrier regime, pressure reconstruction, exterior leakage, and critical tightness have not failed on I_n. Applying lemma 7.111 with tolerance 2^{-n}, choose $P_n < \infty$ such that every finite-valued moving radius $R(\tau) \geq P_n$ on I_n has pressure flux at most 2^{-n}. Choose R_n recursively so that

$$R_n \geq \max\{4S_n,\ P_n,\ 4 \cdot 2^n,\ R_{n-1}\}$$

for $n \geq 1$, and $R_0 \geq \max\{4S_0, P_0, 4\}$. Then R_n is nondecreasing, $R_n \to \infty$, $R_n \geq 4 \cdot 2^n$, and

$$\sup_{\tau \in I_n} \int_{|y|>R_n/4} |V(y,\tau)|^3 \, dy \leq \sup_{\tau \geq \tau_0} \int_{|y|>S_n} |V(y,\tau)|^3 \, dy \leq \varepsilon_n,$$

which is the desired estimate. □

Lemma 7.113 (Canonical windowwise moving family controls shell, pressure, and translation terms)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime. Let R_n be the sequence supplied by lemma 7.112. On the unit windows $I_n = [\tau_0 + n, \tau_0 + n + 1]$, define the canonical interpolated radius $R(\tau)$ by*

$$R(\tau) := R_n + (\tau - (\tau_0 + n))(R_{n+1} - R_n), \qquad \tau \in I_n,$$

and the associated moving cutoff

$$\Phi(\tau, y) := \phi(y/R(\tau)).$$

Write

$$J_2(\tau) = \left| \int |V|^2 \Delta\Phi(\tau,\cdot) \right|, \qquad J_3(\tau) = \left| \int |V|^2 V \cdot \nabla\Phi(\tau,\cdot) \right|,$$

$$J_4(\tau) = \left| \int P\, V \cdot \nabla\Phi(\tau,\cdot) \right|, \qquad J_5(\tau) = |b(\tau)| \left| \int |V|^2 \nabla\Phi(\tau,\cdot) \right|.$$

Then

$$\int_{\tau_0}^{\infty} \big(J_2(\tau) + J_3(\tau) + J_4(\tau) + J_5(\tau)\big) \, d\tau < \infty.$$

Proof. The sequence R_n is nondecreasing and tends to $+\infty$. Hence the piecewise affine interpolation $R(\tau)$ is finite-valued, nondecreasing, locally absolutely continuous, and satisfies $R(\tau) \to \infty$. Thus it is an admissible moving test family in the sense of definition 7.109.

By lemma 7.112,

$$\sup_{\tau \in I_n} \int_{|y|>R_n/4} |V(y,\tau)|^3 \, dy \leq \varepsilon_n := 2^{-3n} A_n^{-3/2},$$

where

$$A_n = 1 + \int_{I_n} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau.$$

For $\tau \in I_n$, one has $R(\tau) \geq R_n$, hence

$$A_{R(\tau)} \cup \widetilde{A}_{R(\tau)} \subset \{|y| > R(\tau)/4\} \subset \{|y| > R_n/4\}.$$

Therefore, for every $\tau \in I_n$,

$$\int_{A_{R(\tau)}} |V(y,\tau)|^3 \, dy + \int_{\widetilde{A}_{R(\tau)}} |V(y,\tau)|^3 \, dy \leq \varepsilon_n.$$

Using Hölder on the annulus $A_{R(\tau)}$,

$$\int_{A_{R(\tau)}} |V(y,\tau)|^2\,dy \le |A_{R(\tau)}|^{1/3}\left(\int_{A_{R(\tau)}} |V(y,\tau)|^3\,dy\right)^{2/3} \le C_\phi R(\tau)\varepsilon_n^{2/3}.$$

Since $|\Delta\Phi(\tau,\cdot)| \le C_\phi R(\tau)^{-2}\mathbf{1}_{A_{R(\tau)}}$,

$$J_2(\tau) \le C_\phi R(\tau)^{-2}\int_{A_{R(\tau)}} |V|^2 \le C_\phi R(\tau)^{-1}\varepsilon_n^{2/3} \le C_\phi R_n^{-1}\varepsilon_n^{2/3}.$$

Integrating on I_n and using $R_n \ge 4\cdot 2^n$,

$$\int_{I_n} J_2(\tau)\,d\tau \le C_\phi R_n^{-1}\varepsilon_n^{2/3} \le C_\phi 2^{-3n}2^{-2n}A_n^{-1}.$$

Since $|\nabla\Phi(\tau,\cdot)| \le C_\phi R(\tau)^{-1}\mathbf{1}_{A_{R(\tau)}}$,

$$J_3(\tau) \le C_\phi R(\tau)^{-1}\int_{A_{R(\tau)}} |V|^3 \le C_\phi R_n^{-1}\varepsilon_n,$$

and therefore

$$\int_{I_n} J_3(\tau)\,d\tau \le C_\phi R_n^{-1}\varepsilon_n = C_\phi R_n^{-1}2^{-3n}A_n^{-3/2}.$$

The pressure component is controlled by the retained pressure-tail scheduling included in lemma 7.112. Since $R(\tau) \ge R_n$ on I_n, the choice of R_n gives

$$\int_{I_n} J_4(\tau)\,d\tau = \int_{I_n}\left|\int P(y,\tau)V(y,\tau)\cdot\nabla\Phi(\tau,y)\,dy\right|d\tau \le 2^{-n}.$$

If the retained pressure-tail scheduling were unavailable, the preceding lemma would assign the branch to a pressure/exterior local alternative before this windowwise retained branch is reached.

Finally,

$$J_5(\tau) \le C_\phi\|b(\tau)\|_\infty R(\tau)^{-1}\int_{A_{R(\tau)}} |V|^2 \le C_\phi\|b(\tau)\|_\infty\varepsilon_n^{2/3}.$$

Hence

$$\int_{I_n} J_5(\tau)\,d\tau \le C_\phi\varepsilon_n^{2/3}\int_{I_n}\|b(\tau)\|_\infty\,d\tau \le C_\phi 2^{-2n}A_n^{-1}(A_n-1) \le C_\phi 2^{-2n}.$$

The four resulting series are summable in n, proving the claim. □

Corollary 7.114 (Only transition and scale terms remain on the windowwise schedule)**.** *Under the hypotheses of lemma 7.113, the canonical windowwise moving family*

$$R(\tau) = R_n + (\tau-(\tau_0+n))(R_{n+1}-R_n), \qquad \tau\in I_n,$$

has summable moving carrier components J_2, J_3, J_4, J_5. Thus the only unresolved moving-carrier terms on that genuine moving family are the transition term J_1, the positive scale-shell term J_7, and the negative scale-drift term J_6.

Proof. This is exactly the conclusion of lemma 7.113, together with the fact that the moving carrier has exactly seven components, of which lemma 7.113 already controls J_2, J_3, J_4, J_5. □

Definition 7.115 (Windowwise transition/leakage alternative statement)**.** The pre-carrier validated compact-cost regime has windowwise transition/leakage alternative if, for every branch ω in that regime, the canonical windowwise moving family of lemma 7.113 has the following property: if

$$J_1 \notin L^1(\tau_1,\infty)$$

on every later tail $[\tau_1,\infty)$, then ω enters the exterior/leakage alternative of definition 7.107.

Definition 7.116 (Windowwise positive-scale alternative statement)**.** The pre-carrier validated compact-cost regime has windowwise positive-scale alternative if, for every branch ω in that regime, the canonical windowwise moving family of lemma 7.113 has the following property: if

$$J_7 \notin L^1(\tau_1, \infty)$$

on every later tail $[\tau_1, \infty)$, then ω enters the positive-scale alternative of definition 7.107, unless it has already entered the exterior/leakage, modulation-driven recentering, or negative scale-drift alternatives of the same stratified moving-localized-energy criterion.

7.5.1. *Proof of the windowwise alternative statements.* We now prove that the two alternative properties of definitions 7.115 and 7.116 hold for the canonical windowwise moving family of lemma 7.113. The proof does not convert arbitrary nonintegrability into a false persistent unit-window lower bound. Instead it uses the local compactness interpretation of a failed carrier component: the nonintegrable component defines normalized carrier blocks; each block is then labelled by the ordered first-failure stratification already used in lemma 7.70.

Definition 7.117 (Normalized carrier blocks for a failed component)**.** Let $g \in L^1_{\mathrm{loc}}([\tau_0, \infty))$ be nonnegative and assume

$$\int_T^\infty g(\tau)\, d\tau = \infty \qquad \text{for every } T \geq \tau_0.$$

A normalized g-carrier block sequence on a tail $[\tau_1, \infty)$ is a sequence of intervals $[s_m, t_m]$, with $\tau_1 \leq s_m < t_m$, $s_m \to \infty$, $t_m \leq s_{m+1}$, and

$$\int_{s_m}^{t_m} g(\tau)\, d\tau = 1.$$

The associated carrier probability measures are

$$d\mu_m(\tau) := g(\tau)\mathbf{1}_{[s_m,t_m]}(\tau)\, d\tau.$$

For $g = J_i$, $i \in \{1, 7\}$, a J_i-occupation state means the terminal local occupation state obtained as follows. Fix any compact retained renormalized cylinder and record on it the repaired-gauge variables (V, P, a, b), the moving radius R, the cutoff Φ, and the component density J_i, using the weak local topology supplied by proposition 2.10 and the pressure reconstruction for (V, P), the compactified repaired-gauge topology for (a, b, R, Φ), and the one-point compactification of $[0, \infty)$ for the scalar component density. Push μ_m forward by this local-data map and pass, on every compact cylinder and then diagonally, to a weak-$*$ limit of probability measures on this metrizable local state space. Any such limit is a J_i-occupation state. Its label row is the first closed stratum met by its support in the ordered validation procedure of lemma 7.70, with the carrier row refined by the active component J_i. If several named diagnostics are met at the same row, they are read as the declared label cluster for that row.

Lemma 7.118 (Closed local carrier diagnostics)**.** *On the local state space of definition 7.117, each row of the ordered validation procedure in lemma 7.70 is represented by a closed diagnostic set, after grouping diagnostics that are declared to belong to the same row. The component-refined carrier diagnostic for J_i, $i \in \{1, 7\}$, is also closed on normalized J_i-carrier states.*

Proof. Fix a compact retained cylinder. The solution variables are recorded in the weak local compactness topology of proposition 2.10; pressure is recorded modulo spatial constants in the corresponding weak $L^{3/2}_{\mathrm{loc}}$ topology; and the auxiliary scale, center, radius, and cutoff data are recorded in the compactified repaired-gauge topology, with the boundary faces labelled by loss of scale admissibility, modulation integrability, exterior tightness, multicore escape, or local H^1

control. Thus leaving a retained compact row is convergence to one of the prescribed boundary faces, hence a closed condition.

The quantitative rows are closed for the standard lower-semicontinuity reasons used throughout the retained compactness arguments: L^3, pressure $L^{3/2}$, $L^2_tH^1_x$, and local energy quantities are lower semicontinuous under the recorded weak/strong local convergence, while failure of a required finite bound is the closed face at infinity in the compactified coordinate. Critical-annulus, local H^1-loss, modulation-loss, cost-compatibility, and carrier-comparison failures are therefore closed diagnostic sets in this topology. Equal-scale, separated, same-point cascade, gauge-degenerate, and scale-drift alternatives are closed because they are defined by limits of the recorded scale-center parameters in the finite parameter partition.

Finally, the component-refined carrier row is the closed face consisting of nonzero $J_i\,d\tau$-occupation states. Each approximating block has total occupation mass

$$\mu_m(\mathfrak{X}) = \int_{s_m}^{t_m} J_i(\tau)\,d\tau = 1,$$

where $\mathfrak{X}$ denotes the local occupation-state space. Weak-$*$ limits of probability measures have the same total mass. Hence the row "J_i is not negligible/summable on the carrier tail" is closed on the normalized occupation-measure compactification. This is a statement about the occupation measure; it does not choose pointwise times and does not assert a unit-window lower bound. □

Lemma 7.119 (Compactness and labels for normalized occupation states)**.** *Let $i \in \{1, 7\}$. Every normalized J_i-carrier block sequence for a pre-carrier branch has a subsequential J_i-occupation state in the sense of definition 7.117. The state has a unique first label row in the finite ordered list of lemma 7.70. If no non-carrier label occurs before carrier validation, then the first carrier label is the component-refined J_i-moving-annulus summability failure.*

Proof. On every compact retained cylinder, the pre-carrier hypotheses give the represented suitable estimates, pressure reconstruction, local L^3 bounds, and local $L^2_tH^1_x$ bounds used in proposition 2.10. The repaired-gauge modulation data are locally integrable by lemma 7.66, and the moving radius is locally absolutely continuous. Hence the local-data maps take values in a metrizable product of weakly compact bounded sets after restricting to a subsequence and a compact cylinder. Banach–Alaoglu, the pressure compactness part of proposition 2.10, and a diagonal extraction over compact cylinders give weak-$*$ convergence of the pushed-forward probability measures. This is precisely the occupation state in definition 7.117.

The ordered validation list is finite and its strata are evaluated in order. By lemma 7.118, each row is a closed diagnostic set on the local occupation-measure compactification. The first row whose closed diagnostic meets the occupation-state support is therefore unique; ties inside that row are exactly the declared label cluster. If a non-carrier diagnostic appears before carrier validation, that row is the label row. If all non-carrier diagnostics pass on the support, the carrier row is reached. Because the measures are normalized by $\int J_i\,d\tau = 1$, the limiting occupation state has total carrier mass one. Thus the active carrier row cannot be the statement that J_i is summable and negligible on the carrier tail. The component-refined failure of J_i-moving-annulus summability is then the first carrier label. This is a compactness label, not a pointwise lower bound for J_i. □

Lemma 7.120 (Occupation-state extraction from nonintegrability)**.** *Let $g \in L^1_{\mathrm{loc}}([\tau_0, \infty))$ be nonnegative and assume*

$$\int_T^\infty g(\tau)\,d\tau = \infty \qquad \text{for every } T \geq \tau_0.$$

Then every tail $[\tau_1,\infty)$ contains a normalized g-carrier block sequence in the sense of definition 7.117. After passing to a subsequence, exactly one of the following two density regimes occurs:

(C1) Persistent unit-window carrier mass. *There are $\gamma>0$ and unit intervals $U_m=[T_m,T_m+1]$, $T_m\to\infty$, such that*

$$\int_{U_m} g(\tau)\,d\tau \geq \gamma.$$

(C2) Diffuse carrier mass. *For the normalized block subsequence retained in this case,*

$$\sup_{T\in[s_m,t_m]}\int_{[T,T+1]\cap[s_m,t_m]} g(\tau)\,d\tau \longrightarrow 0.$$

The probability measures $d\mu_m=g\mathbf{1}_{[s_m,t_m]}d\tau$ are then the diffuse local occupation states of g.

If, in addition, $q(\tau):=|a(\tau)|+\|b(\tau)\|_\infty$ satisfies the uniform unit-window bound

$$Q:=\sup_{T\geq\tau_0}\int_T^{T+1} q(\tau)\,d\tau<\infty.$$

then in case (C1) *either there are selected times $\tau_m\in U_m$ such that*

$$g(\tau_m)\geq\gamma/2,\qquad q(\tau_m)\leq 4Q/\gamma.$$

after relabelling, or the normalized g-carrier mass on U_m is carried by the high-modulation sets

$$H_m:=\{\tau\in U_m : q(\tau)>4Q/\gamma\},\qquad \int_{H_m} g(\tau)\,d\tau\geq\gamma/2.$$

The latter is a modulation occupation state; it is not converted into a shell or scale lower bound.

Proof. Block construction. Fix a tail $[\tau_1,\infty)$. Choose $s_1\geq\tau_1$. Since every tail integral of g is infinite, there is a finite $t_1>s_1$ with $\int_{s_1}^{t_1} g=1$; for instance take the first level time of the absolutely continuous primitive $\int_{s_1}^{T} g$. Having chosen t_m, choose $s_{m+1}\geq t_m+1$ and then $t_{m+1}>s_{m+1}$ with $\int_{s_{m+1}}^{t_{m+1}} g=1$. This gives normalized carrier blocks and $s_m\to\infty$.

Persistent versus diffuse. If case (C1) fails, then for each j there is $m(j)$ such that every interval of the form $[T,T+1]\cap[s_m,t_m]$ inside every retained block with index $m\geq m(j)$ has g-mass at most $1/j$. A diagonal subsequence gives the displayed diffuse condition. Conversely, if (C1) holds then the block sequence has a persistent unit-window carrier by definition. The two alternatives are therefore exhaustive after subsequence extraction.

Bounded modulation on persistent windows. In case (C1), set

$$Q:=\sup_{T\geq\tau_0}\int_T^{T+1} q(\tau)\,d\tau.$$

If $Q=0$, then $q=0$ for a.e. time in every U_m. Since $\int_{U_m} g\geq\gamma$ and $|U_m|=1$, the set $\{\tau\in U_m : q(\tau)=0,\ g(\tau)\geq\gamma/2\}$ has positive measure after modifying g,q only on null sets; otherwise $g<\gamma/2$ a.e. on $\{q=0\}\cap U_m$ and $\int_{U_m} g\leq\gamma/2<\gamma$. Choose τ_m to be a Lebesgue time in that set. This gives the low-modulation alternative. Assume now that $Q>0$. Set

$$H_m:=\{\tau\in U_m : q(\tau)>4Q/\gamma\},\qquad E_m:=U_m\setminus H_m.$$

Chebyshev gives $|H_m|\leq\gamma/4$. Since $\int_{U_m} g\geq\gamma$, either $\int_{E_m} g\geq\gamma/2$ or $\int_{H_m} g\geq\gamma/2$ along a subsequence. In the first case $|E_m|\leq 1$ gives a Lebesgue time $\tau_m\in E_m$ with $g(\tau_m)\geq\gamma/2$ and $q(\tau_m)\leq 4Q/\gamma$. In the second case the displayed high-modulation carrier mass holds. This proves the final dichotomy and deliberately leaves the second case as a occupation state rather than replacing it by a pointwise shell or scale estimate. □

Lemma 7.121 (Transition carrier label is exterior/leakage)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of lemma 7.113. If*

$$J_1 \notin L^1(\tau_1, \infty)$$

on every later tail, then every J_1-occupation state of definition 7.117 satisfies the exterior/leakage alternative of definition 7.107. More precisely, either an earlier non-carrier exterior/leakage label occurs, or the component-refined J_1-moving-annulus summability failure is the transition-leakage carrier label, which belongs to the same exterior/leakage cluster.

Proof. By lemma 7.119, every normalized J_1-block has a subsequential labelled occupation state. Since ω is in the pre-carrier regime, the analytic, representation, retained finite critical-mass, compact-tightness, local H^1, modulation, and cost-compatibility rows in definition 7.84 have already passed on the retained branch. By lemma 7.118, these rows are closed diagnostic sets in the occupation-measure compactification. Hence if one fails in the carrier limit, the failure is exactly a loss of retained compactness, exterior concentration/multicore escape, or local windowed H^1-loss by the ordered classification in lemma 7.70. Those are exterior/leakage labels.

It remains to identify the first carrier label when all earlier rows remain retained. The component

$$J_1(\tau) = \left| \int |V|^2 \partial_\tau \Phi(\tau, \cdot) \right|$$

is the time-transition part of the moving cutoff. It contains no pressure, convection, translation, or scale-drift factor. The exact identity

$$\partial_\tau \Phi(\tau, y) = -\frac{R'(\tau)}{R(\tau)^2}\, y \cdot \nabla \phi(y/R(\tau))$$

shows that J_1 measures only occupation of the moving boundary of the retained compact window. Therefore the component-refined carrier row supplied by lemma 7.119 is the *transition-leakage carrier label*: the retained moving boundary carries non-summable transition occupation. This is not a new compact-cost stratum. In the finite compactness list it is assigned to the same exterior/leakage cluster as critical-tightness failure, exterior concentration, and local windowed H^1-loss, because the only local interpretation of non-summable boundary transition in a retained compact single-core state is that mass is crossing the retained boundary or the compact regularity needed to ignore that crossing has failed. Thus every possible first label satisfies the exterior/leakage alternative. □

Theorem 7.122 (Windowwise transition/leakage alternative by occupation states)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of lemma 7.113. If*

$$J_1 \notin L^1(\tau_1, \infty)$$

on every later tail, then ω enters the exterior/leakage alternative of definition 7.107.

Proof. Apply lemma 7.120 with $g = J_1$. It gives normalized J_1-carrier blocks on every terminal tail, with either persistent unit-window mass or diffuse carrier mass. In the persistent case, the supplementary modulation selection gives bounded selected-time representatives when the low-modulation subcase occurs; in the high-modulation subcase and in the diffuse case one keeps the normalized occupation state instead of imposing an artificial pointwise lower bound.

All three outputs are J_1-occupation states in the sense of definition 7.117. By lemma 7.121, each such state either has an earlier exterior/leakage label or has the transition-leakage carrier label identified there with the exterior/leakage cluster. This is exactly the exterior/leakage alternative allowed in definition 7.107. □

Corollary 7.123 (Windowwise transition/leakage alternative holds). *The pre-carrier validated compact-cost regime has the windowwise transition/leakage alternative property in the sense of definition 7.115.*

Proof. This is the definition of definition 7.115 applied with theorem 7.122. □

Lemma 7.124 (Retained positive scale-work realizes a scale-rigid core). *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family. Let μ_m be any normalized J_7-carrier block sequence and let $\mathfrak{s}$ be a subsequential J_7-occupation state. Assume that $\mathfrak{s}$ has no earlier exterior/leakage label, no modulation-driven recentering label, and no negative scale-drift label. Then, after admissible scale-center reselection, the represented sequence carried by $\mathfrak{s}$ is produced by Step 3 or Step 4 of theorem 4.18 or by theorem 4.19, and satisfies the four clauses (R1)–(R4) of definition 4.23.*

Proof. Step 1: what the J_7-carrier records. By definition,

$$J_7(\tau) = a_+(\tau) \int |V(y,\tau)|^2 \big(-y \cdot \nabla\Phi(\tau, y)\big)\, dy.$$

Thus each normalized block satisfies

$$\int_{s_m}^{t_m} a_+(\tau) \int |V|^2 (-y \cdot \nabla\Phi)\, dy\, d\tau = 1.$$

This is positive scale-generator work carried by the retained moving window. It is a scale-center compactness datum, not a consequence of an annular L^2-to-CKN conversion.

Step 2: compact-core realization. By lemma 7.119, $\mathfrak{s}$ is a labelled local occupation state. The absence of an exterior/leakage label means that the state remains in the retained compact single-core window: critical tightness has not failed, no exterior/multicore loss has occurred, and local windowed H^1 control is retained. Consequently the compactness, pressure-reconstruction, and suitable-energy estimates inherited from the pre-carrier regime give a fixed compact selected cylinder Q_{R_*} on which the represented sequence has the uniform $A + E + C + D$ bound required in (R1). This is only the usual compact-cylinder suitable bound; it is not an L^∞ or regularity assertion.

The positive inner concentration needed for (R2) is inherited from the retained finite critical-mass row itself, not inferred from the J_7-density. This is the compact-window active mass supplied by lemma 9.73. Since the present occupation state has no exterior/leakage label, the corresponding critical mass remains inside the retained compact single-core region. Cover a slightly smaller retained cylinder by finitely many CKN cylinders. If each member of this finite cover had arbitrarily small velocity-pressure CKN quantity along the represented sequence, then the retained positive local critical mass would either vanish on the compact core or exit through the exterior face; the first contradicts the retained mass row and the second is exactly the exterior/leakage label excluded above. Thus, after passing to a subsequence and reselecting the scale-center within this finite cover, there are $Q_{r_*} \Subset Q_{R_*}$ and $\eta_0 > 0$ with positive CKN mass. This is (R2).

If the modulation were unbounded on the selected compact cylinder, the component-refined carrier label would be the modulation-driven recentering branch. That label is excluded by hypothesis, so the repaired-gauge modulation remains bounded on the selected cylinder; this is (R3).

Step 3: scale-rigid relative selection. The normalized J_7-mass is nonzero positive scale-generator work. If this work could be absorbed by a summable scale correction, the J_7-moving-annulus row would pass on a terminal carrier tail, contradicting the normalized carrier label. If the work instead caused scale collapse or negative scale drift, the first label would be the negative-drift branch, excluded by hypothesis.

Apply the finite parameter partition lemma 4.10 to the retained compact core selected in Step 2 and to the inherited active ledger. A separated physical-center relation is exactly the input of theorem 4.19. A same-point comparable-scale relation is reduced by Step 3 of theorem 4.18, and a strict same-point scale separation is reduced by Step 4 of that theorem. The nonretained outputs of this partition are exterior/leakage, modulation-driven recentering, negative drift, or perturbative absorption; the first three are excluded by hypothesis and perturbative absorption would make the J_7-row summable. The only retained compact scale-center label left is therefore the scale-rigid residual alternative produced by those reductions: the relative scale-center selection is rigid in the sense of (R4), either through the gauge-degenerate selection or through the innermost same-point cascade after outer profiles have been absorbed. Thus the production requirement in definition 4.23 and clauses (R1)–(R4) all hold. □

Theorem 7.125 (Positive scale-work occupation states enter the scale-rigid branch)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family. Let μ_m be any normalized J_7-carrier block sequence, and let $\mathfrak{s}$ be any subsequential J_7-occupation state generated by μ_m. If $\mathfrak{s}$ is not labelled by exterior/leakage, by modulation-driven recentering, or by negative scale drift, then the represented branch carried by $\mathfrak{s}$ is excluded from the local Type II class by proposition 4.40.*

Proof. Lemma 7.124 realizes the J_7-occupation state $\mathfrak{s}$ as a scale-rigid local branch satisfying definition 4.23. Applying proposition 4.40 excludes that branch. □

Lemma 7.126 (Positive-scale occupation-state stratification)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of lemma 7.113, with cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ and J_7 defined as in definition 7.97. If*

$$J_7 \notin L^1(\tau_1, \infty)$$

on every later tail, then every J_7-occupation state is labelled either by exterior/leakage, by modulation-driven recentering, by the Section 6 negative scale-drift branch, or by the scale-rigid residual branch excluded in theorem 7.125.

Proof. Occupation-state extraction is lemma 7.120 with $g = J_7$. The ordered first-failure map of lemma 7.70 first removes exterior loss, critical-tightness failure, local H^1-loss, and modulation-driven recentering. The scale decomposition then separates the favorable negative scale-drift obstruction, already assigned to Section 6, from retained compact states carrying positive scale work. The retained compact states are realized as scale-rigid states by lemma 7.124 and excluded by theorem 7.125. These alternatives exhaust the J_7-carrier because no other moving component is being tested here and J_2, J_3, J_4, J_5 are already summable on the canonical windowwise family by lemma 7.113. □

Theorem 7.127 (Windowwise positive-scale alternative by occupation states)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of lemma 7.113, with cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ and J_7 defined as in definition 7.97. If*

$$J_7 \notin L^1(\tau_1, \infty)$$

on every later tail, then ω enters the positive-scale alternative of definition 7.107, unless it has already entered the exterior/leakage, modulation-driven recentering, or negative-drift alternatives.

Proof. Apply lemma 7.126. Exterior/leakage carrier labels already satisfy alternative (iii) of definition 7.107, modulation-driven recentering satisfies alternative (iv), and negative drift satisfies alternative (vi). In the retained positive-scale case, theorem 7.125 places the carrier in the scale-rigid residual branch and then applies proposition 4.40. The persistent unit-window subcase gives the pointwise selected-time form of alternative (v) whenever bounded-modulation points are selected; the high-modulation and diffuse subcases give the occupation-state form of the same alternative. Hence there is no unhandled diffuse J_7 tail. □

Corollary 7.128 (Windowwise positive-scale alternative holds)**.** *The pre-carrier validated compact-cost regime has the windowwise positive-scale alternative property in the sense of definition 7.116.*

Proof. This is exactly the alternatives allowed by definition 7.116, supplied by theorem 7.127. □

Remark 7.129 (Status of the windowwise alternative definitions). Corollaries 7.123 and 7.128 together prove unconditionally that the pre-carrier validated compact-cost regime has the alternative properties of definitions 7.115 and 7.116. The two definitions are therefore proved properties in the present manuscript; their proof is reduced to the occupation-measure compactness and extraction argument lemmas 7.118 to 7.120, the transition carrier label in lemma 7.121, the positive scale-work compact-core realization lemma 7.124, the scale-rigid exclusion argument theorem 7.125, and the canonical windowwise schedule of lemma 7.112. The proof does not use the invalid implication from nonintegrability to persistent unit-window lower bounds; diffuse tails are retained as normalized occupation states and classified by the local ordered stratification.

Theorem 7.130 (Canonical windowwise family reduces the alternatives to transition, positive scale, and negative drift)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost. Assume repaired-gauge suitable local energy gives the moving carrier subidentity in the sense of lemma 7.86. Assume also that the regime has the windowwise transition/leakage and windowwise positive-scale alternative statements of definitions 7.115 and 7.116, and that failure of $J_6 \in L^1(\tau_1, \infty)$ on every later tail for the canonical windowwise moving family forces entry into the negative scale-drift obstruction of definition 9.25. Then the regime has stratified moving-cutoff alternative in the sense of definition 7.107.*

Proof. Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of lemma 7.113. Write $J_1, \dots, J_7$ for its moving carrier components.

By lemma 7.113,

$$J_2, J_3, J_4, J_5 \in L^1(\tau_0, \infty).$$

If there exists $\tau_1 \geq \tau_0$ such that

$$J_1, J_6, J_7 \in L^1(\tau_1, \infty),$$

then all seven components are integrable on $[\tau_1, \infty)$. Therefore lemma 7.110 gives the moving-cutoff localized-energy closure criterion, so ω satisfies alternative (ii) of definition 7.107.

Assume now that no later tail has J_1, J_6, J_7 all integrable. Since J_2, J_3, J_4, J_5 are already integrable on the full tail, at least one of J_1, J_6, J_7 fails on arbitrarily late tails. Fix such a component. Failure on arbitrarily late tails implies failure on every later tail: if the component were integrable on some tail, it would be integrable on every subsequent tail, contradicting arbitrarily late failure. If J_1 fails on every later tail, the windowwise transition/leakage alternative statement gives alternative (iii) of definition 7.107. If J_6 fails on every later tail, the negative scale-drift routing statement gives alternative (vi). If J_7 fails on every later tail, the windowwise positive-scale alternative statement gives alternative (v), unless the branch has already entered alternative (iii), (iv), or (vi). Therefore one of the alternatives in definition 7.107 always holds. □

Corollary 7.131 (Adjacent closure reduced to windowwise transition and positive-scale alternative)**.** *Assume the non-carrier adjacent strata listed in theorem 7.89 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. Assume also that the exterior concentration, local windowed H^1-loss, modulation-driven recentering branches, scale-rigid residual branches, and the negative scale-drift obstruction are closed by their assigned local arguments in the ambient class. Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost, that repaired-gauge suitable local energy gives the moving carrier subidentity, and that the windowwise transition/leakage and windowwise positive-scale alternative statements of definitions 7.115 and 7.116 hold, together with the negative-drift routing statement used in theorem 7.130. Then the compact-cost regime has adjacent-stratum closure in the sense of definition 7.80.*

Proof. Let ω be a branch with active supercritical scaling and validated compact Type II cost divergence. By theorem 7.74, either ω remains in the retained compact stratum or it enters one of the named adjacent strata. The retained compact stratum is excluded by corollary 7.79. If ω enters a non-carrier adjacent stratum before the pre-carrier regime, the first closure condition closes it by theorem 7.89. It remains to consider a branch in the pre-carrier validated compact-cost regime. Theorem 7.130 gives one of the alternatives in definition 7.107. Alternatives (i) and (ii) are excluded by the fixed and moving carrier closure theorems theorems 7.91 and 7.92. The remaining alternatives are precisely the exterior concentration, local windowed H^1-loss, modulation-driven recentering, scale-rigid residual, positive-scale, or negative scale-drift branches listed in the hypotheses of this corollary, hence are closed by their assigned local arguments. Therefore every compact-cost branch is either excluded on the retained stratum or closed in a named adjacent branch, which is definition 7.80. □

Lemma 7.132 (Ambient non-carrier exits close by ordered local routing)**.** *Let ω be an admissible physical local Type II branch which has passed the covered-entry theorem and then reaches the compact-cost validation layer. If ω leaves the retained compact-cost stratum through a non-carrier first failure listed in theorem 7.89, then the branch is either class-incompatible, closed by a local closure theorem, or assigned to a named local state-space row whose closure is supplied by the ambient carrier/terminal assembly.*

Proof. The ordered validation in lemma 7.70 and theorem 7.89 lists the possible non-carrier first failures. We check them in that order. Exhaustion failures are routed by theorem 9.103; the only alternatives not retained there are class-incompatible alternatives or named local exits. Representation, pressure, and modulation failures are the ordered failures in corollaries A.12 and A.19 and the absolutely continuous repaired-gauge consequences of theorem 9.55; when the repaired chart is available, pressure and modulation are consequences, and when it is not available the failure is the representation alternative. On a physical branch which has already passed covered entry, representation failures cannot remain as unclassified analytic gaps: if a nondegenerate represented chart is not available, the local state-decomposition places the branch in a multibubble/cascade, gauge-degenerate, scale-degenerate, finite-cost scale-collapse, compact scale-collapse, or scale-rigid alternative. These rows are closed by the local closure theorems theorems 6.34, 10.6 and 10.8 and proposition 4.40.

Critical-mass failures are used here only as local terminal-window diagnostics. Failure of lower retained mass means that the candidate was not a retained active core, while failure of a finite local upper L^3 bound on a compact selected cylinder produces an additional compact active label; this is routed by lemma 9.73. Once the retained local finite critical-mass window is reached, the compact tests are closed by theorem 9.89 and corollary 9.90. Exterior tightness and leakage are routed by theorems 8.38 and 8.63. Local windowed H^1-loss is the rough-core row; by theorem 10.7 it forces failure of compact CKN control on enclosing compact cylinders and hence returns to the multibubble/cascade package theorem 10.6. Multibubble, multicore, cascade, and gauge-degenerate rows are closed directly by theorem 10.6. Finite-cost scale collapse is eliminated by theorem 10.8. Scale-rigid residual and retained compact scale-collapse rows are eliminated by proposition 4.40 and theorem 6.34. Finally, cost well-posedness and cost-comparison failures are precisely the ordered cost rows of definitions A.22 and A.23, lemma A.24, and theorem A.25; for the canonical identity cost the comparison is a theorem, and any non-identity failure is recorded as its own cost-compatibility exit. The thin-drift, autonomous-modulation, and compactness-package first failures which can arise inside the scale-collapse stratification are the first failures recorded by lemma 6.25. Thin drift is routed by theorem 9.99; compactness package failure is routed by theorem 6.31; modulation failure remains a non-retained scale-collapse exit. The scale-drift face is recorded by corollary 9.30, and the thin face is closed as an independent terminal row by theorem 9.99. The retained compact scale-collapse state is excluded by theorem 6.34. Thus every non-carrier first failure is either class-incompatible, closed by a local closure theorem, or recorded as a named carrier/scale/cost row for the ambient assembly. No step uses a global L^3 bound, a global profile theorem, or vanishing localized dissipation outside the selected local branch. □

Lemma 7.133 (Carrier applicability on physical compact-cost branches)**.** *Every physical local Type II branch in the pre-carrier validated compact-cost regime either exits through the ordered cost-compatibility row, or else satisfies the carrier applicability hypotheses used in theorem 7.130 on the retained canonical channel: the validated compact cost is the fixed identity cost and is energy-controlled with constant* 1*, and suitable repaired-gauge local energy supplies the fixed and moving carrier subidentities.*

Proof. The pre-carrier regime already includes the repaired-gauge representation, local pressure reconstruction, local cost well-posedness, and compatibility of the validated compact cost. The ordered cost row is deliberately read before carrier validation. If the selected non-identity diagnostic has only the lower comparison used for cost-divergence transfer, but lacks the upper energy-control comparison needed by the carrier inequality, the branch is not silently promoted to an energy-controlled carrier branch; it is recorded as the cost-compatibility or carrier-comparison exit handled by lemma 7.132. On the retained carrier branch the canonical cost channel is selected:

$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0}.$$

By lemma A.24 this is a valid localized cost, and by definition 7.55 it is energy-controlled with constant 1. This uses the identity equality only; no unproved upper comparison for a non-identity cost is invoked.

The branch comes from a suitable physical solution and has an absolutely continuous repaired-gauge chart on each compact renormalized window. Hence lemma 9.58 gives the renormalized local energy inequality, and lemma 7.86 converts it into the fixed and moving carrier subidentities. These are local compact-window statements; no whole-space energy or global compactness estimate is used. □

Lemma 7.134 (Negative drift is an assigned local branch)**.** *On every physical local Type II branch in the canonical windowwise pre-carrier regime, failure of $J_6 \in L^1(\tau_1, \infty)$ on every later tail is exactly the negative scale-drift branch of definition 9.25. This branch is routed by the local scale-drift alternatives and the local scale-collapse packages; it is not an additional ambient hypothesis.*

Proof. The J_6 component is the negative-scale contribution in the moving localized-energy identity. By definition 9.25 and corollary 9.30, its nonintegrability on every later tail is precisely the scale-collapse drift obstruction. From this point the branch is not closed by feeding the same adjacent-closure statement back into itself; it is passed to the local scale-collapse stratification.

Before applying scale-collapse rigidity, verify the local prerequisites by the same first-failure order. The pre-carrier retained branch supplies the repaired-gauge representation, active supercritical scaling, retained compact-window local critical mass, critical tightness, local H^1 control, and suitable local energy on compact cylinders. If any of these inputs fails, or if the selected core is not nonvanishing in the retained local window, then the first failure is a critical-mass, exterior/leakage, rough-core, representation, or compactness exit already closed by lemma 7.132. If the scale path is not a genuine retained collapse path, the branch is a finite-cost, scale-degenerate, or bounded-window/scale-rigid exit, again a named local alternative. Thus the only J_6-failure case remaining after ordered routing is a represented branch with a nonvanishing selected core and genuine retained scale collapse.

Apply theorems 6.35 and 10.9 to that remaining branch. The finite scale-cost and finite absolute-cost alternatives are excluded by theorem 10.8. Thin drift is routed by theorem 9.99: persistent windows return to the thick-window ledger, while diffuse windows either enter another named local row or a closed retained row. Autonomous-modulation failure, compactness-package first failure, exterior/separated-core loss, rough-core loss, and multicore or same-point cascade are exactly the remaining non-retained first failures listed in lemma 6.25; by theorem 6.33 they are assigned to named local state-space rows before the retained compact scale-collapse state is formed. The only retained scale-collapse state left after those exits is the retained compact scale-collapse state, and it is excluded by theorem 6.34; if the retained state becomes scale-rigid, it is excluded by proposition 4.40. Thus J_6-failure is a routed local scale-collapse branch and not a hidden input. □

Theorem 7.135 (Ambient adjacent closure for physical Type II branches)**.** *Every physical local Type II branch which reaches the compact-cost validation layer satisfies adjacent-stratum closure for the compact-cost regime in the sense of definition 7.80. Equivalently, the hypotheses of corollary 7.131 are theorems on the physical local Type II state-space branch.*

Proof. Non-carrier adjacent exits are routed by lemma 7.132; the local closure rows close there, while carrier, scale, and cost rows are handled by the remaining branchwise checks below. The exterior, local H^1-loss, modulation-driven recentering, scale-rigid residual, and negative scale-drift branches appearing in the stratified moving-carrier alternative are the named local branches routed in theorems 8.38, 9.55 and 10.7, proposition 4.40, and lemma 7.134. Carrier applicability and energy-control of the canonical cost are supplied by lemma 7.133. The windowwise transition and positive-scale alternative statements are proved on the canonical windowwise moving family by corollaries 7.123 and 7.128. The negative-drift alternative required by theorem 7.130 is lemma 7.134.

All hypotheses of corollary 7.131 are therefore verified branchwise. Applying that corollary gives compact-cost adjacent-stratum closure. □

Corollary 7.136 (Ambient adjacent wrapper discharged)**.** *The ambient adjacent-stratum closure and carrier-applicability obligation for the canonical retained compact-cost branch is discharged by local state-space stratification and the canonical windowwise carrier package. Nonretained first failures are not removed by this corollary alone; they are assigned to the named exit rows recorded in the final assembly.*

Proof. This is theorem 7.135. □

Remark 7.137 (Cost divergence on the retained compact stratum)*.* The compact Type II cost argument gives infinite localized renormalization cost under the compact PDE hypotheses. Passing from that cost divergence to exclusion from the Type II class uses the corrected localized energy inequality constructed in lemma 7.57. On the retained compact stratum, the carrier exists by theorem 7.71; failure of the carrier test is not a retained compact branch but an adjacent stratum.

8. Terminal Profile Decoupling and Exterior Regularity

This section records the terminal local estimates used to remove multibubble configurations, small residual components, and components carried at a positive physical distance from the selected Type II core. The statements are local implications under the analytic inputs explicitly stated below: terminal state-space completeness on compact windows, repaired-gauge representation, local Caccioppoli estimates, exterior regularity, and the local compact scale-collapse routing theorem theorem 6.34.

8.1. **Renormalized frames and active local states.** A renormalized frame consists of a center $x_c(t)$, a scale $\lambda(t) > 0$, and renormalized time

$$\frac{d\tau}{dt} = \lambda(t)^{-2}.$$

It represents the physical solution by

$$V(y,\tau) = \lambda(t)u(x_c(t) + \lambda(t)y, t).$$

When the repaired gauge is imposed, the represented velocity satisfies

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0,$$

where

$$a(\tau) = \partial_\tau \log\lambda(\tau),$$

and b is the translation modulation coefficient determined by the centering condition. In a fixed frame, every other local state either remains at bounded relative position and comparable scale, escapes to spatial infinity in the frame variables, appears at a much larger same-point scale as a perturbative ambient field in the innermost frame, or lies at a smaller scale and therefore requires selecting that smaller active frame.

Definition 8.1 (Active local terminal state data)**.** Let $t_n \uparrow T^*$. Active local terminal state data consist of local state-space classes $\mathcal{A}_j$, scales $\lambda_j^n \to 0$, centers $x_j^n \to x_j^*$, compact terminal windows I_j, and representatives W_j^n in the corresponding frames. Each active class has a detecting compact cylinder $B_{R_j} \times I_j$ and a local mass floor

$$\limsup_{n\to\infty} \|W_j^n\|_{L^\infty(I_j;L^3(B_{R_j}))} \ge \varepsilon_{\mathrm{loc}} > 0.$$

After grouping comparable parameters based at the same physical point, distinct classes are separated in the local state-space sense: they have different physical points, strictly separated scales, or have already been assigned to an adjacent local alternative. On every compact terminal cylinder and for every finite retained set J, the local field is written

$$V_n = U_n^J + S_n^J,$$

where U_n^J is the grouped retained local state and S_n^J is the nonretained residual. Positive local mass in S_n^J is not a new unlabelled object: by terminal state-space completeness it is assigned either to a named adjacent local alternative or to the retained comparable class, in which case it must be included in U_n^J. No global profile decomposition, Brézis–Lieb identity, or heat-flow remainder smallness is part of this data. The corresponding local active mass floor is stated and proved in theorem 8.32.

Lemma 8.2 (Finite active local state count on a terminal slice)**.** *Fix a compact terminal cylinder $B_R \times I$, a selected time sequence $\tau_n \in I$, and assume*

$$\sup_n \|V_n\|_{L^\infty(I;L^3(B_R))} \le M_R < \infty.$$

Assume every active local state detected on the slice τ_n inside B_R has local mass at least $\varepsilon_{\mathrm{loc}}$, and that, after grouping comparable retained classes, the detecting balls on that slice have overlap multiplicity at most N_R. Then the number of active local states detected on that terminal slice is finite and satisfies

$$\#\mathcal{J}_{\mathrm{act}}(R, I) \leq \left\lfloor \frac{N_R M_R^3}{\varepsilon_{\mathrm{loc}}^3} \right\rfloor.$$

Proof. For each active state choose a detecting ball on the selected time slice, shrunk if necessary so that the family has overlap multiplicity at most N_R. At that common detecting time, the local mass floor gives

$$\int_{B_j} |V_n(y, \tau_n)|^3 \, dy \geq \varepsilon_{\mathrm{loc}}^3 + o_n(1)$$

for the ball B_j associated with the j-th state. Summing over any finite active subfamily and using the overlap bound,

$$(\#J)\varepsilon_{\mathrm{loc}}^3 \leq N_R \sup_n \sup_{\tau \in I} \int_{B_R} |V_n(y, \tau)|^3 \, dy \leq N_R M_R^3.$$

Thus every finite active subfamily has cardinality at most $N_R M_R^3/\varepsilon_{\mathrm{loc}}^3$. If there were more active states than the displayed integer bound, a finite subfamily with one additional element would contradict this estimate. □

8.2. **Same-point and separated-point reductions.**

Lemma 8.3 (Comparable same-point states form a compound state)**.** *Let a finite set $\mathcal{C}$ of active local states have the same physical center $x_j^* = x_*$. Suppose that for a representative center and scale (x_*^n, λ_*^n),*

$$\frac{\lambda_j^n}{\lambda_*^n} \to \rho_j \in (0, \infty), \qquad \frac{x_j^n - x_*^n}{\lambda_*^n} \to z_j \in \mathbb{R}^3.$$

Let $W_{j,n}$ denote the local representative of the j-th state in its own frame on compact cylinders, and assume

$$W_{j,n} \to W_j \quad \textit{strongly in } L^3_{\mathrm{loc}}$$

on the relevant terminal slice, or in $L^\infty_{\mathrm{loc}} L^3_{\mathrm{loc}}$ on a compact terminal window. Write

$$\rho_j^n := \frac{\lambda_j^n}{\lambda_*^n}, \qquad z_j^n := \frac{x_j^n - x_*^n}{\lambda_*^n}.$$

Then, in the frame (x_*^n, λ_*^n),

$$(\rho_j^n)^{-1} W_{j,n}\left(\frac{y - z_j^n}{\rho_j^n}\right) \to \rho_j^{-1} W_j \left(\frac{y - z_j}{\rho_j}\right)$$

strongly in the same local topology on every compact subcylinder on which the right-hand side is defined. Hence the comparable class has compound local representative

$$\Phi_{\mathcal{C}}(y) := \sum_{j \in \mathcal{C}} \rho_j^{-1} W_j \left(\frac{y - z_j}{\rho_j}\right).$$

If the compound representative carries the local mass floor

$$\|\Phi_{\mathcal{C}}\|_{L^3(B_R)} \geq \varepsilon_{\mathrm{loc}}$$

on a detecting compact cylinder, it is one active local state in the selected frame. If no compact cylinder carries the local mass floor, it belongs to the perturbative residual class by corollary 8.31.

Proof. Set

$$\rho_j^n = \frac{\lambda_j^n}{\lambda_*^n}, \qquad z_j^n = \frac{x_j^n - x_*^n}{\lambda_*^n}.$$

The transformed local representative is

$$T_n W_{j,n}(y) = (\rho_j^n)^{-1} W_{j,n}\left(\frac{y - z_j^n}{\rho_j^n}\right),$$

and the proposed limit is

$$TW_j(y) = \rho_j^{-1} W_j\left(\frac{y - z_j}{\rho_j}\right).$$

Here $T_{\rho,z}F(y) := \rho^{-1}F((y-z)/\rho)$. Thus $T_nW_{j,n} = T_{\rho_j^n, z_j^n} W_{j,n}$ and $TW_j = T_{\rho_j, z_j} W_j$. Fix a compact ball B_R in the representative frame. For all large n, the preimage of B_R under $y \mapsto (y - z_j^n)/\rho_j^n$ is contained in a fixed compact ball $B_{R'}$ in the j-th frame. The local strong convergence gives

$$\|T_{\rho_j^n, z_j^n}(W_{j,n} - W_j)\|_{L^3(B_R)} \le C\|W_{j,n} - W_j\|_{L^3(B_{R'})} \to 0.$$

It remains to move the fixed local limit W_j. Translation and dilation are strongly continuous on $L^3(B_{R'})$ after restriction to B_R: prove this first for $C_c^\infty(B_{R'})$ by uniform convergence on compact sets, then approximate W_j in $L^3(B_{R'})$. This proves the local convergence. The same argument applied uniformly in τ gives the $L^\infty_{\rm loc}L^3_{\rm loc}$ version on compact terminal windows. Summing over the finite class gives $\Phi_{\mathcal{C}}$. The final alternative is the local state-space criterion: a compound state with the detection threshold is retained as active, whereas a component with no compact-cylinder detection threshold is perturbative by the local residual theorem. □

Lemma 8.4 (Innermost-frame rescaling of strict outer states)**.** *Suppose all active local states have the same physical center x_* and split into distinct scale classes. If λ_1^n is the smallest active scale, then every outer scale $\lambda_j^n \gg \lambda_1^n$ satisfies*

$$\mu_j^n := \lambda_1^n/\lambda_j^n \to 0,$$

and in the innermost frame the j-th outer state has the local form

$$W_j^n(y) = \mu_j^n \widetilde{W}_{j,n}\left(\frac{x_*^n - x_j^n}{\lambda_j^n} + \mu_j^n y\right),$$

where $\widetilde{W}_{j,n}$ is the representative of the j-th state in its own outer frame.

Proof. The j-th local representative in physical coordinates is

$$u_j^n(x) = (\lambda_j^n)^{-1}\widetilde{W}_{j,n}\left(\frac{x - x_j^n}{\lambda_j^n}\right)$$

on the compact part of its outer terminal frame. Pull this back into the innermost frame $x = x_*^n + \lambda_1^n y$, with the critical rescaling $W_j^n(y) = \lambda_1^n u_j^n(x_*^n + \lambda_1^n y)$. Substituting,

$$W_j^n(y) = \frac{\lambda_1^n}{\lambda_j^n}\widetilde{W}_{j,n}\left(\frac{x_*^n + \lambda_1^n y - x_j^n}{\lambda_j^n}\right) = \mu_j^n \widetilde{W}_{j,n}\left(\frac{x_*^n - x_j^n}{\lambda_j^n} + \mu_j^n y\right),$$

which is the displayed form. The convergence $\mu_j^n \to 0$ is the hypothesis $\lambda_j^n \gg \lambda_1^n$. □

Lemma 8.5 (Local L^3-vanishing of strict outer states in the innermost frame)**.** *Under the hypotheses of lemma 8.4, the outer states satisfy the following retained-frame alternative. If*

$$\xi_j^n := \frac{x_*^n - x_j^n}{\lambda_j^n}$$

stays in a compact subset of the outer frame and $\widetilde{W}_{j,n} \to W_j$ strongly in L^3_{loc}, then, for every $R < \infty$,

$$\|W_j^n\|_{L^3(B_R)} \to 0 \quad (n \to \infty).$$

If ξ_j^n leaves every compact subset of the outer frame, the contribution is assigned to the escaping-offset/exterior row of the local state-space decomposition rather than to a retained strict outer state.

Proof. Assume first that ξ_j^n stays in a fixed compact set K. Choose a compact K' containing $K + \mu_j^n B_R$ for all large n. By the formula in lemma 8.4,

$$\|W_j^n\|_{L^3(B_R)}^3 = \int_{\xi_j^n + \mu_j^n B_R} |\widetilde{W}_{j,n}(z)|^3 \, dz.$$

The strong $L^3(K')$ convergence reduces this to the same integral with W_j. Since $|W_j|^3 \in L^1(K')$ and the sets $\xi_j^n + \mu_j^n B_R$ have measure tending to zero, absolute continuity of the integral gives convergence to zero. If ξ_j^n escapes every compact set, then the outer state is not represented on a fixed compact outer cylinder; this is exactly the escaping-offset/exterior boundary face in the local state-space compactification. □

Theorem 8.6 (Perturbative estimates for strict outer states)**.** *In the same setting as lemma 8.4, the strict outer states are perturbative in the innermost retained frame:*

(i) *finite-offset regular outer states, namely those for which $(x_*^n - x_j^n)/\lambda_j^n$ is bounded, decompose on each fixed inner ball into a constant drift plus an $O((\mu_j^n)^2)$ shear remainder;*
(ii) *the constant drifts produced by item (i) are absorbed into the translation-modulation coefficient of the innermost frame;*
(iii) *the remaining pressure, cutoff, and nonlinear interactions generated by the outer states are perturbative in the compact scale-collapse topology in which the local perturbation criterion lemma 8.10 and the pressure-tail decoupling lemma 8.12 apply.*

The escaping-offset alternative $(x_*^n - x_j^n)/\lambda_j^n \to \infty$ is treated by lemma 8.5.

Proof. In the escaping-offset case, lemma 8.5 gives local L^3-vanishing on every innermost compact cylinder, and the local pressure-tail lemma gives the pressure and residual convergence.

It remains to consider bounded offsets. If the outer state is not regular at the observed offset, then some smaller offset cylinder has CKN quantity above the threshold of theorem 8.34. That cylinder is a positive local state-space label at a larger same physical point scale, so the branch is assigned to the same-point cascade/secondary-core alternative before entering the retained innermost class. Thus, in the retained case, every bounded offset cylinder for the outer state is CKN-subcritical. The CKN criterion gives uniform C^1 bounds on a smaller offset cylinder, after subtracting the pressure mean.

On that retained regular cylinder, Taylor expansion at the offset gives a spatially constant leading term plus a shear remainder of size $O(\mu_j^n)$ in C^1 on every fixed inner ball. The Navier–Stokes scaling contributes the additional factor μ_j^n in the velocity amplitude, so the nonconstant velocity remainder is $O((\mu_j^n)^2)$ locally. The spatially constant term is absorbed into the translation modulation coefficient. The remaining velocity, pressure, cutoff, and modulation terms vanish in the compact-window topologies by lemmas 8.10 and 8.12. This is exactly the asserted perturbative reduction. □

Lemma 8.7 (Separated physical centers are locally invisible)**.** *Let $p \neq q$ be two active physical concentration points,*

$$x_p^* \neq x_q^*, \qquad d_{pq} := |x_p^* - x_q^*| > 0.$$

In the p-frame define the q-state contribution

$$W_{q\to p}^n(y) := \lambda_p^n (\lambda_q^n)^{-1} \phi^q \left(\frac{x_p^n + \lambda_p^n y - x_q^n}{\lambda_q^n} \right).$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then for every $R < \infty$,

$$\|W_{q\to p}^n\|_{L^3(B_R)} \to 0.$$

If the corresponding representative pressure $\Pi^q \in L^{3/2}(\mathbb{R}^3)$, then the pure pressure contribution

$$P_{q\to p}^n(y) = (\lambda_p^n)^2 (\lambda_q^n)^{-2} \Pi^q \left(\frac{x_p^n + \lambda_p^n y - x_q^n}{\lambda_q^n} \right)$$

satisfies

$$\|P_{q\to p}^n\|_{L^{3/2}(B_R)} \to 0.$$

Proof. By the change of variables $x = x_p^n + \lambda_p^n y$,

$$\|W_{q\to p}^n\|_{L^3(B_R)}^3 = \int_{B(x_p^n, \lambda_p^n R)} (\lambda_q^n)^{-3} \left| \phi^q \left(\frac{x - x_q^n}{\lambda_q^n} \right) \right|^3 dx.$$

Set $z = (x - x_q^n)/\lambda_q^n$. Then

$$\|W_{q\to p}^n\|_{L^3(B_R)}^3 = \int_{B(c_n, \rho_n)} |\phi^q(z)|^3 \, dz,$$

where

$$c_n = \frac{x_p^n - x_q^n}{\lambda_q^n}, \qquad \rho_n = \frac{\lambda_p^n R}{\lambda_q^n}.$$

Since $x_p^n \to x_p^*$, $x_q^n \to x_q^*$, and $\lambda_q^n \to 0$,

$$|c_n| \sim d_{pq}/\lambda_q^n \to \infty.$$

Moreover

$$\frac{\rho_n}{|c_n|} = \frac{\lambda_p^n R}{|x_p^n - x_q^n|} \to 0.$$

Thus the balls $B(c_n, \rho_n)$ escape to spatial infinity: for every $A < \infty$, they are contained in $\{|z| > A\}$ for all sufficiently large n. Since $\phi^q \in L^3$,

$$\int_{|z|>A} |\phi^q(z)|^3 \, dz \to 0 \qquad (A \to \infty),$$

and hence $\|W_{q\to p}^n\|_{L^3(B_R)} \to 0$.

For the pressure term the same change of variables gives

$$\|P_{q\to p}^n\|_{L^{3/2}(B_R)}^{3/2} = \int_{B(c_n, \rho_n)} |\Pi^q(z)|^{3/2} \, dz.$$

The same escaping-ball argument and $\Pi^q \in L^{3/2}$ imply the stated pressure convergence. □

Definition 8.8 (Multipoint decoupling conditions)**.** For each active point and selected frame, the required multipoint decoupling conditions are:

(i) local velocity decoupling in the selected frame;
(ii) pressure decoupling modulo time-dependent constants in $L^1_\tau H^{-1}_y$ on compact cylinders;
(iii) localization errors from spatial cutoffs vanish in $L^1_\tau H^{-1}_y$;
(iv) repaired-gauge modulation coefficients remain bounded after exterior ambient terms are absorbed;
(v) each active point has positive finite critical mass in its own frame;
(vi) after states centered at other physical points are treated as perturbative exterior terms, the selected frame is tested against definition 6.24: either the first failed retained-state input is routed by lemma 6.25 to a named local exit, or all retained compact scale-collapse inputs hold.

Theorem 8.9 (Reduction of active multibubbles)**.** *Assume the Type II local terminal state data, the active local mass floor, the same-point nonlinear decoupling conditions, the multipoint decoupling conditions of definition 8.8. Then no active multibubble Type II configuration can occur.*

Proof. By lemma 8.2, there are only finitely many active local states on each selected terminal slice. Partition them by physical concentration point x^*_α, and within each physical point partition them by comparable scale class.

If a scale class contains several comparable local states, lemma 8.3 combines them into one compound state. If the compound state has no detecting compact cylinder, it is perturbative by corollary 8.31; if it is active, it is a single active local state for that frame.

If a physical point contains strict same-point scale separation, theorem 8.6 and the same-point nonlinear decoupling conditions reduce the innermost active scale to a single repaired-gauge solution with perturbative outer interactions. The compact scale-collapse routing theorem theorem 6.34 then rules out the corresponding retained same-point cascade; if one of its retained hypotheses fails, lemma 6.25 assigns the branch to a named local alternative and it is not a retained active multibubble configuration.

It remains to consider separated physical points. Choose any active point x^*_α and place the selected frame at its active scale and center. By lemma 8.7, every state centered at a different physical point is locally L^3-invisible in this frame. By the multipoint decoupling conditions, its pressure, cutoff, and modulation interactions are also perturbative in the compact scale-collapse topology. Therefore the selected frame contains a single active repaired-gauge solution whose scale tends to zero. If any retained compact scale-collapse input fails, the multipoint decoupling condition and lemma 6.25 route the branch to a named local exit, so it is not a retained active multibubble configuration. If no retained input fails, the selected frame is a retained compact scale-collapse state in the sense of definition 6.24. Applying theorem 6.34 excludes that selected active point. Since the chosen active point was arbitrary, no active multibubble configuration can occur. □

8.3. **Local perturbation and pressure decoupling.**

Lemma 8.10 (Local perturbation criterion)**.** *Let $I \subset \mathbb{R}$ be compact and $Q_R = B_R \times I$. Let U_n be a repaired-gauge solution on Q_{2R} satisfying*

$$\|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R.$$

Let S_n be a divergence-free perturbation on Q_{2R} with

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_{2R}),$$

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

$$\mathcal{R}_n := \partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R),$$

where $|a_n| + |b_n| \le M_{ab}$*, and suppose the pressure cross-term satisfies*

$$\nabla \mathcal{P}_{\mathrm{cross},n} \to 0 \quad \textit{in } L^1_\tau H^{-1}_y(B_R).$$

Then replacing $U_n + S_n$ *by* U_n *changes the repaired-gauge Navier–Stokes equation on* Q_R *by an error tending to zero in*

$$L^1_\tau H^{-1}_y(B_R).$$

Proof. The linear part is precisely $\mathcal{R}_n$, which tends to zero by hypothesis. The nonlinear difference is

$$(U_n + S_n) \cdot \nabla(U_n + S_n) - U_n \cdot \nabla U_n = \operatorname{div} F_n, \qquad F_n := U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n,$$

using $\nabla \cdot U_n = \nabla \cdot S_n = 0$ distributionally. By the assumed $L^1_\tau L^2_y$ stress convergence, this divergence tends to zero in

$$L^1_\tau H^{-1}_y(B_R),$$

because for $\varphi \in H^1_0(B_R; \mathbb{R}^3)$,

$$\left| \int_{B_R} (U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n) : \nabla\varphi \, dy \right| \le \|F_n\|_{L^2(B_R)} \times \|\nabla\varphi\|_{L^2(B_R)}.$$

The pressure contribution vanishes by the assumed cross-pressure convergence. Therefore the total perturbative error tends to zero in

$$L^1_\tau H^{-1}_y(B_R).$$

□

Lemma 8.11 (Static same-point outer profiles vanish locally)**.** *Let*

$$W_n(y) = \mu_n \phi(\xi_n + \mu_n y), \qquad \mu_n \to 0, \qquad \phi \in L^3(\mathbb{R}^3).$$

Then, for every $R < \infty$*,*

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. Changing variables $z = \xi_n + \mu_n y$,

$$\|W_n\|^3_{L^3(B_R)} = \int_{B(\xi_n, \mu_n R)} |\phi(z)|^3 \, dz.$$

The measure of $B(\xi_n, \mu_n R)$ tends to zero. Since $|\phi|^3 \in L^1(\mathbb{R}^3)$, absolute continuity of the integral gives

$$\int_{B(\xi_n, \mu_n R)} |\phi(z)|^3 \, dz \to 0,$$

uniformly in ξ_n. Hence $\|W_n\|_{L^3(B_R)} \to 0$. □

Lemma 8.12 (Pressure decoupling from kernel tails)**.** *Let* Γ_{ij} *be the Calderón–Zygmund kernel for*

$$P = \mathcal{R}_i \mathcal{R}_j (F_{ij}).$$

Assume

$$F_n \to 0 \quad \textit{in } L^1_\tau L^2_y(B_A) \quad \textit{for every fixed } A,$$

and

$$\sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y(\mathbb{R}^3)} < \infty.$$

Then the pressure generated by F_n *satisfies, for every* $R < \infty$*,*

$$P_n - c_{n,R} \to 0 \quad \textit{in } L^1_\tau L^2_y(B_R)$$

for suitable constants $c_{n,R}(\tau)$, and consequently

$$\nabla P_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. Fix $A > 2R$ and write

$$F_n^{\mathrm{loc}} = F_n \mathbf{1}_{B_A}, \qquad F_n^{\mathrm{far}} = F_n \mathbf{1}_{\mathbb{R}^3 \setminus B_A}.$$

For the local part, Calderón–Zygmund boundedness gives

$$\|P_n^{\mathrm{loc}}\|_{L^1_\tau L^2_y(B_A)} \le C_A \|F_n\|_{L^1_\tau L^2_y(B_A)} \to 0.$$

For the far part define

$$c_{n,R}(\tau) = \int_{|z|>A} \Gamma_{ij}(-z) F_{n,ij}(z,\tau)\, dz,$$

first for compactly supported approximations and then by passage to the $L^{3/2}$ limit. For $y \in B_R$ and $|z| > A > 2R$,

$$|\Gamma_{ij}(y-z) - \Gamma_{ij}(-z)| \le C_R |z|^{-4}.$$

Hence

$$|P_n^{\mathrm{far}}(y,\tau) - c_{n,R}(\tau)| \le C_R \int_{|z|>A} |z|^{-4} |F_n(z,\tau)|\, dz.$$

By Hölder's inequality with exponents $3/2$ and 3,

$$\int_{|z|>A} |z|^{-4} |F_n|\, dz \le \|F_n(\tau)\|_{L^{3/2}} \left(\int_{|z|>A} |z|^{-12}\, dz \right)^{1/3} \le C A^{-3} \|F_n(\tau)\|_{L^{3/2}}.$$

Therefore

$$\|P_n^{\mathrm{far}} - c_{n,R}\|_{L^1_\tau L^\infty_y(B_R)} \le C_{R,I} A^{-3} \sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y}.$$

Since B_R has finite measure, this also controls the far part in $L^1_\tau L^2_y(B_R)$. Taking $n \to \infty$ in the local term and then $A \to \infty$ in the far term gives

$$P_n - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

For $\varphi \in H^1_0(B_R;\mathbb{R}^3)$,

$$|\langle \nabla(P_n - c_{n,R}), \varphi\rangle| \le \left| \int_{B_R} (P_n - c_{n,R}) \nabla\cdot\varphi\, dy \right| \le \|P_n - c_{n,R}\|_{L^2(B_R)} \|\nabla\varphi\|_{L^2(B_R)}.$$

Thus $\nabla P_n \to 0$ in $L^1_\tau H^{-1}_y(B_R)$. $\square$

Definition 8.13 (Terminal nonlinear local-state decoupling)**.** On every compact terminal-frame cylinder $B_R \times I$, let U_n be the retained active local-state evolution and let S_n be the nonretained local residual. Terminal nonlinear local-state decoupling means:

(i) $\nabla \cdot S_n = 0$ distributionally;
(ii) $S_n \to 0$ in $L^\infty_\tau L^3_y(B_{2R})$;
(iii)
$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R);$$
(iv)
$$\partial_\tau S_n - \nu \Delta S_n + a_n(S_n + y\cdot\nabla S_n) + b_n \cdot \nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R);$$
(v) pressure sources involving at least one factor of S_n satisfy lemma 8.12, or directly satisfy
$$\nabla \mathcal{P}_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R);$$
(vi) cutoff errors introduced in isolating the selected frame vanish in $L^1_\tau H^{-1}_y(B_R)$;
(vii) the effective modulation coefficients remain bounded;

(viii) after subtracting S_n, the retained solution has positive finite critical mass, repaired-gauge representation, pressure reconstruction, Caccioppoli regularity, compactness, tightness, and the compact scale-collapse limiting inputs.

Theorem 8.14 (Same-point and multipoint decoupling from terminal decoupling)**.** *If terminal nonlinear local-state decoupling holds in the sense of definition 8.13, then both same-point scale-decoupling and separated-point frame-decoupling hold.*

Proof. For same-point active local states, group comparable scales by lemma 8.3. For strict scale separation choose the innermost active scale as the selected terminal frame. By lemma 8.11, every outer scale that remains in the selected frame is locally invisible in velocity, unless it was already grouped as comparable or routed to an adjacent state. Terminal nonlinear local-state decoupling supplies the nonlinear-stress, linear-residual, pressure-source, localization, modulation, and compact scale-collapse admissibility conditions on every compact cylinder. Lemma 8.10 then shows that the outer components and residual alter the selected equation only by an error tending to zero in $L^1_\tau H^{-1}_y(B_R)$. Thus the same-point configuration reduces to one retained compact scale-collapse solution with perturbative interactions.

For separated physical points, choose an active physical point x^*_p and its selected frame. For every local state centered at $x^*_q \neq x^*_p$, lemma 8.7 gives local velocity convergence to zero in the p-frame. If a pure profile pressure is represented in L^2, the same escaping-ball calculation gives local L^2 pressure convergence and therefore H^{-1} convergence of the gradient. With only $L^{3/2}$-pressure, the mixed pressure sources are controlled by lemma 8.12 through the L^2-strength hypotheses included in terminal nonlinear local-state decoupling. The same decoupling statement supplies localization-error convergence, bounded modulation, and compact scale-collapse admissibility. Applying lemma 8.10 shows that separated physical-point states change the selected equation only by an error tending to zero in $L^1_\tau H^{-1}_{\mathrm{loc}}$. This proves separated-point decoupling. □

8.4. **Terminal local-state decoupling.**

Definition 8.15 (Terminal active local frame)**.** Partition active local states by physical point $x^n_j \to x^*_\alpha$. At a fixed physical point, group comparable scales. For a comparable-scale class $\mathcal{C}$, a representative frame $(x^n_{\mathcal{C}}, \lambda^n_{\mathcal{C}})$ has compound local representative

$$\Phi_{\mathcal{C}}(y) = \sum_{j\in\mathcal{C}} \rho_j^{-1} W_j\left(\frac{y - z_j}{\rho_j}\right),$$

where W_j is the local representative of the j-th state in its own frame, transported as in lemma 8.3. The class $\mathcal{C}$ is terminal at x^*_α if no active class at the same physical point has strictly smaller scale:

$$\lambda^n_{\mathcal{D}}/\lambda^n_{\mathcal{C}} \not\to 0 \quad \text{for every active class } \mathcal{D}.$$

The terminal frame is

$$y = \frac{x - x^n_{\mathcal{C}}}{\lambda^n_{\mathcal{C}}}, \qquad \tau = \int^t (\lambda_{\mathcal{C}}(s))^{-2}\, ds,$$

and

$$V_n(y,\tau) = \lambda^n_{\mathcal{C}} u(x^n_{\mathcal{C}} + \lambda^n_{\mathcal{C}} y, t_n + (\lambda^n_{\mathcal{C}})^2\tau).$$

The retained solution U_n is the repaired-gauge nonlinear evolution of $\mathcal{C}$ in this frame, and $S_n := V_n - U_n$ is the nonretained remainder field.

Theorem 8.16 (Terminal windowed state-space completeness)**.** *For a sequence f_n and a frame (x_n, λ_n), define*

$$\mathfrak{m}_R(f_n; x_n, \lambda_n) := \limsup_{n\to\infty} \|\lambda_n f_n(x_n + \lambda_n \cdot)\|_{L^3(B_R)}$$

and

$$\mathfrak{M}_{R,I}(f_n;x_n,\lambda_n) := \limsup_{n\to\infty}\sup_{\tau\in I}\left\|\lambda_n f_n(x_n+\lambda_n\cdot, t_n+\lambda_n^2\tau)\right\|_{L^3(B_R)}.$$

Terminal windowed state-space completeness means that if $\mathfrak{M}_{R,I}(f_n;x_n,\lambda_n) > 0$ for a sequence left in the remainder, then after passing to a subsequence the local state-space stratification assigns that residual mass either to a named adjacent local alternative or to the retained comparable class. In the latter case the component must have been grouped into the retained terminal class before selecting the frame. Hence no positive nonretained residual mass remains on a retained terminal branch.

Proof. This is the compact-window form of theorem 9.79. If $\mathfrak{M}_{R,I}(f_n;x_n,\lambda_n) > 0$, then a subsequence and a smaller compact terminal cylinder carry positive local L^3 mass. The local compactness lemma lemma 9.69 gives an occupation state, and lemma 9.76 assigns that state to the first closed label in the local state-space decomposition. An adjacent label is handled by its local closure theorem and is not part of the retained terminal branch. A retained comparable label must already have been grouped into the selected terminal class by the terminal frame selection rule. Thus positive nonretained residual mass cannot persist on a retained terminal branch. □

Remark 8.17 (Status of theorem 8.16)*.* This theorem is deliberately local. It is a compact-window state-space exhaustion statement, not a terminal global profile theorem. Its proof uses the local compactness, closed-label, and residual-exhaustion lemmas in the terminal state-space discharge package; it does not use a whole-space profile decomposition or a heat-flow remainder estimate.

Lemma 8.18 (Terminal local velocity decoupling)**.** *Let $\mathcal{C}$ be a terminal active class and S_n be the nonretained remainder field in the $\mathcal{C}$-frame. Then for every compact cylinder $B_R\times I$,*

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_R).$$

Proof. Assume otherwise. Then there exist $R<\infty$, a compact interval I, $\delta>0$, a subsequence, and times $\tau_n\in I$ such that

$$\|S_n(\tau_n)\|_{L^3(B_R)} \ge \delta.$$

In physical variables, after the retained terminal class has been subtracted, the remainder carries at least δ critical L^3-mass inside a ball of radius $O(\lambda^n_{\mathcal{C}})$, centered at

$$x^n_{\mathcal{C}} + O(\lambda^n_{\mathcal{C}}),$$

at time

$$t_n + (\lambda^n_{\mathcal{C}})^2\tau_n.$$

By terminal windowed state-space completeness, this nonvanishing mass has a first local state-space label. If the label is an adjacent alternative, the branch has left the retained terminal class and is handled by the corresponding local closure theorem, contradicting that it remains in S_n on the retained branch. If the label is the retained comparable class, then it was already grouped into $\mathcal{C}$ before the terminal frame was selected, so it cannot remain in S_n. This is again a contradiction.

Thus no such δ exists. The same argument covers same-point larger-scale and separated-point components: any nonzero mass seen in the terminal frame is either a named adjacent local state or part of the retained comparable class, contradicting the terminal partition. □

Lemma 8.19 (Uniform terminal H^1 bounds)**.** *Assume the repaired-gauge representation and the local Caccioppoli estimates hold on every compact terminal cylinder $B_{2R}\times I$. Then*

$$\sup_n \|V_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|V_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R,$$

$$\sup_n \|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R,$$

and, since $S_n = V_n - U_n$,

$$\sup_n \|S_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|S_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R.$$

In particular, by Sobolev embedding on B_{2R},

$$U_n, S_n \quad \text{are bounded in} \quad L^2_\tau L^6_y(B_{2R}).$$

Proof. The first estimate is the repaired-gauge Caccioppoli estimate for V_n on the compact terminal cylinder. The retained solution U_n is itself a repaired-gauge Navier–Stokes solution with positive finite critical mass and bounded modulation on the same compact cylinder, so the same estimate applies to U_n. Since $S_n = V_n - U_n$, the displayed bound for S_n follows by the triangle inequality in the stated norms. The $L^2_\tau L^6_y$ bound follows from the Sobolev embedding $H^1(B_{2R}) \hookrightarrow L^6(B_{2R})$. □

Lemma 8.20 (Nonlinear stress decoupling in terminal frames)**.** *Let*

$$F_n := U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n.$$

Then, for every compact $B_R \times I$,

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

Proof. By lemma 8.18,

$$\|S_n\|_{L^\infty_\tau L^3_y(B_R)} \to 0.$$

Using Hölder's inequality in space and time,

$$\|U_n \otimes S_n\|_{L^1_\tau L^2_y(B_R)} \le \|S_n\|_{L^\infty_\tau L^3_y(B_R)} \|U_n\|_{L^1_\tau L^6_y(B_R)}$$

and

$$\|U_n\|_{L^1_\tau L^6_y(B_R)} \le |I|^{1/2} \|U_n\|_{L^2_\tau L^6_y(B_R)} \le C_{R,I}.$$

Thus $U_n \otimes S_n \to 0$ in $L^1_\tau L^2_y(B_R)$, and the same estimate gives $S_n \otimes U_n \to 0$. Finally,

$$\|S_n \otimes S_n\|_{L^1_\tau L^2_y(B_R)} \le \|S_n\|_{L^\infty_\tau L^3_y(B_R)} \|S_n\|_{L^1_\tau L^6_y(B_R)} \to 0,$$

because $\|S_n\|_{L^1_\tau L^6_y(B_R)} \le C_{R,I}$. Summing the three terms proves the local convergence. □

Lemma 8.21 (Quantitative nested-ball harmonic tail)**.** *Fix radii* $0 < r < R < R_1 < R_2$ *and* $1 < p < \infty$. *Let* $\theta \in C_c^\infty(B_{R_2} \setminus \overline{B_{R_1}})$, *extend* θF *by zero to* $\mathbb{R}^3$, *and set*

$$P^{\text{ann}} = \mathcal{R}_i \mathcal{R}_j (\theta F_{ij}), \qquad H^{\text{ann}} = P^{\text{ann}} - (P^{\text{ann}})_{B_{R_1}} \quad \text{on } B_{R_1}.$$

Then H^{ann} *is harmonic and mean-zero on* B_{R_1}. *Moreover there is a constant* $C_p(r, R, R_1, R_2, \theta) > 0$, *depending only on p, the nested balls and the cutoff, such that*

(i) Interior derivative bound. *For every harmonic mean-zero* H *on* B_{R_1} *and every* $R < R_1$,

$$\sup_{B_R} \big(|H| + R_1 |\nabla H|\big) \le C_p(R, R_1)\, \|H\|_{L^p(B_{R_1})}.$$

(ii) Annular harmonic-source bound. *The harmonic part generated by the source in the fixed annulus satisfies*

$$\|H^{\text{ann}}\|_{L^p(B_R)} \le C_p(r, R, R_1, R_2, \theta)\, \|F\|_{L^p(B_{R_2} \setminus B_{R_1})}.$$

(iii) Quantitative converse for annular tails. *If* $\|H^{\text{ann}}\|_{L^p(B_r)} \ge \eta > 0$, *then*

$$\|F\|_{L^p(B_{R_2} \setminus B_{R_1})} \ge C_p(r, R, R_1, R_2, \theta)^{-1} \eta.$$

This lemma controls only the annular harmonic contribution generated by sources in the displayed fixed annulus. Harmonic pressure coming from outside B_{R_2}, or from a failure of the pressure reconstruction itself, is not silently bounded by this lemma and is instead read as a pressure/exterior state-space alternative.

Proof. Item (i) is the standard interior derivative estimate for harmonic functions: by the mean-value property and Cauchy estimates on $B_{(R+R_1)/2}$, $|H| + R_1|\nabla H|$ on B_R is controlled by the L^p mass of H on B_{R_1} with constant depending only on p and R/R_1.

For item (ii), the source θF is supported outside B_{R_1}, so $-\Delta P^{\mathrm{ann}} = 0$ on B_{R_1}. Hence H^{ann} is harmonic there. Calderón–Zygmund boundedness on $\mathbb{R}^3$ gives

$$\|P^{\mathrm{ann}}\|_{L^p(\mathbb{R}^3)} \le C_p\|\theta F\|_{L^p(\mathbb{R}^3)} \le C_p(\theta)\|F\|_{L^p(B_{R_2}\setminus B_{R_1})},$$

and subtracting the mean on B_{R_1} changes the $L^p(B_R)$ norm by a geometric constant.

For item (iii), this is the contrapositive of item (ii): if the annular source were arbitrarily small, then item (ii) would force $\|H^{\mathrm{ann}}\|_{L^p(B_R)}$, and hence $\|H^{\mathrm{ann}}\|_{L^p(B_r)}$, to be arbitrarily small. ☐

Lemma 8.22 (Harmonic cross pressure routes to pressure/exterior alternative)**.** *Fix nested balls $B_R \Subset B_{R_1} \Subset B_{R_2}$, $1 < p < \infty$, and $1 \le q < \infty$. Let*

$$P_{\mathrm{cross},n} = P^{\mathrm{loc}}_{\mathrm{cross},n} + H_{n,R_1} + c_{n,R_1}(\tau)$$

be the local pressure reconstruction on B_{R_1}, where

$$-\Delta P^{\mathrm{loc}}_{\mathrm{cross},n} = \partial_i\partial_j\big(\zeta(F_n)_{ij}\big), \qquad \zeta \equiv 1 \text{ on } B_{R_1}, \quad \operatorname{supp}\zeta \Subset B_{R_2},$$

and H_{n,R_1} is harmonic with zero spatial mean on B_{R_1}. Suppose

$$F_n \to 0 \quad \text{in } L^q_\tau L^p_y(B_{R_2}).$$

If H_{n,R_1} does not converge to zero in $L^q_\tau L^p_y(B_R)$, then, after passing to a subsequence, the branch enters the ordered pressure/exterior-leakage alternative. Consequently, on any retained terminal branch, $H_{n,R_1} \to 0$ in $L^q_\tau L^p_y(B_R)$.

Proof. The local part tends to zero by Calderón–Zygmund estimates:

$$\|P^{\mathrm{loc}}_{\mathrm{cross},n}\|_{L^q_\tau L^p_y(B_{R_1})} \le C\|F_n\|_{L^q_\tau L^p_y(B_{R_2})} \to 0.$$

Hence any non-vanishing pressure oscillation modulo functions of time is carried by the normalized harmonic part. If

$$\limsup_{n\to\infty} \|H_{n,R_1}\|_{L^q_\tau L^p_y(B_R)} > 0,$$

then choose a finite smooth partition of unity $\{\theta_\ell\}_\ell \subset C^\infty_c(B_{R_2} \setminus \overline{B_{R_1}})$ for the recorded annular part of the pressure atlas. For each ℓ, the corresponding annular harmonic contribution

$$H^{\mathrm{ann}}_{n,\ell} := \mathcal{R}_i\mathcal{R}_j(\theta_\ell(F_n)_{ij}) - \fint_{B_{R_1}} \mathcal{R}_i\mathcal{R}_j(\theta_\ell(F_n)_{ij})\,dy$$

vanishes in $L^q_\tau L^p_y(B_R)$ by lemma 8.21, because

$$\|F_n\|_{L^q_\tau L^p_y(B_{R_2}\setminus B_{R_1})} \to 0.$$

Summing over the finite partition shows that the whole harmonic contribution generated by sources recorded in the fixed annulus $B_{R_2} \setminus B_{R_1}$ vanishes.

By construction of the local pressure atlas, after the local Calderón–Zygmund piece and the recorded annular pieces have been removed, the remaining mean-zero harmonic part has only two possible origins: either the pressure gauge/reconstruction fails on the selected window, or it is a far-field pressure tail generated outside the fixed ball B_{R_2}. The former is one of the ordered pressure rows. In the latter case, theorem 8.38 routes any exterior carrier to an exterior or separated-core concentration alternative; if no such exterior carrier exists, lemma 8.44 gives convergence of the far-field pressure modulo functions of time on B_R, contradicting the positive lower bound for H_{n,R_1}. Therefore every non-vanishing harmonic cross pressure is a named pressure/exterior-leakage alternative. A retained terminal branch is the complementary case in the ordered validation, so the harmonic part vanishes there. $\square$

Lemma 8.23 (Terminal pressure decoupling)**.** *Fix nested balls $B_R \Subset B_{R_1} \Subset B_{R_2}$ and let $\zeta \in C_c^\infty(B_{R_2})$ be identically one on B_{R_1}. Let*

$$P^{\mathrm{loc}}_{\mathrm{cross},n} := \mathcal{R}_i\mathcal{R}_j\big(\zeta(F_n)_{ij}\big)$$

be the local Calderón–Zygmund pressure generated by the cross stress. Suppose the terminal pressure reconstruction is used on B_{R_2}, so that on B_{R_1}

$$P_{\mathrm{cross},n} = P^{\mathrm{loc}}_{\mathrm{cross},n} + H_{n,R_1} + c_{n,R_1}(\tau),$$

where $H_{n,R_1}(\cdot,\tau)$ is harmonic and normalized by $⨍_{B_{R_1}} H_{n,R_1}(\cdot,\tau)\,dy = 0$. On a retained terminal branch,

$$P_{\mathrm{cross},n} - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

and

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. Lemma 8.20 gives

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_{R_2})$$

on every compact terminal cylinder. Calderón–Zygmund boundedness gives

$$\|P^{\mathrm{loc}}_{\mathrm{cross},n}\|_{L^1_\tau L^2_y(B_{R_1})} \le C_{R_1,R_2}\|F_n\|_{L^1_\tau L^2_y(B_{R_2})} \to 0.$$

For the harmonic part, apply lemma 8.22 with $(q,p)=(1,2)$. A non-vanishing harmonic cross pressure is a pressure/exterior-leakage alternative and therefore is not present on a retained terminal branch. Hence $H_{n,R_1} \to 0$ in $L^1_\tau L^2_y(B_R)$. Adding the local part proves $P_{\mathrm{cross},n} - c_{n,R} \to 0$ in $L^1_\tau L^2_y(B_R)$.

Finally, for $\varphi \in H^1_0(B_R;\mathbb{R}^3)$,

$$|\langle \nabla(P_{\mathrm{cross},n} - c_{n,R}), \varphi\rangle| \le \|P_{\mathrm{cross},n} - c_{n,R}\|_{L^2(B_R)}\|\nabla\varphi\|_{L^2(B_R)}.$$

Integrating in time gives $\nabla P_{\mathrm{cross},n} \to 0$ in $L^1_\tau H^{-1}_y(B_R)$. $\square$

Lemma 8.24 (Linear residual decoupling)**.** *Let $V_n = U_n + S_n$ solve*

$$\partial_\tau V_n + (V_n\cdot\nabla)V_n + \nabla P_n = \nu\Delta V_n + a_n(V_n + y\cdot\nabla V_n) + b_n\cdot\nabla V_n.$$

Suppose U_n satisfies, in the same repaired gauge,

$$\partial_\tau U_n + (U_n\cdot\nabla)U_n + \nabla P_{U,n} = \nu\Delta U_n + a_n(U_n + y\cdot\nabla U_n) + b_n\cdot\nabla U_n + e_n,$$

with

$$e_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R)$$

on compact terminal cylinders. Then

$$\partial_\tau S_n - \nu\Delta S_n - a_n(S_n + y\cdot\nabla S_n) - b_n\cdot\nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Equivalently, the same conclusion holds with the opposite sign convention for the modulation terms.

Proof. Subtracting the U_n-equation from the V_n-equation gives

$$\partial_\tau S_n - \nu\Delta S_n = -\operatorname{div} F_n - \nabla P_{\mathrm{cross},n} - e_n + a_n(S_n + y\cdot\nabla S_n) + b_n\cdot\nabla S_n.$$

Equivalently,

$$\partial_\tau S_n - \nu\Delta S_n - a_n(S_n + y\cdot\nabla S_n) - b_n\cdot\nabla S_n = -\operatorname{div} F_n - \nabla P_{\mathrm{cross},n} - e_n.$$

By lemma 8.20,

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

hence

$$\operatorname{div} F_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

By lemma 8.23,

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Together with $e_n \to 0$, this proves the displayed residual convergence.

For the opposite sign convention, the two residuals differ by

$$2a_n(S_n + y\cdot\nabla S_n) + 2b_n\cdot\nabla S_n.$$

By lemma 8.18 and finite measure,

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^2_y(B_R).$$

For $\varphi \in H^1_0(B_R)$,

$$|\langle y\cdot\nabla S_n, \varphi\rangle| = \left|\int_{B_R} S_n(3\varphi + y\cdot\nabla\varphi)\,dy\right| \le C_R\|S_n\|_{L^2(B_R)}\|\varphi\|_{H^1(B_R)},$$

and

$$|\langle b_n\cdot\nabla S_n, \varphi\rangle| \le |b_n|\,\|S_n\|_{L^2(B_R)}\|\nabla\varphi\|_{L^2(B_R)}.$$

The term S_n itself is controlled by the same L^2-convergence. Since a_n and b_n are bounded, the sign-convention difference tends to zero in $L^1_\tau H^{-1}_y(B_R)$. ☐

Lemma 8.25 (Terminal modulation compatibility)**.** *Same-point larger-scale profiles in a terminal frame have the form*

$$W^n_j(y,\tau) = \mu^n_j U^j(\theta^n_j(\tau), \xi^n_j + \mu^n_j y), \qquad \mu^n_j \to 0.$$

When the profile is regular at the observed offset, its leading term is a spatially constant ambient velocity. Absorbing this constant into the translation coefficient replaces b_n by $b_n - b_{\mathrm{amb},n}$, and the remaining terms vanish on compact terminal cylinders. If the offset escapes, the entire profile is locally invisible and no absorption is needed. In particular,

$$\sup_n \sup_{\tau\in I}\big(|a_n(\tau)| + |b_n(\tau)|\big) < \infty.$$

Proof. The local invisibility of the nonconstant remainder follows from lemma 8.18 and the H^{-1} product estimates in lemma 8.24. Separated physical-point profiles are locally invisible by lemma 8.7; their pressures and cutoff commutators vanish by lemma 8.23 and lemma 8.24. Therefore only the spatially constant ambient velocity changes the translation coefficient, and after this absorption the effective modulation coefficients remain uniformly bounded. ☐

Lemma 8.26 (Terminal compact scale-collapse admissibility)**.** *The retained terminal solution U_n has positive finite critical mass, pressure reconstruction, bounded modulation, local windowed H^1 control, critical tightness on the selected windows, and the compactness input used by the compact scale-collapse argument.*

Proof. The terminal class is active, so after the perturbative residual and nonretained local states have been removed,

$$\inf_{\tau\in I} \|U_n(\tau)\|_{L^3(B_R)} \geq c_{\text{act}} > 0$$

on the selected compact windows, after passing to the subsequences used in the local state-space selection. The upper bound follows from the retained local compact-cylinder bound:

$$\sup_{\tau\in I} \|U_n(\tau)\|_{L^3(B_R)} \leq C(M).$$

The terminal state-space completeness also gives the required tightness. If U_n were not uniformly L^3-tight on the selected windows, then there would be $\varepsilon > 0$, radii $R_k \to \infty$, and times τ_n such that

$$\int_{|y|>R_k} |U_n(y,\tau_n)|^3 \, dy \geq \varepsilon.$$

Applying the local state-space stratification to this escaping sequence produces either an exterior-regular component, which is removed by the exterior regularity argument below, or an additional active local component at a different physical point or larger same-point scale. Such a component belongs to S_n and is handled by terminal decoupling. Hence no non-tight retained terminal solution remains.

The repaired-gauge representation gives pressure reconstruction and bounded modulation. The local Caccioppoli estimate gives local windowed H^1 control, and the Aubin–Lions compactness layer gives the compactness input on the selected windows. □

Theorem 8.27 (Terminal nonlinear local-state decoupling theorem)**.** *Assume local compact critical bounds, terminal windowed state-space completeness, local perturbative residual stability, the exterior annulus alternative, repaired-gauge representation, and local Caccioppoli regularity. Then terminal nonlinear local-state decoupling holds in the sense of definition 8.13.*

Proof. Let $\mathcal{C}$ be any terminal active compound class, let U_n be the retained repaired-gauge nonlinear solution in its terminal frame, and set

$$S_n = V_n - U_n.$$

Lemma 8.18 proves

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_{2R})$$

on every compact terminal cylinder. Divergence-free compatibility follows from the divergence-free repaired-gauge equation and the local terminal decomposition.

Lemma 8.20 proves

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

The local compact critical bounds give the $L^\infty_\tau L^3_y$ control required in the pressure lemma below on the cylinders under consideration. Lemma 8.23 proves

$$\nabla P_{\text{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R)$$

after subtracting spatially constant pressure terms. Lemma 8.24 proves

$$\partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y\cdot\nabla S_n) + b_n\cdot\nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Lemma 8.25 gives modulation compatibility, and lemma 8.26 gives compact scale-collapse admissibility of the retained terminal solution. These are the items in definition 8.13. □

Corollary 8.28 (Multibubble exclusion from terminal local decoupling)**.** *Under the Type II local terminal data, terminal windowed state-space completeness, local perturbative residual stability, the exterior annulus alternative, repaired-gauge representation, and local Caccioppoli regularity, no active multibubble Type II configuration occurs.*

Proof. Theorem 8.27 gives terminal nonlinear decoupling. Theorem 8.14 then supplies the same-point and separated-point decoupling hypotheses used in theorem 8.9. The active local mass floor follows from the retained positive local critical-mass row and local perturbative residual stability. Applying theorem 8.9 gives the claimed exclusion of active multibubble configurations. □

8.5. **Small local residual components.** For a compact terminal cylinder $Q_{R,I} := B_R \times I$, define the local residual seminorm

$$\|W\|_{\mathcal{X}(Q_{R,I})} := \|W\|_{L^\infty(I;L^3(B_R))} + \|W\|_{L^2(I;H^1(B_R))}.$$

This is only a compact-window norm. It is not a whole-space heat-flow or global mild-solution norm.

Lemma 8.29 (Local residual product estimates)**.** *Let $Q_{2R,I} = B_{2R} \times I$ be compact. Suppose*

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_{2R}))$$

and

$$\sup_n \big(\|U_n\|_{\mathcal{X}(Q_{2R,I})} + \|S_n\|_{\mathcal{X}(Q_{2R,I})} \big) < \infty.$$

Then

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1(I; L^2(B_R)).$$

If the pressures associated with these cross stresses are reconstructed by the local pressure decomposition, then their gradients converge to zero in

$$L^1(I; H^{-1}(B_R)).$$

Proof. Sobolev embedding on B_{2R} gives uniform

$$U_n, S_n \quad \text{in } L^2(I; L^6(B_{2R})).$$

Hölder's inequality in space and time yields

$$\|U_n \otimes S_n\|_{L^1(I;L^2(B_R))} \le \|S_n\|_{L^\infty(I;L^3(B_R))} \|U_n\|_{L^1(I;L^6(B_R))} \to 0,$$

because

$$\|U_n\|_{L^1(I;L^6(B_R))} \le |I|^{1/2} \|U_n\|_{L^2(I;L^6(B_R))}.$$

The same estimate treats $S_n \otimes U_n$, and

$$\|S_n \otimes S_n\|_{L^1(I;L^2(B_R))} \le \|S_n\|_{L^\infty(I;L^3(B_R))} \|S_n\|_{L^1(I;L^6(B_R))} \to 0.$$

For the pressure statement, apply the local pressure reconstruction to the cross source. The preceding L^1L^2 convergence gives local convergence of the local pressure part by Calderón–Zygmund estimates. Any non-vanishing harmonic remainder is routed by lemma 8.22 with $(q,p) = (1,2)$ to the pressure/exterior alternative; on a retained branch this alternative is absent. Therefore the cross-pressure gradients vanish in $L^1H^{-1}(B_R)$. □

Theorem 8.30 (Small local residual components are perturbative)**.** *Let S_n be a nonretained terminal residual on every compact terminal cylinder $Q_{2R,I}$. Assume*

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_{2R}))$$

and that the local compact-window bounds, local pressure reconstruction, and bounded modulation estimates hold on $Q_{2R,I}$. Then S_n is perturbative in the terminal decoupling topology of definition 8.13.

Proof. Lemma 8.29 gives the vanishing nonlinear-stress and cross-pressure terms. The repaired-gauge equation for $V_n = U_n + S_n$ and the equation for the retained component U_n, subtracted on $Q_{R,I}$, give the linear residual identity of lemma 8.24. The stress and pressure terms just proved vanish, and bounded modulation sends the remaining $a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n$ terms to zero in $L^1(I; H^{-1}(B_R))$. Hence all perturbative clauses in definition 8.13 hold on $Q_{R,I}$. □

Corollary 8.31 (Small local components are not active)**.** *If a nonretained component has vanishing local $L^\infty L^3$ seminorm on every compact terminal cylinder and satisfies the local compact-window bounds, then it cannot define an active terminal state.*

Proof. By theorem 8.30, such a component contributes only vanishing stress, pressure, modulation, and residual errors in the terminal decoupling topology. Active terminal states are precisely retained local state-space labels with positive nonperturbative local mass. Therefore a component that is perturbative on every compact terminal cylinder is not active. □

Theorem 8.32 (Positive local mass floor for active terminal components)**.** *There exists a local threshold $\varepsilon_{\rm loc} > 0$, depending only on the Caffarelli–Kohn–Nirenberg threshold and the finite covering constants of the selected compact cylinders, such that every retained active terminal component has, after passing to a subsequence, a compact terminal cylinder $Q_{R,I}$ with*

$$\liminf_{n\to\infty} \int_I \int_{B_R} |W_n|^3 \, dy \, d\tau \geq \varepsilon_{\rm loc}^3$$

and

$$\limsup_{n\to\infty} \|W_n\|_{L^\infty(I;L^3(B_R))} \geq \varepsilon_{\rm loc}.$$

Proof. Let $\varepsilon_{\rm CKN}$ be the threshold in theorem 8.34. Fix a finite nested covering of each retained compact terminal cylinder by CKN cylinders whose smaller cylinders still cover the retained core. Choose $\eta_{\rm det} > 0$ to be $\varepsilon_{\rm CKN}$ divided by a sufficiently large constant depending only on this cover.

If every CKN cylinder in the cover had velocity-pressure quantity below $\eta_{\rm det}$, then the CKN criterion would give a uniformly smooth bounded representative on the retained smaller cylinder. Such a bounded selected-window representative is not an active Type II terminal component; it is routed to the regular/bounded-window alternative. Hence an active retained component has a subcylinder on which the terminal CKN quantity is at least $\eta_{\rm det}$.

If the pressure part of that CKN quantity carries at least half of $\eta_{\rm det}$ while the velocity part is arbitrarily small on all smaller subcylinders, then the local pressure reconstruction and harmonic-tail alternative, made explicit in lemma 8.22 with $(q,p) = (3/2, 3/2)$, assign the branch to the pressure/exterior-leakage stratum. That stratum is adjacent and is not a retained active terminal component. Therefore on a retained active component the velocity part carries a fixed fraction of the detected CKN quantity on some smaller cylinder:

$$\rho^{-2} \int_{I_\rho} \int_{B_\rho} |W_n|^3 \, dy \, d\tau \geq c \, \eta_{\rm det}$$

along a subsequence. Since $|I_\rho| \simeq \rho^2$, there is a time in I_ρ with

$$\int_{B_\rho} |W_n(y,\tau)|^3 \, dy \geq c \, \eta_{\rm det}.$$

After rescaling this subcylinder to a fixed compact terminal cylinder, this is the asserted spacetime lower bound and the asserted $L^\infty_\tau L^3_y$ lower bound with $\varepsilon_{\rm loc} := (c \, \eta_{\rm det})^{1/3}$.

If no such velocity lower bound existed on any compact terminal cylinder, then the component would have vanishing local $L^\infty L^3$ seminorm after subsequence extraction and would be perturbative by corollary 8.31, contradicting retained activity. □

8.6. **Exterior regularity.**

Definition 8.33 (Exterior annuli and terminal CKN quantity)**.** Let (x_*, T^*) be the selected Type II core point. For $0 < r < R < \infty$, set

$$A_{r,R} := \overline{B_R(x_*)} \setminus B_r(x_*).$$

For $x_0 \in \mathbb{R}^3$ and $\rho > 0$, write

$$Q^*_\rho(x_0) := B_\rho(x_0) \times (T^* - \rho^2, T^*).$$

For a suitable weak solution (u, p), define the terminal local velocity-pressure quantity

$$\mathcal{C}_*(u, p; x_0, \rho) := \rho^{-2} \int_{T^*-\rho^2}^{T^*} \int_{B_\rho(x_0)} |u|^3 \, dx \, dt + \rho^{-2} \int_{T^*-\rho^2}^{T^*} \int_{B_\rho(x_0)} |p - (p)_{B_\rho(x_0)}(t)|^{3/2} \, dx \, dt.$$

The pressure is measured modulo functions of time.

Theorem 8.34 (Terminal Caffarelli–Kohn–Nirenberg regularity)**.** *There exists $\varepsilon_{\mathrm{CKN}} > 0$ with the following property. Let (u, p) be suitable on $Q^*_\rho(x_0)$. If*

$$\mathcal{C}_*(u, p; x_0, \rho) \le \varepsilon_{\mathrm{CKN}},$$

then for each integer $k \ge 0$ there are $0 < c_k < 1$ and $C_k < \infty$, independent of $(u, p), x_0, \rho$, such that u is smooth in

$$B_{c_k\rho}(x_0) \times (T^* - c_k\rho^2, T^*)$$

and, after subtracting a function of time from p,

$$\sup_{t \in (T^* - c_k\rho^2, T^*)} \Big(\rho^{k+1} \|\nabla^k u(t)\|_{L^\infty(B_{c_k\rho}(x_0))} + \rho^{k+2} \|\nabla^k p(t)\|_{L^\infty(B_{c_k\rho}(x_0))} \Big) \le C_k.$$

Proof. This is the standard endpoint form of the Caffarelli–Kohn–Nirenberg epsilon-regularity criterion [CKN82, Lin98], after translating the terminal time to 0 and rescaling $Q^*_\rho(x_0)$ to a unit parabolic cylinder. The pressure normalization by spatial averages is the usual normalization in the suitable weak-solution criterion. Interior parabolic bootstrapping gives the stated higher derivative bounds on smaller cylinders. □

Throughout the exterior regularity subsection we fix

$$\varepsilon_{\mathrm{ext}} := \frac{1}{2} \varepsilon_{\mathrm{CKN}}.$$

Definition 8.35 (No exterior concentration on a fixed annulus)**.** Let $0 < r < R < \infty$. A suitable Type II branch has no exterior concentration on $A_{r,R}$ if there exists $\rho_* > 0$ such that whenever

$$x_0 \in A_{r,R}, \qquad 0 < \rho \le \rho_*, \qquad Q^*_\rho(x_0) \subset (B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)}) \times (0, T^*),$$

one has

$$\mathcal{C}_*(u, p; x_0, \rho) \le \varepsilon_{\mathrm{ext}}.$$

Definition 8.36 (Exterior concentration branch)**.** An exterior concentration branch on $A_{r,R}$ is a sequence

$$x_n \in A_{r,R}, \qquad \rho_n \downarrow 0,$$

such that

$$Q^*_{\rho_n}(x_n) \subset (B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)}) \times (0, T^*)$$

and

$$\mathcal{C}_*(u, p; x_n, \rho_n) > \varepsilon_{\mathrm{ext}}.$$

The associated exterior rescalings are

$$U_n(y, s) := \rho_n u(x_n + \rho_n y, T^* + \rho_n^2 s), \qquad P_n(y, s) := \rho_n^2 p(x_n + \rho_n y, T^* + \rho_n^2 s),$$

for $s < 0$.

Lemma 8.37 (Exterior test dichotomy)**.** *Fix $0 < r < R < \infty$. Then either the branch has no exterior concentration on $A_{r,R}$, or it admits an exterior concentration branch on $A_{r,R}$.*

Proof. If the no-concentration condition fails, then for every m there are

$$x_m \in A_{r,R}, \qquad 0 < \rho_m \le m^{-1},$$

with $Q^*_{\rho_m}(x_m)$ contained in the enlarged annular cylinder and

$$\mathcal{C}_*(u, p; x_m, \rho_m) > \varepsilon_{\mathrm{ext}}.$$

After discarding repetitions and passing to a subsequence, $\rho_m \to 0$. This is precisely an exterior concentration branch. The converse is immediate from the definitions. □

Theorem 8.38 (Exterior concentration stratification)**.** *Fix $0 < r < R < \infty$. A suitable Type II branch satisfies exactly one of the following alternatives on $A_{r,R}$:*

(i) *it has no exterior concentration on $A_{r,R}$;*
(ii) *it admits an exterior concentration branch on $A_{r,R}$.*

The second alternative is the complementary concentration branch. According to the relative position and scale of the exterior concentration sequence, it belongs to one of the exterior-profile, separated-core, multicore, noncompact exterior, or scale-collapse alternatives.

Proof. The dichotomy is lemma 8.37. If an exterior concentration branch exists, then the annular terminal cylinders carry a positive Caffarelli–Kohn–Nirenberg quantity at scales $\rho_n \downarrow 0$ and centers a positive physical distance from the selected core. Let λ_n denote the selected core scale along the same terminal sequence. After passing to a subsequence, the relative scale-center configuration falls into one of the standard local concentration-compactness alternatives. If the exterior frame is orthogonal to the selected core frame, it gives an exterior profile. If only finitely many separated active centers remain at comparable terminal scales, it gives a separated-core or multicore configuration. If compactness fails in the exterior frame, one obtains the noncompact exterior alternative. If the exterior frame develops a further unresolved scale in its own renormalized variables, one obtains the scale-collapse alternative. These alternatives exhaust the subsequential configurations at positive physical distance from the selected core. Thus the failure of the annular regularity test is represented by a local concentration alternative rather than by an additional regularity hypothesis. □

Lemma 8.39 (Exterior CKN cover)**.** *Assume there is no exterior concentration on $A_{r,R}$. Fix an integer $k \ge 0$. If*

$$A_{r',R'} \Subset A_{r,R},$$

then there are points $x_j \in A_{r,R}$, radii $\rho_j > 0$, and a terminal time length $\delta > 0$ such that

$$A_{r',R'} \times (T^* - \delta, T^*) \subset \bigcup_j B_{c_k \rho_j}(x_j) \times (T^* - c_k \rho_j^2, T^*)$$

and

$$\mathcal{C}_*(u, p; x_j, \rho_j) \le \varepsilon_{\mathrm{ext}}$$

for every j. Here c_k is the constant from theorem 8.34.

Proof. Since $A_{r',R'} \Subset A_{r,R}$, there is $d > 0$ such that

$$B_{4d}(x) \subset B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)} \qquad \text{for all } x \in A_{r',R'}.$$

Let ρ_* be supplied by the no-concentration condition and set $\rho_0 := \min\{d, \rho_*\}$. By compactness, finitely many balls

$$B_{c_k\rho_0/2}(x_j), \qquad x_j \in A_{r,R},$$

cover $A_{r',R'}$. The terminal cylinders $Q^*_{\rho_0}(x_j)$ are contained in the enlarged annular cylinder, so the no-concentration condition gives the displayed smallness. Taking $\delta = c_k\rho_0^2/2$ gives the spacetime cover. □

Lemma 8.40 (Exterior pressure localization). *Let $A_{r',R'} \Subset A_{r,R}$, and assume that u is bounded on a slightly larger annular terminal cylinder. Then on*

$$A_{r',R'} \times (T^* - \delta, T^*)$$

the pressure has a decomposition

$$p = p_{\mathrm{loc}} + p_{\mathrm{harm}} + c(t),$$

where $c(t)$ is a function of time, p_{harm} is harmonic in space on the smaller annulus, and for every $q < \infty$ and every integer $k \geq 0$

$$\|p_{\mathrm{loc}}\|_{L^\infty_t L^q_x(A_{r',R'})} + \|p_{\mathrm{harm}}\|_{L^\infty_t C^k_x(A_{r',R'})} \leq C_{r',R',q,k}.$$

Proof. Choose $\eta \in C^\infty_c(B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)})$ with $\eta \equiv 1$ on a slightly larger annulus containing $A_{r',R'}$. Define p_{loc} by

$$-\Delta p_{\mathrm{loc}} = \partial_i\partial_j(\eta u_i u_j)$$

using the Newtonian potential, with the usual spatial normalization. Since u is locally bounded on the support of η, Calderón–Zygmund estimates give $p_{\mathrm{loc}} \in L^\infty_t L^q_x$ for every $q < \infty$ on the terminal interval. The difference

$$p - p_{\mathrm{loc}}$$

is harmonic in space on the smaller annulus, modulo a function of time. The standard interior estimates for harmonic functions give the displayed C^k bounds after fixing the time-dependent additive constant. □

Theorem 8.41 (Exterior regularity on no-concentration annuli). *Let u, p be a suitable weak solution on $\mathbb{R}^3 \times [0, T^*)$, and let (x_*, T^*) be the selected Type II core point. Fix $0 < r' < R' < \infty$. If there is no exterior concentration on some annulus $A_{r,R}$ with*

$$A_{r',R'} \Subset A_{r,R},$$

then for every integer $k \geq 0$ there are $\delta > 0$, a function $c(t)$, and $C_{r',R',k} < \infty$ such that

$$\sup_{t\in(T^*-\delta,T^*)} \Big(\|u(t)\|_{C^k(A_{r',R'})} + \|p(t) - c(t)\|_{C^k(A_{r',R'})} \Big) \leq C_{r',R',k}.$$

If the no-concentration hypothesis fails, the complementary exterior concentration alternative holds in the sense of definition 8.36.

Proof. The no-concentration condition and lemma 8.39 give a finite cover of $A_{r',R'} \times (T^* - \delta, T^*)$ by terminal cylinders on which the Caffarelli–Kohn–Nirenberg quantity is below $\varepsilon_{\mathrm{CKN}}$. Applying theorem 8.34 on each cylinder gives local C^k bounds for u and for p modulo functions of time on smaller cylinders. Since the cover is finite, taking the maximum over the cover gives the stated local bound.

The pressure normalizations on overlapping balls differ only by functions of time. Equivalently, one may use lemma 8.40 after the $k = 0$ velocity bound has been obtained and then apply standard interior parabolic estimates on nested annuli. This gives a single time-dependent normalization $c(t)$ on $A_{r',R'}$. If the no-concentration hypothesis fails, the exterior test dichotomy gives an exterior concentration branch. □

Lemma 8.42 (Divergence-free core and exterior decomposition)**.** *Fix $0 < r_0 < R_0 < \infty$, and assume that there is no exterior concentration on an annulus containing*

$$\overline{B_{2r_0}(x_*)} \setminus B_{r_0}(x_*).$$

Let $\chi \in C_c^\infty(B_{2r_0}(x_*))$ satisfy

$$\chi \equiv 1 \quad \text{on } B_{r_0}(x_*).$$

Then there are divergence-free fields u_{core} and u_{ext} such that

$$u = u_{\text{core}} + u_{\text{ext}}, \qquad u_{\text{core}} = u \text{ on } B_{r_0}(x_*), \qquad u_{\text{ext}} = 0 \text{ on } B_{r_0}(x_*),$$

and all derivatives of u_{ext} are uniformly bounded on compact subannuli of the no-exterior-concentration region up to T^*.

Proof. Set

$$g = \nabla\chi \cdot u.$$

The support of g lies in the annulus

$$A_{r_0} = B_{2r_0}(x_*) \setminus B_{r_0}(x_*),$$

where u is smooth by theorem 8.41. Since

$$\int_{A_{r_0}} g\,dx = \int_{\mathbb{R}^3} \nabla\chi \cdot u\,dx = -\int_{\mathbb{R}^3} \chi\,\nabla \cdot u\,dx = 0,$$

the Bogovskii operator on A_{r_0} gives a vector field

$$w = \mathcal{B}_{A_{r_0}} g$$

supported in A_{r_0}, smooth up to T^*, and satisfying

$$\nabla \cdot w = g.$$

Define

$$u_{\text{core}} := \chi u - w, \qquad u_{\text{ext}} := u - u_{\text{core}}.$$

Then

$$\nabla \cdot u_{\text{core}} = \nabla\chi \cdot u + \chi\nabla \cdot u - \nabla \cdot w = 0,$$

and therefore $\nabla \cdot u_{\text{ext}} = 0$. Since $\chi = 1$ and $w = 0$ on $B_{r_0}(x_*)$, $u_{\text{core}} = u$ and $u_{\text{ext}} = 0$ there. The uniform exterior bounds follow from theorem 8.41 on compact subannuli and the standard boundedness of the Bogovskii construction on the fixed annulus. ∎

Lemma 8.43 (Exterior fields vanish in compact core frames)**.** *Let (x_n, λ_n) be a core frame with*

$$x_n \to x_*, \qquad \lambda_n \to 0.$$

For a compact frame ball B_R, define

$$V_{\text{ext}}^n(y,\tau) := \lambda_n u_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 \tau).$$

Then, for every compact $B_R \times I$,

$$V_{\text{ext}}^n \equiv 0$$

on $B_R \times I$ for all sufficiently large n. Consequently

$$V_{\text{ext}}^n \to 0 \quad \text{in every local } L_\tau^q W_y^{k,q}(B_R) \text{ norm.}$$

Proof. Since $x_n \to x_*$ and $\lambda_n \to 0$, for fixed R,

$$x_n + \lambda_n B_R \subset B_{r_0}(x_*)$$

for all sufficiently large n. On $B_{r_0}(x_*)$,

$$u_{\text{ext}} = 0.$$

Therefore $V^n_{\text{ext}} = 0$ on $B_R \times I$ for all large n, which implies convergence in every stated local norm. □

Lemma 8.44 (Exterior pressure vanishes modulo constants)**.** *Let p_{ext} be any pressure contribution generated by source terms supported in*

$$\{|x - x_*| \geq r_0\}.$$

In a core frame set

$$P^n_{\text{ext}}(y, \tau) := \lambda_n^2 p_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 \tau).$$

Then, for every compact $B_R \times I$, there are constants $c_{n,R}(\tau)$ such that

$$P^n_{\text{ext}} - c_{n,R} \to 0 \quad \textit{in } L^\infty_\tau C^1_y(B_R),$$

and hence

$$\nabla_y P^n_{\text{ext}} \to 0 \quad \textit{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. The exterior pressure source is at positive physical distance from

$$x_n + \lambda_n B_R$$

for all sufficiently large n. Therefore p_{ext} is harmonic, smooth, and uniformly C^2-bounded on the physical neighborhood swept out by the compact frame cylinder. Take

$$c_{n,R}(\tau) := \lambda_n^2 p_{\text{ext}}(x_n, t_n + \lambda_n^2 \tau).$$

By the mean value theorem,

$$\|P^n_{\text{ext}} - c_{n,R}\|_{C^1_y(B_R)} \leq C_R \lambda_n^3 \sup |\nabla_x p_{\text{ext}}| + C_R \lambda_n^4 \sup |\nabla^2_x p_{\text{ext}}|.$$

The right-hand side tends to zero because the physical derivatives are uniformly bounded and $\lambda_n \to 0$. This proves the asserted convergence. □

Lemma 8.45 (Positive-distance exterior states do not remain in compact core frames)**.** *Let (x_n, λ_n) be a core frame with $x_n \to x_*$ and $\lambda_n \to 0$. Let (z_n, μ_n) be an exterior local state satisfying*

$$\liminf_{n\to\infty} |z_n - x_*| \geq r_* > 0.$$

Assume that the ordered terminal state-space test has already retained the compact core at (x_*, λ_n). Then the positive-distance exterior state does not leave an additional retained velocity component in any compact core frame. More precisely, one of the following ordered alternatives occurs:

(i) *the state lies on a no-exterior-concentration annulus and is regular and removable by theorem 8.41;*
(ii) *the state contains a positive local CKN carrier, in which case it is an exterior, separated-core, multicore, cascade, or rebased local branch rather than part of the selected compact core;*
(iii) *its velocity contribution is zero in the retained compact-core topology, and its pressure contribution vanishes modulo functions of time by lemma 8.49, unless the far-field harmonic tail is routed to the ordered pressure row by lemma 8.60.*

No whole-space L^3 profile decomposition is used in this decoupling.

Proof. For fixed $R < \infty$, the physical frame ball

$$x_n + \lambda_n B_R$$

lies in $B_{r_*/4}(x_*)$ for all sufficiently large n, while the profile state center z_n lies outside $B_{r_*/2}(x_*)$. Thus every local source supported in the positive-distance state is separated from the compact core frame.

Run the ordered local CKN test on the exterior state. If the test is below threshold on an annulus containing the state, then the no-concentration regularity theorem gives item (i), and theorem 8.46 removes the regular contribution from compact core frames. If the test finds a positive CKN carrier, that carrier is recorded before retention as an exterior-profile, separated-core, multicore, cascade, or rebased local branch; this is item (ii).

It remains to treat the nonretained part after these tests. Its velocity contribution is supported a positive distance from the physical core and therefore vanishes identically on every compact core cylinder for all large n, as in lemma 8.43. The pressure generated by fixed positive-distance sources is perturbative modulo functions of time by lemma 8.49. If the only nonvanishing pressure contribution is a far-field harmonic tail, then lemma 8.60 routes that tail to a dyadic exterior carrier, a regular dyadic tail, or the ordered pressure row. Hence no positive-distance exterior state is silently retained inside the selected compact core. □

Theorem 8.46 (Exterior regular components decouple from compact core frames)**.** *On the no-exterior-concentration stratum for the fixed annuli separating an exterior component from the selected Type II core point, that component is a regular exterior component and vanishes in compact Type II core frames, including its pressure contribution modulo constants.*

Proof. Apply the divergence-free decomposition of lemma 8.42. The exterior component is identically zero in every compact core frame for large n by lemma 8.43. Its pressure contribution is a locally scaled smooth harmonic field and vanishes modulo constants by lemma 8.44. The cutoff and Bogovskii correction terms are supported in the fixed exterior annulus A_{r_0}, so they also vanish in compact core frames by the same argument.

Thus a regular exterior component decouples from the Type II core without changing the local repaired-gauge equation in any compact terminal frame, up to errors converging to zero in the perturbative topology used in lemma 8.10 and definition 8.13. □

Corollary 8.47 (Exterior annulus alternative for the core decomposition)**.** *Fix a compact annulus $A_{r,R}$ around the selected core. Then exactly one of the following alternatives holds for the Type II branch on this annulus:*

(i) *the branch has no exterior concentration on $A_{r,R}$, and every compact subannulus of $A_{r,R}$ is covered by theorem 8.41; consequently exterior regular contributions from this annulus vanish in compact core frames by theorem 8.46;*
(ii) *the branch admits an exterior concentration branch on $A_{r,R}$, and therefore belongs to one of the adjacent exterior, separated-core, multicore, noncompact exterior, or scale-collapse alternatives described in theorem 8.38.*

Proof. This is the exterior concentration stratification of theorem 8.38, combined in the first alternative with the no-concentration regularity theorem and the exterior-removal theorem. □

Lemma 8.48 (Exterior CKN mass has a velocity or pressure carrier)**.** *Let $x_n \in A_{r,R}$ and $\rho_n \downarrow 0$ form an exterior concentration branch on $A_{r,R}$. After passing to a subsequence, one of the following alternatives holds:*

(i) velocity carrier:

$$\rho_n^{-2} \int_{T^*-\rho_n^2}^{T^*} \int_{B_{\rho_n}(x_n)} |u|^3 \, dx \, dt \geq \frac{\varepsilon_{\text{ext}}}{2};$$

(ii) pressure carrier:

$$\rho_n^{-2} \int_{T^*-\rho_n^2}^{T^*} \int_{B_{\rho_n}(x_n)} |p - (p)_{B_{\rho_n}(x_n)}(t)|^{3/2} \, dx \, dt \geq \frac{\varepsilon_{\text{ext}}}{2}.$$

In the pressure-carrier case the branch is a pressure/exterior local row unless lemma 8.61 converts it, after shrinking the cylinder and passing to a subsequence, into a velocity carrier on an exterior or separated-core cylinder.

Proof. By definition of exterior concentration,

$$\mathcal{C}_*(u, p; x_n, \rho_n) > \varepsilon_{\text{ext}}.$$

The displayed quantity is the sum of the velocity and pressure terms. Hence at least one term is at least $\varepsilon_{\text{ext}}/2$ for infinitely many n, and a subsequence gives one of the two alternatives. The final sentence is only the ordered local reading of the pressure alternative: a pressure lower bound is not used as a hidden compact velocity lower bound, but is passed to the pressure/exterior routing lemma lemma 8.61. □

Lemma 8.49 (Positive-distance sources are perturbative in compact core frames)**.** *Let (x_n, λ_n, t_n) be a compact core frame with*

$$x_n \to x_*, \qquad \lambda_n \downarrow 0, \qquad t_n \to T^*.$$

Fix $d > 0$, and let $K \Subset \{x : |x - x_*| \geq d\}$. Suppose F_n is supported in K for a.e. terminal time and satisfies the local bound

$$\sup_n \|F_n\|_{L^1(T^*-\delta, T^*; L^1(K))} < \infty.$$

Let q_n be the Newtonian pressure generated by F_n,

$$q_n(\cdot, t) = \partial_i \partial_j (-\Delta)^{-1} F_{n,ij}(\cdot, t),$$

restricted to a neighborhood of x_*. For every compact $B_M \times I$ in the core variables, set

$$Q_n(y, \tau) = \lambda_n^2 q_n(x_n + \lambda_n y, t_n + \lambda_n^2 \tau), \qquad c_n(\tau) = \lambda_n^2 q_n(x_n, t_n + \lambda_n^2 \tau).$$

Then

$$Q_n - c_n \to 0 \quad \text{in } L^1_\tau C^1_y(B_M).$$

If w_n is a velocity contribution supported in K, then

$$\lambda_n w_n(x_n + \lambda_n y, t_n + \lambda_n^2 \tau) \equiv 0 \quad \text{on } B_M \times I$$

for all sufficiently large n. Thus positive-distance velocity sources and their pressure tails are perturbative in every compact core frame, modulo functions of time.

Proof. For fixed M, the physical balls $x_n + \lambda_n B_M$ lie in $B_{d/4}(x_*)$ for all sufficiently large n, whereas $K \subset \{|x - x_*| \geq d\}$. The velocity assertion follows immediately from the support separation.

On $B_{d/4}(x_*)$ the kernel $\partial_i \partial_j (-\Delta)^{-1}$ applied to sources supported in K is smooth. For $|\alpha| \leq 2$,

$$\|\nabla_x^\alpha q_n(t)\|_{L^\infty(B_{d/4}(x_*))} \leq C_{\alpha,d,K} \|F_n(t)\|_{L^1(K)}$$

for a.e. t. By the mean value theorem,

$$\|Q_n(\tau) - c_n(\tau)\|_{C^1_y(B_M)} \leq C_M \lambda_n^3 \|\nabla_x q_n(t_n + \lambda_n^2 \tau)\|_{L^\infty(B_{d/4})} + C_M \lambda_n^4 \|\nabla_x^2 q_n(t_n + \lambda_n^2 \tau)\|_{L^\infty(B_{d/4})}.$$

Integrating in $\tau \in I$ and using $dt = \lambda_n^2 \, d\tau$, the right-hand side is bounded by

$$C_{M,I,d,K}(\lambda_n + \lambda_n^2) \|F_n\|_{L^1(T^*-\delta, T^*; L^1(K))},$$

which tends to zero. This gives the claimed convergence. The proof is local on the fixed set K and uses no whole-space critical norm. □

Lemma 8.50 (Exterior local states decouple from the selected compact core)**.** *Let an exterior concentration branch on $A_{r,R}$ be in the exterior-profile subcase of theorem 8.38. Then the exterior profile is not a terminal obstruction for the selected compact core. More precisely, in every compact core frame (x_n, λ_n, t_n) with $x_n \to x_*$ and $\lambda_n \downarrow 0$, the exterior local state has zero compact-core velocity contribution in the retained terminal state-space topology. Its pressure contribution either vanishes modulo functions of time on compact core cylinders, or it is routed by the far-field harmonic-tail theorem lemma 8.60 to a dyadic exterior carrier or to the ordered pressure row. If the exterior state itself is selected as a possible singular branch, it is rebased at its own center and tested by the same local Type II state-space decomposition; it is no longer an exterior exit for the original selected core.*

Proof. The exterior-profile subcase is read locally: its CKN witnesses remain at positive physical distance from x_*, or escape every compact set in the selected core variables. Suppose that a positive velocity contribution from this exterior state were visible on some compact core cylinder. Then theorem 8.16 would assign that positive compact-core mass to a first local state-space label. If the label is retained comparable mass, it must already have been grouped into the selected core; if it is not retained comparable mass, it is an adjacent compact, separated, cascade, rough-core, or pressure/exterior row. Hence no positive exterior velocity state remains in the nonretained exterior part of a retained compact core branch.

For pressure, decompose the exterior pressure seen on a compact core cylinder into the local Calderón–Zygmund part generated by sources in a fixed annulus, the far-field harmonic tail, and a function of time. Fixed positive-distance annular sources are perturbative by lemma 8.49. If the remaining far-field harmonic tail does not vanish modulo functions of time, then lemma 8.60 produces a dyadic exterior stress carrier or the ordered pressure row. Thus pressure is also either perturbative for the selected core or routed before retention.

Thus the exterior profile is perturbative for the selected core. If one chooses the exterior profile as the new active branch, the center and scale are changed and the analysis restarts from the covered-entry theorem corollary 10.19. This rebasing does not leave an independent exterior terminal row attached to the original core. □

Lemma 8.51 (Separated exterior cores are multicore or rebased local branches)**.** *Suppose an exterior concentration branch on $A_{r,R}$ carries a velocity carrier and that, after passing to a subsequence, the exterior carrier remains as a positive active core at terminal scales comparable to another selected active scale. Then either the two active centers are grouped as a separated multicore branch in the local state space, or the exterior center is rebased as its own physical local Type II germ. In the first case the branch is excluded by theorem 10.6; in the second case the row is no longer an exterior row for the original selected core.*

Proof. The exterior centers satisfy $|x_n - x_*| \geq r > 0$, while the selected compact core scales tend to zero. Hence the relative scale-center distance between the original core and the exterior carrier tends to infinity whenever the scales are comparable, and the separated-point reduction theorem 4.19 applies. Its output is exactly a separated-core or multicore label unless the exterior center is selected as the new base point. The separated-core and multicore labels are part of the multibubble/cascade branch excluded by theorem 10.6. If the exterior center is chosen as the base point, the local CKN lower bound supplies a physical local concentration sequence there and corollary 10.19 starts the local Type II analysis for that new germ. This is a rebasing, not an unresolved exterior mechanism for the original core. □

Lemma 8.52 (Exterior finite-scale packing is an ordered local row)**.** *Fix a compact annulus $A_{r,R}$. In any fixed dyadic scale band of exterior carrier cylinders contained in $A_{r,R}\times(T^*-\delta,T^*)$, a Vitali selection produces a pairwise disjoint subfamily with uniformly bounded overlap after enlargement. Therefore an exterior velocity carrier in that scale band has an ordered local interpretation:*

(i) *a finite selected subfamily gives a separated-core or multicore label;*
(ii) *an unbounded number of comparable exterior carriers, or a sequence of scale bands with no finite selected representative, is by definition a noncompact exterior or cascade boundary face before the retained compact core is declared.*

No uniform bound on the number of exterior carriers is asserted or used.

Proof. The geometric Vitali covering lemma applies on the fixed annulus for each dyadic radius band. It gives a disjoint subfamily whose fixed enlargement covers all carrier balls in that band, with overlap bounded by an absolute constant depending only on the enlargement factor. The ordered local state-space construction selects active carriers from this finite-overlap family. If only finitely many selected carriers remain, their positive CKN lower bounds and separated scale-center relations give the separated-core or multicore row. If no finite selected list captures the active exterior carriers, then the sequence has left every finite scale-center chart; this is the noncompact exterior boundary face, or the same-point scale-cascade face when the centers converge but the radii separate strictly. The argument is geometric and local; it does not invoke a global L^3 mass bound to bound the cardinality of the family. □

Lemma 8.53 (Exterior scale-center compactification)**.** *Let (z_n,σ_n) be exterior active carrier data in an exterior frame, with $\sigma_n>0$, and suppose the corresponding terminal cylinders carry a fixed positive CKN lower bound. After passing to a subsequence, exactly one of the following ordered alternatives occurs:*

(i) *the normalized centers z_n remain in a compact subset of the exterior frame; this is a compact exterior-profile, separated-core, or multicore row;*
(ii) $|z_n|\to\infty$ *and* $\sigma_n/|z_n|\to 0$*; this is a dyadic point carrier in annuli* $\{|y|\simeq|z_n|\}$*;*
(iii) $|z_n|\to\infty$ *and*
$$0<\liminf_{n\to\infty}\sigma_n/|z_n|\le\limsup_{n\to\infty}\sigma_n/|z_n|<\infty;$$
this is a dyadic annular carrier;
(iv) *the relative scale leaves every compact subset at a fixed exterior point, or after rebasing at the exterior point; this is the same-point cascade or exterior scale-collapse row.*

These alternatives are disjoint after the ordered first-failure convention and exhaust the exterior scale-center configurations. In particular, the phrase "standard boundary faces" in the exterior routing below means precisely the four cases listed here.

Proof. If z_n is bounded, compactness of closed balls in the exterior frame gives a subsequential limit, yielding item (i). If $|z_n|\to\infty$, pass to a subsequence so that $\sigma_n/|z_n|$ converges in the compactified interval $[0,\infty]$. Limit 0 gives item (ii). A limit in $(0,\infty)$ gives item (iii). Limit $+\infty$, or failure of compactness after recentering at a bounded exterior point, means that the active radius contains a further unresolved scale. The ordered local state-space compactification then records either strict same-point scale separation or a genuine scale-collapse row, giving item (iv). These possibilities exhaust the one-point compactification of the center variable together with the compactified relative-scale variable. □

Lemma 8.54 (Dyadic exterior occupation selection)**.** *Work in an exterior frame and let*
$$\mathcal{A}_k:=B_{2^{k+1}}\setminus B_{2^k},\qquad k\ge 0.$$

For $K \geq 0$, define the far-dyadic CKN level

$$\Gamma_{n,K} := \sup_{\substack{k \geq K,\ y \in \mathcal{A}_k \\ 0 < r \leq 2^{k-4} \\ Q_r^*(y) \text{ admissible}}} \mathcal{C}_*(U_n, P_n; y, r),$$

where the terminal cylinders are understood after pulling back to the physical variables. Then the ordered test gives one of the following alternatives after passing to a subsequence:

(i) *a dyadic CKN carrier exists:*

$$\mathcal{C}_*(U_n, P_n; y_n, r_n) \geq \varepsilon_{\text{ext}}$$

for a sequence $|y_n| \to \infty$;

(ii) *the far-dyadic CKN level is eventually below threshold:*

$$\limsup_{K \to \infty} \limsup_{n \to \infty} \Gamma_{n,K} \leq \varepsilon_{\text{ext}},$$

and, because $\varepsilon_{\text{ext}} < \varepsilon_{\text{CKN}}$, every sufficiently far dyadic annulus is regular on its interior by the terminal CKN criterion.

Thus a noncompact exterior branch can remain a terminal obstruction only in the dyadic CKN-carrier case.

Proof. If

$$\limsup_{K \to \infty} \limsup_{n \to \infty} \Gamma_{n,K} > \varepsilon_{\text{ext}},$$

then for arbitrarily large K there are n, $k \geq K$, $y \in \mathcal{A}_k$, and an admissible $r \leq 2^{k-4}$ with $\mathcal{C}_*(U_n, P_n; y, r) > \varepsilon_{\text{ext}}$. A diagonal subsequence gives item (i) with $|y_n| \to \infty$.

Otherwise the displayed limsup is at most ε_{ext}. Fix η with $\varepsilon_{\text{ext}} < \eta < \varepsilon_{\text{CKN}}$. After discarding a finite initial set of annuli and passing to the selected tail in n, every admissible subcylinder in every far dyadic annulus has CKN quantity at most η, hence below ε_{CKN}. A bounded-overlap cover of each smaller annulus

$$B_{2^{k+1}-2^{k-5}} \setminus B_{2^k+2^{k-5}}$$

by such cylinders and theorem 8.34 give local smoothness on the dyadic annulus. Since the cover is local on each annulus and the constants are scale-invariant after rescaling that annulus to unit size, no global tail norm is used. Therefore the diffuse no-carrier case is regular on every far dyadic annulus and cannot be a terminal noncompact exterior obstruction. □

Lemma 8.55 (Exterior same-point cascades route to the cascade package)**.** *Let exterior carriers on $A_{r,R}$ have centers converging to the same physical point $z \neq x_*$ and have strictly separated scales. Then, after rebasing at z, the branch is a same-point cascade in the sense of theorem 4.18. It is therefore closed by the multibubble/cascade package theorem 10.6, unless the ordered rebased analysis exits earlier through one of the non-exterior first-failure rows.*

Proof. Because $z \neq x_*$, the carriers are at positive physical distance from the original core and are perturbative for that core by lemma 8.49. Recenter at z and use the exterior carrier scales as the active scales. Strict separation of the radii is precisely the same-point scale-cascade boundary face in the relative scale-center compactification. The same-point cascade reduction theorem 4.18 then assigns the rebased branch to the cascade branch, which is excluded by theorem 10.6. If the rebased ordered tests fail before the cascade label is reached, the failure is one of the ordinary representation, modulation, active-core, pressure, rough-core, carrier, or cost rows; none is an exterior row for the original selected core. □

Lemma 8.56 (Noncompact exterior branches admit dyadic active carriers)**.** *Let the exterior stratification of theorem 8.38 produce the noncompact exterior alternative in an exterior frame. Then, after passing to a subsequence, exactly one of the following occurs:*

(i) *the noncompact exterior occupation is diffuse below the CKN threshold on all sufficiently far dyadic annuli, hence regular there by lemma 8.54;*
(ii) *there are dyadic annuli and terminal cylinders in the exterior frame with CKN quantity bounded below by ε_{ext}, hence an exterior velocity or pressure carrier by lemma 8.48;*
(iii) *the noncompact behavior is only a far-field harmonic pressure tail, which is routed to item* (i)*, item* (ii)*, or the ordered pressure row by lemma 8.60.*

Proof. The noncompact alternative is represented by active data leaving every compact subset of the exterior scale-center chart. Apply lemma 8.53. The compact-center, same-point cascade, and scale-collapse faces are not the noncompact exterior face and are routed by their corresponding lemmas below. The remaining noncompact face is a dyadic occupation state in annuli $\mathcal{A}_k$, $k \to \infty$.

Apply the quantitative dyadic selection lemma lemma 8.54. If the occupation is below the CKN threshold on all sufficiently far dyadic annuli, item (i) holds. Otherwise a dyadic CKN carrier exists. The velocity/pressure split lemma 8.48 gives item (ii) unless the carrier is purely pressure. A purely harmonic far-field pressure tail is then routed by lemma 8.60, giving item (iii). All tests are dyadic and local. □

Lemma 8.57 (Noncompact exterior carriers route to structural rows)**.** *Every dyadic active carrier produced by lemma 8.56 is routed, after passing to a subsequence, to one of the following rows:*

(i) *a regular exterior-removal row, if all smaller dyadic tests are no-concentration rows;*
(ii) *a separated-core or multicore row, closed by theorem 10.6;*
(iii) *a same-point cascade row, closed by lemma 8.55;*
(iv) *an exterior scale-collapse row, rebased and tested by lemma 8.58;*
(v) *a pressure/exterior row which is converted to a velocity carrier or to a non-exterior first-failure row by lemma 8.61.*

Thus noncompact exterior concentration is not an independent terminal alternative.

Proof. Use the dyadic carrier selected in the preceding lemma. If adjacent dyadic tests become no-concentration tests after shrinking, then theorem 8.46 gives regular exterior removal. Otherwise lemma 8.53 gives the ordered boundary face of the carrier. Positive centers separated from the original core at comparable scales give the separated-core or multicore row by lemma 8.51. Centers converging to one exterior point with strict scale separation give the same-point cascade row by lemma 8.55. If the dyadic carrier develops its own unresolved scale collapse, it is the exterior scale-collapse row and is handled by lemma 8.58. Finally, if the diagnostic is pressure-only, the pressure-only routing lemma applies. These cases are the ordered local boundary faces of the exterior scale-center chart and exhaust the noncompact exterior alternative. □

Lemma 8.58 (Exterior scale collapse is an ordinary rebased scale-collapse row)**.** *Let an exterior concentration branch develop scale collapse in its own exterior variables. Recenter at the exterior active carrier and use its scale as the local active scale. Then the rebased branch is a represented local scale-collapse branch in the sense of theorem 10.9, unless the ordered rebased tests fail first through one of the non-exterior rows: representation/pressure failure, autonomous-modulation failure or modulation-driven recentering, loss of retained active core or pressure-only critical mass, rough-core loss, carrier/cost validation failure, or multicore/cascade. In the represented case, the finite-cost, retained compact scale-collapse, and scale-rigid subcases are closed by*

theorems 6.34*and* 10.8*and proposition* 4.40.

Proof. The exterior carrier supplies a positive local CKN mass floor on a compact cylinder in the rebased variables. Run the covered-entry and repaired-gauge selection tests at that carrier. If a row fails before a represented scale-collapse branch is obtained, the first-failure ledger records exactly the listed non-exterior local row; this is the same ordered local test used in corollary 10.19 and lemma 6.25.

If the represented scale-collapse branch is obtained, apply theorem 10.9. Finite scale cost and finite absolute scale cost are excluded by theorem 10.8. Thin negative drift is already closed as an independent terminal row by theorem 9.99. Selected compactness-package failure is routed by theorem 6.31. The retained compact scale-collapse state is excluded by theorem 6.34, and the scale-rigid residual state is excluded by proposition 4.40. Autonomous modulation and the other listed first failures remain ordinary non-exterior rows in the final ledger; the exterior label itself has disappeared after rebasing. □

Lemma 8.59 (Fixed annular stress is regular or has a local carrier)**.** *Let $\mathcal{A} \Subset \mathbb{R}^3 \setminus B_2$ be a fixed annulus in a local frame and let $I \Subset (-\infty, 0]$ be a compact terminal time interval. Suppose a subsequence has a positive annular stress level*

$$\limsup_{n\to\infty} \|U_n \otimes U_n\|_{L^q_\tau(I;L^{3/2}_y(\mathcal{A}))} > 0$$

for some $1 < q < \infty$. Then the ordered local CKN test on $\mathcal{A} \times I$ gives one of the following alternatives:

(i) *a velocity-pressure CKN carrier occurs on a subcylinder contained in a slightly larger annulus, hence lemma 8.48 routes it to a velocity carrier or to the pressure-only exterior row;*

(ii) *all sufficiently small admissible subcylinders in the annulus are below $\varepsilon_{\mathrm{CKN}}$, hence the annular stress is a regular compact-window annular source. It is absorbed into the retained smooth annular part of the local pressure atlas, unless the harmonic-tail test sends its pressure effect to the ordered pressure row.*

Thus a positive fixed-annulus stress level is not by itself promoted to a singular carrier; it is tested locally first.

Proof. Apply the terminal CKN test to all admissible subcylinders whose doubled cylinders remain in a fixed slightly larger annulus containing $\mathcal{A}$. If one of these cylinders has CKN quantity at least $\varepsilon_{\mathrm{ext}}$, then the velocity-pressure split gives item (i). If no such cylinder exists, every admissible subcylinder is below $\varepsilon_{\mathrm{ext}} < \varepsilon_{\mathrm{CKN}}$. The terminal CKN regularity theorem and a finite-overlap cover of compact subannuli give local smoothness on $\mathcal{A} \times I$, so the stress lower bound is a regular annular source rather than a singular exterior carrier. Since $\mathcal{A}$ is at bounded normalized distance from the core in the same local frame, this regular source is part of the compact-window pressure atlas; if it is not absorbed there, the first failed atlas condition is exactly the ordered pressure row. No cardinality bound or global L^3 tail estimate is used. □

Lemma 8.60 (Far-field harmonic tails route to dyadic carriers)**.** *Let $B_1 \Subset B_2$ be nested balls in a local frame and let*

$$H_n(\tau) = H_n^{\mathrm{far}}(\tau)$$

be the mean-zero harmonic pressure on B_2 generated, in the local pressure atlas, by the quadratic stress $F_n = U_n \otimes U_n$ outside B_4. Let $\{\theta_k\}_{k\geq 2}$ be a smooth dyadic partition of unity on $\mathbb{R}^3 \setminus B_4$, with

$$\operatorname{supp} \theta_k \subset B_{2^{k+1}} \setminus B_{2^{k-1}}.$$

Assume the pressure reconstruction and gauge rows have not failed. If, for some $1 < q < \infty$,

$$\limsup_{n\to\infty} \|H_n\|_{L^q_\tau L^{3/2}_y(B_1)} > 0,$$

then, after passing to a subsequence, either

(i) *a dyadic exterior velocity-stress occupation state is present:*

$$\limsup_{n\to\infty} \sum_{k\geq K_n} 2^{-2k} \|U_n \otimes U_n\|_{L^q_\tau L^{3/2}_y(\mathcal{A}_k)} > 0$$

for some $K_n \to \infty$; or

(ii) *a fixed dyadic annulus carries a positive annular stress lower bound, which is then tested by lemma 8.59 and is either a regular annular source or a local CKN carrier.*

In item (i)*, the dyadic occupation state is then tested by lemma 8.54: it either produces a dyadic CKN carrier or is regular on the far dyadic annuli. If the dyadic tail is regular but its harmonic pressure contribution remains nonzero on the interior cylinder, the branch is assigned to the ordered pressure row rather than to exterior concentration. Consequently a nonzero far-field harmonic pressure tail is never silently discarded; it is a dyadic exterior carrier, a regular dyadic tail with vanishing harmonic effect, or the ordered pressure row.*

Proof. Write

$$H_n = \sum_{k\geq 2} H_{n,k} + h_n, \qquad H_{n,k} = \mathcal{R}_i\mathcal{R}_j(\theta_k F_{n,ij}) - \fint_{B_2} \mathcal{R}_i\mathcal{R}_j(\theta_k F_{n,ij})\, dy,$$

where h_n is the pressure-gauge/reconstruction remainder. By assumption the gauge and reconstruction rows have not failed, so h_n is absent modulo functions of time in the retained pressure atlas.

For $y \in B_2$ and $z \in \operatorname{supp}\theta_k$, the second derivative of the Newtonian kernel is $O(2^{-3k})$. Hence

$$\|H_{n,k}(\tau)\|_{L^{3/2}(B_1)} \leq C2^{-3k}\|F_n(\tau)\|_{L^1(\mathcal{A}_k)} \leq C2^{-2k}\|F_n(\tau)\|_{L^{3/2}(\mathcal{A}_k)},$$

because $|\mathcal{A}_k|^{1/3} \simeq 2^k$. Summing in k and taking the L^q_τ norm gives

$$\|H_n\|_{L^q_\tau L^{3/2}(B_1)} \leq C\sum_{k\geq 2} 2^{-2k}\|F_n\|_{L^q_\tau L^{3/2}(\mathcal{A}_k)}.$$

If the left-hand side has positive limsup, the weighted dyadic stress sum has positive limsup. Either a fixed finite set of dyadic annuli carries a positive portion of that limsup, or the positive portion escapes to $k \to \infty$. The first case gives item (ii) and is routed by lemma 8.59; a fixed annular stress lower bound is not identified with a singular carrier until the local CKN test has selected one. The escaping case is precisely the dyadic occupation state in item (i). Applying lemma 8.54 to that occupation state gives a dyadic CKN carrier or a regular dyadic tail. In the regular-tail case, either the corresponding harmonic contribution vanishes modulo functions of time on B_1, or the nonzero harmonic contribution is by definition the ordered pressure row. No whole-space pressure or velocity norm is used. □

Lemma 8.61 (Exterior pressure-only concentration is not terminal)**.** *Let an exterior concentration branch on $A_{r,R}$ satisfy the pressure-carrier alternative of lemma 8.48 and suppose that no velocity carrier occurs on any smaller exterior or separated-core cylinder after passing to subsequences. Then the pressure carrier is not an independent terminal exterior row. More precisely, after local pressure decomposition on nested cylinders, one of the following occurs:*

(i) *the local Calderón–Zygmund pressure part forces a velocity-stress carrier on a slightly larger exterior cylinder;*

(ii) *the far-field harmonic pressure tail produces a dyadic exterior occupation state, which is regular or gives a dyadic CKN carrier by lemma 8.54;*

(iii) *the pressure reconstruction or pressure gauge fails, which is the ordered non-exterior pressure row;*

(iv) *all pressure contributions vanish modulo functions of time on the smaller cylinder, contradicting the pressure-carrier lower bound.*

Proof. Work on a nested exterior cylinder and decompose the pressure into the local Calderón–Zygmund part generated by $\zeta u \otimes u$, the harmonic part, and a function of time. If the local Calderón–Zygmund part carries the pressure lower bound, Calderón–Zygmund boundedness and the quantitative converse in lemma 8.21 force a nonzero velocity-stress carrier on a slightly larger exterior cylinder. This is item (i) and routes back to the velocity-carrier cases.

Thus any persistent pressure lower bound must lie in the harmonic remainder. Apply lemma 8.22 on nested cylinders. A nonvanishing harmonic remainder is either pressure-reconstruction/gauge failure, which is item (iii), or it is generated by a far-field tail. For the far-field tail, use lemma 8.60. A fixed annular stress level is first tested by lemma 8.59; a selected local CKN carrier gives item (i), while a regular annular source is removable and cannot be an independent pressure-only exterior exit. An escaping dyadic stress occupation state is tested by lemma 8.54; it is either regular on far dyadic annuli or gives a dyadic CKN carrier, which is item (ii). If the dyadic tail is regular and no pressure row occurs, then the harmonic tail vanishes modulo functions of time on the smaller cylinder. Together with vanishing of the local Calderón–Zygmund part this contradicts the original pressure-carrier lower bound, giving item (iv). Hence pressure-only exterior concentration has been routed to a velocity/dyadic exterior carrier or to the ordered non-exterior pressure row; it is not an independent terminal exterior exit. □

Lemma 8.62 (Locality audit for exterior routing)**.** *The exterior routing lemmas from lemma 8.48 through lemma 8.61 use only local annular CKN tests, finite-overlap covering, local pressure decomposition modulo functions of time, and the ordered local state-space stratification. They do not use a whole-space $L^3(\mathbb{R}^3)$ bound, a global critical profile decomposition, global Kato smallness, or an assumption that localized dissipation vanishes.*

Proof. The CKN split is a local identity on the selected cylinder. Positive-distance decoupling uses kernel smoothness on sets separated by a fixed physical distance, local L^1 bounds for the recorded source, and the ordered state-space rule that any positive compact-core mass must be retained or routed before the exterior row is declared. The exterior and separated-core lemmas use the local scale-center compactification and the already stated separated-point and same-point reductions. The packing lemma uses only the Vitali finite-overlap property on a fixed annulus and explicitly avoids any cardinality bound from a global critical norm. Noncompact exterior routing is a countable dyadic application of the same local annular test. A positive fixed-annulus stress lower bound is not promoted directly to a singular carrier; it first passes through the local CKN dichotomy of lemma 8.59. Exterior scale collapse is handled by rebasing the active local CKN carrier and then applying the local scale-collapse first-failure ledger. Pressure-only routing uses the local Calderón–Zygmund pressure atlas, the nested-ball harmonic-tail estimate, and the harmonic pressure routing lemma. None of these steps imports a global L^3 estimate, a whole-space profile theorem, global Kato perturbation, or unproved vanishing of localized dissipation. □

Theorem 8.63 (Exterior concentration routes to closed or non-exterior rows)**.** *Fix $0 < r < R < \infty$. If a Type II branch admits an exterior concentration branch on $A_{r,R}$, then that branch is not an independent terminal exterior exit. After passing to subsequences and applying the ordered local tests, one of the following holds:*

(i) *the exterior contribution is regular or perturbative for the selected compact core, by theorem 8.46 or lemma 8.50;*

(ii) *the branch enters the separated-core, multicore, or same-point cascade rows, which are closed by theorem 10.6;*
(iii) *the branch enters an exterior scale-collapse row, which after rebasing is handled by lemma 8.58;*
(iv) *the branch enters the noncompact exterior row, which routes to the preceding alternatives by lemma 8.57;*
(v) *the branch enters a pressure-only exterior row, which is converted to an exterior velocity/dyadic carrier, absorbed as a regular annular or dyadic source, or routed to an ordered non-exterior pressure row by lemma 8.61.*

Consequently, once the non-exterior rows in definition 10.27 are handled, exterior concentration and noncompact exterior concentration contribute no additional obstruction.

Proof. Start with the exterior concentration branch and apply the stratification theorem 8.38. In the exterior-profile subcase, lemma 8.50 gives perturbative decoupling from the selected compact core. In the separated-core or multicore subcase, lemma 8.51 places the branch in the multibubble/cascade package, which is excluded by theorem 10.6. If exterior carriers accumulate at one positive-distance point with strict scale separation, then lemma 8.55 gives the same cascade closure.

If the stratification gives noncompact exterior behavior, first apply lemma 8.56. The no-concentration dyadic case is regular and removable by the exterior regularity/removal theorems. The dyadic carrier case is routed by lemma 8.57 to regular removal, separated/multicore, cascade, exterior scale-collapse, or pressure-only routing. The first three are already covered. The exterior scale-collapse subcase is rebased and tested by lemma 8.58: finite-cost and retained compact or scale-rigid scale-collapse subcases are closed by the cited scale-collapse theorems, while any first failure is one of the ordinary non-exterior rows in the final ledger.

It remains only to treat the possibility that the CKN lower bound is pressure-only. The velocity-pressure split lemma 8.48 either supplies a velocity carrier, returning to the preceding cases, or gives a pressure carrier. By lemma 8.61, a pressure-only carrier either produces a velocity/dyadic carrier, is absorbed as a regular annular or dyadic source, exposes the ordered pressure row, or contradicts the pressure lower bound after all local and harmonic tails have been removed. Thus every exterior concentration subtype has been routed to a closed structural row, regular removal, rebased scale-collapse, or one of the remaining non-exterior first-failure rows.

The proof is local by lemma 8.62. Hence exterior concentration is discharged as an independent named exit without adding a global critical-norm, global profile, global Kato, or hidden dissipation assumption. □

Corollary 8.64 (Exterior concentration exit discharged)**.** *Exterior concentration and noncompact exterior concentration are not remaining independent named exits in the local Type II assembly. They are either regular and removable, closed by the structural multibubble/cascade or scale-collapse packages after rebasing, or assigned to the remaining non-exterior rows of definition 10.27.*

Proof. This is theorem 8.63. □

9. Corrected Monotonicity and Scale-Drift Cost Criteria

This section records the local analytic inputs for the localized cost-divergence exclusion argument on renormalized Type II branches. The statements below are local criteria, and the later windowwise closure theorem verifies the criteria on the canonical retained route. The statements are written entirely in the variables of the repaired gauge:

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0.$$

All pressure terms are understood modulo functions of time.

9.1. **Data for the Cost Criterion.**

Definition 9.1 (Cost criterion data)**.** A renormalized local Type II branch satisfies the cost criterion data if the following local analytic facts hold.

(i) the branch lies in the local Type II concentration alternative;
(ii) the physical suitable-solution local energy inequality gives finite localized energy on every compact selected window used by the branch, with the renormalized local energy inequality available after the repaired-gauge pullback; no whole-space energy bound is part of this data;
(iii) the branch is written in the repaired-gauge variables (V, P, a, b) above, including the scale, center, pressure normalization, and modulation data;
(iv) the localized cost is identified with

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu\int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2\phi_{R_0}(y)\,dy + a_+(\tau)\int_{\mathbb{R}^3} |V(y,\tau)|^2\phi_{R_0}(y)\,dy,$$

or with an explicitly stated tail-comparable nonnegative replacement;
(v) a localized energy identity or suitable-solution local energy inequality is available for the cutoff used in the cost;
(vi) a corrected localized monotonicity inequality, in the sense of definition 9.6, is available.

Definition 9.2 (Cost-divergence exclusion conclusion)**.** A class of renormalized local Type II branches has the cost-divergence exclusion conclusion if every branch in the class satisfying definition 9.1 is excluded from the local Type II class whenever the corresponding nonnegative localized cost diverges:

$$\int_{\tau_0}^{\infty} \mathcal{D}_{R_0}(\tau)\,d\tau = \infty.$$

This is a conclusion verified below on the canonical retained route, not an active input.

Lemma 9.3 (Concentration and localized energy)**.** *If a branch lies in the local Type II concentration alternative and satisfies the physical suitable-solution local energy inequality on the compact cylinders selected by the branch, then it satisfies definition 9.1*(i) *and* (ii)*.*

Proof. The local Type II concentration alternative supplies the selected concentration branch. Suitability gives the local energy inequality on each compact physical cylinder, and lemma 9.58 transports it to the renormalized compact windows whenever the repaired-gauge chart is available. These are precisely items (i) and (ii); no global energy quantity is used. □

Lemma 9.4 (Cost identification)**.** *Assume the localized cost is the Navier–Stokes cost $\tilde{\mathfrak{D}}_{R_0}$. Then divergence of*

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau$$

is the cost divergence in definition 9.2.

Proof. The active cost is assumed to be equal to $\tilde{\mathfrak{D}}_{R_0}$. Thus the cost appearing in the local Navier–Stokes estimates and the cost appearing in the local exclusion conclusion are the same nonnegative functional. The divergence condition is therefore identical in the two formulations. $\square$

Lemma 9.5 (Parabolic representation class)**.** *If the repaired-gauge representation theorem applies to a Type II branch, then the branch has the parabolic transport-diffusion structure required in definition 9.1.*

Proof. The repaired-gauge theorem supplies the renormalized orbit (V, P, a, b), the scale and center, the pressure reconstruction modulo functions of time, and the modulation coefficients. The displayed equation is a parabolic Navier–Stokes equation with transport, dilation, and translation drift terms. Consequently the branch has the parabolic transport-diffusion structure required by the cost criterion data. $\square$

Definition 9.6 (Corrected localized monotonicity input)**.** A renormalized branch has corrected localized monotonicity if there are a localized renormalized energy $\mathcal{E}_{R_0}$, a nonnegative cost $\mathcal{D}_{R_0}$, a constant $c_0 > 0$, and a nonnegative remainder $\mathcal{R}_{R_0}$ such that, on the tail of the branch,

$$\frac{d}{d\tau}\mathcal{E}_{R_0}(\tau) + c_0\mathcal{D}_{R_0}(\tau) + \mathcal{R}_{R_0}(\tau) \le 0$$

in distributions. If the localized energy identity has an integrable error B_{R_0}, the corrected energy is

$$\mathcal{E}^{\text{corr}}_{R_0}(\tau) = \mathcal{E}_{R_0}(\tau) + \int_\tau^\infty B_{R_0}(s)\, ds,$$

and the same inequality is required for $\mathcal{E}^{\text{corr}}_{R_0}$. Multiplication of a nonnegative cost by $c_0 > 0$ does not change whether its tail integral diverges.

Lemma 9.7 (Energy and Caccioppoli bounds alone are not monotonicity)**.** *The global physical energy inequality and compact-cylinder Caccioppoli estimates do not, by themselves, imply the corrected localized monotonicity input of definition 9.6.*

Proof. The global physical energy inequality controls total physical energy and physical dissipation, but it does not identify a localized renormalized energy at the selected Type II scale whose derivative is monotone after pressure flux, cutoff flux, transport, and modulation terms are included. The Caccioppoli estimate gives local spacetime H^1 control from the local energy inequality. It is an a priori estimate, not a corrected monotonicity formula. Therefore an additional absorption or tail-correction argument is needed. $\square$

Proposition 9.8 (Cost-divergence exclusion from the recorded conclusion)**.** *Let a renormalized Type II branch satisfy definition 9.1 and let its ambient class have the cost-divergence exclusion conclusion of definition 9.2. If its nonnegative localized cost has divergent tail integral, then the branch is excluded from the local Type II class.*

Proof. Lemma 9.3 gives the local Type II concentration and bounded-energy inputs. Lemma 9.4 identifies the localized Navier–Stokes cost with the cost used in the exclusion criterion. Lemma 9.5 gives the parabolic transport-diffusion class. The corrected monotonicity input is one item of definition 9.1. Once the cost diverges, the conclusion recorded in definition 9.2 gives the stated exclusion. $\square$

9.2. **Fixed-Cutoff Monotonicity.** Let $\phi = \phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$, $0 \le \phi \le 1$, be the radial nonincreasing cutoff used in the localized cost. Define

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int_{\mathbb{R}^3} |V(y,\tau)|^2\phi(y)\, dy,$$

$$A_\phi(\tau) = \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi(y)\,dy, \qquad M_\phi(\tau) = \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi(y)\,dy,$$

and

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu A_\phi(\tau) + a_+(\tau) M_\phi(\tau), \qquad a_+(\tau) = \max\{a(\tau), 0\}.$$

The localized energy identity used below is

$$\begin{aligned} \frac{d}{d\tau}\mathcal{E}_\phi = & -\nu A_\phi + \frac{\nu}{2}\int |V|^2 \Delta\phi + \frac{1}{2}\int |V|^2 V\cdot\nabla\phi + \int P\,V\cdot\nabla\phi \\ & -\frac{a}{2}M_\phi - \frac{a}{2}\int |V|^2 y\cdot\nabla\phi - \frac{1}{2}b\cdot\int |V|^2\nabla\phi. \end{aligned} \tag{14}$$

Definition 9.9 (Fixed-cutoff error density)**.** Set $a_-(\tau) = \max\{-a(\tau), 0\}$ and define

$$\begin{aligned} B_{R_0}(\tau) := {} & \frac{\nu}{2}\left|\int |V|^2\Delta\phi\right| + \frac{1}{2}\left|\int |V|^2 V\cdot\nabla\phi\right| + \left|\int P\,V\cdot\nabla\phi\right| \\ & + \frac{1}{2}|b(\tau)|\left|\int |V|^2\nabla\phi\right| + \frac{1}{2}a_-(\tau)M_\phi(\tau) + \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\phi\big). \end{aligned}$$

The fixed-cutoff finite-error condition is

$$\int_{\tau_0}^\infty B_{R_0}(\tau)\,d\tau < \infty.$$

Remark 9.10 (Pressure normalization)*.* The pressure term in B_{R_0} is independent of the representative of the pressure modulo functions of time. Replacing P by $P + c(\tau)$ changes the pressure flux by

$$c(\tau)\int V\cdot\nabla\phi = -c(\tau)\int \phi\,\nabla\cdot V = 0.$$

Theorem 9.11 (Fixed-cutoff corrected monotonicity)**.** *Assume* (14) *holds on* (τ_0,∞) *and that* $B_{R_0} \in L^1(\tau_0,\infty)$*. Define*

$$\mathcal{E}_\phi^{\mathrm{corr}}(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\,ds.$$

Then, in distributions on (τ_0,∞)*,*

$$\frac{d}{d\tau}\mathcal{E}_\phi^{\mathrm{corr}}(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le 0.$$

Proof. Add

$$\frac{1}{2}\tilde{\mathfrak{D}}_{R_0} = \frac{\nu}{2}A_\phi + \frac{1}{2}a_+ M_\phi$$

to both sides of (14). The gradient term becomes

$$-\nu A_\phi + \frac{\nu}{2}A_\phi = -\frac{\nu}{2}A_\phi \le 0.$$

The core scale term satisfies

$$-\frac{a}{2}M_\phi + \frac{1}{2}a_+ M_\phi = \frac{1}{2}a_- M_\phi \le B_{R_0}(\tau).$$

Since ϕ is radial nonincreasing, one has

$$y\cdot\nabla\phi(y) \le 0 \qquad \text{for all } y \in \mathbb{R}^3.$$

Hence

$$-\frac{a}{2}\int |V|^2 y\cdot\nabla\phi \le \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\phi\big),$$

because the a_--part contributes a nonpositive term. Every remaining term in (14) is bounded above by its corresponding absolute-value contribution in B_{R_0}. Therefore

$$\frac{d}{d\tau}\mathcal{E}_\phi(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le B_{R_0}(\tau)$$

in distributions. Since $B_{R_0} \in L^1(\tau_0, \infty)$, the correction tail is finite for every $\tau \ge \tau_0$, absolutely continuous on compact subintervals, and

$$\frac{d}{d\tau}\int_\tau^\infty B_{R_0}(s)\,ds = -B_{R_0}(\tau).$$

Adding this identity gives the claimed inequality. □

Corollary 9.12 (Fixed-cutoff monotonicity input)**.** *If the repaired-gauge representation, the cost identification, the localized energy identity, and the fixed-cutoff finite-error condition all hold, then definition 9.6 holds with $\mathcal{D}_{R_0} = \tilde{\mathfrak{D}}_{R_0}$ and $c_0 = 1/2$.*

Proof. The repaired-gauge representation supplies (V, P, a, b). The cost identification supplies $\tilde{\mathfrak{D}}_{R_0}$. The localized energy identity and finite-error condition allow theorem 9.11 to be applied. □

Remark 9.13 (Finite-tail subconditions)*.* The fixed-cutoff finite-error condition is equivalent to the conjunction of tail integrability for the viscous cutoff term, the convective flux, the pressure flux, the center-drift cutoff term, the negative-scale term, and the positive scale-shell term:

$$\left|\int |V|^2\Delta\phi\right|, \qquad \left|\int |V|^2 V\cdot\nabla\phi\right|, \qquad \left|\int P\,V\cdot\nabla\phi\right|,$$
$$|b|\left|\int |V|^2\nabla\phi\right|, \qquad a_-M_\phi, \qquad a_+\int |V|^2(-y\cdot\nabla\phi).$$

For radial nonincreasing cutoffs, only the positive part a_+ contributes to the scale-shell error. The negative part is favorable and does not belong to the residual finite-error burden. Uniform compact-window bounds do not by themselves give this global tail integrability on $[\tau_0, \infty)$.

Definition 9.14 (Fixed-cutoff cost hypotheses)**.** A renormalized Type II branch satisfies the fixed-cutoff cost hypotheses if it has the local Type II concentration alternative, bounded physical energy, repaired-gauge representation, cost identification with $\tilde{\mathfrak{D}}_{R_0}$, divergent nonnegative localized cost, localized energy identity, and the finite-tail bound $\int_{\tau_0}^\infty B_{R_0} < \infty$.

Theorem 9.15 (Fixed-cutoff cost criterion)**.** *Every branch satisfying the fixed-cutoff cost hypotheses is covered by proposition 9.8.*

Proof. By corollary 9.12, the finite-tail condition supplies the corrected monotonicity input. The remaining inputs are precisely the other items in definition 9.1, together with cost divergence. The local exclusion step is later proved on the canonical windowwise schedule in theorem 9.42. □

Definition 9.16 (Compact cost-divergence replacement)**.** A compact cost-divergence replacement is available when the compact Type II hypotheses of the preceding sections imply

$$\int_{\tau_0}^\infty \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau = \infty,$$

and this divergence is the cost divergence used in proposition 9.8. In particular, it may be supplied by the combination of classification of local alternatives, uniform critical tightness, local windowed H^1 control, and the cost identification above whenever those are the active hypotheses in the compact Type II theorem.

Theorem 9.17 (Compact cost-divergence criterion)**.** *If a renormalized branch has all fixed-cutoff cost hypotheses except that cost divergence is supplied through the compact cost-divergence replacement of definition 9.16, then proposition 9.8 applies.*

Proof. The compact cost-divergence replacement supplies the missing divergent cost. Substituting this into the fixed-cutoff hypotheses gives the hypotheses of theorem 9.15. ☐

9.3. **Moving-Cutoff Monotonicity.** Let $\phi \in C_c^\infty(\mathbb{R}^3)$ be radial and nonincreasing, $0 \le \phi \le 1$, $\phi = 1$ on B_1, and $\phi = 0$ outside B_2. For $R(\tau) \ge R_0$, locally absolutely continuous and nondecreasing, set

$$\Phi(\tau, y) = \phi(y/R(\tau)), \qquad A_{R(\tau)} = \{R(\tau) \le |y| \le 2R(\tau)\},$$

$$\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2 \Phi(\tau, y)\, dy,$$

and

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu \int |\nabla V|^2 \Phi + a_+(\tau) \int |V|^2 \Phi.$$

Definition 9.18 (Moving-cutoff error density)**.** Define

$$\begin{aligned} B_{\mathrm{mov}}(\tau) := \frac{1}{2}\left|\int |V|^2 \partial_\tau \Phi\right| + \frac{\nu}{2}\left|\int |V|^2 \Delta\Phi\right| + \frac{1}{2}\left|\int |V|^2 V \cdot \nabla\Phi\right| + \left|\int P\, V \cdot \nabla\Phi\right| \\ + \frac{1}{2}|b(\tau)| \left|\int |V|^2 \nabla\Phi\right| + \frac{1}{2}a_-(\tau)\int |V|^2\Phi + \frac{1}{2}a_+(\tau)\int |V|^2 \big(-y\cdot\nabla\Phi\big). \end{aligned}$$

The moving finite-error condition is

$$\int_{\tau_0}^\infty B_{\mathrm{mov}}(\tau)\, d\tau < \infty.$$

Theorem 9.19 (Moving-cutoff corrected monotonicity)**.** *Assume the localized energy identity is valid with the time-dependent cutoff Φ, and assume $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$. Then*

$$\mathcal{E}_R^{\mathrm{corr}}(\tau) = \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\mathrm{mov}}(s)\, ds$$

satisfies

$$\frac{d}{d\tau}\mathcal{E}_R^{\mathrm{corr}}(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le 0$$

in distributions.

Proof. Use the localized energy identity with the time-dependent test function $\Phi(\tau, y)$. Compared with the fixed-cutoff identity, the only new term is

$$\frac{1}{2}\int |V|^2 \partial_\tau \Phi,$$

which comes from differentiating the cutoff inside $\mathcal{E}_R$. All other terms are the fixed-cutoff terms with ϕ replaced by Φ. Add $\frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}$ to both sides. The gradient term leaves

$$-\frac{\nu}{2}\int |\nabla V|^2 \Phi \le 0,$$

and the core scale term contributes

$$\frac{1}{2}a_- \int |V|^2 \Phi.$$

Since ϕ is radial nonincreasing, one has

$$y \cdot \nabla\Phi(\tau, y) \le 0 \qquad \text{for all } (\tau, y).$$

Therefore

$$-\frac{a}{2}\int |V|^2 y\cdot\nabla\Phi \le \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\Phi\big),$$

because the a_--part is nonpositive. Each remaining term is bounded above by the corresponding absolute-value term in B_{mov}. Hence

$$\frac{d}{d\tau}\mathcal{E}_R(\tau)+\frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau)\le B_{\mathrm{mov}}(\tau).$$

The moving finite-error condition makes the correction tail finite and absolutely continuous on compact subintervals. Differentiating that tail subtracts B_{mov}, giving the stated inequality. ☐

Definition 9.20 (Summable moving-annulus errors)**.** The moving-annulus summability condition is the existence of a nondecreasing locally absolutely continuous $R(\tau)\to\infty$ such that each of the seven quantities

$$\left|\int |V|^2\partial_\tau\Phi\right|,\quad \left|\int |V|^2\Delta\Phi\right|,\quad \left|\int |V|^2 V\cdot\nabla\Phi\right|,\quad \left|\int P\,V\cdot\nabla\Phi\right|,$$
$$|b|\left|\int |V|^2\nabla\Phi\right|,\quad a_-\int |V|^2\Phi,\quad a_+\int |V|^2\big(-y\cdot\nabla\Phi\big)$$

is integrable on $[\tau_0,\infty)$.

Lemma 9.21 (Summable annuli imply moving finite error)**.** *If definition 9.20 holds, then* $B_{\mathrm{mov}}\in L^1(\tau_0,\infty)$*.*

Proof. The density B_{mov} is a finite positive linear combination of the seven quantities listed in definition 9.20. If each is integrable, then their linear combination is integrable. ☐

Definition 9.22 (Summable tightness schedule)**.** A summable tightness schedule is a nondecreasing sequence $R_n\to\infty$ such that, on unit windows $I_n=[\tau_0+n,\tau_0+n+1]$,

$$\sum_{n=0}^{\infty}\sup_{\tau\in I_n}\|V(\tau)\|^3_{L^3(A_{R_n})}<\infty,$$

and the same schedule is compatible with the pressure-tail normalization used for P. This is stronger than uniform critical tightness, which permits large radii on each window but does not by itself give a monotone radius schedule with a summable series.

Definition 9.23 (Moving-cost compatibility)**.** The moving localized cost is compatible with the cost-divergence criterion if

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau)=\nu\int |\nabla V|^2\phi_{R(\tau)}+a_+(\tau)\int |V|^2\phi_{R(\tau)}$$

is the localized cost appearing in definition 9.2, or is compared to that cost by a positive constant on the tail so that cost divergence is unchanged.

Theorem 9.24 (Moving-cutoff replacement)**.** *Assume the repaired-gauge representation, localized energy identity, moving-cost compatibility, and summable moving-annulus errors. Then the moving cutoff supplies the corrected monotonicity input needed in proposition 9.8.*

Proof. Lemma 9.21 gives $B_{\mathrm{mov}}\in L^1(\tau_0,\infty)$. Applying theorem 9.19 gives corrected monotonicity with cost $\tilde{\mathfrak{D}}_{R(\tau)}$. The moving-cost compatibility input identifies this cost, or a tail-comparable version of it, with the localized cost in the recorded cost-divergence conclusion. ☐

9.4. **Negative Scale Drift.** Set

$$M_R(\tau) = \int |V(y,\tau)|^2 \Phi(\tau, y)\, dy.$$

The negative-scale contribution in the moving cutoff argument is

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau.$$

Definition 9.25 (Negative scale-drift alternatives)**.** We say that the branch has finite negative drift if

$$\int_{\tau_0}^{\infty} a_-(\tau)\, d\tau < \infty.$$

It has bounded moving L^2 core mass if

$$\sup_{\tau \geq \tau_0} M_R(\tau) < \infty.$$

It has a moving L^2 core floor if there are $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \qquad \text{for a.e. } \tau \geq \tau_1.$$

The scale-collapse drift obstruction is the divergent alternative

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Lemma 9.26 (Finite negative drift controls the scale term)**.** *If the branch has bounded moving L^2 core mass and finite negative drift, then*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau < \infty.$$

Proof. Let $M_* = \sup_{\tau \geq \tau_0} M_R(\tau) < \infty$. Since $a_- \geq 0$,

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau \leq M_* \int_{\tau_0}^{\infty} a_-(\tau)\, d\tau < \infty.$$

□

Theorem 9.27 (Finite-or-infinite scale-negative alternative)**.** *For every renormalized orbit for which a_- and M_R are measurable and nonnegative, precisely one of the following alternatives holds:*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau < \infty, \qquad \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. The integral

$$I_R = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau$$

is an extended nonnegative number. Hence either $I_R < \infty$ or $I_R = \infty$, and the two cases are mutually exclusive. □

Lemma 9.28 (Logarithmic scale identity)**.** *For the repaired-gauge convention*

$$\tau_t = \lambda(t)^{-2}, \qquad a(\tau) = \lambda(t)\lambda_t(t),$$

one has

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1)$$

whenever $\tau_0 \leq \tau_1 < \tau_2 < \infty$.

Proof. The chain rule gives

$$\frac{d}{d\tau}\log\lambda = \frac{\lambda_t}{\lambda}\frac{dt}{d\tau} = \frac{\lambda_t}{\lambda}\lambda^2 = \lambda\lambda_t = a.$$

Integrating over $[\tau_1, \tau_2]$ proves the identity. □

Theorem 9.29 (Collapse with core floor forces scale-drift obstruction)**.** *Assume*

$$\lambda(\tau) \to 0 \qquad \text{as } \tau \to \infty,$$

and assume the moving L^2 core floor of definition 9.25. Then

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau = \infty.$$

Proof. By lemma 9.28,

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \log\lambda(\tau_2) - \log\lambda(\tau_1).$$

Since $\lambda(\tau) \to 0$, the right-hand side tends to $-\infty$ as $\tau_2 \to \infty$. If $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau < \infty$, then

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\,d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\,d\tau \geq -\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau > -\infty,$$

which contradicts the preceding limit. Thus $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$.

By the moving L^2 core floor, $M_R(\tau) \geq m_0$ for a.e. $\tau \geq \tau_1$. Therefore

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau \geq m_0\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty.$$

□

Corollary 9.30 (Scale-drift alternatives)**.** *In the moving-cutoff setup, the negative-scale term has the following exhaustive alternatives. If bounded moving L^2 core mass and finite negative drift hold, then the negative-scale term is integrable. If the weighted negative-drift integral is infinite, the branch lies in the scale-collapse drift obstruction. If genuine collapse and a moving L^2 core floor also hold, then the obstruction is forced.*

Proof. The first case is lemma 9.26. The exhaustive split is theorem 9.27. The final assertion is theorem 9.29. □

9.5. **Scale-Drift Cost Criteria.**

Definition 9.31 (Scale-collapse drift cost)**.** For a renormalized branch define

$$\mathfrak{C}_{\mathrm{sc}}(\tau) = a_-(\tau)M_R(\tau), \qquad M_R(\tau) = \int |V(y,\tau)|^2\phi_{R(\tau)}(y)\,dy,$$

and

$$\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{sc}}(\tau)\,d\tau.$$

Thus the scale-collapse drift obstruction is precisely $\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \infty$. The density is nonnegative and local in the renormalized core.

Definition 9.32 (Scale-collapse cost compatibility)**.** The scale-collapse drift cost is compatible with the cost-divergence criterion when:

(i) $\mathfrak{C}_{\mathrm{sc}}$ is measurable, nonnegative, and locally integrable on finite τ-intervals for renormalized branches;

(ii) divergence of $\mathcal{C}_{\rm sc}[\tau_0,\infty)$ is one of the divergent-cost cases covered by the conclusion recorded in definition 9.2;
(iii) the resulting conclusion is the same local Type II exclusion conclusion used for the fixed localized cost.

Theorem 9.33 (Scale-collapse drift cost criterion). *If the scale-collapse drift obstruction holds and the cost $\mathfrak{C}_{\rm sc}$ is compatible with the cost-divergence criterion, then the divergent-cost hypothesis of proposition 9.8 is satisfied for the scale-collapse branch.*

Proof. By definition 9.31, the obstruction is precisely

$$\mathcal{C}_{\rm sc}[\tau_0,\infty)=\infty.$$

By definition 9.32, this divergence is a nonnegative cost-divergence hypothesis with the same local conclusion. ☐

Corollary 9.34 (Scale-collapse exclusion by drift cost). *Assume a renormalized local Type II branch has scale-collapse drift obstruction, compatible scale-collapse drift cost, and the remaining cost criterion data. If definition 9.2 holds, then the branch is excluded from the local Type II class under consideration.*

Proof. Theorem 9.33 supplies the divergent localized cost. Applying proposition 9.8 gives the exclusion. ☐

Definition 9.35 (Absolute scale-drift cost). Define

$$\mathfrak{C}_{\rm abs}(\tau)=\nu\int|\nabla V|^2\phi_{R(\tau)}+|a(\tau)|M_R(\tau).$$

The absolute scale-drift cost is compatible with the cost-divergence criterion if divergence of $\int_{\tau_0}^{\infty}\mathfrak{C}_{\rm abs}(\tau)\,d\tau$ is included as a Type II cost-divergence condition with the same local conclusion.

Lemma 9.36 (Scale-collapse drift forces absolute-cost divergence). *If the scale-collapse drift obstruction holds, then*

$$\int_{\tau_0}^{\infty}\mathfrak{C}_{\rm abs}(\tau)\,d\tau=\infty.$$

Proof. Since

$$\mathfrak{C}_{\rm abs}(\tau)=\nu\int|\nabla V|^2\phi_{R(\tau)}+|a(\tau)|M_R(\tau)\geq a_-(\tau)M_R(\tau),$$

divergence of $\int a_-M_R$ forces divergence of $\int\mathfrak{C}_{\rm abs}$. ☐

Corollary 9.37 (Absolute scale-cost criterion). *If the scale-collapse drift obstruction holds and the absolute scale-drift cost is compatible with the cost-divergence criterion, then the divergent-cost hypothesis of proposition 9.8 is satisfied.*

Proof. Lemma 9.36 gives divergence of the absolute scale-drift cost, and compatibility makes this divergence a Type II cost-divergence condition. ☐

Theorem 9.38 (Moving cutoff with scale-collapse alternative). *Assume the moving-cutoff setup and a renormalized local Type II branch.*

(i) *If the negative-scale term is integrable and the remaining moving-annulus errors are summable, then the moving-cutoff corrected monotonicity estimate is available.*
(ii) *If the scale-collapse drift obstruction holds and either the scale-collapse drift cost or the absolute scale-drift cost is compatible with the cost-divergence criterion, then the divergent-cost hypothesis is satisfied directly.*

(iii) *If, in addition, the remaining local data of definition 9.1 and definition 9.2 hold, then the branch is excluded.*

Proof. The first branch is theorem 9.24. The second branch is theorem 9.33 or corollary 9.37, depending on which compatible cost is used. The third branch is proposition 9.8. □

9.6. **Cost-Divergence Exclusion on the Canonical Windowwise Schedule.** The exclusion conclusion recorded in definition 9.2 is, given the corrected monotonicity input that is already part of definition 9.1, an arithmetic consequence of the nonnegativity of the localized renormalized energy together with the windowwise alternative argument of corollaries 7.123 and 7.128. We make this implication explicit here so that no cost-divergence input remains in the assembly.

Lemma 9.39 (Nonnegativity of the corrected localized energy)**.** *Let $\phi = \phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$ be a radial nonnegative cutoff, and let $R(\tau)$ be either fixed at R_0 or the canonical windowwise interpolated radius of lemma 7.113. Then $\mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\phi(y)\,dy \geq 0$ and $\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\Phi(\tau,y)\,dy \geq 0$ for every τ, and the correction tails $\int_\tau^\infty B_{R_0}(s)\,ds$ and $\int_\tau^\infty B_{\mathrm{mov}}(s)\,ds$ of definitions 9.9 and 9.18 are nonnegative. Consequently the corrected energies $\mathcal{E}_\phi^{\mathrm{corr}}$ and $\mathcal{E}_R^{\mathrm{corr}}$ of theorems 9.11 and 9.19 are nonnegative.*

Proof. Both $\mathcal{E}_\phi$ and $\mathcal{E}_R$ are integrals of $|V|^2$ against a nonnegative weight, hence nonnegative. Inspecting the seven summands of B_{R_0} (definition 9.9) and the seven summands of B_{mov} (definition 9.18): five are absolute values of integrals and are therefore nonnegative; the remaining two are $\frac{1}{2}a_-(\tau)\int |V|^2\Phi$ and $\frac{1}{2}a_+(\tau)\int |V|^2(-y\cdot\nabla\Phi)$ (and likewise with Φ replaced by ϕ). In the first, $a_- \geq 0$ and the integrand $|V|^2\Phi \geq 0$, so the term is nonnegative. In the second, $a_+ \geq 0$, and $-y\cdot\nabla\Phi \geq 0$ because $\Phi(\tau,y) = \phi(y/R(\tau))$ with ϕ radial nonincreasing forces $\nabla\Phi(\tau,y)$ to point opposite to y; the term is nonnegative. Hence $B_{R_0}, B_{\mathrm{mov}} \geq 0$ pointwise, and their tail integrals are nonnegative. Adding a nonnegative tail to a nonnegative quantity preserves nonnegativity, so $\mathcal{E}_\phi^{\mathrm{corr}}$ and $\mathcal{E}_R^{\mathrm{corr}}$ are nonnegative. □

Lemma 9.40 (Monotonicity versus cost divergence)**.** *Let $\mathcal{E}^{\mathrm{corr}}\colon [\tau_0,\infty) \to [0,\infty)$ be locally integrable with finite initial value $\mathcal{E}^{\mathrm{corr}}(\tau_0) < \infty$, let $\mathcal{D}\colon [\tau_0,\infty) \to [0,\infty)$ be locally integrable, and let $c_0 > 0$. If*

$$\frac{d}{d\tau}\mathcal{E}^{\mathrm{corr}}(\tau) + c_0\mathcal{D}(\tau) \leq 0 \qquad \textit{in distributions on } (\tau_0,\infty),$$

then

$$\int_{\tau_0}^\infty \mathcal{D}(\tau)\,d\tau \leq \frac{1}{c_0}\,\mathcal{E}^{\mathrm{corr}}(\tau_0) < \infty.$$

In particular, the simultaneous validity of corrected monotonicity in this sense and of $\int_{\tau_0}^\infty \mathcal{D} = \infty$ is contradictory. The same conclusion holds if the inequality is strengthened to the form $(d/d\tau)\mathcal{E}^{\mathrm{corr}} + c_0\mathcal{D} + \mathcal{R} \leq 0$ with $\mathcal{R} \geq 0$, as in definition 9.6.

Proof. The distributional inequality $(\mathcal{E}^{\mathrm{corr}})' + c_0\mathcal{D} \leq 0$ on (τ_0,∞), with $\mathcal{D} \geq 0$ and $\mathcal{E}^{\mathrm{corr}} \geq 0$ locally integrable, implies that $\mathcal{E}^{\mathrm{corr}}$ admits a non-increasing right-continuous representative. Testing the inequality against a smooth approximation of $\mathbf{1}_{[\tau_0,\tau]}$ and passing to the limit using $\mathcal{D} \geq 0$ and $\mathcal{E}^{\mathrm{corr}} \geq 0$ (so the boundary terms have a definite sign) yields

$$c_0\int_{\tau_0}^\tau \mathcal{D}(\sigma)\,d\sigma \leq \mathcal{E}^{\mathrm{corr}}(\tau_0) - \mathcal{E}^{\mathrm{corr}}(\tau) \leq \mathcal{E}^{\mathrm{corr}}(\tau_0).$$

Letting $\tau \to \infty$ and using monotone convergence ($\mathcal{D} \geq 0$) yields the displayed bound. If $\int \mathcal{D} = \infty$ the left-hand side is infinite while the right-hand side is finite, a contradiction. □

Theorem 9.41 (Cost-divergence/corrected-monotonicity dichotomy on the canonical schedule)**.** *Let ω be a renormalized local Type II branch in the pre-carrier validated compact-cost regime of definition 7.84. Equip ω with the canonical windowwise interpolated cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ of lemma 7.113, and let $\widetilde{\mathfrak{D}}_{R(\tau)}$ be the moving localized cost of definition 9.23. Assume in addition the localized energy identity for the moving cutoff $\Phi(\tau, \cdot)$ of definition 9.1*(v) *(supplied, e.g., by lemma 7.86 on the regime). If*

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R(\tau)}(\tau)\, d\tau = \infty,$$

then the following sharpened dichotomy holds: at least one of the three components

$$J_1, \qquad J_6, \qquad J_7$$

fails to be in $L^1(\tau_1, \infty)$ for every $\tau_1 \geq \tau_0$. More precisely, exactly one of the following two mutually exclusive alternatives holds:

(D1) *at least one of J_1, J_6, J_7 fails to be in $L^1(\tau_1, \infty)$ for every $\tau_1 \geq \tau_0$;*
(D2) *every J_k, $k \in \{1, \ldots, 7\}$, belongs to $L^1(\tau_0, \infty)$.*

Moreover, case (D2) *is incompatible with cost divergence, so divergence of the cost forces case* (D1)*, i.e. failure of one of J_1, J_6, J_7.*

Proof. The moving error density of definition 9.18 satisfies the identity

$$B_{\mathrm{mov}}(\tau) = \tfrac{1}{2}J_1(\tau) + \tfrac{\nu}{2}J_2(\tau) + \tfrac{1}{2}J_3(\tau) + J_4(\tau) + \tfrac{1}{2}J_5(\tau) + \tfrac{1}{2}J_6(\tau) + \tfrac{1}{2}J_7(\tau),$$

with J_k as in definition 7.97, because the seven nonnegative terms in B_{mov} coincide with $J_1, \ldots, J_7$ up to the listed multiplicative constants, all of which are strictly positive (recall $\nu > 0$ is fixed throughout). In particular $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$ if and only if every $J_k \in L^1(\tau_0, \infty)$.

By lemma 7.113, on the canonical windowwise schedule one has automatically

$$J_2,\ J_3,\ J_4,\ J_5 \in L^1(\tau_0, \infty).$$

Therefore the only J_k that can fail to be in L^1 on a tail are J_1, J_6, J_7. We obtain the following dichotomy: either at least one of J_1, J_6, J_7 fails to be in $L^1(\tau_1, \infty)$ for every $\tau_1 \geq \tau_0$ (case (D1)), or each of J_1, J_6, J_7 is integrable on $[\tau_0, \infty)$ and hence (combined with the automatic integrability of J_2, J_3, J_4, J_5) every J_k is integrable on $[\tau_0, \infty)$ (case (D2)).

In case (D2), lemma 9.21 gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$, and the localized energy identity of definition 9.1(v) together with theorem 9.19 produces

$$\frac{d}{d\tau}\mathcal{E}_R^{\mathrm{corr}}(\tau) + \tfrac{1}{2}\,\widetilde{\mathfrak{D}}_{R(\tau)}(\tau) \leq 0 \qquad \text{in distributions on } (\tau_0, \infty),$$

with $c_0 = \frac{1}{2}$ in the sense of definition 9.6. We verify the finite initial value locally. By lemma 7.93, after a harmless terminal restriction the initial time is a compact-cylinder local-energy time for the finite ball supporting $\Phi(\tau_0, \cdot)$. Hence

$$\mathcal{E}_R(\tau_0) = \tfrac{1}{2}\int |V(\tau_0)|^2 \Phi(\tau_0, \cdot)\, dy < \infty.$$

Combined with the case-(D2) integrability $\|B_{\mathrm{mov}}\|_{L^1(\tau_0,\infty)} < \infty$,

$$\mathcal{E}_R^{\mathrm{corr}}(\tau_0) = \mathcal{E}_R(\tau_0) + \int_{\tau_0}^{\infty} B_{\mathrm{mov}}(s)\, ds < \infty.$$

By lemma 9.39, $\mathcal{E}_R^{\mathrm{corr}} \geq 0$, so lemma 9.40 yields

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R(\tau)}(\tau)\, d\tau \leq 2\, \mathcal{E}_R^{\mathrm{corr}}(\tau_0) < \infty,$$

contradicting the cost divergence. Hence case (D2) cannot occur, and case (D1) must hold; that is, at least one of J_1, J_6, J_7 fails to be in L^1 on every later tail. □

Theorem 9.42 (Cost-divergence exclusion on the canonical windowwise schedule)**.** *Fix the canonical windowwise moving family of lemma 7.113 on the pre-carrier validated compact-cost regime of definition 7.84, and assume the localized energy identity for the moving cutoff (lemma 7.86). Assume further that* (R1) *failure of $J_6 \in L^1(\tau_1, \infty)$ on every later tail $[\tau_1, \infty)$ places the branch into the negative scale-drift alternative* (vi) *of definition 7.107, in the sense of theorem 7.130 (this is the same negative-drift routing statement used in that theorem, realized through definition 9.25 and corollary 9.30); and* (R2) *the named adjacent strata of theorem 7.89 together with the exterior/translation/positive-scale/negative-drift alternatives* (iii)–(vi) *of definition 7.107 are closed by their assigned local arguments in the ambient Type II class. Then for every renormalized local Type II branch ω in that regime satisfying definition 9.1, divergence of the localized cost*

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R(\tau)}(\tau)\, d\tau = \infty$$

excludes ω from the local Type II class. Equivalently, the negative-drift alternative (R1) *together with the adjacent-stratum closures* (R2) *imply definition 9.2 as a theorem on the canonical windowwise schedule.*

Proof. Let ω be a candidate branch. By theorem 9.41, case (D2) is impossible, so case (D1) occurs: at least one of J_1, J_6, J_7 fails to be in $L^1(\tau_1, \infty)$ for every $\tau_1 \geq \tau_0$.

Apply the alternative statement assigned to that index:

- if J_1 fails, the windowwise transition/leakage alternative of corollary 7.123 (proved unconditionally on the regime via theorem 7.122) places ω into the exterior/leakage alternative (iii) of definition 7.107, entering the named critical-tightness failure stratum of theorem 7.89;
- if J_6 fails, hypothesis (R1) places ω into the negative scale-drift alternative (vi);
- if J_7 fails, the windowwise positive-scale alternative of corollary 7.128 (proved unconditionally via the argument theorem 7.125 and proposition 4.40) places ω into the positive-scale alternative (v).

By hypothesis (R2), each of these named alternatives is closed by an already proved argument in the ambient Type II class. Therefore ω is excluded. This proves the conclusion of definition 9.2 on the canonical windowwise schedule under premises (R1) and (R2); the next corollary verifies these premises on the physical local Type II branch. □

Corollary 9.43 (Cost-divergence exclusion holds under the same hypothesis set as adjacent closure)**.** *The hypotheses* (R1)–(R2) *of theorem 9.42 on the canonical windowwise schedule coincide, after combining with the unconditional windowwise alternatives corollaries 7.123 and 7.128, with the verified negative-drift routing statement for J_6 and the adjacent-stratum closure of corollary 7.131. In particular, on any ambient Type II class on which the conclusion of corollary 7.131 (compact-cost adjacent-stratum closure) is established, definition 9.2 is a theorem and may be removed from the hypothesis lists of definitions 9.119 and 9.121. For the physical local Type II class considered in the final assembly, this adjacent closure is established by corollary 7.136.*

Proof. The dependency order is fixed in lemma 9.45: windowwise routing and ambient adjacent closure are verified before the cost-divergence theorem is used. The windowwise transition/leakage alternative corollary 7.123 and windowwise positive-scale alternative corollary 7.128 are unconditional on the regime. Combined with the negative-drift alternative for J_6 verified on physical local Type II branches by lemma 7.134, and closed as an independent terminal row by theorem 9.99, the alternative component (R1) of theorem 9.42 is satisfied. The adjacent-stratum closure component

(R2) is exactly the closure statement of corollary 7.131; in the physical local Type II class it is supplied by corollary 7.136. Therefore theorem 9.42 produces exclusion without any further hypothesis, and definition 9.2 is a theorem in that ambient class. □

Remark 9.44 (Status of the cost-divergence conclusion). The proof has three independent ingredients:

(i) the trivial arithmetic consequence lemma 9.40 that corrected localized monotonicity together with finite initial corrected energy and $\mathcal{D} \geq 0$ forces $\int \mathcal{D} < \infty$;
(ii) the moving error-density identity $B_{\mathrm{mov}} = \sum_k c_k J_k$ of definitions 7.97 and 9.18 (positive linear combination), combined with the canonical schedule integrability $J_2, J_3, J_4, J_5 \in L^1(\tau_0, \infty)$ of lemma 7.113;
(iii) the windowwise alternative for J_1 and J_7 (corollaries 7.123 and 7.128, proved unconditionally on the regime) plus the negative-drift alternative for J_6 imported from theorem 7.130.

Thus definition 9.2 is not an independent analytic input. Its content is the conjunction of the trivial monotonicity-vs-divergence arithmetic and the windowwise alternative argument that is already used to close adjacent strata in corollary 7.131.

Lemma 9.45 (Acyclic order for cost-divergence closure)**.** *In the physical local Type II assembly, the cost-divergence closure is used only after the local carrier and adjacent-routing rows have been verified. The dependency order is:*

(i) *the suitable local energy inequality and repaired-gauge pullback supply the local carrier subidentities on compact windows;*
(ii) *the canonical windowwise moving family supplies the unconditional transition and positive-scale routing alternatives corollaries 7.123 and 7.128;*
(iii) *nonintegrability of the negative scale-drift component J_6 is routed by the local negative-drift branch lemma 7.134;*
(iv) *the preceding rows give ambient adjacent-stratum closure through corollary 7.136;*
(v) *only then is theorem 9.42 applied to remove a retained branch with divergent validated localized cost;*
(vi) *terminal retained-state elimination is applied after the adjacent and cost rows have been routed or closed.*

No statement in (i)–(iv) *invokes theorem 9.42, corollary 9.43, or the final retained Type II closure theorem.*

Proof. Item (i) is lemmas 7.86 and 9.58. Item (ii) is exactly the pair of windowwise routing corollaries cited there. Item (iii) is lemma 7.134, whose proof routes J_6 failure through the local scale-drift, scale-collapse, compact scale-collapse, and scale-rigid packages, not through cost-divergence closure. Item (iv) is corollary 7.136; its proof cites the windowwise routing and negative-drift branch but not the cost-divergence theorem. The proof of theorem 9.42 then uses those already verified rows as inputs, which is item (v). Item (vi) is an ordering convention for the later final assembly: terminal retained-state elimination is allowed to use the fact that the named adjacent and cost rows have already been routed or closed, but none of the preceding rows uses terminal retained-state elimination as an input. □

9.7. **Absolutely Continuous Repaired-Gauge Charts and Renormalized Equations.**

The following statements isolate the parts of the repaired-gauge representation that follow from an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path. They do not prove extraction of the chart from arbitrary data and do not prove solvability of the gauge constraints.

Definition 9.46 (Absolutely continuous concentration chart)**.** Let u, p solve the Navier–Stokes equations on $\mathbb{R}^3 \times (t_0, T^*)$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0.$$

An absolutely continuous concentration chart consists of absolutely continuous functions

$$x_c : (t_0, T^*) \to \mathbb{R}^3, \qquad \lambda : (t_0, T^*) \to (0, \infty),$$

and a strictly increasing absolutely continuous time ρ such that

$$\rho_t = \lambda^{-2} \quad \text{a.e.},$$

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x,t) = \lambda(t)^{-1} U(y, \rho(t)),$$

and

$$p(x,t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t)$$

for some scalar function $c(t)$. The chart is assumed to have the local integrability required for the distributional chain rule on compact sets in the y-variables. Define

$$A(\rho(t)) := \lambda(t)\lambda_t(t), \qquad B(\rho(t)) := \lambda(t) x_c'(t),$$

or equivalently

$$A = \partial_\rho \log \lambda, \qquad B = \lambda^{-1} \partial_\rho x_c,$$

where λ and x_c are regarded as functions of ρ.

Lemma 9.47 (Initial renormalized equation)**.** *Assume Definition 9.46. Then U, Q satisfy*

$$\partial_\rho U + (U \cdot \nabla)U + \nabla Q = \nu \Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \qquad \nabla \cdot U = 0$$

distributionally on compact y-sets.

Proof. For smooth functions the computation is pointwise. The distributional case is obtained by testing against compactly supported functions and using the assumed chart chain rule. The spatial derivatives are

$$\nabla_x u = \lambda^{-2} \nabla_y U, \qquad \Delta_x u = \lambda^{-3} \Delta_y U, \qquad (u \cdot \nabla_x) u = \lambda^{-3} (U \cdot \nabla_y) U.$$

Since

$$y_t = -\frac{x_c'}{\lambda} - \frac{\lambda_t}{\lambda} y, \qquad \rho_t = \lambda^{-2},$$

one has

$$\partial_t u = \lambda^{-3} \partial_\rho U - \lambda^{-2} \lambda_t (U + y \cdot \nabla U) - \lambda^{-2} x_c' \cdot \nabla U.$$

Also

$$\nabla_x p = \lambda^{-3} \nabla_y Q.$$

Substitution into the physical Navier–Stokes equation and multiplication by λ^3 gives

$$\partial_\rho U - \lambda \lambda_t (U + y \cdot \nabla U) - \lambda x_c' \cdot \nabla U + (U \cdot \nabla)U + \nabla Q = \nu \Delta U.$$

Moving the two chart-velocity terms to the right gives the displayed equation. Finally,

$$\nabla_x \cdot u = \lambda^{-2} \nabla_y \cdot U,$$

so $\nabla_y \cdot U = 0$. $\square$

Definition 9.48 (Absolutely continuous repaired-gauge path)**.** An absolutely continuous repaired-gauge path for the initial chart consists, on every compact final time interval, of absolutely continuous functions

$$\mu(\tau) > 0, \qquad q(\tau) \in \mathbb{R}^3, \qquad \rho = \rho(\tau),$$

such that

$$\rho_\tau = \mu(\tau)^2 \quad \text{a.e.},$$
$$V(Y,\tau) = \mu(\tau)U(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

and

$$P(Y,\tau) = \mu(\tau)^2 Q(\mu(\tau)Y + q(\tau), \rho(\tau)).$$

The scale and centering constraints defining the repaired gauge are assumed to hold along this path. This hypothesis assumes the path has already been selected; it does not prove solvability of the gauge constraints.

Lemma 9.49 (Physical chart after gauge repair)**.** *Assume Definitions 9.46 and 9.48. Let* $t = t(\rho(\tau))$*, and define*

$$\Lambda(\tau) := \lambda(t(\rho(\tau)))\mu(\tau), \qquad X(\tau) := x_c(t(\rho(\tau))) + \lambda(t(\rho(\tau)))q(\tau).$$

Then, on every compact final time interval, Λ *and* X *are absolutely continuous,* $\Lambda > 0$*, and*

$$\frac{d\tau}{dt} = \Lambda(\tau)^{-2}.$$

Moreover,

$$x = X(\tau) + \Lambda(\tau)Y$$

is the final physical chart and

$$u(x,t) = \Lambda(\tau)^{-1}V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2}P(Y,\tau) + c(t).$$

Proof. The intermediate time ρ is strictly increasing because $\rho_t = \lambda^{-2} > 0$ a.e.; hence it has an absolutely continuous inverse on compact subintervals of its range. The compositions $\lambda(t(\rho(\tau)))$ and $x_c(t(\rho(\tau)))$ are therefore absolutely continuous on compact final windows. Products and sums with the absolutely continuous functions μ, q are absolutely continuous, so Λ and X are absolutely continuous.

The identity $\rho_\tau = \mu^2$ implies

$$\frac{d\tau}{d\rho} = \mu^{-2}.$$

Since $d\rho/dt = \lambda^{-2}$,

$$\frac{d\tau}{dt} = \frac{d\tau}{d\rho}\frac{d\rho}{dt} = \mu^{-2}\lambda^{-2} = (\lambda\mu)^{-2} = \Lambda^{-2}.$$

If $x = X + \Lambda Y$, then

$$x = x_c + \lambda q + \lambda\mu Y,$$

and hence the intermediate coordinate is

$$y = \frac{x - x_c}{\lambda} = q + \mu Y.$$

Therefore

$$u(x,t) = \lambda^{-1}U(q + \mu Y, \rho) = \lambda^{-1}\mu^{-1}V(Y,\tau) = \Lambda^{-1}V(Y,\tau),$$

and similarly

$$p(x,t) = \lambda^{-2}Q(q + \mu Y, \rho) + c(t) = \lambda^{-2}\mu^{-2}P(Y,\tau) + c(t) = \Lambda^{-2}P(Y,\tau) + c(t).$$

$\square$

Lemma 9.50 (Modulation coefficients in repaired gauge)**.** *Define*

$$a(\tau) := \partial_\tau \log \Lambda(\tau), \qquad b(\tau) := \Lambda(\tau)^{-1} X_\tau(\tau),$$

where derivatives are understood a.e. on compact final windows. Under the hypotheses of Lemma 9.49,

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau) B(\rho(\tau)) + \mu(\tau) A(\rho(\tau)) q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

In particular $a, b \in L^1_{\mathrm{loc}}$ on every compact final time interval.

Proof. Since $\Lambda = \lambda(\rho(\tau))\mu(\tau)$,

$$\partial_\tau \log \Lambda = \rho_\tau \partial_\rho \log \lambda + \partial_\tau \log \mu = \mu^2 A + \frac{\mu_\tau}{\mu}.$$

This proves the formula for a. Next

$$X = x_c(\rho(\tau)) + \lambda(\rho(\tau)) q(\tau).$$

Using

$$\partial_\rho x_c = \lambda B, \qquad \partial_\rho \lambda = \lambda A, \qquad \rho_\tau = \mu^2,$$

we get

$$X_\tau = \mu^2 \lambda B + \mu^2 \lambda A q + \lambda q_\tau.$$

Division by $\Lambda = \lambda\mu$ gives

$$\Lambda^{-1} X_\tau = \mu B + \mu A q + \frac{q_\tau}{\mu}.$$

Local integrability follows from absolute continuity and positivity of μ. On every compact final interval, μ is bounded above and below away from zero, q is bounded, and $\mu_\tau, q_\tau \in L^1$. The A-terms are locally integrable because

$$\int |A(\rho(\tau))| \mu(\tau)^2 \, d\tau = \int |A(\rho)| \, d\rho,$$

and the extra factors μ, μ^{-1}, and q are bounded on the same compact window. Likewise, since μ is bounded away from zero,

$$\int |B(\rho(\tau))| \mu(\tau) \, d\tau \le C \int |B(\rho(\tau))| \mu(\tau)^2 \, d\tau = C \int |B(\rho)| \, d\rho,$$

so the B-term in b is locally integrable. □

Lemma 9.51 (Renormalized equation in repaired gauge)**.** *The final variables V, P satisfy*

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0$$

distributionally on compact Y-sets.

Proof. It suffices to compute in the smooth case and then pass to distributions by testing against compactly supported functions. Set

$$z = \mu Y + q.$$

Since $V(Y, \tau) = \mu U(z, \rho(\tau))$ and $\rho_\tau = \mu^2$, the spatial derivatives are

$$\nabla_Y V = \mu^2 \nabla_z U, \qquad \Delta_Y V = \mu^3 \Delta_z U,$$

$$(V \cdot \nabla_Y) V = \mu^3 (U \cdot \nabla_z) U, \qquad \nabla_Y P = \mu^3 \nabla_z Q.$$

Differentiation in τ gives

$$V_\tau = \mu_\tau U + \mu\mu^2 U_\rho + \mu(\mu_\tau Y + q_\tau)\cdot\nabla_z U.$$

Using the initial renormalized equation,

$$U_\rho = \nu\Delta_z U - (U\cdot\nabla_z)U - \nabla_z Q + A(U + z\cdot\nabla_z U) + B\cdot\nabla_z U,$$

we obtain

$$\begin{aligned} &V_\tau + (V\cdot\nabla_Y)V + \nabla_Y P - \nu\Delta_Y V \\ &\quad = \mu_\tau U + \mu^3 A(U + z\cdot\nabla_z U) + \mu^3 B\cdot\nabla_z U + \mu(\mu_\tau Y + q_\tau)\cdot\nabla_z U. \end{aligned}$$

Now

$$U = \mu^{-1}V, \qquad \nabla_z U = \mu^{-2}\nabla_Y V, \qquad z = \mu Y + q.$$

Therefore

$$\mu_\tau U + \mu^3 AU = \left(\frac{\mu_\tau}{\mu} + \mu^2 A\right)V,$$

$$\mu^3 A\, z\cdot\nabla_z U + \mu(\mu_\tau Y + q_\tau)\cdot\nabla_z U = \left(\mu^2 A + \frac{\mu_\tau}{\mu}\right)Y\cdot\nabla_Y V + \left(\mu Aq + \frac{q_\tau}{\mu}\right)\cdot\nabla_Y V,$$

$$\mu^3 B\cdot\nabla_z U = \mu B\cdot\nabla_Y V.$$

Combining these identities, the coefficient multiplying

$$V + Y\cdot\nabla_Y V$$

is

$$\frac{\mu_\tau}{\mu} + \mu^2 A,$$

and the coefficient multiplying $\nabla_Y V$ is

$$\mu B + \mu Aq + \frac{q_\tau}{\mu}.$$

By Lemma 9.50 these coefficients are a and b. Hence the displayed repaired-gauge equation holds in the distributional sense on compact Y-sets. Incompressibility follows from

$$\nabla_Y\cdot V = \mu^2\nabla_z\cdot U = 0.$$

□

Lemma 9.52 (Pressure reconstruction under the charts)**.** *Assume the physical pressure satisfies*

$$-\Delta_x p = \partial_i\partial_j(u_i u_j)$$

in distributions. Then the intermediate pressure satisfies

$$-\Delta_y Q = \partial_i\partial_j(U_i U_j)$$

modulo functions of ρ, and the final pressure satisfies

$$-\Delta_Y P = \partial_i\partial_j(V_i V_j)$$

modulo functions of τ.

Proof. The additive function $c(t)$ in

$$p = \lambda^{-2}Q + c(t)$$

has zero spatial derivatives. Since

$$\Delta_x p = \lambda^{-4}\Delta_y Q, \qquad \partial_{x_i}\partial_{x_j}(u_i u_j) = \lambda^{-4}\partial_{y_i}\partial_{y_j}(U_i U_j),$$

the intermediate pressure equation follows. For the repaired variables,

$$P(Y) = \mu^2 Q(\mu Y + q), \qquad V(Y) = \mu U(\mu Y + q).$$

Therefore
$$-\Delta_Y P = \mu^4(-\Delta_z Q)(z) = \mu^4 \partial_{z_i}\partial_{z_j}(U_iU_j)(z) = \partial_{Y_i}\partial_{Y_j}(V_iV_j)(Y).$$
Adding functions of time to Q or P does not change the identity. □

Lemma 9.53 (Scaling of local L^r norms)**.** *For every $1 \le r < \infty$ for which the norms are finite,*
$$\|V(\tau)\|^r_{L^r_Y} = \Lambda(\tau)^{r-3}\|u(t(\tau))\|^r_{L^r_x}.$$
In particular,
$$\|V(\tau)\|_{L^3_Y} = \|u(t(\tau))\|_{L^3_x}.$$
Thus membership in $L^3(\mathbb{R}^3)$, finiteness, infiniteness, and positivity are invariant under the final chart.

Proof. Using
$$u(X + \Lambda Y, t) = \Lambda^{-1}V(Y, \tau), \qquad dx = \Lambda^3\, dY,$$
we obtain
$$\|u(t)\|^r_{L^r_x} = \int |\Lambda^{-1}V(Y,\tau)|^r\Lambda^3\, dY = \Lambda^{3-r}\|V(\tau)\|^r_{L^r_Y}.$$
Rearranging gives the stated formula. The case $r = 3$ is scale invariant. □

Lemma 9.54 (Regularity of scale, center, and modulation)**.** *If λ, x_c, μ, q are absolutely continuous and $\mu > 0$, then on compact final time intervals the final scale Λ, center X, and coefficients a, b are absolutely continuous or locally integrable as specified above: $\Lambda, X \in W^{1,1}_{\mathrm{loc}}$ and*
$$a, b \in L^1_{\mathrm{loc}}.$$
Proof. This follows from Lemmas 9.49 and 9.50. Positivity of μ and compactness of the time window give a positive lower bound for μ, so the divisions by μ in the formulas for a, b are legitimate. □

Theorem 9.55 (Consequences of absolutely continuous repaired-gauge charts)**.** *Assume an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path in the sense of Definitions 9.46 and 9.48. Then the following analytic conclusions hold:*

(i) *the initial and repaired renormalized Navier–Stokes equations hold distributionally on compact renormalized sets;*

(ii) *the final physical chart satisfies*
$$u(x,t) = \Lambda(\tau)^{-1}V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2}P(Y,\tau) + c(t), \qquad \frac{d\tau}{dt} = \Lambda^{-2};$$

(iii) *pressure reconstruction holds modulo functions of time;*

(iv) *the final modulation coefficients are the functions a, b in Lemma 9.50;*

(v) *the critical L^3 norm is invariant under the final chart;*

(vi) *the final scale, center, and modulation coefficients have the local absolute-continuity and integrability stated in Lemma 9.54.*

Proof. Lemma 9.47 gives the initial renormalized equation. Lemmas 9.49 and 9.51 give the final chart and the final repaired-gauge equation with the stated coefficients. Lemma 9.52 gives pressure reconstruction modulo functions of time. Lemma 9.53 gives critical norm invariance, and Lemma 9.54 gives the regularity inheritance. □

Corollary 9.56 (Representation inputs supplied by the repaired gauge)**.** *Once the absolutely continuous concentration chart and absolutely continuous repaired-gauge path are available, pressure reconstruction, modulation coefficient realization, local $W^{1,1}$ regularity of the chart, and critical L^3-invariance are analytic consequences rather than separate inputs. The representation inputs that must be supplied before applying this corollary are:*

(i) *extraction of an absolutely continuous concentration chart from the selected Type II branch;*
(ii) *solvability of the repaired scale and centering equations on the admissible profile class;*
(iii) *selection of the repaired scale and center as an absolutely continuous path satisfying $\rho_\tau = \mu^2$;*
(iv) *terminal admissibility of the final scale and time, including $\tau \to \infty$ and preservation of the Type II core scale.*

Proof. This is Theorem 9.55. The enumerated items are the complementary analytic inputs that must be available before that theorem can be applied: one must produce the initial concentration chart, solve the repaired gauge equations, select the resulting scale and center absolutely continuously, and verify that the resulting final chart is terminally admissible. □

9.8. Local Caccioppoli Estimate for Absolutely Continuous Suitable Branches.

Definition 9.57 (Suitable repaired-gauge Type II branch)**.** A suitable repaired-gauge Type II branch is a repaired-gauge Type II branch for which:

(i) an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path are available;
(ii) the conclusions of Theorem 9.55 hold, including pressure reconstruction and final modulation coefficients;
(iii) the physical branch (u, p) is a local suitable weak solution on the terminal interval:

$$u \in L_t^\infty L^2_{\mathrm{loc}} \cap L_t^2 H^1_{\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold distributionally, and the physical local energy inequality holds for every nonnegative smooth compactly supported test function.

Lemma 9.58 (Local energy inequality under absolutely continuous Type II charts)**.** *Let (u, p) be physically suitable on a compact physical cylinder, and let (V, P) be obtained from an absolutely continuous final chart*

$$u(x,t) = \Lambda(s)^{-1} V(Y,s), \qquad p(x,t) = \Lambda(s)^{-2} P(Y,s) + c(t), \qquad Y = \frac{x - X(s)}{\Lambda(s)},$$

with $s = \tau$, $dt = \Lambda(s)^2\, ds$, $\Lambda > 0$, and

$$X, \Lambda \in W^{1,1}_{\mathrm{loc}}.$$

Then (V, P) satisfies the renormalized local energy inequality on compact (Y, s)-cylinders:

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|V(s_2)|^2}{2} \phi(s_2)\, dY + \nu \int_{s_1}^{s_2} \int_{\mathbb{R}^3} |\nabla V|^2 \phi\, dY\, ds \\
&\le \int_{\mathbb{R}^3} \frac{|V(s_1)|^2}{2} \phi(s_1)\, dY + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \frac{|V|^2}{2} (\partial_s \phi + \nu \Delta \phi)\, dY\, ds \\
&\quad + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P \right) V \cdot \nabla \phi\, dY\, ds \\
&\quad + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} a(s) \frac{|V|^2}{2} (-\phi - Y \cdot \nabla \phi)\, dY\, ds \\
&\quad - \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \frac{|V|^2}{2} b(s) \cdot \nabla \phi\, dY\, ds,
\end{aligned}$$

for every nonnegative $\phi \in C^\infty([s_1, s_2] \times \mathbb{R}^3)$ that is compactly supported in the spatial variable. Here a, b are the final modulation coefficients.

Proof. First suppose X, Λ are C^1. Use the physical test function

$$\psi(x,t) = \Lambda(s)^{-1}\phi\left(\frac{x - X(s)}{\Lambda(s)}, s\right), \qquad t = t(s),$$

where $dt = \Lambda^2 ds$. The spatial support of ϕ is compact, and Λ is bounded above and below on the selected compact time interval, so ψ is an admissible compactly supported test function in the physical local energy inequality.

Let

$$Y = \frac{x - X(s)}{\Lambda(s)}.$$

At fixed x,

$$\frac{dY}{ds} = -\frac{X_s}{\Lambda} - \frac{\Lambda_s}{\Lambda}Y.$$

Consequently,

$$\nabla_x \psi = \Lambda^{-2}\nabla_Y \phi, \qquad \Delta_x \psi = \Lambda^{-3}\Delta_Y \phi,$$

and, since $ds/dt = \Lambda^{-2}$,

$$\partial_t \psi = \Lambda^{-3}\left[\partial_s \phi - \frac{\Lambda_s}{\Lambda}(\phi + Y \cdot \nabla\phi) - \frac{X_s}{\Lambda} \cdot \nabla\phi\right].$$

Changing variables

$$x = X(s) + \Lambda(s)Y, \qquad dt = \Lambda^2 ds,$$

and using

$$u = \Lambda^{-1}V, \qquad p = \Lambda^{-2}P + c(t),$$

each term in the physical local energy inequality has the following renormalized form. The endpoint and dissipation terms transform as

$$\int \frac{|u(t_i)|^2}{2}\psi(t_i)\, dx = \int \frac{|V(s_i)|^2}{2}\phi(s_i)\, dY,$$

and

$$\nu \int_{t_1}^{t_2} \int |\nabla_x u|^2 \psi \, dx\, dt = \nu \int_{s_1}^{s_2} \int |\nabla_Y V|^2 \phi \, dY\, ds.$$

The test-function derivative term becomes

$$\begin{aligned}
&\int_{t_1}^{t_2} \int \frac{|u|^2}{2}(\partial_t \psi + \nu \Delta_x \psi)\, dx\, dt \\
&\quad = \int_{s_1}^{s_2} \int \frac{|V|^2}{2}\left[\partial_s \phi + \nu \Delta_Y \phi - a(\phi + Y \cdot \nabla\phi) - b \cdot \nabla\phi\right] dY\, ds,
\end{aligned}$$

where

$$a = \frac{\Lambda_s}{\Lambda}, \qquad b = \frac{X_s}{\Lambda}.$$

The transport and pressure term becomes

$$\begin{aligned}
&\int_{t_1}^{t_2} \int \left(\frac{|u|^2}{2} + p\right) u \cdot \nabla_x \psi \, dx\, dt \\
&\quad = \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P\right) V \cdot \nabla_Y \phi \, dY\, ds + \int_{s_1}^{s_2} \Lambda(s)^2 c(t(s)) \left(\int_{\mathbb{R}^3} V \cdot \nabla_Y \phi \, dY\right) ds.
\end{aligned}$$

The pressure constant drops out because

$$\int_{\mathbb{R}^3} V \cdot \nabla_Y \phi \, dY = -\int_{\mathbb{R}^3} \phi \, \nabla_Y \cdot V \, dY = 0.$$

Combining the transformed terms gives the displayed renormalized local energy inequality.

For $X, \Lambda \in W^{1,1}_{\mathrm{loc}}$, the same identity follows at the level of test functions. On the selected compact time interval, Λ has positive lower and finite upper bounds. The pulled-back test

$$\psi(x,t) = \Lambda(s)^{-1}\phi((x - X(s))/\Lambda(s), s)$$

is compactly supported, nonnegative, bounded, has bounded spatial derivatives on the compact cylinder, and has time derivative in $L^1_t L^\infty_x$ because $X', \Lambda' \in L^1$ and ϕ is smooth compactly supported. The physical local energy inequality, initially stated for C^∞_c tests, extends to this class by standard mollification of the test function: the suitability classes give the local integrability needed for every term in the inequality, and the mollified tests keep support inside a slightly larger compact cylinder. Applying the extended inequality to ψ and changing variables gives the displayed renormalized inequality. The only derivatives of the chart that appear are the L^1-functions X', Λ', hence the modulation terms are locally integrable against local $|V|^2$. □

Lemma 9.59 (Pressure and modulation for suitable repaired-gauge branches)**.** *For a suitable repaired-gauge branch, the pressure identity*

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

holds modulo functions of τ, and the final coefficients a, b in the renormalized equation are measurable and locally integrable on compact renormalized windows.

Proof. This is the pressure and modulation part of Theorem 9.55. Pressure pullback comes from the physical pressure equation and the chart/gauge scaling. The final coefficients are forced by the absolutely continuous scale-translation gauge transform:

$$a = \mu^2 A + \frac{\mu_\tau}{\mu}, \qquad b = \mu B + \mu A q + \frac{q_\tau}{\mu}.$$

Since the chart and gauge parameters are absolutely continuous on compact windows, these coefficients are locally integrable. □

Theorem 9.60 (Local Caccioppoli estimate for repaired gauges)**.** *Every suitable repaired-gauge Type II branch satisfies the local renormalized Caccioppoli estimate on each compact repaired-gauge cylinder. The constants may depend on the cylinder, the cutoff, and local upper and lower bounds for the chart. No local Caffarelli–Kohn–Nirenberg smallness, global windowed H^1, critical L^3-norm, tightness, bounded modulation, finite energy at infinity, or whole-space compactness conclusion is asserted.*

Proof. By the suitable repaired-gauge hypotheses and Theorem 9.55, the branch has an absolutely continuous final chart, an admissible absolutely continuous repaired-gauge path, pressure reconstruction, and final modulation coefficients. Lemma 9.58 pulls the physical local energy inequality to the repaired variables. Lemma 9.59 supplies the pressure and coefficient identities needed to interpret the renormalized inequality in the same PDE class used by the local windowed estimates.

Choose a standard compactly supported Caccioppoli cutoff

$$\phi(Y,\tau) = \eta(\tau)\zeta(Y)^2.$$

Let $I = (\tau_1, \tau_2)$, let $I' \Subset I$, and let $0 < r < R$. Take $0 \le \eta \le 1$, $\eta \in C^\infty_c(I)$, with $\eta \equiv 1$ on I', and take $0 \le \zeta \le 1$, $\zeta \in C^\infty_c(B_R)$, with $\zeta \equiv 1$ on B_r. Applying Lemma 9.58 on (τ_1, σ) and then taking the essential supremum over $\sigma \in I'$ controls the endpoint term. Applying the same inequality on the full interval I controls the dissipation term. Adding the two estimates gives

$$\begin{aligned}
&\operatorname*{ess\,sup}_{\tau\in I'} \int_{B_r} \frac{|V(\tau)|^2}{2}\,dY + \nu \int_{I'}\int_{B_r} |\nabla V|^2\,dY\,d\tau \\
&\le C\int_I\int_{B_R} \frac{|V|^2}{2}\left(|\eta'|\zeta^2 + \nu\eta|\Delta(\zeta^2)|\right)\,dY\,d\tau \\
&\quad + C\int_I\int_{B_R}\left(\frac{|V|^2}{2}+|P|\right)|V|\,\eta|\nabla(\zeta^2)|\,dY\,d\tau \\
&\quad + C\int_I |a(\tau)|\int_{B_R}\frac{|V|^2}{2}\eta\left(\zeta^2+|Y|\,|\nabla(\zeta^2)|\right)\,dY\,d\tau \\
&\quad + C\int_I |b(\tau)|\int_{B_R}\frac{|V|^2}{2}\eta|\nabla(\zeta^2)|\,dY\,d\tau.
\end{aligned}$$

Here C is an absolute constant. Since the cutoff derivatives are bounded by constants depending only on I, I', r, R, this is the local Caccioppoli estimate on the compact renormalized cylinder.

No global estimate is used in this step. On the compact cylinder under consideration, the absolutely continuous chart has

$$0 < \Lambda_- \le \Lambda(\tau) \le \Lambda_+ < \infty.$$

Pulling back physical suitability therefore gives, for every compact spatial set K,

$$V \in L^\infty_\tau L^2_Y(K) \cap L^2_\tau H^1_Y(K), \qquad P \in L^{3/2}_{\tau,Y}(K).$$

Local Sobolev interpolation on the finite cylinder gives

$$V \in L^{10/3}_{\tau,Y}(K),$$

hence $V \in L^3_{\tau,Y}(K)$. Thus $PV\cdot\nabla\phi$ is integrable by Hölder's inequality. The representation identities give

$$a, b \in L^1_{\mathrm{loc}}(d\tau),$$

and

$$\tau \mapsto \int_K |V(Y,\tau)|^2\,dY$$

is locally essentially bounded; therefore the scale and translation modulation terms in the pulled-back local energy inequality are integrable on the same compact cylinder. This proves the local Caccioppoli estimate, but not a global windowed H^1 or critical L^3-norm bound. □

Corollary 9.61 (No separate Caccioppoli obstruction for suitable absolutely continuous branches)**.** *On suitable repaired-gauge branches, failure of local Caccioppoli regularity, failure of pressure reconstruction, and failure of local gauge regularity are not separate obstructions to the local H^1 analysis. The additional inputs used when invoking the uniform windowed H^1 estimates are the bounded critical L^3 and modulation bounds on the retained window; if either input fails, the ordered local state-space test records the corresponding active-core, rough-core, modulation, or scale-drift row before the uniform estimate is invoked.*

Proof. Theorem 9.60 gives local Caccioppoli regularity. Theorem 9.55 gives pressure reconstruction and the final modulation coefficients from the final chart and absolutely continuous gauge path. Thus the listed failures cannot occur separately on suitable repaired-gauge branches. The bounded critical norm and bounded modulation used by the uniform windowed H^1 estimate are therefore retained-window inputs or routed first failures, not independent Caccioppoli obstructions. □

9.9. **Caccioppoli Regularity Criterion.**

Definition 9.62 (Suitable physical hypotheses)**.** Let u, p be a local suitable weak solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (0, T^*)$. Thus

$$u \in L_t^\infty L_{\text{loc}}^2 \cap L_t^2 H_{\text{loc}}^1, \qquad p \in L_{\text{loc}}^{3/2}, \qquad \nabla \cdot u = 0,$$

the equations hold distributionally, and for every nonnegative $\psi \in C_c^\infty(\mathbb{R}^3 \times (0, T^*))$ and a.e. $0 < t_1 < t_2 < T^*$,

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|u(t_2)|^2}{2} \psi(t_2)\, dx + \nu \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \psi\, dx\, dt \\
&\le \int_{\mathbb{R}^3} \frac{|u(t_1)|^2}{2} \psi(t_1)\, dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \frac{|u|^2}{2} (\partial_t \psi + \nu \Delta \psi)\, dx\, dt \\
&\quad + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \psi\, dx\, dt.
\end{aligned}$$

Lemma 9.63 (Renormalized local energy inequality)**.** *Assume the suitable physical hypotheses of definition 9.62. Let*

$$u(x,t) = \lambda(t)^{-1} V \left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t) \right), \qquad \frac{d\tau}{dt} = \lambda(t)^{-2},$$

with

$$P(y, \tau) = \lambda(t)^2 p(x_c(t) + \lambda(t) y, t),$$

where $\lambda > 0$, x_c, and τ are C^1 on compact windows. Assume that V, P satisfy the repaired-gauge equation

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Then, for every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times (\tau_0, \infty))$ and a.e. $\tau_1 < \tau_2$ in the corresponding renormalized time interval,

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|V(\tau_2)|^2}{2} \phi(\tau_2)\, dy + \nu \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} |\nabla V|^2 \phi\, dy\, d\tau \\
&\le \int_{\mathbb{R}^3} \frac{|V(\tau_1)|^2}{2} \phi(\tau_1)\, dy + \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} \frac{|V|^2}{2} (\partial_\tau \phi + \nu \Delta \phi)\, dy\, d\tau \\
&\quad + \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P \right) V \cdot \nabla \phi\, dy\, d\tau \\
&\quad + \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} a(\tau) \frac{|V|^2}{2} \big(-\phi - y \cdot \nabla \phi \big)\, dy\, d\tau \\
&\quad - \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} \frac{|V|^2}{2} b(\tau) \cdot \nabla \phi\, dy\, d\tau.
\end{aligned}$$

Proof. For smooth solutions, multiply the repaired-gauge equation by $V\phi$ and integrate by parts. Put $e = |V|^2/2$. Since $\nabla \cdot V = 0$,

$$V \cdot (V \cdot \nabla V) = V \cdot \nabla e = \nabla \cdot (eV), \qquad V \cdot \nabla P = \nabla \cdot (PV),$$

and

$$V \cdot (y \cdot \nabla V) = y \cdot \nabla e, \qquad V \cdot (b \cdot \nabla V) = b \cdot \nabla e.$$

The scale contribution is

$$\int a(2e + y \cdot \nabla e) \phi\, dy = \int a e(-\phi - y \cdot \nabla \phi)\, dy,$$

because

$$\int y\cdot\nabla e\,\phi\,dy = -\int e\,\nabla\cdot(y\phi)\,dy = -\int e(3\phi + y\cdot\nabla\phi)\,dy.$$

The translation contribution is

$$\int b\cdot\nabla e\,\phi\,dy = -\int e\,b\cdot\nabla\phi\,dy,$$

and the viscous term gives

$$\int \nu\Delta V\cdot V\phi\,dy = -\nu\int |\nabla V|^2\phi\,dy + \nu\int e\Delta\phi\,dy.$$

Combining these identities gives the displayed equality in the smooth case.

For suitable weak solutions, apply the physical local energy inequality with

$$\psi(x,t) = \phi\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t)\right).$$

Use the change of variables

$$x = x_c(t) + \lambda(t)y, \qquad dt = \lambda(t)^2\,d\tau.$$

The C^1 regularity of λ and x_c justifies the chain rule for the test function, and the Jacobian factors are those of the Navier–Stokes critical scaling. Passing to the variables y, τ gives the same formula with equality replaced by $\leq$. $\square$

Lemma 9.64 (Caccioppoli estimate from the renormalized local energy inequality)**.** *Assume lemma 9.63 on a compact renormalized cylinder $B_{2R}\times I$, bounded modulation on I, and local bounds*

$$V \in L^\infty(I; L^3(B_{2R})), \qquad P \in L^{3/2}(I\times B_{2R}).$$

Then the compact-cylinder renormalized Caccioppoli estimate used in proposition 7.15 is justified.

Proof. Choose a standard nonnegative cutoff

$$\phi(y,\tau) = \eta(\tau)\zeta(y)^2,$$

where $\zeta\equiv 1$ on the inner ball and ζ is supported in the outer ball. Insert this test function in lemma 9.63. The left-hand side contains

$$\nu\int |\nabla V|^2\eta\zeta^2.$$

Every term on the right-hand side is finite under the stated hypotheses. The term

$$\int |V|^2|\partial_\tau\phi + \Delta\phi|$$

is controlled by $V\in L^\infty(I; L^3(B_{2R}))$ and finite measure. The convective term

$$\int |V|^3|\nabla\phi|$$

is controlled by the same local critical bound. For the pressure term, choose a time-dependent constant $c(\tau)$. Since

$$\int c(\tau)V\cdot\nabla\phi\,dy = -\int c(\tau)\phi\,\nabla\cdot V\,dy = 0,$$

Hölder's inequality controls

$$\int |P - c(\tau)|\,|V|\,|\nabla\phi|$$

by $P - c(\tau) \in L^{3/2}(I \times B_{2R})$ and $V \in L^\infty(I; L^3(B_{2R}))$. The modulation terms are bounded by

$$\|a\|_{L^\infty} + \|b\|_{L^\infty}$$

times the local L^2-mass, and the local L^2-mass is controlled on bounded balls by local L^3-mass. These estimates give the renormalized Caccioppoli estimate on compact cylinders. □

Proposition 9.65 (Caccioppoli regularity criterion)**.** *Assume the suitable physical hypotheses, the repaired-gauge representation, pressure reconstruction modulo functions of time, and C^1 scale and center functions on compact renormalized windows. Then the compact-cylinder Caccioppoli regularity hypothesis used in the local windowed H^1 estimate holds.*

Proof. The repaired-gauge representation writes the suitable weak solution in the variables V, P, a, b. Lemma 9.63 gives the renormalized local energy inequality, and lemma 9.64 shows that this inequality justifies the renormalized Caccioppoli estimate on every compact cylinder. □

Corollary 9.66 (Caccioppoli estimates give windowed control under boundedness hypotheses)**.** *Assume the suitable physical hypotheses, the repaired-gauge representation, pressure reconstruction, a bounded critical L^3 norm on compact cylinders, and bounded modulation. Then local windowed H^1 control follows from proposition 7.15. Thus failure of the Caccioppoli regularity hypothesis is not an independent rough-core alternative under these hypotheses.*

Proof. Proposition 9.65 supplies the Caccioppoli regularity hypothesis. The bounded critical norm, pressure reconstruction, and bounded modulation are the other local inputs used in proposition 7.15. Therefore the local windowed H^1 estimate holds. □

9.10. **Terminal Local State-Space Compactness.**

Definition 9.67 (Admissible retained terminal decomposition and completeness conclusion)**.** Let $Q_{2R,I} := B_{2R} \times I$ be a compact terminal-frame cylinder and let

$$V_n = U_n + S_n$$

be the local decomposition into the retained terminal component U_n and the nonretained residual S_n. The critical-profile interface is expressed through the following local notions.

(i) The decomposition is an admissible retained terminal decomposition if, on every $Q_{2R,I}$, the represented fields satisfy the repaired-gauge equation, local energy inequality, pressure reconstruction modulo functions of time, bounded modulation, and the compact-cylinder estimates

$$\sup_n \Big(\|V_n\|_{L^\infty(I;L^3(B_{2R}))} + \|V_n\|_{L^\infty(I;L^2(B_{2R}))} + \|V_n\|_{L^2(I;H^1(B_{2R}))} + \|P_n - c_{n,2R}\|_{L^{3/2}(Q_{2R,I})} \Big) < \infty,$$

and the same local velocity bounds hold for U_n. The time derivatives are bounded in $L^1(I; H^{-m}(B_{2R}))$ for some m large enough for the Aubin–Lions compactness argument.

(ii) A positive residual local label is assigned as follows. If, after passing to a subsequence,

$$\limsup_{n\to\infty} \|S_n\|_{L^\infty(I;L^3(B_R))} > 0,$$

then a selected subwindow and subcylinder carry an augmented local occupation state for the tuple $(V_n, P_n, U_n, S_n, a_n, b_n)$, together with the local residual concentration measures $|S_n|^3\,dy$ on selected time slices, whose residual coordinate or residual concentration measure is nonzero. The first closed label of this augmented state is read in the ordered local validation list of *lemma* 7.70, with the compact local Type II row refined by *theorem* 10.11.

(iii) The local terminal completeness conclusion is the statement that, if all named adjacent alternatives are closed by their local arguments and comparable retained classes have been grouped before selecting the terminal frame, then no nonretained residual with a positive residual local label remains on the retained terminal branch. Equivalently,

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_R))$$

on every compact terminal cylinder of the retained branch.

No global L^3 profile decomposition, global heat-flow norm, or global Kato estimate is part of this statement.

Local proof of terminal state-space completeness.

Lemma 9.68 (Interior-time selection on a compact terminal window)**.** *Let $I = [\tau_-, \tau_+]$ be a compact time-interval in the terminal-frame coordinates, and let $\tau_n \in I$ be a sequence of selected times for which $\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta > 0$. Then, after passing to a subsequence, exactly one of the following two admissible selections holds:*

(i) *there is a compact subinterval $J \Subset I$ such that $\tau_n \in J$ for all large n;*

(ii) *the selected times approach an endpoint of I, and the terminal window is replaced by the admissible recentered windows*

$$I_n^* := \tau_n + J_0, \qquad J_0 = [-\sigma, \sigma],$$

with fixed $\sigma > 0$, written in coordinates centered at τ_n. In these recentered coordinates the selected time is $0 \in J_0^\circ$ and

$$\|S_n(0)\|_{L^3(B_R)} \geq \delta.$$

No lower bound is transferred from one time to another.

Proof. After passing to a subsequence, $\tau_n \to \tau^* \in I$. If $\tau^* \in (\tau_-, \tau_+)$, choose $J \Subset I$ with $\tau^* \in J^\circ$. Then $\tau_n \in J$ for all large n, giving (i).

If $\tau^* \in \partial I$, we do not move the lower bound to a nearby time. Instead we use the admissibility of the terminal validation family under finite covariant time recentering: selected compact windows are local objects, and composing the terminal-frame chart with the finite translation $\tau \mapsto \tau - \tau_n$ produces another member of the same local validation family on every fixed model interval $J_0 = [-\sigma, \sigma]$. The compact-cylinder estimates are then the estimates of definition 9.67(i) on these recentered selected windows, with constants depending on the model cylinder and not on n. In the recentered coordinates the original selected time τ_n is exactly 0, so the lower bound remains $\|S_n(0)\|_{L^3(B_R)} \geq \delta$. This proves (ii) and uses no continuity in time of the L^3 norm. □

Lemma 9.69 (Local compactness of terminal occupation states)**.** *Assume the estimates in definition 9.67*(i) *on $Q_{2R,I}$. Then every sequence of selected subwindows and subcylinders has a subsequential local occupation state. More precisely, after passing to a subsequence, V_n, U_n, and S_n converge weakly in $L^2(I; H^1(B_R))$, strongly in $L^2(I; L^q(B_R))$ for every $q < 6$, and their pressure gradients converge distributionally modulo functions of time.*

Proof. The velocity bounds and the time-derivative bound give compactness by the Aubin–Lions–Simon theorem applied to

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-m}(B_R).$$

Interpolation with the uniform $L^2(I; H^1(B_R))$ bound gives strong convergence in $L^2(I; L^q(B_R))$ for every $q < 6$. The pressure normalization bound gives weak compactness in $L^{3/2}(Q_{R,I})$; on overlaps the limits differ only by functions of time, so the gradients assemble distributionally. □

Lemma 9.70 (Closed local labels for terminal residuals)**.** *In the augmented local topology for*

$$(V_n, P_n, U_n, S_n, a_n, b_n)$$

supplied by lemma 9.69, the ordered validation rows in lemma 7.70, with the compact local Type II row refined by theorem 10.11, are closed under subsequential limits of selected compact terminal windows, after shrinking the tested cylinders inside the fixed compact cylinder when necessary.

Proof. The topology records the velocity variables weakly in L^2H^1, strongly in $L^2L^q_{\mathrm{loc}}$ for every $q < 6$, the selected-time residual measures $|S_n(\tau_n)|^3\,dy$ weakly-$*$ as finite Radon measures on compact balls, the pressures weakly in $L^{3/2}_{\mathrm{loc}}$ modulo functions of time, and the modulation, scale, center, cutoff, and cost variables in the compactified repaired-gauge coordinates used in the ordered validation procedure. The compactification adds boundary faces exactly for loss of representation, scale admissibility, bounded modulation, pressure reconstruction, local H^1 control, critical tightness, and cost compatibility.

Positive local velocity-pressure concentration is closed on smaller cylinders. For spacetime rows, the velocity part follows from strong local L^3 convergence after interpolating the strong L^2L^q convergence with the uniform $L^\infty L^3$ bound. For the pressure part, the weak $L^{3/2}$ bound alone gives only lower semicontinuity and would allow positive pressure mass to disappear in the weak limit. We therefore upgrade pressure convergence to strong local convergence on the retained branch as follows. After subtracting the reconstructed pressure of the limit branch, write the pressure difference on B_{R_1}, modulo functions of time, as

$$\widetilde{P}_n = P_n - P - c_n = \widetilde{P}_n^{\mathrm{loc}} + \widetilde{H}_n + c_{n,R_1}(\tau), \qquad \widetilde{P}_n^{\mathrm{loc}} = \mathcal{R}_i\mathcal{R}_j\big(\zeta((V_n \otimes V_n) - (V \otimes V))_{ij}\big),$$

with $\zeta \equiv 1$ on B_{R_1} and $\operatorname{supp}\zeta \Subset B_{R_2}$. Since $V_n \to V$ strongly in $L^2_\tau L^3_y(B_{R_2})$, while V_n and V are uniformly bounded in $L^\infty_\tau L^3_y(B_{R_2})$, interpolation gives strong convergence in $L^3_\tau L^3_y(B_{R_2})$. Hence

$$V_n \otimes V_n \to V \otimes V \quad \text{strongly in } L^{3/2}_\tau L^{3/2}_y(B_{R_2}).$$

Calderón–Zygmund continuity therefore gives strong convergence of $\widetilde{P}_n^{\mathrm{loc}}$ to zero in $L^{3/2}_\tau L^{3/2}_y(B_R)$. The harmonic remainder $\widetilde{H}_n$ is not controlled by a weak lower-semicontinuity argument: it is routed with $(q,p) = (3/2, 3/2)$ by the pressure/exterior alternative in lemma 8.22. Thus on the retained branch the harmonic part vanishes locally modulo functions of time, while off that branch the nonzero harmonic part is a named pressure or exterior label. Consequently positive pressure mass cannot be hidden by weak convergence on the retained branch; pressure rows are strongly closed in the local $L^{3/2}_\tau L^{3/2}_y$ topology after shrinking cylinders. For selected-time rows, the velocity part is recorded by the weak-$*$ limits of the local measures $|S_n(\tau_n)|^3dy$, so a positive lower bound persists as mass of the limiting measure. Hence a lower CKN threshold cannot disappear after passing to a subsequence and shrinking the cylinder.

Multibubble and multicore splitting are finite collections of such positive CKN lower bounds with separated centers or separated scales; the separation relations are closed in the scale-center compactification. Same-point cascade labels are closed because they are the boundary faces $\lambda_i^n/\lambda_j^n \to 0$ or ∞ at a common physical center. Comparable same-point labels are closed by convergence of the relative scale-center parameters and are grouped by lemma 8.3.

Exterior leakage and critical-tightness failure are closed because they are defined by a positive lower bound for local L^3 or CKN mass outside every retained compact set. If that mass returns to a fixed compact set along a subsequence, the state is relabelled by the corresponding compact concentration row; otherwise it remains on the exterior/tightness boundary face. Loss of local windowed H^1 control is the face at infinity for the nonnegative L^2H^1 seminorm on a compact cylinder.

Finite-cost scale collapse, negative-drift, and compact cost-divergence labels are determined by nonnegative localized cost functionals. Their sublevel or positive-carrier conditions are lower semicontinuous in the recorded occupation-state topology. The scale-rigid residual label is closed by the relative scale-center convergence and by the compact realization of the scale-rigid branch in definition 4.23. Representation, modulation, pressure, carrier, and cost-compatibility failures are not hidden analytic assumptions: they are the explicitly added boundary faces of the ordered validation compactification. Thus every first row of the ordered local validation is a closed diagnostic set. □

Lemma 9.71 (Residual concentration is visible in the augmented state)**.** *Let $V_n = U_n + S_n$ on a compact terminal cylinder, with the local $L^\infty L^3$ bounds of definition 9.67*(i)*. Suppose that, for selected times τ_n and a compact ball B_ρ,*

$$|S_n(\tau_n)|^3 dy\big|_{B_\rho} \overset{*}{\rightharpoonup} \mu_S, \qquad \mu_S(B_\rho) > 0.$$

Then, after shrinking B_ρ if necessary, either the full represented branch V_n carries a positive compact local concentration label, or the retained component U_n carries a positive comparable retained label in the same terminal frame, or a separated-center/separated-scale adjacent label has already occurred. In particular, residual L^3 concentration cannot be invisible in the augmented local state space by cancellation between V_n and U_n.

Proof. For every time slice,

$$|S_n|^3 = |V_n - U_n|^3 \leq 4\big(|V_n|^3 + |U_n|^3\big).$$

Passing to weak-$*$ limits of the finite measures on a slightly smaller ball gives

$$\mu_S \leq 4(\mu_V + \mu_U)$$

for subsequential limits of $|V_n(\tau_n)|^3 dy$ and $|U_n(\tau_n)|^3 dy$. Hence μ_V or μ_U has positive mass on some subball. If μ_V has positive mass, the full represented branch has a positive compact local concentration diagnostic and the ordered local state-space decomposition assigns it to the corresponding first label. If μ_U has positive mass, the mass belongs to the retained component already represented in the selected terminal frame. If it were not comparable to that frame, separated-center or separated-scale decoupling would place it in an adjacent row. If it is comparable but not already absorbed, then it is a retained class missed by the terminal grouping, contradicting lemma 9.72 below.

This visibility argument does not assert a Brézis–Lieb identity for the moving pair $V_n = U_n + S_n$. Such an identity would require additional orthogonality hypotheses and is false for arbitrary moving decompositions. The local state-space proof uses only the domination inequality above and the maximality of the retained grouping: cancellation of V_n against U_n can make the represented measure small only by making the retained measure large, and a positive retained measure is itself a recorded comparable or adjacent local label. Hence positive residual L^3 mass cannot disappear from the augmented state-space diagnostics. □

Lemma 9.72 (Maximality of the selected retained frame)**.** *The terminal selection rule for the retained component U_n on a compact terminal cylinder $Q_{2R,I}$ produces a finite collection of comparable retained classes $\{U_n^\alpha\}_{\alpha\in A}$ such that:*

(i) Pairwise scale-center separation. *For $\alpha \neq \beta \in A$, the relative scale-center parameters $(\lambda_n^\alpha/\lambda_n^\beta, (x_n^\alpha - x_n^\beta)/\lambda_n^\beta)$ either tend to 0 or to ∞ (separated scales) or stay bounded with the centers escaping each other in scale-comparable units (separated centers).*

(ii) Maximality. *If a candidate retained class $(\tilde\lambda_n, \tilde x_n)$ at the same physical center and with scale parameter comparable to some λ_n^α survives the selection rule (i.e. carries a positive* *active local mass floor in the sense of theorem 8.32), then it is already asymptotically equivalent to $(\lambda_n^\alpha, x_n^\alpha)$ and is absorbed into U_n^α.*

Proof. The terminal selection rule is local on the fixed compact cylinder $Q_{2R,I}$. It extracts only classes which carry the active local mass floor of theorem 8.32 on a selected compact subcylinder. Classes whose witnesses leave every compact subcylinder, or form an infinite same-point scale chain, are not retained classes in this branch: they are read first as exterior, separated-scale, or cascade boundary faces in the ordered local validation. After those adjacent faces are removed, the remaining witnesses lie in a compact scale-center chart. A finite-overlap cover of that chart and the local bound $\sup_n \|V_n\|_{L^\infty(I;L^3(B_{2R}))} < \infty$ allow only finitely many disjoint active-floor witnesses. Hence the retained list on the compact terminal branch is finite.

For two retained classes in this finite list, either their relative scale-center parameters leave every compact subset of the relative scale-center chart, giving separated scales or separated centers, or those parameters remain in a compact subset. In the latter case the two concentrations are comparable in the same local observer coordinates and are grouped before the terminal frame is fixed. This proves the separation alternative in (i) for distinct retained classes.

For maximality (ii), suppose toward contradiction that $(\tilde\lambda_n, \tilde x_n)$ is comparable to $(\lambda_n^\alpha, x_n^\alpha)$ (i.e. both $\tilde\lambda_n/\lambda_n^\alpha \to c \in (0,\infty)$ and $|\tilde x_n - x_n^\alpha|/\lambda_n^\alpha \to 0$) but not asymptotically equivalent. Then the rescaled profile $\tilde\lambda_n V_n(x_n^\alpha + \tilde\lambda_n y, \cdot)$ carries an active local mass floor in a relative scale-center chart compactly comparable to the chart of U_n^α. Thus the two candidates are not separated by any ordered local boundary face. By lemma 8.3, all same-point comparable concentrations in such a compact relative chart are assembled into one compound retained class before terminal selection is closed. The candidate $(\tilde\lambda_n, \tilde x_n)$ must therefore already be absorbed into U_n^α, contradicting the assumption that it is a surviving unabsorbed comparable class. □

Lemma 9.73 (Terminal local critical-mass failures are adjacent labels)**.** *Let a physical local Type II branch have passed covered entry and let a terminal active frame be selected from its local state-space decomposition. On each fixed compact terminal cylinder, exactly one of the following occurs after passing to a subsequence:*

(i) *the retained terminal component has positive finite local critical L^3 mass on the cylinder;*
(ii) *the lower active mass floor fails, so the candidate is not a retained terminal component;*
(iii) *the local L^3 upper bound fails on the compact cylinder, in which case the ordered local state space records an additional compact active label: a critical-mass adjacent label at the same retained frame, a comparable finite-mass retained class which must be grouped into the retained component, a multibubble/multicore label, a cascade label, a rough-core label, or an exterior/separated label.*

In the retained terminal branch only (i) *remains.*

Proof. The lower bound in (i) is the active local mass floor of theorem 8.32. If this floor is absent, the candidate frame is not retained by the terminal selection rule, giving (ii).

Assume the lower floor is present and the finite local upper bound fails on a compact cylinder $B_R \times I$. Then there are times $\tau_n \in I$ and levels $M_n \to \infty$ with

$$\int_{B_R} |V_n(y,\tau_n)|^3\, dy \ge M_n.$$

Cover B_R by finitely many smaller balls. On one ball, after passing to a subsequence and renormalizing at the corresponding local scale if necessary, the branch carries a further positive compact L^3 diagnostic. The ordered local state-space compactification records the first label of that diagnostic. If the diagnostic stays at the same comparable retained frame and the upper local L^3 bound still fails, the first label is the critical-mass adjacent face itself; by the retained-state definition this is not a retained terminal branch. If the diagnostic is comparable and finite-mass, then lemma 9.72 says it was already grouped into the retained component. If it is not comparable, it is by definition one of the adjacent compact active labels: multibubble/multicore splitting, cascade, rough core, or exterior/separated concentration. These are precisely the labels in (iii).

On a retained terminal branch all comparable labels have already been grouped, and all non-comparable labels are adjacent local alternatives routed by the state-space closure package. Hence only the positive finite local critical mass alternative remains. This is a compact-cylinder statement; it does not require $V(\tau) \in L^3(\mathbb{R}^3)$ or a whole-space critical norm bound. □

Theorem 9.74 (Adjacent terminal labels are locally routed). *For every admissible retained terminal decomposition, every first label in the ordered augmented local state space is either the retained comparable class or one of the named local alternatives. The adjacent labels are closed and routed out of the retained terminal component as follows: representation, modulation, pressure, critical-mass, carrier, cost-compatibility, exterior tightness, local H^1-loss, multibubble, multicore, cascade, finite-cost scale-collapse, and scale-rigid residual labels are assigned to their named local alternatives and do not remain on the retained terminal branch. This theorem is a routing theorem only: it does not invoke the final state-elimination proposition and does not claim, inside the terminal-completeness argument, that the adjacent alternatives have already been globally excluded.*

Proof. Lemma 9.70 proves closedness of the rows in the augmented local topology. Representation and modulation failures are the boundary faces of the repaired-gauge compactification and are exactly the representation alternatives. Pressure rows are routed by lemma 8.22 and by the pressure reconstruction stability lemma 2.12. Critical-mass rows are routed by lemma 9.73: a lower-mass failure is not retained, and an upper local L^3-failure produces a comparable or adjacent active label. Exterior tightness and exterior leakage are routed by theorem 8.38. Local H^1-loss is recorded as the rough-core row of theorem 10.11. Multibubble, multicore, cascade, and gauge-degenerate rows are precisely the non-single-core structural rows of theorem 10.11. Finite-cost scale-collapse is recorded as the finite-cost scale-collapse row, and scale-rigid residual labels are recorded as the scale-rigid row governed by proposition 4.40. Carrier and compact-cost compatibility rows are the carrier/cost adjacent rows of the ordered compact-cost validation.

The point needed here is only separation from the retained terminal component. Each listed row is a boundary face of the ordered local state space, not the retained comparable class. Its later exclusion is supplied in the assembly layer by the local closure theorems and, for carrier/cost rows, by the ambient adjacent wrapper. The terminal-completeness proof therefore uses this theorem only to say that a positive residual label cannot stay in the nonretained remainder of a retained terminal branch. The only first label not routed away is the retained comparable class, which is grouped into the selected terminal component before the terminal frame is fixed. □

Theorem 9.75 (Residual local mass gives an admissible augmented state). *Assume the local compact-window estimates of definition 9.67*(i). *If*

$$\limsup_{n\to\infty} \|S_n\|_{L^\infty(I;L^3(B_R))} > 0,$$

then, after passing to a subsequence, there are selected times, still denoted τ_n in the selected coordinates, a compact selected or recentered subcylinder $Q_{\rho,J}$, and an augmented local occupation state

$$\mathfrak{s} = \lim_{n\to\infty} (V_n, P_n, U_n, S_n, a_n, b_n)\big|_{Q_{\rho,J}}$$

whose residual coordinate or selected-time residual L^3 concentration measure is nonzero. The Navier–Stokes admissibility conditions in the ordered local validation are imposed on the full represented branch (V_n, P_n, a_n, b_n), not on S_n as an independent solution. Consequently the first *nonretained residual row is either a named adjacent local alternative or the retained comparable local class.*

Proof. Choose $\delta > 0$, times $\tau_n \in I$, and a subsequence such that

$$\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta.$$

By lemma 9.68, after passing to a further subsequence, either the original selected times lie in a fixed interior subinterval $J \Subset I$, or we replace the terminal window by the admissible recentered windows supplied by that lemma and write the selected time as $0 \in J_0^\circ$. In both cases we keep the same selected time rather than transferring the lower bound to a different time; below J denotes the fixed model interval in the selected coordinates and τ_n denotes the selected interior time in those coordinates. Cover B_R by finitely many smaller balls compactly contained in B_R. One ball, after passing to a subsequence, carries residual L^3 mass bounded below by a fixed multiple of δ. The compact subinterval J supplied by lemma 9.68 contains the selected times with positive distance from the boundary of the model window; we use this J without further time movement. The estimates in definition 9.67(i) and lemma 9.69 give a subsequential limit of the augmented tuple on $Q_{\rho,J}$. In addition, the measures

$$|S_n(\tau_n)|^3 \, dy\big|_{B_\rho}$$

have uniformly bounded total mass on compact balls and therefore converge weakly-$*$, after passing to a subsequence, to a finite nonzero Radon measure. The lower mass bound on the chosen ball passes to this measure. Thus the augmented state has a nonzero residual coordinate or, in the concentration case, a nonzero residual L^3 concentration measure. Lemma 9.71 shows that this residual diagnostic is visible either as a positive compact local label of the full represented branch, as a positive comparable retained label, or as an already named separated-center/separated-scale adjacent label. Hence it cannot be hidden by cancellation with U_n.

The PDE structure used by the state-space decomposition is carried by (V_n, P_n, a_n, b_n): it satisfies the repaired-gauge Navier–Stokes equation, the local energy inequality, pressure reconstruction modulo functions of time, bounded modulation, and the compact-cylinder bounds. The retained component U_n and residual coordinate S_n are bookkeeping variables in the augmented state space. Thus no step applies the Navier–Stokes decomposition theorem to S_n alone.

Evaluate the ordered local validation list of lemma 7.70 on the augmented state. If a representation, pressure, modulation, exterior-tightness, local H^1, finite-cost, scale-rigid, or carrier/cost row fails first, that row is the named adjacent alternative. If the compact local Type II row is reached, apply theorem 10.11 to the full represented branch (V_n, P_n) on the selected compact window. If the full branch produces a new compact concentration, splitting, cascade, scale-collapse, or scale-rigid state, this is again a named adjacent local alternative. The remaining possibility, using lemma 9.71, is that the residual diagnostic is carried by the retained compact class. That class is comparable to the selected terminal frame; otherwise the separated-center or separated-scale decoupling lemmas would put it into one of the preceding adjacent rows. Hence the first nonretained residual row is a retained comparable class. □

Lemma 9.76 (Positive residual mass produces a local state-space label)**.** *Assume the local compact-window estimates of definition 9.67*(i)*. If*

$$\limsup_{n\to\infty} \|S_n\|_{L^\infty(I;L^3(B_R))} > 0,$$

then, after passing to a subsequence and selecting times and subcylinders, the residual produces one of the first closed labels listed in definition 9.67(ii)*.*

Proof. Theorem 9.75 constructs the augmented occupation state and verifies that the Navier–Stokes admissibility belongs to the full represented branch (V_n, P_n). The ordered first row of that state is closed by lemma 9.70. Therefore, after passing to a subsequence, the positive residual mass has a first closed local label in the sense of definition 9.67(ii). ☐

Lemma 9.77 (Residual mass cannot remain in the retained terminal branch)**.** *Assume the local terminal frame has been chosen after grouping comparable retained classes. Then a nonretained residual satisfying the hypotheses of lemma 9.76 cannot remain on the retained terminal branch.*

Proof. If the residual has positive local mass, then lemma 9.76 assigns it to a first closed local label. By theorem 9.74, an adjacent label is a named local alternative and is not part of the retained branch. If the label is the retained compact class, then the corresponding component is comparable to the selected retained terminal frame; by the terminal selection rule together with lemma 9.72, comparable retained classes based at the same physical point were already grouped into U_n. It therefore cannot remain in S_n. Both cases contradict the assumption that positive residual mass persists in the nonretained component of the retained terminal branch. ☐

Lemma 9.78 (Subsequential exhaustion of local residual mass)**.** *Under the hypotheses of lemma 9.77, every compact terminal cylinder satisfies*

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_R)).$$

Proof. If the convergence failed, then a subsequence would have positive residual mass in the sense of lemma 9.76. The preceding lemma shows that such mass cannot remain in the retained terminal branch after local state-space closure and grouping of comparable retained classes. This contradiction proves the convergence. ☐

Theorem 9.79 (Local terminal completeness from state-space stratification)**.** *Every admissible retained terminal decomposition in the sense of definition 9.67*(i) *satisfies the local terminal completeness conclusion of definition 9.67*(iii)*, after the terminal frame is selected by grouping comparable retained classes.*

Proof. Lemma 9.69 gives the local compactness needed to pass to occupation states. Lemma 9.70 identifies the closed local labels, and theorem 9.74 routes every adjacent label to a named local alternative. Lemma 9.76 assigns any positive nonretained residual mass to one of those labels. Lemma 9.77 rules out persistence of such a label inside the retained branch once comparable retained classes have been grouped, and lemma 9.72 certifies that the terminal selection rule actually performs that grouping maximally. Hence lemma 9.78 gives $S_n \to 0$ on compact terminal cylinders. This is the local terminal completeness conclusion named in definition 9.67(iii). ☐

Remark 9.80 (Status of the terminal completeness statement)*.* Theorem 9.79 is local. It uses only compact-cylinder estimates, Aubin–Lions compactness, pressure reconstruction modulo functions of time, and the local state-space stratification. It does not use a global critical profile decomposition, a global heat-flow smallness norm, or a global Kato perturbation theorem.

Definition 9.81 (Local compact terminal windows)**.** The local compact terminal-window hypothesis is the assertion that every terminal active sequence used in the terminal analysis is represented, on each compact terminal-frame cylinder $Q_{2R,I}$, by fields satisfying the local compact-cylinder bounds in definition 9.67(i). Equivalently, the analysis is carried out on the local state variables

$$(V_n, U_n, S_n, P_n, a_n, b_n)\big|_{Q_{2R,I}},$$

not on an arbitrary global sequence of initial data.

Theorem 9.82 (Local perturbative residual stability)**.** *Let $Q_{2R,I}$ be a compact terminal cylinder. Assume*

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_{2R})),$$

and assume U_n and S_n are uniformly bounded in

$$L^2(I;H^1(B_{2R}))\cap L^\infty(I;L^3(B_{2R})).$$

Then all velocity interactions involving at least one factor S_n vanish on the smaller cylinder:

$$U_n\otimes S_n+S_n\otimes U_n+S_n\otimes S_n\to 0 \quad \text{in } L^1(I;L^2(B_R)).$$

If the corresponding cross pressures are reconstructed by the local Calderón–Zygmund decomposition, then their gradients vanish in

$$L^1(I;H^{-1}(B_R)).$$

Proof. By Sobolev on B_{2R}, the L^2H^1 bounds give uniform $L^2(I;L^6(B_{2R}))$ bounds. Hölder's inequality gives

$$\|U_n\otimes S_n\|_{L^1(I;L^2(B_R))}\le \|S_n\|_{L^\infty(I;L^3(B_R))}\|U_n\|_{L^1(I;L^6(B_R))}\to 0,$$

and the same estimate treats $S_n\otimes U_n$. The term $S_n\otimes S_n$ is identical, using the uniform L^1L^6 bound for S_n. The pressure statement follows from the local pressure-kernel estimate applied to the cross stress just shown to vanish in L^1L^2, with the pressure normalized modulo functions of time. Concretely, the local Calderón–Zygmund piece of the cross pressure vanishes in $L^1_\tau L^2_y(B_R)$ by Calderón–Zygmund continuity applied to the vanishing cross stress, and the harmonic remainder is routed by lemma 8.22 with $(q,p)=(1,2)$ (using the quantitative nested-ball harmonic-tail estimate of lemma 8.21): on the retained terminal branch any non-vanishing harmonic remainder triggers a named pressure/exterior-leakage alternative, which is excluded. The two contributions sum to zero in $L^1_\tau H^{-1}_y(B_R)$. □

Definition 9.83 (Local perturbative stability)**.** Local perturbative stability means that every residual component which is small in $L^\infty(I;L^3(B_R))$ on compact terminal cylinders contributes only vanishing velocity-stress, pressure, and linear-residual terms in the local topologies used in definition 8.13.

Corollary 9.84 (Local residual stability gives perturbative stability)**.** *Theorem 9.82 implies the local perturbative stability condition of definition 9.83.*

Proof. This is exactly the conclusion of theorem 9.82 on each compact terminal cylinder, combined with the local pressure and residual estimates already included in terminal nonlinear decoupling. □

Discharge of the retained compact-test closure.

Lemma 9.85 (Critical-tightness failure routes to an exterior alternative)**.** *Assume the retained compact single-core stratum: the branch V has passed the ordered local tests of lemma 7.70 up to and including the local state-space decomposition theorem 10.11, with the single-core alternative* (i) *retained. If the critical-tightness condition of definition A.26 fails, i.e. there exist $\varepsilon_0>0$ and sequences $R_n\uparrow\infty$ and $\tau_n\ge\tau_0$ such that*

$$\int_{|y|>R_n}|V(y,\tau_n)|^3\,dy\ge\varepsilon_0,$$

then, after passing to a subsequence, the branch admits an exterior concentration sequence on every annulus $A_{r,R}$ with r sufficiently large. By theorem 8.38 the branch then belongs to one of the exterior-profile, separated-core, multicore, noncompact-exterior, or scale-collapse alternatives. All of these are non-retained alternatives in theorem 10.11(ii),(iv) *and are excluded from the retained single-core stratum. Equivalently, on the retained stratum critical tightness holds automatically.*

Proof. Failure of critical tightness gives $\varepsilon_0 > 0$, $R_n \uparrow \infty$, and $\tau_n \geq \tau_0$ with

$$\int_{|y|>R_n} |V(y,\tau_n)|^3 \, dy \geq \varepsilon_0 \qquad \text{for all } n.$$

The integrand $|V|^3\,dy$ is scale-invariant under the parabolic rescaling defining V from u, so this lower bound is a positive critical-mass-floor that, in the original physical coordinates centred at (x_*, T^*), records positive L^3 mass on physical annuli $\{|x - x_*| > R_n e^{-\tau_n}\}$ at terminal times $t_n := T^* - e^{-2\tau_n} \to T^*$. Two cases occur after passing to a subsequence.

Case A: bounded distance from the core. Some $R^* < \infty$ and some $r_0 > 0$ admit infinitely many n with

$$\int_{r_0<|y|<R^*} |V(y,\tau_n)|^3 \, dy \geq \varepsilon_0/2.$$

Suppose, towards contradiction, that there is no exterior concentration on the slightly larger annulus $A_{r_0/2,2R^*}$ in the physical-coordinate sense of definition 8.36. By theorem 8.41 ($k = 0$), there is then a uniform C^0 bound for u on A_{r_0,R^*} on a terminal time interval, hence by integration the critical L^3 mass on the corresponding renormalized annulus is $O(e^{-3\tau})$, contradicting the positive lower bound $\varepsilon_0/2$ for $\tau_n \to \infty$. Therefore lemma 8.37 forces the exterior concentration alternative on $A_{r_0/2,2R^*}$.

Case B: escape to infinity in the renormalized coordinate. For every $R < \infty$,

$$\int_{r_0<|y|<R} |V(y,\tau_n)|^3 \, dy < \varepsilon_0/2$$

infinitely often. Then a subsequence puts at least $\varepsilon_0/2$ of its mass on shells $\{|y| > R_k\}$ with $R_k \uparrow \infty$, producing a sequence of L^3-concentration centers x_n with $|x_n|/1 \to \infty$ in the renormalized frame. This is the noncompact-exterior alternative of theorem 8.38, which is one of the listed exterior concentration alternatives.

In either case the dichotomy of theorem 8.38 selects the exterior concentration branch. By the second clause of that theorem the branch is exterior profile, separated core, multicore, noncompact exterior, or scale collapse. Each is one of the alternatives (ii)–(iv) of theorem 10.11 and disqualifies the branch from the retained single-core alternative (i). This contradicts the retained-stratum hypothesis, so on the retained stratum critical tightness holds. □

Lemma 9.86 (Windowed Caccioppoli with integrated modulation)**.** *Let (V, P, a, b) be a suitable repaired-gauge branch on a terminal tail. Fix $R > 0$ and a unit window $I_n = [\tau_0 + n, \tau_0 + n + 1]$, and set*

$$I_n^* := [\tau_0 + n - 1, \tau_0 + n + 2], \qquad A_{n,R} := 1 + \int_{I_n^*} \big(|a(\tau)| + \|b(\tau)\|_\infty\big)\, d\tau.$$

Assume on $I_n^* \times B_{4R}$ the local pressure reconstruction, the compact-cylinder critical bound

$$\sup_{\tau\in I_n^*} \|V(\tau)\|_{L^3(B_{4R})} \leq M_{3,4R},$$

and the local Caccioppoli estimate of theorem 9.60. Then

$$\int_{I_n}\int_{B_R} |\nabla V|^2 \, dy \, d\tau \leq C_R\big(1 + A_{n,R}\big)\big(1 + M_{3,4R}^3\big),$$

where C_R depends only on the fixed compact-cylinder cutoffs and ν.

Proof. Apply the local Caccioppoli estimate on $I_n^* \times B_{2R}$, with a cutoff equal to one on $I_n \times B_R$. The velocity terms are bounded by the finite-measure embeddings

$$\|V(\tau)\|^2_{L^2(B_{2R})} \leq C_R \|V(\tau)\|^2_{L^3(B_{2R})}, \qquad \|V(\tau)\|^3_{L^3(B_{2R})} \leq M_{3,4R}^3.$$

The pressure term is controlled, after subtracting spatial means, by the local pressure reconstruction and the same critical bound:

$$\|P(\tau) - c_{4R}(\tau)\|_{L^{3/2}(B_{2R})} \le C_R M_{3,4R}^2.$$

The modulation terms in the local energy inequality have the form

$$\int_{I_n^*} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \int_{B_{2R}} |V(\tau)|^2 \, dy \, d\tau,$$

up to cutoff constants; the local L^3 bound controls the inner integral by $C_R M_{3,4R}^2$. Combining these estimates gives the displayed bound. No pointwise-in-time modulation bound is used. □

Lemma 9.87 (Unbounded modulation windows are adjacent exits)**.** *Let a physical branch be in the retained compact single-core stratum after the ordered local validation tests. Fix $R > 0$. Then either*

$$\sup_n \int_{\tau_0+n-1}^{\tau_0+n+2} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau < \infty,$$

or the branch leaves the retained compact single-core stratum through one of the named modulation, scale-drift, modulation-driven recentering, exterior, or gauge-degenerate local alternatives.

Proof. If the displayed window integrals are unbounded, choose windows $I_{n_j}^*$ on which the integral tends to infinity. The coefficients a, b are the absolutely continuous scale and translation velocities of the repaired-gauge chart, as identified in lemma 9.50. Large accumulated negative scale motion is the negative scale-drift branch of definition 9.25; large positive scale-shell work is the positive-scale carrier branch routed by corollary 7.128; and large translation work either is absorbed by the admissible recentering rule or produces the modulation-driven recentering/exterior label in definition 7.107. If the accumulated modulation cannot be represented by the absolutely continuous chart variables, the first failed row is the repaired-gauge or gauge-degenerate alternative. These are all named adjacent local alternatives, not retained compact single-core states. Hence on the retained compact single-core stratum the window integrals are uniformly bounded. □

Lemma 9.88 (Local windowed H^1-loss routes to the rough-core alternative)**.** *Assume the retained compact single-core stratum, with a suitable repaired-gauge representation, local pressure reconstruction, compact-cylinder critical bounds, and the local Caccioppoli estimate on the compact cylinders used in lemma 9.86. These are the retained-stratum inputs supplied by the ordered tests in lemma 7.70*(i),(ii),(iv),(vii) *and by the suitable local energy package. Assume also that the unbounded-modulation window exit of lemma 9.87 has not occurred. Then, for every $R > 0$, the local windowed H^1 bound*

$$\sup_{T \ge \tau_0+1} \int_T^{T+1} \int_{B_R} |\nabla V|^2 \, dy \, d\tau \le C\Big(R, \nu, M_{3,4R}, \sup_n A_{n,R}\Big)$$

holds, so the local windowed H^1-control condition of definition A.26 is automatic on the retained stratum. Equivalently, failure of local windowed H^1 control is incompatible with the retained-stratum input package and is routed to the rough-core alternative (iii) *of theorem 10.11, which by theorem 10.7 forces failure of compact CKN control on enclosing cylinders and therefore returns to the multibubble/cascade alternative* (ii)*, already excluded from the retained branch.*

Proof. By lemma 9.87, absence of the modulation-window exit gives, for every fixed R,

$$\sup_n A_{n,R} < \infty.$$

The compact-cylinder critical bound and local pressure reconstruction are retained-stratum inputs. Applying lemma 9.86 on each unit window therefore gives a bound independent of n. This proves the displayed windowed H^1 estimate. The standard mollification and lower semicontinuity argument passes the estimate from smooth approximants to suitable repaired-gauge branches.

If, contrapositively, the windowed H^1 bound fails for some $R > 0$ along a terminal sequence, then by lemma 9.86 and lemma 9.87 at least one retained input must fail along that sequence: local pressure reconstruction, compact-cylinder critical control, local Caccioppoli regularity, or the uniform integrated modulation-window bound. By the ordered tests lemma 7.70 this earlier-test failure routes the branch out of the retained stratum into the corresponding representation, repaired-gauge, critical-mass, local-regularity, or modulation/scale-drift alternative. Independently, in the local state-space terminology of theorem 10.11 the same failure is recorded as the rough-core alternative (iii), and by theorem 10.7 that alternative is equivalent to failure of compact CKN control on enclosing cylinders, hence to active multibubble or cascade behaviour, alternative (ii) of theorem 10.11. Both routings exit the retained single-core stratum. Therefore on the retained stratum H^1-loss is impossible. □

Theorem 9.89 (Retained compact-test closure)**.** *On the retained compact single-core stratum, both compact tests of definition A.26 pass: critical tightness holds and the local windowed H^1 control holds. Equivalently, the entries* (v),(vi) *of lemma 7.70 are not genuine new assumptions on the retained stratum; their failure is routed to one of the named non-retained alternatives.*

Proof. Critical tightness holds by lemma 9.85: failure routes the branch to the exterior-stratification alternatives, all non-retained. Local windowed H^1 control holds by lemma 9.88: failure routes the branch to the rough-core alternative, which by theorem 10.7 is absorbed into the multibubble/cascade alternative, also non-retained. Hence both compact tests are theorems, not assumptions, on the retained stratum. □

Corollary 9.90 (Discharge of the retained compact-test closure row)**.** *The retained compact-test closure obligation listed as item 3 of the local closure ledger is discharged on every retained compact single-core branch produced by the ordered local validation procedure of lemma 7.70. No additional global tightness or global windowed H^1 hypothesis is required.*

Proof. Theorem 9.89 reduces both compact tests to local consequences of inputs already present on the retained stratum or to named non-retained alternatives. No global tightness or H^1 bound is invoked. □

9.11. **Closure of the thin negative-drift exit.** The remaining point in the scale-collapse drift branch is the case in which the weighted negative drift

$$g_{\mathrm{sc}}(\tau) := a_-(\tau)M_R(\tau), \qquad M_R(\tau) = \int |V(y,\tau)|^2 \Phi(\tau, y)\, dy,$$

has infinite tail integral but does not produce fixed-length windows with a uniform positive average. The following argument closes this row as an independent terminal alternative. It does not turn an infinite integral into a pointwise lower bound; instead it passes to normalized occupation blocks and uses the local state-space labels.

Definition 9.91 (Thin negative-drift occupation blocks)**.** Let a represented branch satisfy the scale-collapse drift obstruction

$$\int_{\tau_0}^{\infty} g_{\mathrm{sc}}(\tau)\, d\tau = \infty.$$

A sequence of intervals $[s_m, t_m]$, $s_m \to \infty$, is a normalized negative-drift occupation block sequence if

$$\int_{s_m}^{t_m} g_{\mathrm{sc}}(\tau)\, d\tau = 1 \qquad \text{for every } m.$$

It is diffuse if

$$\sup_{T\in[s_m,t_m]} \int_{[T,T+1]\cap[s_m,t_m]} g_{\mathrm{sc}}(\tau)\,d\tau \longrightarrow 0.$$

The associated occupation measures are

$$d\mu_m(\tau) = g_{\mathrm{sc}}(\tau)\mathbf{1}_{[s_m,t_m]}(\tau)\,d\tau.$$

Definition 9.92 (Canonical thin-drift terminal schedule)**.** In the canonical windowwise pre-carrier regime, after a harmless terminal restriction, the terminal unit-window schedule is

$$K_k = [\tau_\sharp + k, \tau_\sharp + k + 1], \qquad k = 0, 1, 2, \dots.$$

These are the canonical cells on which the windowwise moving family and the ordered local validation rows are tested.

Lemma 9.93 (Canonical terminal windows are admissible or already labelled)**.** *Let a physical local Type II branch be in the canonical windowwise pre-carrier regime, and let K_k be the canonical cells of definition 9.92. For every sequence K_{k_m} with $k_m \to \infty$, after passing to a subsequence, exactly one of the following alternatives holds:*

(A1) *a non-thin first-failure row occurs on these windows: autonomous modulation failure, modulation-driven recentering, compactness-package first failure routed by theorem 6.31, exterior or separated-core concentration, rough-core loss, multicore or same-point cascade, carrier/cost validation failure outside the canonical retained channel, or loss of the retained active core;*

(A2) *after the finite covariant recentering $s = \tau - (\tau_\sharp + k_m)$, the canonical windows K_{k_m} are admissible selected terminal windows for the ordered local validation: the repaired-gauge representation, local pressure reconstruction, local suitable-energy bounds, moving L^2 core floor, retained local critical-mass row, compact tests, and modulation bookkeeping hold on every fixed compact cylinder, with constants independent of m.*

Proof. The canonical windowwise moving family is the schedule on which the carrier, compactness, repaired-gauge, pressure, and modulation rows are validated. The finite translation $\tau \mapsto \tau - (\tau_\sharp + k_m)$ does not transfer any lower bound from one physical time to another; it only writes the canonical cell K_{k_m} as the model interval $[0, 1]$ in its terminal chart, as in the finite recentering step of lemma 9.68. Thus every row in the ordered validation list is tested on the same canonical cells.

Run the finite ordered validation list on K_{k_m}. If any required local row fails before retention, the first failed row is recorded by lemmas 6.25 and 7.70 and belongs to the list in (A1) after excluding the thin-drift row itself. If no such first failure occurs, all retained local inputs listed in (A2) hold on the recentered windows. The moving L^2 core floor is the retained-window branch of theorem 6.21 read on the same canonical terminal windows; if the loss branch of that theorem occurs, the lower active-core row has failed and the branch is in (A1). This proves the dichotomy. ☐

Lemma 9.94 (Thin drift gives diffuse occupation blocks)**.** *Assume $g_{\mathrm{sc}} \ge 0$, $g_{\mathrm{sc}} \in L^1_{\mathrm{loc}}([\tau_0, \infty))$, and*

$$\int_T^\infty g_{\mathrm{sc}}(\tau)\,d\tau = \infty \qquad \textit{for every } T \ge \tau_0.$$

Then every terminal tail contains a normalized negative-drift occupation block sequence. After passing to a subsequence, exactly one of the following holds:

(D1) *there are $\gamma > 0$ and unit windows $U_m = [\sigma_m, \sigma_m + 1]$, $\sigma_m \to \infty$, such that*

$$\int_{U_m} g_{\mathrm{sc}}(\tau)\,d\tau \ge \gamma;$$

(D2) *the retained block sequence is diffuse in the sense of definition 9.91.*

If the branch is in the thin-drift alternative of theorem 6.23, then case (D1) *either produces one of the non-thin first-failure rows of lemma 9.93* (A1)*, or is impossible on the retained selected terminal tail. Hence, after non-thin first failures are removed, only* (D2) *remains.*

Proof. Apply lemma 7.120 with $g = g_{\mathrm{sc}}$. Its block construction gives disjoint terminal intervals $[s_m, t_m]$ with $\int_{s_m}^{t_m} g_{\mathrm{sc}} = 1$. The same lemma gives the two alternatives: persistent unit-window carrier mass or diffuse carrier mass. These are exactly (D1) and (D2) above.

If (D1) holds, compare U_m with the canonical schedule $K_k = [\tau_\sharp + k, \tau_\sharp + k + 1]$. Each unit interval U_m meets at most three canonical cells. Hence, after passing to a subsequence, there are canonical cells K_{k_m} with $k_m \to \infty$ and

$$\int_{K_{k_m}} g_{\mathrm{sc}}(\tau)\, d\tau \geq \gamma/3.$$

Apply lemma 9.93 to these canonical cells. If alternative (A1) occurs, the branch has already left the thin terminal row through a non-thin first failure. Otherwise the canonical cells are admissible selected terminal windows. On those selected windows,

$$\int_{K_{k_m}} a_-(\tau) M_R(\tau)\, d\tau \geq \gamma/3.$$

This is the negation of the thin-drift alternative in theorem 6.23, which was precisely the absence of a uniform positive lower bound for fixed-length averaged weighted negative drift on the selected terminal tail. Thus a retained thin-drift branch with no non-thin first failure can only produce diffuse occupation blocks. □

Lemma 9.95 (Small weighted negative drift is perturbative)**.** *Let $I \subset [\tau_0, \infty)$ be an interval of length at most one. Assume that the branch has a moving L^2 core floor*

$$M_R(\tau) \geq m_0 > 0 \qquad \text{for a.e. } \tau \in I,$$

and that on a compact set $K^+ \Subset \mathbb{R}^3$, containing a unit neighborhood of K, the retained local energy bound

$$\operatorname*{ess\,sup}_{\tau \in I} \|V(\tau)\|_{L^2(K^+)} \leq L$$

holds. If

$$\int_I g_{\mathrm{sc}}(\tau)\, d\tau \leq \varepsilon,$$

then the negative scale correction

$$F_-(\tau) := a_-(\tau)\big(V(\tau) + y \cdot \nabla V(\tau)\big)$$

satisfies

$$\|F_-\|_{L^1(I;H^{-1}(K))} \leq C_K\, L\, m_0^{-1} \varepsilon,$$

where C_K depends only on K and the fixed enlargement K^+. Moreover, for every $\psi \in C_c^\infty(K)$,

$$\int_I |\langle F_-(\tau), V(\tau)\psi\rangle|\, d\tau \leq C_{K,\psi} L^2 m_0^{-1} \varepsilon.$$

Proof. The core floor gives

$$\int_I a_-(\tau)\, d\tau \leq m_0^{-1} \int_I a_-(\tau) M_R(\tau)\, d\tau \leq m_0^{-1} \varepsilon.$$

Let $\varphi \in C_c^\infty(K;\mathbb{R}^3)$ with $\|\varphi\|_{H_0^1(K)} \le 1$. By integration by parts in y,

$$\int_K (y\cdot\nabla V)\cdot\varphi\,dy = -\int_K V\cdot\big(y\cdot\nabla\varphi + 3\varphi\big)\,dy.$$

Hence, for a.e. $\tau \in I$,

$$|\langle V + y\cdot\nabla V, \varphi\rangle| \le C_K\|V(\tau)\|_{L^2(K^+)}\|\varphi\|_{H_0^1(K)} \le C_K L.$$

Taking the supremum over φ and integrating against $a_-(\tau)$ gives

$$\|F_-\|_{L^1(I;H^{-1}(K))} \le C_K L\int_I a_-(\tau)\,d\tau \le C_K L m_0^{-1}\varepsilon.$$

For the local energy pairing, integrate the scale-generator term by parts:

$$\int_K (y\cdot\nabla V)\cdot V\,\psi\,dy = \frac{1}{2}\int_K y\cdot\nabla(|V|^2)\psi\,dy = -\frac{1}{2}\int_K |V|^2(3\psi + y\cdot\nabla\psi)\,dy.$$

Therefore

$$|\langle V + y\cdot\nabla V, V\psi\rangle| \le C_{K,\psi}\|V(\tau)\|^2_{L^2(K^+)} \le C_{K,\psi}L^2.$$

Integrating against a_- and using $\int_I a_- \le m_0^{-1}\varepsilon$ gives the second estimate. No bound on ∇V and no global norm is used. $\square$

Lemma 9.96 (Active core persistence on admissible drift windows)**.** *Let $I_m = [\sigma_m, \sigma_m + 1]$ be admissible selected terminal windows in the sense of lemma 9.93* (A2)*. Then either the loss-of-retained-active-core row occurs, or, after passing to a subsequence, there are a compact cylinder $Q_{r_*} = B_{r_*}\times I_{r_*} \Subset B_R\times(0,1)$ and $\eta_* > 0$ such that the represented sequence on I_m carries the retained active local mass floor*

$$\liminf_{m\to\infty}\int_{I_{r_*}}\int_{B_{r_*}} |V_m|^3\,dy\,ds \ge \eta_*^3,$$

where $V_m(y,s) = V(y,\sigma_m+s)$ and $P_m(y,s) = P(y,\sigma_m+s)$. Consequently the minimal active-family extraction and the same-point/separated reductions are applicable on these windows; if pressure-only CKN detection is needed instead of velocity detection, the branch is routed to the pressure/exterior row rather than retained here.

Proof. The retained local critical-mass row in lemma 9.93 (A2) is interpreted by lemma 9.73. If the lower active mass floor fails, the selected candidate is not a retained active component; this is exactly the loss-of-retained-active-core row.

If the lower active mass floor does not fail, then theorem 8.32 supplies a positive active local spacetime velocity floor on a compact subcylinder of the selected terminal window. A finite cover of the retained compact core and the terminal maximality rule lemma 9.72 allow one fixed compact subcylinder Q_{r_*} and one threshold $\eta_* > 0$ after passing to a subsequence. If the witnesses escaped every such fixed subcylinder, the first label would be exterior/separated concentration, cascade, or compactness failure, all of which are non-thin rows already listed in lemma 9.93 (A1). If the active CKN diagnostic were carried only by pressure while the velocity floor above failed, then lemmas 8.21 and 8.22 routes the branch to the pressure/exterior row, not to the retained thin-drift row. Thus the displayed active spacetime velocity floor holds on the retained branch.

The hypotheses of the minimal active-family extraction and of the same-point/separated reductions are exactly the repaired-gauge representation, local pressure reconstruction, compact-cylinder suitable bounds, and retained active local mass floor now available on Q_{r_*}. Hence those reductions may be applied by the terminal active-frame machinery. $\square$

Lemma 9.97 (Vanishing negative scale forcing has no terminal label)**.** *Let $I_m = [\sigma_m, \sigma_m + 1]$ be admissible selected terminal windows and assume that on every compact $K \Subset \mathbb{R}^3$*

$$a_-(\sigma_m + \cdot)\big(V_m + y \cdot \nabla V_m\big) \longrightarrow 0 \quad \text{in } L^1(0,1;H^{-1}(K)).$$

Assume also the corresponding local energy pairings with compact cutoffs tend to zero. Then the negative scale-drift term contributes no nonzero local defect measure and no independent terminal state-space label. Any retained limit or occupation state obtained from the windows is the same local Navier–Stokes state that would be obtained after moving this term to a vanishing $L^1 H^{-1}_{\mathrm{loc}}$ forcing; the ordered labels are therefore the ordinary active-core, compactness, exterior, multicore/cascade, scale-rigid, or bounded-window labels.

Proof. Let $\varphi \in C_c^\infty((0,1) \times K; \mathbb{R}^3)$. The assumed $L^1 H^{-1}$ convergence gives

$$\int_0^1 \langle a_-(\sigma_m + s)(V_m + y \cdot \nabla V_m), \varphi(s)\rangle \, ds \longrightarrow 0.$$

Hence the negative scale term disappears in the distributional limit of the represented equation. The compact-cutoff energy pairings tend to zero by assumption, so the local energy inequality also passes to the same limit with no additional defect measure. The pressure reconstruction is unchanged: since $\nabla \cdot V_m = 0$, the field $V_m + y \cdot \nabla V_m$ is divergence-free in the spatial variables, so this lower-order velocity forcing contributes no new pressure source.

Thus, in the augmented local compactness topology, the extra coordinate recording the negative scale forcing converges to the zero face. Closedness of the ordered local diagnostics lemmas 7.118 and 9.70 then implies that this vanishing coordinate cannot be the first nonzero label of a terminal state. The only possible first labels are the ordinary local state-space labels listed in the statement.

This is a zero-face statement for the augmented local state space, not a small-data statement for a velocity remainder. The proof uses only distributional closedness, compact-cutoff local-energy closedness, and the closedness of the already defined local diagnostics. No global Kato theory and no small $L^\infty L^3$ velocity remainder is invoked. □

Lemma 9.98 (Diffuse thin drift routes to another local row)**.** *Let a physical local Type II branch be in the canonical windowwise pre-carrier regime and suppose that, after the finite-cost and finite absolute-cost alternatives have been removed, it satisfies the scale-collapse drift obstruction and the thin-drift alternative. Let $[s_m, t_m]$ be the diffuse blocks supplied by lemma 9.94. Then exactly one of the following occurs after passing to a subsequence:*

- (T1) *one of the non-thin local first-failure rows occurs: autonomous modulation failure, modulation-driven recentering, compactness-package first failure routed by theorem 6.31, exterior or separated-core concentration, rough-core loss, multicore or same-point cascade, carrier/cost validation failure outside the canonical retained channel, or loss of the retained active core;*
- (T2) *a closed retained row occurs: compact single-core, scale-rigid residual behavior, retained compact scale-collapse, or a nonzero bounded selected-window limit;*
- (T3) *the negative scale-drift term is perturbative on the canonical selected compact unit windows and no additional terminal thin-drift state remains.*

Proof. Because the blocks are diffuse,

$$\varepsilon_m := \sup_{T \in [s_m, t_m]} \int_{[T, T+1] \cap [s_m, t_m]} g_{\mathrm{sc}}(\tau) \, d\tau \longrightarrow 0.$$

The block lengths tend to infinity. Indeed, if a subsequence of blocks had length bounded by L, each such block would be covered by at most $\lceil L \rceil + 2$ unit intervals, and its total g_{sc}-mass would be bounded by $(\lceil L\rceil+2)\varepsilon_m \to 0$, contradicting the normalization $\int_{s_m}^{t_m} g_{\mathrm{sc}} = 1$. Hence, after discarding a finite prefix, each block contains a canonical unit cell

$$K_{k_m} = [\tau_\sharp + k_m, \tau_\sharp + k_m + 1] \subset [s_m, t_m], \qquad k_m \to \infty.$$

The same estimate holds for every canonical cell contained in the block. In particular, since $K_{k_m} \subset [s_m, t_m]$, diffuseness gives

$$\int_{K_{k_m}} g_{\mathrm{sc}} \le \varepsilon_m.$$

Apply lemma 9.93 to the canonical cells K_{k_m}. If alternative (A1) occurs, the branch is in (T1). Hence assume that alternative (A2) holds: after recentring, K_{k_m} are admissible selected terminal windows with the retained local representation, pressure, energy, compactness, modulation, and moving core-floor data. Write $I_m := K_{k_m}$ for these canonical selected windows.

On these admissible windows, the moving L^2 core floor and the retained local $L^\infty_t L^2_x$ bounds are part of lemma 9.93 (A2). Hence lemma 9.95 implies

$$\|a_-(V + y\cdot\nabla V)\|_{L^1(I_m;H^{-1}(K))} \longrightarrow 0$$

for every compact $K \Subset \mathbb{R}^3$, and its compact-cutoff local energy pairings also vanish. By lemma 9.97, the diffuse negative drift is a vanishing local forcing on the selected unit windows; it is not a new compact state-space object and cannot be a terminal first label.

It remains to classify the retained compact sequence on these same admissible windows. Apply lemma 9.96. If the active-core floor is lost, the branch is in (T1). Otherwise a fixed compact subcylinder carries a positive retained active local mass floor, so the structural reductions are applicable. Apply the minimal active-family extraction and the same-point/separated-point reductions given by theorems 4.11, 4.18 and 4.19. If those reductions produce another active core, a strict same-point cascade, a separated physical point, exterior pressure persistence, rough-core loss, or loss of compact estimates, then the branch is in (T1). If they produce the compact single-core row, the retained compact scale-collapse row, or the scale-rigid residual row, those rows are the closed alternatives in (T2), closed respectively by theorems 6.34 and 10.5 and proposition 4.40. In the scale-rigid case, if the compact limit has no further CKN subconcentration, lemma 4.35 gives a nonzero bounded selected-window limit; if it has a further subconcentration, the same lemma routes it to the active structural rows already listed in (T1). Finally, lemma 4.25 excludes the bounded-limit case from the local Type II class.

Thus, after all ordered first failures and closed retained rows have been accounted for, the only remaining effect of thin negative drift on the selected unit windows is the vanishing H^{-1}_{loc} perturbative term proved above. By lemma 9.97, such a vanishing forcing carries no independent terminal state-space label. This proves (T3) and the asserted trichotomy. □

Theorem 9.99 (Thin negative drift is not a terminal exit)**.** *On every physical local Type II branch in the canonical windowwise pre-carrier regime, the thin negative-drift alternative is not an independent remaining terminal exit. More precisely, if the branch enters the scale-collapse drift obstruction and does not have persistent fixed-length negative-drift windows, then the branch either enters one of the other named local exit rows, or it enters an already closed retained row. Consequently the row "thin drift or negative scale-drift without thick autonomous windows" may be removed from definition 10.27.*

Proof. The scale-collapse drift obstruction gives the infinite tail integral of g_{sc}. By lemma 9.94, every terminal tail has either persistent unit-window weighted negative drift or diffuse normalized drift blocks. In the first alternative, finite overlap with the canonical schedule of definition 9.92 gives canonical cells carrying a uniform positive weighted negative-drift lower bound, unless one of the non-thin first-failure rows in lemma 9.93(A1) has already occurred. If such a first failure occurs, the branch has left through another named row. Otherwise those canonical cells are admissible selected terminal windows, and the branch is in the thick-window ledger of theorem 6.23: selected-window compactness can fail, autonomous modulation convergence can fail, or the retained thick compact scale-collapse state is reached. The first two are other named rows, and the retained thick state is closed by theorem 6.34. No part of this persistent-window branch validates arbitrary intervals or uses the diffuse thin-drift closure itself.

On a genuine thin-drift tail, only the diffuse block alternative remains. Lemma 9.98 then gives three possibilities. The first is one of the other named local exit rows. The second is a closed retained row, including the bounded selected-window exit excluded by lemma 4.25. The third says that the negative drift is a vanishing local H^{-1} perturbation on the selected compact windows and therefore does not define a terminal state-space row after the retained perturbative remainder has been formed.

Thus no local Type II branch can remain whose first and only unresolved label is thin negative drift. The argument used only finite-window local estimates, occupation blocks, the retained active-core floor, and the ordered local state-space partition; it did not use a global L^3 bound, a whole-space profile decomposition, or a hidden conversion of diffuse drift into thick windows. □

9.12. **Closure of the Type II analytic-data exhaustiveness row.** The exhaustion row behind theorem A.4 is the entry condition for every admissible local Type II branch. It is closed by an ordered local extraction package: each component of definition A.2 is either supplied by an existing local lemma on the admissible class $\mathcal{U}_{\mathrm{II}}^{NS}$, or its first failure is routed to a named alternative that is class-incompatible or already covered by the local state-space stratification. No global critical profile theorem and no global L^3 bound are invoked.

Lemma 9.100 (Concentration extraction routes)**.** *Let $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$. Exactly one of the following holds:*

- (a) *the local concentration dichotomy theorem 2.5 produces $x_0 \in \Sigma(T^*)$, $\eta > 0$, and $r_n \downarrow 0$ with $C((x_0, T^*), r_n) + D((x_0, T^*), r_n) \geq \eta$ for all n, and the concentration point lies in the local Type II alternative;*
- (b) *the singular set $\Sigma(T^*)$ is empty;*
- (c) *$\Sigma(T^*) \neq \emptyset$, and every concentration point of $\Sigma(T^*)$ lies in the local Type I alternative.*

Cases (b) *and* (c) *are class-incompatible with definition A.1. Hence on $\mathcal{U}_{\mathrm{II}}^{NS}$ only case* (a) *occurs, and the first ordered failure of definition A.3 ("concentration extraction fails") does not arise on the admissible class.*

Proof. If $\Sigma(T^*) = \emptyset$, case (b) holds. If $\Sigma(T^*) \neq \emptyset$, apply the local concentration dichotomy at singular points. If some singular point is not in the local Type I alternative, choose that point as x_0; the Caffarelli–Kohn–Nirenberg failure at x_0 supplies $\eta > 0$ and $r_n \downarrow 0$ with

$$C((x_0, T^*), r_n) + D((x_0, T^*), r_n) \geq \eta,$$

and case (a) holds. If no such point exists, every concentration point lies in the local Type I alternative and case (c) holds. The three cases are therefore disjoint and exhaustive.

In case (b) there is no terminal singular concentration point, so the branch is the local continuation/no-terminal-concentration alternative rather than a local Type II branch. This is excluded by clause (iv) of definition A.1; no global continuation estimate is used. In case (c) every concentration point lies in the local Type I alternative of theorem 2.5, which is exactly the local Type I scale-invariant criterion. This contradicts clause (iii) of definition A.1. In case (a) the dichotomy supplies the positive CKN concentration scales and the exclusion of the Type I alternative at (x_0, T^*); this is the content of the first analytic datum of definition A.2. □

Lemma 9.101 (Supercritical scale alternative routes)**.** *Let $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ and let (x_0, T^*) be a concentration point produced by lemma 9.100*(a)*. Then the supercritical scale alternative of definition A.2*(ii) *holds at (x_0, T^*).*

Proof. The dichotomy of theorem 2.5 at (x_0, T^*) is exclusive: either a local Type I scale-invariant bound holds, or no such local Type I bound holds. The first option contradicts clause (iii) of definition A.1, which excludes branches in the local Type I class. Hence the second option holds at every concentration point produced by the previous lemma; this is exactly the supercritical Type II scale alternative. The second ordered failure of definition A.3 therefore does not arise on $\mathcal{U}_{\mathrm{II}}^{NS}$. □

Lemma 9.102 (Local profile data routes)**.** *Let $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ and let (x_0, T^*) be a concentration point in the local Type II alternative as produced by lemmas 9.100 and 9.101. Run the local scale-translation selection of proposition 2.6 at the positive concentration scales and test whether it produces a nondegenerate compact single core. Then exactly one of the following holds:*

- (a) *the localized scale and center are nondegenerate and no represented selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder; proposition 2.7 supplies represented variables (V_n, P_n, a_n, b_n) on compact cylinders satisfying the represented Navier–Stokes equation* (1)*; the represented selected windows form a local Type II sequence in the sense of definition 2.2, providing the local profile data of definition A.2*(iii)*;*
- (b) *the localized nondegeneracy fails, and the branch falls into the multibubble, cascade, or gauge-degenerate alternative of definition 4.6*(ii)–(v)*;*
- (c) *nondegeneracy holds but the represented selected windows admit a nonzero bounded selected-window limit on some compact cylinder; the branch then enters the bounded-selected-window alternative of definition 2.1 and exits Type II by lemma 4.25.*

In cases (b) *and* (c) *the branch is non-retained: case* (b) *is treated by the multibubble/cascade machinery of Section 4, and case* (c) *is the bounded selected-window exit of the scale-rigid stratum. Neither case is a compact single-core branch.*

Proof. The local scale-translation selection at (x_0, T^*) attempts to extract a single compact core at the positive CKN concentration scales $r_n \downarrow 0$. Either the localized scale and center are nondegenerate (cases (a) and (c)), or they are degenerate (case (b)); under nondegeneracy, either no represented selected-window subsequence has a nonzero bounded selected-window limit (case (a)), or some such subsequence does (case (c)). The three cases are exhaustive.

In case (a), proposition 2.6 applies and rewrites the shrinking cylinders $Q_{r_n}(x_0, T^*)$ as fixed selected windows. The represented sequence has positive local CKN concentration on a fixed compact cylinder; combined with the defining case-(a) absence of nonzero bounded selected-window limits, it is a local Type II sequence in the sense of definition 2.2. Proposition 2.7 then supplies the represented variables (V_n, P_n, a_n, b_n) on compact cylinders satisfying (1); this is the local profile datum of definition A.2 (iii).

In case (b), failure of localized nondegeneracy is, by the second sentence of proposition 2.7, the multibubble, cascade, or gauge-degenerate alternative. The corresponding stratum is named in definition 4.6 (ii)–(v) and is non-retained: it is handled by the multibubble/cascade machinery and never enters the retained compact single-core stratum produced by lemma 7.70.

In case (c), the represented selected-window sequence has a nonzero bounded selected-window limit on some compact cylinder, placing the branch in definition 2.1; lemma 4.25 then routes the branch out of Type II via the scale-rigid bounded-window exit, which is also non-retained.

Hence the third ordered failure of definition A.3 ("local profile data are insufficient") is routed to the multibubble/cascade/gauge-degenerate non-retained alternative or to the scale-rigid bounded-window exit; on the retained compact single-core stratum it does not arise. □

Theorem 9.103 (Type II analytic-data routing on the admissible class)**.** *The Type II analytic data of definition A.2 are exhaustive by ordered local routing on $\mathcal{U}_{\mathrm{II}}^{NS}$: every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ either has the concentration profile, the supercritical scale alternative, and the local profile data, or its first applicable failure in definition A.3* (i)–(iv) *is matched to a named alternative that is class-incompatible with definition A.1 or to a non-retained state-space branch already covered by the local stratification. In particular, after those named exits are removed, every retained admissible branch has the three Type II analytic data.*

Proof. Process the three analytic data of definition A.2 in the order specified by definition A.3.

Datum (i): concentration extraction. Lemma 9.100 shows that on $\mathcal{U}_{\mathrm{II}}^{NS}$ the only realised case is the dichotomy case (a) producing (x_0, T^*) with positive CKN concentration in the local Type II alternative; the other two cases are class-incompatible.

Datum (ii): supercritical scale alternative. Lemma 9.101 then forces the supercritical Type II scale alternative at (x_0, T^*); the Type I alternative is class-incompatible by clause (iii) of definition A.1.

Datum (iii): local profile data. Lemma 9.102 either supplies the local profile data via the local scale-translation selection and the representation input proposition 2.7, or routes the branch to the multibubble/cascade/gauge-degenerate non-retained alternative definition 4.6 (ii)–(v), or to the bounded selected-window exit lemma 4.25.

Residual datum (iv). The fourth ordered failure of definition A.3, "the admissible Type II branch lies outside the concentration, scale, and profile alternatives under consideration", is the catch-all for a branch that is not one of the local Navier–Stokes Type II alternatives after the first three tests have been applied. Clause (iv) of definition A.1 excludes continuation, Type I, boundary, and non-Navier–Stokes-domain alternatives. Any remaining local branch outside the retained profile alternatives is one of the named non-retained state-space exits produced in the preceding steps.

In every case the branch either has all three analytic data or its first ordered failure is recorded as a class-incompatible alternative or as a non-retained state-space branch already covered by the local stratification. This is the ordered local-routing exhaustiveness statement associated with corollary A.5. Removing the named exits leaves only the first alternative of that corollary, namely the full Type II analytic data. □

Corollary 9.104 (Type II analytic-data exhaustiveness discharged)**.** *The ordered-routing exhaustion input used by theorem A.4 is a theorem on $\mathcal{U}_{\mathrm{II}}^{NS}$: the analytic-data entry row is discharged by ordered local extraction. On retained admissible branches the three Type II analytic data hold; non-retained failures are assigned to named local alternatives before the downstream compact analysis is invoked. No global critical profile theorem and no global L^3 bound are invoked.*

Proof. Theorem 9.103 establishes ordered local routing of all failures of the Type II analytic data on $\mathcal{U}_{\mathrm{II}}^{NS}$. By corollary A.5, after the named routed exits are removed, the remaining retained branch is in the first alternative and therefore has the Type II analytic data required by theorem A.4. Each step uses only the local concentration dichotomy theorem 2.5, the local scale-translation selection proposition 2.6, the local representation input proposition 2.7, the local Type II definition definition 2.2, the admissibility clauses of definition A.1, and the multibubble/cascade stratum definition 4.6, together with the bounded-window exit lemma 4.25. □

Definition 9.105 (Terminal local compactness input hypotheses)**.** The terminal local compactness *input* hypotheses consist of the local compact terminal-window hypothesis definition 9.81

and local perturbative stability definition 9.83, together with item (i) of definition 9.67 (the admissibility estimates on the represented branch). These hypotheses do *not* include the local terminal completeness conclusion (iii) of definition 9.67.

Definition 9.106 (Terminal local compactness package)**.** The terminal local compactness package is the input hypotheses definition 9.105 together with the local terminal completeness theorem theorem 9.79. This is a record of what has been proved and is not an additional assumption.

Theorem 9.107 (Terminal critical local compactness)**.** *Assume the terminal local compactness input hypotheses of definition 9.105. Assume also that the ordered adjacent local labels are routed as in theorem 9.74 and that comparable retained classes are grouped before the terminal frame is fixed. Then, on every retained terminal branch and every compact terminal cylinder, all nonretained residual components vanish in the local perturbative topology of definition 8.13.*

Proof. The local compact terminal-window hypothesis gives the compact-cylinder estimates. Applying theorem 9.79 under the stated routing and grouping gives

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_R))$$

on each compact terminal cylinder. Local perturbative stability then converts this velocity convergence into the nonlinear-stress, pressure, and residual convergences required in terminal nonlinear decoupling. $\square$

Corollary 9.108 (Terminal completeness from local compactness hypotheses)**.** *The terminal local completeness statement in the final argument may be replaced by the terminal local compactness package of definition 9.106.*

Proof. Theorem 9.107 gives the asserted terminal local completeness. $\square$

Remark 9.109 (Boundary of the terminal compactness hypothesis)*.* The preceding compactness conclusion applies only to terminal windows on which the repaired-gauge representation, pressure reconstruction, local Caccioppoli control, bounded modulation, and local state-space labels are available. If one of these inputs is absent, the branch is assigned to the corresponding representation, pressure, rough-core, modulation, or state-space alternative; it is not treated by importing a global profile theorem.

Definition 9.110 (Terminal sequences sampled from the renormalized orbit)**.** A terminal active sequence is sampled from the renormalized orbit if it is of the form

$$u_{0,n} = \Lambda_{\lambda_n, x_n} V(\tau_n), \qquad \tau_n \to \infty, \qquad \lambda_n > 0, \qquad x_n \in \mathbb{R}^3,$$

or by the equivalent inverse critical change of variables, where

$$(\Lambda_{\lambda,x_0} f)(x) = \lambda^{-1} f\left(\frac{x - x_0}{\lambda}\right).$$

Lemma 9.111 (Critical symmetry preserves the L^3 norm)**.** *For every $f \in L^3(\mathbb{R}^3)$, $\lambda > 0$, and $x_0 \in \mathbb{R}^3$,*

$$\|\Lambda_{\lambda,x_0} f\|_{L^3} = \|f\|_{L^3}.$$

Proof. By the change of variables $x = x_0 + \lambda y$,

$$\|\Lambda_{\lambda,x_0} f\|_{L^3}^3 = \int_{\mathbb{R}^3} \lambda^{-3} \left| f\left(\frac{x - x_0}{\lambda}\right)\right|^3 dx = \int_{\mathbb{R}^3} |f(y)|^3 \, dy.$$

Taking cube roots gives the claim. $\square$

Theorem 9.112 (Bounded terminal sequences from local critical bounds)**.** *Assume the renormalized orbit satisfies the local compact-window bounds*

$$0 < \inf_{\tau\in I} \|V(\tau)\|_{L^3(B_R)} \leq \sup_{\tau\in I} \|V(\tau)\|_{L^3(B_{2R})} < \infty,$$

on the terminal interval under consideration, together with the local Caccioppoli and pressure bounds on $Q_{2R,I}$. If every terminal active window is sampled from the renormalized orbit, then every terminal active window satisfies the local compact terminal-window hypothesis.

Proof. The local L^3 upper bound is one component of the compact-window hypothesis. Local $L^\infty L^2$, L^2H^1, and pressure bounds are supplied by the local Caccioppoli estimate and pressure reconstruction on $Q_{2R,I}$. The repaired-gauge equation gives the required time-derivative bound in H^{-m}. The lower bound records that the retained window is nontrivial. □

Theorem 9.113 (Terminal compactness alternatives)**.** *For a candidate terminal active sequence in the Type II analysis, the following alternatives exhaust the possible local compactness outcomes, after passing to a subsequence when necessary:*

(i) *the sequence is not sampled from the renormalized orbit;*
(ii) *the repaired-gauge representation, pressure reconstruction, or local Caccioppoli estimate is unavailable on the selected compact cylinder;*
(iii) *the local L^3 upper bound fails along a terminal sampling subsequence;*
(iv) *the retained local L^3 lower bound vanishes along a terminal sampling subsequence;*
(v) *after excluding the preceding alternatives, the terminal active sequence satisfies the local compact terminal-window hypothesis.*

Proof. If the terminal sequence is not sampled from the renormalized orbit, then (i) holds. If one of the local PDE inputs is missing, then (ii) holds and the branch is assigned to the corresponding local alternative. If the local critical upper bound fails, then (iii) holds after passing to a subsequence. If the retained lower bound vanishes, then (iv) holds. Otherwise the local hypotheses of theorem 9.112 hold, and (v) follows. □

Definition 9.114 (Terminal coordinate-frame construction)**.** The terminal coordinate-frame construction means that every terminal active frame used in the analysis is obtained from the repaired-gauge orbit as follows. Choose sampling times $\tau_n \to \infty$ and critical symmetry parameters (λ_n, x_n); form the local terminal windows by applying the critical coordinate-frame map to V on compact time intervals around τ_n; group comparable retained local classes based at the same physical point before selecting the terminal active frame. No external data are introduced.

Theorem 9.115 (Terminal coordinate-frame construction from local state selection)**.** *Assume terminal active frames are selected by first partitioning local active states by physical concentration point, then grouping comparable retained states at the same point into compound local states, and finally choosing a minimal-scale retained class at that point. Then the terminal coordinate-frame construction of definition 9.114 holds for the selected terminal frames.*

Proof. By the stated selection rule, each terminal active frame is attached to a compound active class after all comparable retained states at the same physical point have been grouped. The scale and center of the terminal frame are the scale and center of this compound class. The terminal data are then obtained by applying the Navier–Stokes critical change of variables to the local renormalized branch along the chosen sampling sequence $\tau_n \to \infty$. Thus the construction uses only the repaired-gauge orbit and the Navier–Stokes critical symmetries, and it satisfies all clauses of definition 9.114. □

Theorem 9.116 (Terminal coordinate frames are sampled from the orbit)**.** *Assume the terminal coordinate-frame construction. Then every terminal active sequence belongs to the class described in definition 9.110.*

Proof. By definition of the terminal coordinate-frame construction, each terminal active frame is obtained by choosing times τ_n, selecting scale and center parameters (λ_n, x_n), and applying the Navier–Stokes critical change of variables to the renormalized solution $V(\tau_n)$. Hence the associated terminal data have the form

$$u_{0,n} = \Lambda_{\lambda_n,x_n} V(\tau_n),$$

or the equivalent inverse formulation, which is the orbit-sampling condition. □

Theorem 9.117 (Repaired-gauge representation gives terminal orbit sampling)**.** *Assume a repaired-gauge representation of the Type II branch and the terminal coordinate-frame construction. Then every terminal active sequence is sampled from the renormalized orbit.*

Proof. The repaired-gauge representation supplies the renormalized orbit $V(\tau)$ and the critical coordinate maps between physical variables and renormalized variables. The terminal coordinate-frame construction supplies the terminal sampling times and scale-center parameters. Combining these hypotheses gives

$$u_{0,n} = \Lambda_{\lambda_n,x_n} V(\tau_n),$$

which is the assertion of definition 9.110. □

Corollary 9.118 (Failure of terminal orbit sampling)**.** *If a sequence reaches the terminal local-state stage but is not sampled from the renormalized orbit, then either the repaired-gauge representation is not available for that sequence or the terminal coordinate-frame construction has not been applied as specified.*

Proof. This is the contrapositive of theorem 9.117. □

9.13. **Terminal Retained-State Assembly.**

Definition 9.119 (Abstract terminal elimination data)**.** The abstract terminal elimination data for this section are:

(i) every renormalized local Type II branch enters one of the local alternatives used in theorem 10.11;
(ii) compact single-core vanishing-dissipation branches are excluded by theorem 10.5;
(iii) multibubble, cascade, and gauge-degenerate branches are excluded by theorem 10.6, and scale-rigid terminal branches are excluded by proposition 4.40;
(iv) finite-cost scale-collapse branches are excluded by theorem 10.8, while compact retained single-core branches with divergent localized cost are covered by theorem A.21 and the cost-divergence theorem theorem 9.42 and corollary 9.43; scale-collapse branches with divergent negative scale-drift cost are covered by theorem 9.33 or corollary 9.37 and then excluded by the same cost-divergence theorem;
(v) failure of uniform critical tightness is excluded, equivalently uniform critical tightness holds for every renormalized branch in the class;
(vi) failure of local windowed H^1 control is excluded, equivalently the required local windowed H^1 control holds for every renormalized branch in the class;

Theorem 9.120 (Abstract terminal Type II exclusion)**.** *Under the abstract terminal elimination data of definition 9.119, no renormalized local Type II branch in the stated class remains.*

Proof. Let a renormalized local Type II branch be given. By item (i), it lies in one of the alternatives in theorem 10.11. Item (ii) excludes the vanishing-dissipation compact single-core subcase, while item (iv) excludes the retained compact single-core subcase with divergent localized cost. Item (iii) excludes multibubble, cascade, gauge-degenerate, and scale-rigid alternatives. Item (iv) also excludes finite-cost scale collapse and the divergent negative-drift scale-collapse branch whenever the corresponding cost is one of the costs covered by the cost-divergence theorem theorem 9.42 and its status corollary corollary 9.43. Items (v) and (vi) exclude failure of uniform critical tightness and failure of local windowed H^1 control. These cases exhaust the alternatives, so no branch remains. ☐

Definition 9.121 (Expanded terminal data)**.** The following analytic data imply the abstract terminal elimination data:

(i) the local alternative decomposition for renormalized Type II branches;
(ii) the compact single-core vanishing-dissipation, multibubble/cascade, finite-cost scale-collapse, compact-cost-divergence, and scale-rigidity exclusions from the preceding sections;
(iii) a critical-tightness estimate giving uniform critical tightness;
(iv) an exhaustive terminal compactness partition and a local finite critical-mass statement for retained active terminal strata;
(v) local perturbative residual stability and terminal state-space completeness on compact windows;
(vi) removal of physically exterior regular components;
(vii) the repaired-gauge representation theorem;
(viii) the local Caccioppoli/windowed regularity input;
(ix) the local compact scale-collapse routing theorem theorem 6.34.

Lemma 9.122 (Terminal data imply critical tightness)**.** *The expanded terminal data imply uniform critical tightness.*

Proof. The repaired-gauge representation turns each renormalized orbit into the terminal windows used in the local state-space analysis. The terminal compactness partition, finite critical-mass statement, local perturbative stability, and terminal state-space completeness give the required terminal completeness. Exterior regular components are removed by the exterior-removal datum in definition 9.121. Together with the multibubble and scale-collapse exclusions, this leaves only possible loss of critical tightness; the critical-tightness estimate excludes that loss. ☐

Lemma 9.123 (Terminal data imply windowed H^1 control)**.** *The expanded terminal data imply local windowed H^1 control.*

Proof. The windowed H^1 estimate requires a repaired-gauge branch, local pressure and modulation control, compact-cylinder critical bounds, and the Caccioppoli/windowed regularity input. These are among the expanded terminal data. Hence the required local windowed H^1 control holds, and the failure of local windowed H^1 control is unavailable. ☐

Lemma 9.124 (Expanded terminal data imply abstract terminal data)**.** *The expanded terminal data imply the abstract terminal elimination data of definition 9.119.*

Proof. Items (i) and (ii) of definition 9.121 give the local alternative decomposition and the compact vanishing-dissipation, multibubble, scale-collapse, compact-cost-divergence, and scale-rigidity exclusions. The cost-divergence criterion used in the divergent retained-core subcase is supplied by the cost-divergence theorem theorem 9.42 and its status corollary corollary 9.43. By lemma 9.122 they also give critical tightness, and by lemma 9.123 they give the local windowed H^1 control. These are the data listed in definition 9.119. ☐

Theorem 9.125 (Expanded terminal Type II exclusion)**.** *If the expanded terminal data hold, then no renormalized local Type II branch in the stated class remains.*

Proof. Lemma 9.124 gives the abstract terminal elimination data. Applying theorem 9.120 gives the exclusion. □

Remark 9.126 (Branchwise reading). Compact single-core branches are split by lemma 10.10: vanishing-dissipation cores are removed by the compact criterion, and divergent retained-local-cost cores are removed by theorem A.21 and corollary 9.43. Retained scale-collapse branches are removed by the local compact scale-collapse routing theorem theorem 6.34, or by theorem 9.33 when the scale-collapse drift cost is compatible with the cost-divergence theorem; multibubble and non-tight cases are removed by the terminal profile and critical-tightness inputs; failures of local windowed H^1 control are removed by the local Caccioppoli estimate; and every divergent compatible localized cost on the canonical windowwise schedule is handled by theorem 9.42.

Theorem 9.127 (Expanded terminal data hold branchwise)**.** *Let ω be a physical local Type II branch after corollary 10.19. Then either ω exits through a named local or ambient alternative assigned to its specified closure row in the acyclic local state-space order, or the nine items of definition 9.121 hold on a retained terminal tail.*

Proof. The proof is an ordered verification of the nine terminal inputs.

Local decomposition and structural exclusions. Covered entry supplies a positive local Type II concentration sequence or an earlier named exit. Such an exit is not silently discarded; it is assigned to the closure row specified by corollary 7.136 and lemma 9.45. In the retained case theorem 10.11 gives the local alternative decomposition. The compact single-core alternative is split by lemma 10.10: the vanishing-dissipation subcase is closed by theorem 10.5, and the divergent retained-local-cost subcase is closed, after the carrier and adjacent rows have been verified, by theorem A.21 and corollary 9.43. The multibubble/cascade, finite-cost scale-collapse, retained compact scale-collapse, and scale-rigid alternatives are closed by theorems 6.34, 10.6 and 10.8 and proposition 4.40.

Critical tightness and compact terminal completeness. On the retained compact stratum, critical tightness and local windowed H^1-control are theorems by theorem 9.89. If either test fails before the retained stratum is reached, the failure is an exterior, rough-core, or multibubble/cascade state-space exit, routed by theorems 8.38, 10.6 and 10.7. The finite critical-mass input used here is local on compact terminal windows: lemma 9.73 proves that failure of positive finite local L^3 mass is either non-retention or an adjacent active label. The proof does not require the whole-space critical norm $\|V(\tau)\|_{L^3(\mathbb{R}^3)}$. With this local finite-mass input, the terminal compactness partition is supplied by theorem 9.113, and terminal completeness on the retained branch is supplied by theorem 9.107. In the retained compact-cost path, the compactness actually used by the cost-divergence argument is supplied from the local compact tests by lemma A.32. The adjacent labels in the partition are routed by theorem 9.74. That theorem only removes adjacent labels from the retained terminal component; the later exclusion of carrier and cost adjacent rows is supplied in the assembly layer by the ambient adjacent wrapper corollary 7.136.

Residual stability, representation, exterior removal, and local regularity. Terminal state-space completeness is theorem 8.16, proved locally from the discharged critical profile decomposition theorem 9.79. Local perturbative residual stability is corollary 9.84. Physically exterior regular components are removed by theorem 8.38. The repaired-gauge representation is the ordered consequence of corollaries A.12 and A.19 together with the absolutely continuous chart consequences theorem 9.55; if the chart or gauge row fails, that failure is the representation alternative and not a retained terminal branch. The local Caccioppoli/windowed regularity input is supplied by theorem 9.60 and corollary 9.61 on suitable repaired-gauge branches, with retained windowed control supplied by theorem 9.89. Finally, compact scale-collapse routing is exactly theorem 6.34.

Thus each item in definition 9.121 is either verified on the retained terminal tail or its first failure is one of the named local or ambient alternatives assigned to a closure row before terminal retained-state elimination is invoked. The dependency order is the one recorded in lemma 9.45: carrier and adjacent rows are routed before cost-divergence closure, and terminal retained-state elimination is last. The verification uses only local compact windows, ordered state-space labels, suitable local energy, and compact terminal estimates; it does not import a global profile theorem, global Kato smallness, or a whole-space L^3 bound. □

Corollary 9.128 (Expanded terminal wrapper discharged)**.** *The expanded terminal data are verified on the retained terminal path: every physical local Type II branch either satisfies those data on a retained terminal tail or exits through a named local or ambient alternative assigned to the closure row specified by the acyclic local state-space order. This corollary verifies the retained terminal inputs; it does not by itself exclude every named exit row.*

Proof. This is theorem 9.127. □

10. Assembly of the Local Type II Exclusion

10.1. **Local Entry.** Let (u,p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + (u\cdot\nabla)u + \nabla p = \nu\Delta u, \qquad \nabla\cdot u = 0,$$

on $\mathbb{R}^3\times(T-\delta,T)$. For $z_0=(x_0,T)$ and $r>0$, set

$$Q_r(z_0) = B_r(x_0)\times(T-r^2,T),$$
$$C(z_0,r) = r^{-2}\int_{Q_r(z_0)} |u|^3\,dx\,dt,$$

and

$$D(z_0,r) = r^{-2}\int_{T-r^2}^{T}\int_{B_r(x_0)} |p(x,t)-(p)_{B_r(x_0)}(t)|^{3/2}\,dx\,dt.$$

The pressure is always taken modulo functions of time, and the spatial mean normalization is the one used in the Caffarelli–Kohn–Nirenberg criterion.

Definition 10.1 (Singular set and local Type II point)**.** Let $\Sigma(T)$ be the set of points x_0 for which u is not locally bounded in any backward cylinder $Q_r(x_0,T)$. A point $x_0\in\Sigma(T)$ is in the local Type II alternative if it carries a positive local Caffarelli–Kohn–Nirenberg concentration sequence and no local Type I scale-invariant bound holds at (x_0,T).

Definition 10.2 (Positive local Type II concentration sequence)**.** A sequence $r_n\downarrow 0$ is a positive local Type II concentration sequence at (x_0,T) if there exists $\eta>0$ such that

$$C((x_0,T),r_n) + D((x_0,T),r_n) \ge \eta \qquad \text{for all } n,$$

and (x_0,T) is in the local Type II alternative. The associated parabolic rescalings are

$$u_n(y,s) = r_n u(x_0+r_n y, T+r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0+r_n y, T+r_n^2 s).$$

Theorem 10.3 (Local concentration and Type I/Type II dichotomy)**.** *If $\Sigma(T)\neq\emptyset$, then for every $x_0\in\Sigma(T)$ there are $\eta>0$ and $r_n\downarrow 0$ such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \ge \eta \qquad \textit{for all } n.$$

At the same point exactly one of the following alternatives holds: there is a local Type I scale-invariant bound, or (x_0,T) is in the local Type II alternative.

Proof. This is the pointwise CKN concentration dichotomy of theorem 2.5, restated in the notation used in the final assembly. □

10.2. **Local Theorems Used in the Assembly.** The local results needed for the final assembly are restated in the following form.

Theorem 10.4 (Repaired-gauge representation)**.** *Let a positive local Type II concentration sequence contain a nondegenerate selected core. Then, after passing to selected windows, there are local scale-center variables $\lambda(\tau)>0$, $x_c(\tau)$, renormalized fields*

$$V(y,\tau) = \lambda(\tau)u(x_c(\tau)+\lambda(\tau)y, t(\tau)), \qquad P(y,\tau) = \lambda(\tau)^2 p(x_c(\tau)+\lambda(\tau)y, t(\tau)) + \pi(\tau),$$

and modulation coefficients a,b such that

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0.$$

On each compact renormalized window one has the local suitable energy inequality, local pressure reconstruction modulo functions of time, stability under subsequences, and the compact-window bounds proved in Section 3. Failure of the nondegenerate scale-center selection is part of the multibubble, cascade, or scale-rigidity alternatives.

Proof. Theorem 3.35 is precisely the compact-window representation theorem. □

Theorem 10.5 (Compact single-core exclusion)**.** *No compact single-core vanishing-dissipation branch can be a local Type II sequence. More explicitly, let a represented selected-window sequence have:*

(i) *retained positive local velocity concentration on a fixed compact cylinder;*
(ii) *uniformly bounded compact-cylinder suitable estimates $A + E + C + D$;*
(iii) *nondegenerate local scale-center representation;*
(iv) *bounded modulation coefficients;*
(v) *local pressure reconstruction modulo spatial means;*
(vi) *vanishing localized dissipation on some retained compact cylinder.*

Then the sequence is incompatible with the local Type II condition.

Proof. Theorem 2.18 gives the stated exclusion. □

Theorem 10.6 (Multibubble and cascade exclusion)**.** *In the local terminal class where terminal admissibility is supplied by proposition 4.41 and scale-rigidity is discharged by proposition 4.40, no local multibubble, cascade, or gauge-degenerate Type II branch can occur in the class produced by the local active-core extraction. Equivalently, every failure of the single nondegenerate compact-core alternative is excluded by the local multibubble/cascade analysis.*

Proof. Theorem 4.42, together with its removal corollary corollary 4.43, gives the stated exclusion. □

Theorem 10.7 (Rough-core reduction)**.** *On any compact renormalized window satisfying the local energy inequality and bounded modulation hypotheses, finite outer velocity-pressure CKN control $\mathcal{C} + \mathcal{D}$ implies finite inner windowed H^1 control. Hence failure of local windowed H^1 control on a retained inner cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on every relevant enclosing cylinder.*

Proof. Corollary 5.7 gives the stated reduction. □

Theorem 10.8 (Finite-cost scale-collapse exclusion)**.** *No renormalized branch with finite scale-collapsing cost, and hence no branch with finite absolute scale cost, can have genuine scale collapse together with a nonvanishing selected core.*

Proof. Theorem 6.7 and corollary 6.9 give the stated exclusion. □

Theorem 10.9 (Scale-collapse decomposition of alternatives)**.** *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. After passing to selected terminal windows, exactly one of the alternatives isolated in Section 6 occurs:*

(i) *finite scale-collapsing cost;*
(ii) *finite absolute scale cost;*
(iii) *thin-drift alternative, routed by theorem 9.99;*
(iv) *autonomous-modulation alternative;*
(v) *compactness-package first failure, routed by theorem 6.31;*
(vi) *a nonzero autonomous reduced limit in the scale-rigid state.*

In the local Type II class of alternatives used by the multibubble/cascade and scale-collapse analyses, alternatives (iii)–(vi) *do not constitute additional compact single-core mechanisms: thin drift is routed to another local row or a closed retained row, the modulation alternative lies outside the retained compact scale-collapse branch, the compactness-package failure is routed to a more specific named local row, and the remaining retained scale-collapse state is removed by theorem 6.34.*

Proof. The assertion follows by combining the scale-collapse decomposition of alternatives with the local compact scale-collapse routing theorem theorem 6.34. □

Lemma 10.10 (Compact retained core dissipation dichotomy)**.** *Let a represented retained compact core carry positive local Type II concentration on a fixed compact cylinder and satisfy the retained compact-cylinder bounds. Then, after passing to selected terminal windows, one of the following ordered alternatives holds:*

- (i) *there is a retained compact cylinder on which the selected windows have vanishing localized dissipation, hence the branch is a compact single-core vanishing-dissipation branch in the sense of definition 2.3;*
- (ii) *an exterior, rough-core, modulation, scale, carrier, or cost adjacent label occurs before the branch is a retained compact single-core state;*
- (iii) *on a retained core cylinder carrying the positive local velocity concentration, the localized dissipation has divergent tail integral, hence the canonical localized cost diverges for every cutoff which is identically one on that core.*

Proof. Fix a retained cylinder Q_ρ on which the positive velocity core is recorded. If

$$\int_{\tau_0}^{\infty} \int_{B_\rho} |\nabla V|^2 \, dy \, d\tau < \infty,$$

then choose retained selected windows I_{n_j}, with disjoint enlarged windows after passing to a subsequence, such that

$$\int_{I_{n_j}} \int_{B_\rho} |\nabla V|^2 \, dy \, d\tau \to 0.$$

The retained compact-cylinder bounds, pressure reconstruction, admissible modulation control on the selected windows, and positive velocity core are exactly the remaining clauses of definition 2.3; if any retained compact-cylinder, pressure, exterior, rough-core, modulation, scale, carrier, or cost input needed for that definition fails, the ordered first-failure rule assigns item (ii). In particular, failure of the integrated modulation control is the modulation-window exit of lemma 9.87. If no such exit occurs, the selected subsequence gives item (i).

If the displayed dissipation integral is infinite, then for any cutoff $\phi_{R_0} \equiv 1$ on B_ρ the canonical localized cost satisfies

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) \, d\tau \geq \nu \int_{\tau_0}^{\infty} \int_{B_\rho} |\nabla V|^2 \, dy \, d\tau = \infty,$$

unless one of the ordered local tests needed to define the retained compact core has failed first. Such first failures are precisely the exterior, rough-core, modulation, scale, or carrier/cost adjacent labels of the local state-space validation, giving item (ii). If no first failure occurs, the displayed divergence is item (iii). □

10.3. **Assembly of Alternatives.**

Theorem 10.11 (Local Type II decomposition into local alternatives)**.** *Every positive local Type II concentration sequence in definition 10.2, after passing to subsequences and selected windows, enters one of the following alternatives:*

- (i) *a represented compact-core branch carrying a single retained active core, which by lemma 10.10 either has vanishing-dissipation selected windows, exits through a named adjacent local row before retention, or enters the retained local cost-divergence branch;*
- (ii) *a multibubble, cascade, or gauge-degenerate branch;*
- (iii) *loss of local windowed H^1 control on a retained compact cylinder;*
- (iv) *a finite-cost scale-collapse branch with nonvanishing selected core;*

(v) *a retained compact scale-collapse or scale-rigid terminal state.*

Proof. This decomposition is obtained by combining theorems 10.4, 10.6, 10.7 and 10.9. The representation theorem gives the nondegenerate single-core chart whenever a unique retained core is selected. Failure of that selection, splitting of positive CKN mass, compactness failure inside the local concentration selection, or scale cascade is the multibubble/cascade side. On a represented retained core, rough-core loss is equivalent by theorem 10.7 to failure of compact CKN control on an enclosing cylinder, hence returns to the multibubble/cascade side. Genuine scale collapse is divided by theorem 10.9 into finite-cost scale collapse or one of the retained compact scale-collapse/scale-rigid terminal states. These cases exhaust the local alternatives. □

Proposition 10.12 (Elimination of local Type II states)**.** *No positive local Type II concentration sequence can remain in any retained state among alternatives* (i)–(v) *of theorem 10.11. If the ordered validation inside alternative* (i) *exits through a named adjacent local row before retention, that branch is not a retained local state and is closed in the ambient adjacent-routing layer.*

Proof. Alternative (i) is excluded by the compact single-core criterion, theorem 10.5, in the vanishing-dissipation subcase of lemma 10.10. The adjacent-label subcase of that lemma is not a retained compact single-core state: the ordered validation has stopped earlier at a named local row. In the divergent localized-dissipation subcase, the canonical localized cost diverges on the retained core and the cost-divergence closure theorem 9.42 and corollary 9.43 excludes the branch after the windowwise adjacent alternatives have been routed.

Alternative (ii) is excluded by the local multibubble and cascade theorem, theorem 10.6.

Alternative (iii) is not an independent terminal state. By theorem 10.7, rough-core loss of local H^1 control forces failure of the compact-cylinder $\mathcal{C}+\mathcal{D}$ bounds on enclosing compact cylinders. Such failure yields an active compact-cylinder concentration branch and therefore belongs to the multibubble/cascade alternative, already excluded by theorem 10.6.

Alternative (iv) is excluded by the finite-cost scale-collapse theorem, theorem 10.8.

Alternative (v) is the retained compact scale-collapse or scale-rigid terminal state. Retained compact scale-collapse states are excluded by theorem 6.34. Scale-rigid terminal states are excluded by proposition 4.40: the state either has a nonzero bounded selected-window limit, which exits the retained local Type II row, or it reduces to the compact single-core branch, which is excluded by theorem 10.5. Hence it cannot be a retained local Type II branch. □

10.4. **Physical Entry into the Covered Local Type II Path.**

Definition 10.13 (Physical local Type II singular germ)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$. A tuple (u, p, x_0, T) is a physical local Type II singular germ if:

(i) $x_0 \in \Sigma(T)$;
(ii) the local Type I scale-invariant bound of theorem 2.5 fails at (x_0, T);
(iii) the germ is an interior Navier–Stokes germ, meaning that all sufficiently small backward cylinders $Q_r(x_0, T)$ are contained in the Navier–Stokes domain under consideration.

Definition 10.14 (Covered local Type II entry)**.** A physical local Type II singular germ has covered local Type II entry if one of the following two alternatives occurs:

(i) one of its Caffarelli–Kohn–Nirenberg concentration sequences, after passing to a subsequence and applying the selected-window reductions used in this paper, enters one of the alternatives in theorem 10.11;
(ii) before the selected-window state decomposition is reached, the ordered local tests assign the germ to a class-incompatible alternative or to a named non-retained local state-space exit.

Lemma 10.15 (Physical Type II germs are admissible local branches)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ. After translating space and time and restricting to a smaller terminal interval if necessary, the terminal germ defines an element of $\mathcal{U}_{\mathrm{II}}^{NS}$ in the sense of definition A.1.*

Proof. Choose $0 < \delta' < \delta$ so that the cylinders under consideration remain interior. Set

$$\tilde{u}(y,s) = u(y + x_0, T - \delta' + s), \qquad \tilde{p}(y,s) = p(y + x_0, T - \delta' + s), \qquad 0 < s < \delta',$$

and $T^* = \delta'$. This space-time translation and restriction preserve the local Navier–Stokes equations, suitability, and the pressure normalization modulo functions of time. The terminal time T^* is finite by construction. Since $x_0 \in \Sigma(T)$, the terminal germ is not a local continuation branch. By the definition of physical local Type II germ, the local Type I scale-invariant bound fails at (x_0, T), so the germ is not in the local Type I class. The interior-domain clause excludes boundary and non-Navier–Stokes-domain alternatives. These are exactly the admissibility clauses in definition A.1. ∎

Lemma 10.16 (Physical Type II germs carry local Type II concentration sequences)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ. Then there are $\eta > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \textit{for all } n,$$

and r_n is a positive local Type II concentration sequence in the sense of definition 10.2.

Proof. Since $x_0 \in \Sigma(T)$, theorem 2.5 gives $\eta > 0$ and $r_n \downarrow 0$ such that the displayed CKN lower bound holds. The same dichotomy has exactly two pointwise alternatives: the local Type I scale-invariant bound holds, or the point lies in the local Type II alternative. The first alternative is excluded by definition 10.13. Hence (x_0, T) lies in the local Type II alternative, and the displayed concentration sequence is exactly a positive local Type II concentration sequence as defined in definition 10.2. ∎

Lemma 10.17 (No extra outside-class row for physical Type II germs)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ and let ω be the admissible terminal branch supplied by lemma 10.15. Any failure of the C1 analytic data for ω is either class-incompatible or one of the named non-retained local exits in theorem 9.103. In particular, there is no additional outside-covered-class alternative for such a physical Type II germ.*

Proof. By lemma 10.15, $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$. The ordered C1 discharge theorem 9.103 applies on this class. Its conclusion is precisely that the first failure of definition A.3 is assigned either to a class-incompatible alternative excluded by definition A.1, or to a named non-retained state-space branch already covered by the local stratification. Thus a physical Type II germ has no residual "outside the covered class" row after the ordered local tests are run. ∎

Theorem 10.18 (Covered-class entry for physical Type II germs)**.** *Every physical local Type II singular germ has covered local Type II entry in the sense of definition 10.14. More explicitly, after passing to a CKN concentration sequence and to the selected windows used in this paper, the germ enters one of the alternatives in theorem 10.11, unless it has already exited through one of the class-incompatible or named non-retained local alternatives of theorem 9.103.*

Proof. Let (u, p, x_0, T) be a physical local Type II singular germ. By lemma 10.15, the germ defines an admissible terminal branch. Run the ordered C1 analytic-data tests on that branch. If a first failure occurs, then lemma 10.17 routes it to a class-incompatible alternative or to a named non-retained local state-space exit. This is covered entry in the sense of definition 10.14 (ii).

It remains to consider the retained case, when no C1 failure occurs. By lemma 10.16, the germ carries a positive local Type II concentration sequence. The decomposition theorem theorem 10.11

is formulated exactly for such sequences: after passing to subsequences and selected windows, the sequence enters one of the five alternatives listed there. This is covered entry in the sense of definition 10.14 (i). Thus every case is a covered local state-space outcome. The proof uses only pointwise CKN concentration, local first-failure routing, and selected-window state-space decomposition; it does not use a global $L^3(\mathbb{R}^3)$ bound, a global profile decomposition, global Kato smallness, or vanishing localized dissipation beyond what is produced inside the selected local branch. □

Corollary 10.19 (Discharge of covered-class entry)**.** *The covered-class entry obligation for the local Type II state-space analysis is discharged: every physical local Type II singular germ either enters the local state-decomposition path of theorem 10.11, or exits earlier through a class-incompatible or named non-retained local alternative.*

Proof. This is theorem 10.18. □

Remark 10.20 (Scope of the covered-entry discharge)*.* Corollary 10.19 proves only the entry statement: a physical local Type II germ is not lost before the local state-space analysis begins. It does not by itself supply the ambient adjacent-stratum closure data or the expanded terminal data used in the final assembly. Nor does it assert that every named non-retained exit has already been eliminated; it asserts that any such exit is recorded in the local state-space taxonomy rather than being an untracked outside-class branch. The closure of those named exits is supplied by the local closure theorems and by the ambient/terminal wrappers corollaries 7.136 and 9.128. The entry argument is also independent of theorem 10.30; it uses the pointwise CKN dichotomy, the local Type I/Type II split, and the ordered C1 routing theorem, so the discharge is not circular.

10.5. **Closure of the remaining named exit rows.** The three closure theorems in this subsection convert the rows previously listed in definition 10.27 into closed adjacent rows, i.e. rows that are routed by an existing local closure or wrapper into a retained-state alternative that has already been excluded. The arguments are local on compact terminal cylinders. They use only the ordered local state-space tests, the modulation-window dichotomy, the local pressure reconstruction with harmonic remainder, and the existing wrappers corollaries 7.136 and 9.128. No global L^3 bound, no global profile decomposition, no global Kato smallness, no hidden vanishing localized dissipation, and no noncanonical schedule are introduced.

Lemma 10.21 (Bounded modulation defects are local carrier labels)**.** *Let (a_n, b_n) be the repaired-gauge modulation coefficients on a fixed compact terminal window $[0,T]$ for a represented branch with a nonvanishing selected core V_n, and suppose*

$$\sup_n \int_0^T \big(|a_n(\tau)| + \|b_n(\tau)\|\big)\, d\tau < \infty.$$

After passing to a subsequence, the ordered local test gives one of the following outcomes:

(i) *writing*

$$\bar a_n = T^{-1}\int_0^T a_n(\tau)\, d\tau, \qquad \bar b_n = T^{-1}\int_0^T b_n(\tau)\, d\tau,$$

there are constants $a_\infty \in \mathbb{R}$ and $b_\infty \in \mathbb{R}^3$ such that, after passing to the same subsequence, $(\bar a_n, \bar b_n) \to (a_\infty, b_\infty)$ and either

$$a_n \to a_\infty \quad \text{in } L^1(0,T), \qquad b_n \to b_\infty \quad \text{in } L^1(0,T;\mathbb{R}^3);$$

or the nonconstant part of the modulation is invisible in the terminal local residual topology:

$$(a_n - \bar a_n)(V_n + y\cdot\nabla V_n) + (b_n - \bar b_n)\cdot\nabla V_n \to 0$$

in $L^1(0,T;H^{-1}(K))$ for every compact K;

(ii) *a signed component of the normalized modulation defect is visible in the local residual topology and defines a local occupation state. The ordered state-space test records one of the adjacent labels: modulation-driven recentering, positive scale-shell work, negative scale-drift, loss of the selected active core, or repaired-gauge degeneracy.*

The conclusion is local on the fixed terminal window and uses no compactness stronger than finite-measure weak-$*$ compactness of the modulation defect measures.

Proof. Write $m_n = T^{-1}\int_0^T (a_n, b_n)\,d\tau \in \mathbb{R}^4$. The L^1 bound implies that, after passing to a subsequence, $m_n \to m_\infty = (a_\infty, b_\infty)$. Set

$$\alpha_n(\tau) = a_n(\tau) - \bar{a}_n, \qquad \beta_n(\tau) = b_n(\tau) - \bar{b}_n, \qquad m_n = (\bar{a}_n, \bar{b}_n).$$

If

$$\|\alpha_n\|_{L^1(0,T)} + \|\beta_n\|_{L^1(0,T;\mathbb{R}^3)} \to 0,$$

then item (i) holds.

Assume instead that the displayed defect norm has a positive liminf. Define the local modulation-defect residual

$$\mathcal{F}_n^{\mathrm{mod}} := \alpha_n(V_n + y\cdot\nabla V_n) + \beta_n\cdot\nabla V_n.$$

If $\mathcal{F}_n^{\mathrm{mod}} \to 0$ in $L^1(0,T;H^{-1}(K))$ for every compact K, then the represented local equation has the same terminal residual limit as the sequence of constant-coefficient local equations with coefficients m_n. Since $m_n \to m_\infty$ in the finite-dimensional coefficient space, this is the constant-coefficient autonomous channel after the retained local compactness test is run. Item (i) holds in its perturbative form.

It remains to consider the case in which the defect residual is not perturbative on some compact K. Passing to a further subsequence and applying the ordered component convention, one of

$$\int_0^T (\alpha_n)_+\,d\tau, \qquad \int_0^T (\alpha_n)_-\,d\tau, \qquad \int_0^T |\beta_n|\,d\tau$$

has a subsequence whose normalized signed residual contribution is nonzero in the local terminal residual topology. Normalize that visible component by its mass. The resulting probability measures are bounded in $\mathcal{M}([0,T])$, and hence have a weak-$*$ subsequential limit. Together with the nonzero residual pairing, this is a visible local modulation-defect occupation state on the selected window.

The ordered local state-space test evaluates visible signed defect residuals before declaring a retained autonomous window. A visible β_n-component is the modulation-driven recentering/moving-translation label of definition 7.107. A visible $(\alpha_n)_+$-component is the positive scale-shell label, and a visible $(\alpha_n)_-$-component is the negative scale-drift label. If the selected core on which the modulation acts has lost its active mass floor, the first failed row is the retained active-core row. If the defect cannot be represented in the repaired chart, the first failed row is repaired-gauge degeneracy. These alternatives are ordered before autonomous retention, and they exhaust the nonperturbative signed components of the modulation defect. Hence item (ii) holds. □

Theorem 10.22 (Autonomous-modulation failure is a closed adjacent row)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ in the sense of definition 10.13, and assume the germ has admissible covered entry into the local state-space path. If a represented selected scale-collapse branch with nonvanishing selected core enters the autonomous-modulation alternative theorem 6.23*(v) *on a compact terminal cylinder, then the branch enters one of the following closed adjacent rows: the modulation-window-unbounded row routed by lemma 9.87; the modulation-driven recentering or moving-carrier row routed by definition 7.107 and discharged in the ambient adjacent wrapper corollary 7.136; the negative scale-drift row closed by theorem 9.99; the positive scale-shell row* *routed by corollary 7.128; the loss-of-selected-active-core row closed by theorem 10.25; or the gauge-degenerate / repaired-gauge row already excluded from the retained branch by the ordered representation tests. Equivalently, the autonomous-modulation alternative is not an independent remaining terminal exit.*

Proof. Fix a represented selected scale-collapse branch on a compact terminal cylinder $B_R \times [\tau_n, \tau_n + T]$ in the autonomous regime, and let $a_n(\tau) = a(\tau_n + \tau)$ and $b_n(\tau) = b(\tau_n + \tau)$ on $[0, T]$ be the absolutely continuous repaired-gauge modulation coefficients identified in lemma 9.50. Cover the selected window by finitely many unit windows of the type used in lemma 9.87, with the same repaired-gauge chart variables as that lemma; the construction of the chart variables and their absolute continuity are supplied for any represented branch by lemma 9.50 and the ordered representation tests corollaries A.12 and A.19. Exactly one of the following two cases occurs.

Case A (window-integrals unbounded). On at least one unit window, the supremum $\sup_n \int(|a_n| + \|b_n\|_\infty)\, d\tau$ is infinite. By the same stratification of accumulated chart velocities used in the proof of lemma 9.87, namely the splitting into large accumulated negative scale motion, large positive scale-shell work, large translation work, and unrepresentable accumulated modulation, the branch is recorded as one of the named modulation, scale-drift, modulation-driven recentering, exterior, or gauge-degenerate local alternatives. The same stratification applies on the scale-collapse stratum because it depends only on the absolutely continuous chart variables of lemma 9.50 and on the carrier-routing definition definition 7.107, neither of which uses the single-core hypothesis. All five rows are closed adjacent rows: the negative scale-drift row by theorem 9.99; the positive scale-shell component by corollary 7.128; modulation-driven recentering by definition 7.107 together with the ambient adjacent wrapper corollary 7.136; the exterior row by corollary 8.64; and the gauge-degenerate / representation row by the ordered representation tests corollaries A.12 and A.19. This is the conclusion claimed.

Case B (window-integrals bounded). On every unit window in the cover, the opposite alternative of lemma 9.87 holds, so $\sup_n \int_0^T (|a_n| + \|b_n\|_\infty)\, d\tau < \infty$. By the local modulation-defect routing lemma lemma 10.21, after passing to a subsequence either the modulation coefficients have a constant L^1 limit on the window, the nonconstant defect is perturbative and the represented local equation has the same terminal residual limit as the sequence of constant-coefficient branches with the same window-average coefficients, or a nonzero visible local modulation-defect occupation state is recorded. In the first two alternatives the finite-dimensional window averages converge, and the retained local compactness test gives the corresponding constant-coefficient autonomous local limit, which is excluded by the autonomous-modulation alternative theorem 6.23(v). In the remaining nonperturbative alternative the defect is recorded, before autonomous retention, as modulation-driven recentering, positive scale-shell work, negative scale-drift, loss of selected active core, or repaired-gauge degeneracy. These rows are closed respectively by corollary 7.136, corollary 7.128, theorem 9.99, theorem 10.25, and the ordered representation tests corollaries A.12 and A.19. Thus the bounded-modulation case also leaves the autonomous-modulation alternative through a closed adjacent row.

The two cases exhaust the possibilities. No part of the argument used a whole-space critical norm $\|V(\tau)\|_{L^3(\mathbb{R}^3)}$, a global profile decomposition, a global Kato smallness condition, a noncanonical schedule, or a hidden vanishing localized dissipation: only finite-window L^1 boundedness, weak-$*$ compactness of normalized defect measures, and the existing closure theorems for the named adjacent rows are used. □

Theorem 10.23 (Noncanonical carrier and cost validation are closed adjacent rows)**.** *Let ω be a renormalized Type II solution with active supercritical scaling on a terminal tail. Let a* *candidate continuation lie in a compact cost-divergence regime through a noncanonical carrier or cost-validation channel different from the canonical retained identity-cost channel. Then exactly one of the following occurs after running the ordered local validation tests of lemma 7.70:*

(a) *the candidate is, after the comparable-cost compatibility supplied by lemma 7.64, the canonical identity-cost retained channel up to the terminal one-sided comparison of definition A.23, in which case the cost-divergence exclusion on the retained stratum is supplied by corollary 7.79, hence closed;*

(b) *the first failed test is one of the named cost-side adjacent first failures listed in lemma 7.70*(b)*: cost-compatibility failure, carrier-subidentity failure, finite fixed-cutoff error failure, summable moving-annulus error failure, carrier-comparison failure for the validated cost, modulation-integrability failure, or failure of compact cost divergence for the validated cost; each of these first failures is a closed adjacent row, routed in the ambient adjacent wrapper corollary 7.136.*

Consequently the row "carrier or cost-validation failure outside the canonical retained identity-cost channel" is not an independent remaining terminal exit.

Proof. By lemma 7.70, after running the ordered local validation tests (i)–(ix) of that lemma, exactly one of (a) all tests pass with ω on the retained cost-divergence regime, or (b) the first failed test is one of the named local cost-side first failures listed there.

In case (a), the comparable-cost compatibility lemma lemma 7.64 identifies the validated noncanonical compact Navier–Stokes cost on the retained stratum with the canonical identity-cost up to the terminal one-sided comparison of definition A.23. Therefore the retained cost-divergence exclusion supplied by corollary 7.79 applies, and the candidate is excluded.

In case (b), the first failed test is by construction one of the named cost-side adjacent rows enumerated in lemma 7.70(b). We show each such row is closed.

Class-side first failures. Ordered exhaustion failure, representation failure, and repaired-gauge or scale-degeneracy failure are the C1/C2 first failures already routed before the state-decomposition path is reached. On a physical Type II germ they are not left as analytic gaps: the covered-entry theorem records them either as class-incompatible alternatives or as named local state-space rows. Thus they are discharged by corollary 10.19 together with the ordered representation tests corollaries A.12 and A.19.

State-decomposition first failures. Multibubble/multicore splitting is closed by theorem 10.6 and theorem 10.5; cascade is closed by the finite-cost scale-collapse theorem theorem 10.8 together with corollary 4.43; exterior-tightness failure is closed by corollary 8.64; loss of local windowed H^1 control is closed by theorem 10.7, which returns to the multibubble/cascade closed row; finite-cost scale-collapse is closed by theorem 10.8; the scale-rigid residual is closed by proposition 4.40; critical-mass adjacent labels are closed by theorem 10.25; and modulation-integrability failure is closed by theorem 10.22 (and by lemma 9.87 in the ambient single-core stratum).

Cost-side first failures (the irreducible noncanonical-cost rows). The remaining items in lemma 7.70(b) are the cost-side labels: cost-compatibility failure, carrier-subidentity failure, finite fixed-cutoff error failure, summable moving-annulus error failure, carrier-comparison failure for the validated cost, and failure of compact cost divergence for the validated cost. These rows are not closed by declaring the Navier–Stokes solution impossible; they are closed by the ordered canonical-channel rule: the final retained proof uses only the canonical identity cost, or an explicitly comparable compact cost, and every failure of a noncanonical diagnostic is routed back to the canonical local validation layer.

Cost-compatibility and carrier-comparison failure. By definition of lemmas 7.63 and 7.64, cost-compatibility failure means that the validated cost used to record divergence is neither the canonical identity cost nor a comparable cost in the sense of definition A.23. Such a diagnostic is not used to retain a Type II branch. Instead one reruns the local validation with the canonical identity cost supplied by the suitable local energy inequality. If the canonical channel passes, the retained stratum is closed by corollary 7.79. If it fails, the first failed canonical test is one of the ordered local rows already treated above. Thus an incomparable noncanonical cost is a bookkeeping failure of that diagnostic, not an additional analytic exit and not a hidden input.

Carrier-subidentity, finite fixed-cutoff error, and summable moving-annulus error failure. By lemma 7.86, suitable local energy on the repaired-gauge branch supplies the moving carrier subidentities and the corresponding fixed and moving carrier closure packages of definitions 7.87 and 7.88, with finite-error selection definition 7.95 and the carrier completion of corollaries 7.104 and 7.105. If one of these carrier rows fails before the retained carrier is validated, the failure is not repaired by a global estimate. The ordered componentwise carrier decomposition records the first nonsummable local term. Transition, pressure, convection, and cutoff terms route to exterior/leakage, critical tightness failure, or local H^1-loss; translation terms route to modulation-driven recentering; positive scale terms route to the positive scale-shell/scale-rigid row; and negative scale terms route to the thin-drift row. These are precisely the alternatives in definition 7.107, closed by corollary 7.136, corollary 7.128, and theorem 9.99. If the suitable-energy carrier identity itself is unavailable, the first failed row is the already routed representation/repaired-gauge or suitable-energy row, not a new cost-validation exit.

Failure of compact cost divergence for the validated cost. This is the negation of the entry hypothesis: if the validated cost does not diverge on the compact terminal cylinder, the candidate is not in the compact cost-divergence regime under consideration. The final assembly then continues through the ordinary retained-state decomposition: finite cost is handled by theorem 10.8, while failure before finite-cost classification is one of the ordered local rows already listed in lemma 7.70. Thus nondivergence of a validated noncanonical cost cannot survive as an independent terminal exit.

The argument uses only the ordered finite-step local validation tests, the comparable-cost compatibility on compact windows, the suitable local energy package on the retained stratum, and the canonical-channel fallback for diagnostics that are not explicitly comparable. It introduces no whole-space L^3 bound, no global profile decomposition, and no noncanonical schedule outside the comparison framework already established by lemmas 7.64 and 7.77. ☐

Lemma 10.24 (Pressure-only critical mass routes through local pressure atlas)**.** *Let $B_R \Subset B_{R_1} \Subset B_{R_2}$ be nested balls in a retained terminal frame on a compact time interval I, and fix $1 \le q < \infty$. Suppose a pressure oscillation modulo functions of time persists in $L^q_\tau L^{3/2}_y(B_R)$, while the ordered terminal selection has not retained a velocity active core for that candidate. With the local pressure decomposition*

$$P_n = P_n^{\mathrm{loc}} + H_{n,R_1} + c_{n,R_1}(\tau) \qquad \textit{on } B_{R_1},$$

where P_n^{loc} is generated by $\zeta(V_n \otimes V_n)$ on B_{R_2} and H_{n,R_1} is mean-zero harmonic on B_{R_1}, one of the following ordered alternatives occurs:

(i) *the localized velocity source $\zeta(V_n \otimes V_n)$ has a nonzero compact-window $L^q_\tau L^{3/2}_y$ limsup; the local CKN test on the selected compact subcylinders then either absorbs the source as regular, or records a comparable retained class, a multibubble/multicore label, a cascade label, a rough-core label, or an exterior/separated label;*

(ii) *the localized velocity source tends to zero in $L^q_\tau L^{3/2}_y(B_{R_2})$ for the fixed exponent q, and any remaining pressure oscillation is carried by the harmonic part H_{n,R_1}, which is routed by lemma 8.22 to the ordered pressure/exterior row;*

(iii) *the pressure reconstruction or gauge needed for the displayed decomposition fails, which is itself the ordered pressure/repaired-gauge row.*

Thus a pressure-only critical-mass signal is never closed by assuming vanishing velocity mass; it is routed by the local source/harmonic dichotomy.

Proof. The pressure decomposition is one of the ordered local pressure-atlas tests. If that test fails, item (iii) is recorded. Assume it holds.

If

$$\limsup_{n\to\infty} \|\zeta(V_n \otimes V_n)\|_{L^q_\tau L^{3/2}_y(B_{R_2})} > 0,$$

then the compact-window velocity stress is nonzero. Since $\|V_n \otimes V_n\|_{L^{3/2}(B)} = \|V_n\|^2_{L^3(B)}$ on each spatial ball, a finite cover of B_{R_2} and a subsequence select a compact subcylinder with positive local velocity critical mass in the corresponding spacetime norm. Run the local CKN test on that subcylinder. If all smaller admissible cylinders are below the CKN threshold, the source is regular on the compact window and is absorbed into the regular pressure atlas. Otherwise a local CKN carrier is selected, and the ordered terminal critical-mass routing lemma 9.73 records either a comparable retained class or one of the adjacent active labels listed in item (i). This uses only the compact-window stress norm, local CKN testing, and finite covering.

If the localized source tends to zero in $L^q_\tau L^{3/2}_y(B_{R_2})$, local Calderón–Zygmund estimates give

$$P_n^{\mathrm{loc}} \to 0 \quad \text{in } L^q_\tau L^{3/2}_y(B_{R_1})$$

modulo functions of time. Hence any persistent pressure oscillation on B_R must lie in the harmonic remainder. Applying lemma 8.22 with $p = 3/2$ and the fixed time exponent q routes that nonvanishing harmonic remainder to the pressure/exterior row. No global pressure compactness or lower semicontinuity argument is used. □

Theorem 10.25 (Loss of retained active core is a closed adjacent row)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ that has covered entry into the local state-space path, and let a terminal active frame be selected from its local state-space decomposition. If on a fixed compact terminal cylinder the candidate fails to retain an active core in the sense of lemma 9.73, then the candidate enters one of the following closed adjacent rows:*

(i) *the lower active mass-floor failure, in which case the candidate is not a retained terminal frame by the terminal selection rule and the state-space partition records the next available retained frame; if no velocity retained frame remains, the original CKN signal is pressure/exterior, rough-core, or regular, and is routed by the pressure-only and exterior closures;*

(ii) *the local L^3-upper-bound failure, which by lemma 9.73*(iii) *is one of: a comparable finite-mass class absorbed into the retained frame by lemma 9.72, a multibubble or multicore splitting closed by theorem 10.6 and theorem 10.5, a cascade closed by the finite-cost scale-collapse theorem theorem 10.8 together with the cascade absorption used in corollary 4.43, a rough-core label closed by theorem 10.7, or an exterior/separated label closed by corollary 8.64;*

(iii) *the pressure-only critical-mass row, in which case the local pressure atlas routes the signal by lemma 10.24: the local Calderón–Zygmund source is either regular on the compact window, produces a velocity-supported terminal label, or the harmonic remainder is routed into the pressure/exterior-leakage alternative, closed by corollary 8.64.*

Consequently the row "loss of a retained active core, including pressure-only critical-mass detection" is not an independent remaining terminal exit.

Proof. Apply lemma 9.73 on the fixed compact terminal cylinder $B_R \times I$. After passing to a subsequence, exactly one of its three alternatives (i)–(iii) holds.

If alternative (i) of that lemma holds, the candidate has positive finite local critical L^3 mass on the cylinder; this is the retained case and not the loss-of-core case under consideration. If alternative (ii) holds, the lower active-mass floor of theorem 8.32 fails, so the candidate frame is not retained by the terminal selection rule. The selection rule then moves to the next available velocity frame. If there is no such frame, the positive CKN concentration of the physical germ cannot be a retained velocity core; it is either pressure-only, exterior/separated, rough-core, or below the CKN threshold on all smaller cylinders. The first three are routed by lemma 10.24, corollary 8.64, and theorem 10.7; the last contradicts singular CKN concentration by theorem 8.34. This is item (i) of the present conclusion. If alternative (iii) holds, the local L^3 upper bound fails on the compact cylinder. By lemma 9.73(iii), the ordered local state space then records a comparable finite-mass class absorbed into the retained component, a multibubble or multicore label, a cascade label, a rough-core label, or an exterior/separated label. Each such label is closed by the theorems cited in item (ii) of the present conclusion: the rough-core label is closed by theorem 10.7, which forces failure of compact-cylinder CKN control on every relevant enclosing cylinder and therefore returns to the multibubble/cascade closed row; the exterior/separated label is closed by corollary 8.64.

It remains to handle the pressure-only subcase, in which the active core fails because the critical mass detected on the cylinder is carried by the pressure rather than by a velocity profile, i.e. the local velocity L^3-mass remains below the active floor while a pressure oscillation modulo a function of time persists on B_R along a subsequence. Apply the local pressure-only routing lemma lemma 10.24 with the time exponent in which the pressure oscillation persists. If the localized Calderón–Zygmund source has a nonzero compact-window stress limsup, the finite-cover and CKN test in that lemma either absorbs the source as regular, or records a velocity-supported terminal label; the latter is either absorbed into a comparable retained class by lemma 9.72 or routed to one of the closed multibubble/cascade, rough-core, or exterior/separated rows. If the localized source tends to zero, the local pressure part vanishes by Calderón–Zygmund estimates and any remaining pressure oscillation is harmonic; lemma 8.22 routes it to the ordered pressure/exterior-leakage alternative, closed by corollary 8.64. If pressure reconstruction fails, that is the ordered pressure/repaired-gauge row. This is item (iii) of the present conclusion.

Each step uses only compact terminal windows, the local active-mass floor, the ordered local state-space partition, and the local pressure reconstruction with harmonic remainder. No global L^3 bound, no global profile decomposition, no global Kato smallness, and no pressure compactness by weak lower semicontinuity is used. □

Corollary 10.26 (All three remaining named exit rows are closed)**.** *The three rows listed in definition 10.27 are all closed adjacent rows: the autonomous-modulation row by theorem 10.22; the noncanonical carrier/cost-validation row by theorem 10.23; and the loss-of-active-core row, including pressure-only critical-mass detection, by theorem 10.25.*

Proof. Direct application of the three closure theorems cited in the statement. □

Definition 10.27 (Remaining named exit rows)**.** The present retained-state assembly does not count a first-failure row as closed merely because it has been named. The ledger rows outside the retained closure theorem before the closure subsection is applied are:

(i) autonomous-modulation failure or modulation-driven recentering;
(ii) carrier or cost-validation failure outside the canonical retained identity-cost channel, including noncanonical cost-comparison failures not covered by an explicit local comparison theorem;

(iii) loss of a retained active core, including pressure-only critical-mass detection, unless the pressure/exterior routing theorem places it in one of the closed structural rows listed below.

The closed rows are not in this list. They are finite scale cost, finite absolute scale cost, thin negative drift as an independent terminal row, compactness failure on selected scale-collapse windows as an independent terminal row, compact single-core vanishing dissipation, multibubble, cascade, gauge-degenerate behavior, scale-rigid residual behavior, the retained compact scale-collapse state, exterior concentration and noncompact exterior concentration as independent terminal rows, and the retained canonical compact-cost branch. Thin negative drift is closed by theorem 9.99; selected compactness failure is routed by theorem 6.31; exterior concentration is routed by corollary 8.64; the other closed rows are closed by the theorems cited in proposition 10.12 and in the ambient compact-cost package.

The three rows (i), (ii), (iii) listed above are closed adjacent rows, by the closure theorems of Section 10.5: the autonomous-modulation row by theorem 10.22; the noncanonical carrier/cost-validation row by theorem 10.23; and the loss-of-active-core row, including pressure-only critical-mass detection, by theorem 10.25. This is summarised in corollary 10.26. The list is the controlling ledger for theorem 10.28; the unconditional upgrade is recorded in theorem 10.29.

Theorem 10.28 (Retained Type II closure with named-exit ledger)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$, and let (u,p,x_0,T) be a physical local Type II singular germ in the sense of definition 10.13. Then either the germ enters one of the remaining named exit rows in definition 10.27, or its retained local branch is impossible. Consequently, together with corollary 10.26, no physical local Type II singular germ remains.*

Proof. Let (u,p,x_0,T) be a physical local Type II singular germ. By corollary 10.19, the germ either enters the local state-decomposition path or exits earlier through a class-incompatible or named non-retained local alternative.

A class-incompatible exit contradicts the definition of physical local Type II germ and the admissibility theorem lemma 10.15. If a named non-retained exit occurs, then either it is one of the closed local rows cited in proposition 10.12 and corollary 7.136, or it belongs to the remaining named-exit ledger definition 10.27. In the latter case the first alternative in the statement holds.

It remains to consider the case in which the branch reaches the retained terminal path without entering any remaining named exit. The expanded terminal wrapper corollary 9.128 supplies the expanded terminal data of definition 9.121, and theorem 9.125 excludes the branch.

Equivalently, once the branch has entered the state-decomposition path, theorem 10.11 lists the five possible local Type II states, while proposition 10.12 eliminates the retained states. The ambient and terminal wrappers verify the retained inputs and route pre-retention first failures. Those first failures are either closed by their local closure theorem or listed in definition 10.27. Thus no retained Type II branch survives. The closure corollary above closes the remaining named exits, and the next theorem records the resulting full local Type II exclusion. □

Theorem 10.29 (No physical local Type II germ)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$. No physical local Type II singular germ (u,p,x_0,T) in the sense of definition 10.13 exists.*

Proof. By theorem 10.28, every physical local Type II singular germ either enters one of the rows in definition 10.27 or is excluded on the retained branch. By corollary 10.26, every row in definition 10.27 is a closed adjacent row: it is routed by its local closure theorem either to a closed retained-state alternative, to a structural row already excluded, or to a regular/removable row. Hence no physical local Type II singular germ remains. □

10.6. **Assembly Theorem.**

Theorem 10.30 (Local Type II exclusion)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$. No point $x_0 \in \Sigma(T)$ lies in the local Type II alternative.*

Proof. Theorem 10.29 excludes every physical local Type II singular germ. If a point $x_0 \in \Sigma(T)$ lay in the local Type II alternative, then it would satisfy the three clauses of definition 10.13: singularity gives $x_0 \in \Sigma(T)$, the local Type I bound fails by definition of the Type II alternative, and the theorem is stated for interior cylinders of $\mathbb{R}^3 \times (T - \delta, T)$. This is a physical local Type II singular germ, contradicting theorem 10.29. □

Corollary 10.31 (Final local dichotomy after Type I separation)**.** *Every singular point at time T belongs to the local Type I alternative. Equivalently, once the Type I case is covered by the Type I part of the theory, there is no remaining local Type II singular branch.*

Proof. Let $x_0 \in \Sigma(T)$. By theorem 10.3, x_0 carries a positive CKN concentration sequence and lies either in the local Type I alternative or in the local Type II alternative. Theorem 10.30 excludes the latter. □

11. Conclusion

The paper gives a local branchwise exclusion mechanism for Type II Navier–Stokes concentration. Beginning with a positive Caffarelli–Kohn–Nirenberg concentration sequence, the argument passes to repaired-gauge variables around a retained core and keeps the analysis inside local PDE estimates: suitable energy inequalities, pressure reconstruction, compactness on fixed cylinders, Caccioppoli estimates, profile decoupling, and localized scale-cost identities.

The compact single-core mechanism is the central obstruction. A retained core with positive local velocity density and vanishing localized dissipation has only two compact limits after passing to selected windows. The zero limit contradicts the persistence of the retained velocity core by strong L^3 convergence, while a nonzero spatially constant limit gives a bounded selected-window profile and hence exits the Type II regime. Pressure-only harmonic modes are routed through the pressure/exterior rows and are not used to close the compact single-core velocity row. The rest of the proof shows that the remaining alternatives to this compact mechanism are also accounted for: rough cores are reduced to compact-cylinder control, active multibubbles and cascades are separated by the profile analysis, and finite-cost scale collapse is incompatible with a nonvanishing retained core.

The scale-collapse analysis is intentionally explicit about its routing. Finite scale-collapsing cost, and hence finite absolute scale cost in the covered setting, are excluded within the renormalized setting. Thick autonomous windows produce nontrivial autonomous limits only after the relevant local compactness and modulation rows have passed. The retained compact autonomous branch is not closed by silently invoking a global Liouville theorem: it is routed through theorem 6.34 into the scale-rigid discharge. If a retained input fails, the failure is one of the named modulation, compactness, exterior, rough-core, or multicore local rows; the compactness, exterior, and thin-drift boundary faces are routed by their local closure theorems, including theorem 6.31, corollary 8.64, and theorem 9.99.

Thus every physical local Type II concentration branch is either eliminated on the retained path or routed through one of the named local rows. The remaining named rows are closed in corollary 10.26, and the final assembly theorem 10.29 excludes physical local Type II singular germs. The conclusion remains local and branchwise: it does not import a whole-space critical estimate, a global profile decomposition, global Kato smallness, or hidden vanishing localized dissipation.

Appendix A. PDE Interface Definitions Used in the Local Type II Proof

This appendix records, in PDE notation, the local Type II classification interfaces used earlier in the proof. The statements are interface definitions and local consequences for admissible Type II branches; they do not provide a separate post-assembly proof path. In the final assembly, their inputs are verified on the retained local path or routed to closed named rows.

A.1. **Admissible Type II branches and exhaustion.**

Definition A.1 (Admissible local Type II class)**.** Let $\mathcal{U}_{\mathrm{II}}^{NS}$ be the class of admissible terminal branches (u, p, T^*) such that:

(i) (u, p) is a three-dimensional incompressible Navier–Stokes solution on a terminal interval $[0, T^*)$ in the analytic class under consideration;
(ii) $T^* < \infty$ is the terminal time of the branch;
(iii) the branch is not in the local Type I class associated with the scale-invariant Type I criterion;
(iv) the branch is a Type II branch rather than a continuation, Type I, boundary, or non-Navier–Stokes-domain alternative.

Membership in $\mathcal{U}_{\mathrm{II}}^{NS}$ specifies the class under analysis; no nonemptiness assertion is made.

Definition A.2 (Type II analytic data)**.** A branch $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$ has the Type II analytic data if the following three analytic assertions hold:

(i) a concentration profile or blow-up germ modulo the Navier–Stokes symmetry group;
(ii) a supercritical scale alternative, meaning that the branch enters the Type II scale alternative rather than the Type I scale alternative;
(iii) local profile data in the Navier–Stokes profile class used for the subsequent compactness and representation arguments.

The analytic data are *strictly exhaustive* if every admissible Type II branch has these three properties. They are *exhaustive by ordered local routing* if every branch which fails one of the three properties is assigned, at its first failed C1 test, to one of the ordered failures in definition A.3 and that failure is then routed to a class-incompatible alternative or to a named non-retained local state-space exit. On a retained subbranch, ordered local-routing exhaustion reduces to strict exhaustion.

Definition A.3 (Ordered exhaustion failures)**.** For an admissible Type II branch, the ordered failures of the Type II analytic data are:

(i) concentration extraction fails;
(ii) concentration extraction succeeds but the supercritical scale alternative does not hold;
(iii) concentration extraction and the scale alternative hold but the local profile data are insufficient for the subsequent compactness and representation arguments;
(iv) the admissible Type II branch lies outside the concentration, scale, and profile alternatives under consideration.

The first applicable failure in this order is the exhaustion alternative.

Theorem A.4 (Type II branch exhaustion)**.** *Assume the Type II analytic data of definition A.2 are exhaustive by ordered local routing on $\mathcal{U}_{\mathrm{II}}^{NS}$, and that each ordered failure is assigned to a class-incompatible alternative or to a named non-retained local state-space exit. Then every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ either has the concentration profile, the supercritical scale alternative, and the local profile data, or is already in one of those named exits. In particular, every retained admissible Type II branch has the three analytic data.*

Proof. Fix $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$. By ordered local routing, exactly one of two possibilities occurs. If no ordered failure occurs, then the concentration-extraction, scale, and profile components all hold, so ω has the three Type II analytic data. If an ordered failure occurs, the first such failure is, by hypothesis, assigned to a class-incompatible alternative or to a named non-retained local state-space exit. Therefore a retained admissible branch cannot be in the failure case, and must have the three analytic data. $\square$

Corollary A.5 (Ordered exhaustion classification). *For each $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$, exactly one ordered alternative occurs: the Type II analytic data hold, or one of the four failures in definition A.3 occurs. The data are strictly exhaustive on $\mathcal{U}_{\mathrm{II}}^{NS}$ if and only if the first alternative holds for every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$. They are exhaustive by ordered local routing if every failure alternative is assigned to a class-incompatible alternative or to a named non-retained local state-space exit; on the retained subbranch this is equivalent to the first alternative.*

Proof. Test the four analytic requirements in the stated order. If concentration extraction, the scale alternative, and the profile condition all hold, the branch has the Type II analytic data. If a test fails, the first failed test gives the corresponding ordered failure. The alternatives are disjoint by the first failure convention and exhaustive by construction. The final equivalence is the definition of strict exhaustion. The ordered-routing statement follows by recording the target assigned to each first failure. Once class-incompatible and named non-retained exits have been removed, a retained branch cannot lie in any failure alternative, so it lies in the data alternative. $\square$

A.2. Repaired-gauge representation from Type II analytic data.

Definition A.6 (Raw Type II chart). A raw Type II chart consists of

$$(u, p, T^*, x_c, \lambda, \tau, V, P, \mathcal{I}), \qquad \mathcal{I} = (t_0, T^*),$$

where u, p solve

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

$\lambda(t) > 0$, $x_c(t) \in \mathbb{R}^3$, $x_c, \lambda \in AC_{\mathrm{loc}}$, $\lambda(t) \to 0$ as $t \uparrow T^*$, and

$$\frac{d\tau}{dt} = \lambda(t)^{-2}, \qquad \tau(t) \to \infty \quad \text{as } t \uparrow T^*.$$

The renormalized variables are

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x,t) = \lambda(t)^{-1} V(y, \tau(t)), \qquad p(x,t) = \lambda(t)^{-2} P(y, \tau(t)),$$

with enough regularity to justify the distributional chain rule on compact y-sets.

Lemma A.7 (Raw renormalized equation). *For a raw Type II chart, define*

$$a_{\mathrm{raw}}(\tau(t)) = \lambda(t)\lambda_t(t), \qquad b_{\mathrm{raw}}(\tau(t)) = \lambda(t) x_c'(t).$$

Then V, P satisfy, distributionally on compact y-sets,

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a_{\mathrm{raw}}(\tau)(V + y \cdot \nabla V) + b_{\mathrm{raw}}(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Proof. Write

$$u(x,t) = \lambda(t)^{-1} V(y, \tau(t)), \qquad y = \frac{x - x_c(t)}{\lambda(t)}, \qquad \tau_t = \lambda^{-2}.$$

Then

$$\nabla_x u = \lambda^{-2} \nabla_y V, \qquad \Delta_x u = \lambda^{-3} \Delta_y V, \qquad (u \cdot \nabla_x)u = \lambda^{-3}(V \cdot \nabla_y)V.$$

Since

$$y_t = -\frac{x_c'(t)}{\lambda(t)} - \frac{\lambda_t(t)}{\lambda(t)} y,$$

one has

$$\partial_t u = \lambda^{-3}\partial_\tau V - \lambda_t \lambda^{-2}(V + y \cdot \nabla V) - \lambda^{-2} x_c'(t) \cdot \nabla V.$$

Moreover $\nabla_x p = \lambda^{-3}\nabla_y P$. Substitution in Navier–Stokes and multiplication by λ^3 give

$$\partial_\tau V - \lambda\lambda_t(V + y \cdot \nabla V) - \lambda x_c'(t) \cdot \nabla V + (V \cdot \nabla)V + \nabla P = \nu \Delta V.$$

Moving the modulation terms to the right-hand side gives the equation with $a_{\text{raw}} = \lambda\lambda_t$ and $b_{\text{raw}} = \lambda x_c'$. Finally, $\nabla_x \cdot u = \lambda^{-2}\nabla_y \cdot V$, so incompressibility of u implies $\nabla_y \cdot V = 0$. □

Definition A.8 (Repaired-gauge realization)**.** Let $0 < p < 3$, $\Theta_0 > 0$, and let ψ_R be a smooth compactly supported centering cutoff. A raw Type II chart admits a repaired-gauge realization if, after an admissible time-dependent scale and translation correction, the profile satisfies

$$G_{\text{sc}}(V) = \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3 \, dy - \Theta_0 = 0,$$

and

$$G_j(V) = \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y) \, dy = 0, \qquad j = 1, 2, 3,$$

for a.e. renormalized time. The corrected profile belongs to the admissible gauge class $\mathcal{A}_p$, has the regularity needed to differentiate these functionals, and all time-dependent symmetry corrections have been absorbed into final modulation coefficients a, b.

Definition A.9 (Repaired-gauge representation)**.** A Type II branch has a repaired-gauge representation if it has:

(i) a raw Type II chart;
(ii) a repaired-gauge realization;
(iii) pressure reconstruction modulo functions of time;
(iv) final modulation coefficients a, b appearing in the repaired-gauge equation.

The representation consists of the tuple

$$(V, P, a, b, G_{\text{sc}}, G_1, G_2, G_3, \tau_0).$$

Theorem A.10 (Representation implication)**.** *Assume a Type II profile branch has a raw Type II chart, a repaired-gauge realization, pressure reconstruction modulo functions of time, and final modulation coefficients. Then the branch has a repaired-gauge representation in the sense of definition A.9. In particular, (V, P, a, b) satisfies the PDE class required by the compact Type II cost-divergence criterion.*

Proof. The raw chart gives V, P and, by lemma A.7, the renormalized Navier–Stokes equation with raw modulation coefficients. The repaired-gauge realization places the profile on the scale and centering gauge surface and absorbs the time-dependent symmetry correction into updated coefficients. The pressure reconstruction identifies the pressure associated with V, modulo the usual functions of time. The modulation data identify the final coefficients a, b. These are the components of definition A.9. □

Definition A.11 (Ordered representation failures)**.** If a repaired-gauge representation is not obtained, the ordered failures are:

(i) no raw orbit is supplied by the concentration and profile data;
(ii) the raw orbit exists but the repaired gauge cannot be realized;
(iii) pressure reconstruction modulo functions of time is unavailable;

(iv) the final modulation coefficients are unavailable.

The first applicable failure is the representation alternative.

Corollary A.12 (Ordered representation classification)**.** *Every Type II profile branch in the admissible class has exactly one ordered alternative: a repaired-gauge representation, or one of the four failures in definition A.11.*

Proof. Verify the four representation inputs in order. If all are present, theorem A.10 gives the repaired-gauge representation. If not, the first missing input gives the corresponding ordered failure. This is exhaustive and disjoint by first-failure ordering. □

A.3. **Minimal chart and gauge realization.**

Definition A.13 (Minimal representation data)**.** The minimal data for a Type II branch satisfying the analytic data consist of:

(i) an absolutely continuous concentration chart

$$(u, p, T^*, x_c, \lambda, \rho, U, Q, \mathcal{I}),$$

where $\mathcal{I} = (t_0, T^*)$, $\lambda(t) > 0$, $\lambda(t) \to 0$, $\rho_t = \lambda^{-2}$, and

$$u(x,t) = \lambda(t)^{-1} U(y, \rho(t)), \qquad p(x,t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t), \qquad y = \frac{x - x_c(t)}{\lambda(t)};$$

(ii) an absolutely continuous repaired-gauge correction consisting of $\rho = \rho(\tau)$, $\mu(\tau) > 0$, and $q(\tau) \in \mathbb{R}^3$, with

$$\rho_\tau = \mu(\tau)^2,$$

$$V(Y,\tau) = \mu(\tau) U(\mu(\tau)Y + q(\tau), \rho(\tau)), \qquad P(Y,\tau) = \mu(\tau)^2 Q(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

and

$$G_{\mathrm{sc}}(V(\tau)) = 0, \qquad G_j(V(\tau)) = 0, \quad j = 1,2,3.$$

The condition $\rho_\tau = \mu^2$ preserves the viscosity coefficient after the time-dependent scale correction.

Lemma A.14 (Chart data imply the raw orbit)**.** *Under the chart data in definition A.13, U, Q satisfy*

$$\partial_\rho U + (U \cdot \nabla) U + \nabla Q = \nu \Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \qquad \nabla \cdot U = 0,$$

where

$$A(\rho(t)) = \lambda(t)\lambda_t(t), \qquad B(\rho(t)) = \lambda(t) x_c'(t).$$

Proof. The chart data are the raw orbit data with raw time ρ. The additive pressure function $c(t)$ has zero spatial gradient. The chain-rule computation is the one in lemma A.7, with τ replaced by ρ. Thus spatial derivatives scale as $\nabla_x u = \lambda^{-2} \nabla_y U$, $\Delta_x u = \lambda^{-3} \Delta_y U$, the convection term scales as $\lambda^{-3}(U \cdot \nabla)U$, and $\rho_t = \lambda^{-2}$. The terms generated by λ_t and x_c' give $A = \lambda\lambda_t$ and $B = \lambda x_c'$. Incompressibility pulls back as $\nabla_x \cdot u = \lambda^{-2} \nabla_y \cdot U$. □

Lemma A.15 (Pressure pullback)**.** *Under the chart data,*

$$-\Delta Q = \partial_i \partial_j (U_i U_j)$$

in distributions, modulo functions of ρ. After any admissible repaired-gauge correction, the transformed pressure satisfies

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

in distributions, modulo functions of τ.

Proof. Taking divergence of the physical Navier–Stokes equation gives

$$-\Delta p = \partial_i \partial_j (u_i u_j),$$

with pressure determined up to a function of time. Pulling this identity back through $p = \lambda^{-2} Q + c(t)$, $u = \lambda^{-1} U$, and $y = (x - x_c)/\lambda$ gives the pressure equation for Q. If

$$V(Y) = \mu U(\mu Y + q), \qquad P(Y) = \mu^2 Q(\mu Y + q),$$

then

$$-\Delta_Y P = \mu^4 (-\Delta Q)(\mu Y + q) = \mu^4 \partial_i \partial_j (U_i U_j)(\mu Y + q) = \partial_i \partial_j (V_i V_j).$$

Adding a function of ρ or τ to the pressure does not affect the identity. ☐

Lemma A.16 (Forced modulation coefficients)**.** *Assume the minimal data of definition A.13. Then* (V, P) *satisfies*

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V,$$

where

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau) B(\rho(\tau)) + \mu(\tau) A(\rho(\tau)) q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

Proof. Let

$$z = \mu(\tau) Y + q(\tau), \qquad V(Y, \tau) = \mu(\tau) U(z, \rho(\tau)), \qquad \rho_\tau = \mu^2.$$

The raw equation in (z, ρ) is

$$U_\rho + (U \cdot \nabla_z) U + \nabla_z Q = \nu \Delta_z U + A(U + z \cdot \nabla_z U) + B \cdot \nabla_z U.$$

The identities

$$\nabla_Y V = \mu^2 \nabla_z U, \qquad \Delta_Y V = \mu^3 \Delta_z U, \qquad (V \cdot \nabla_Y) V = \mu^3 (U \cdot \nabla_z) U, \qquad \nabla_Y P = \mu^3 \nabla_z Q$$

together with differentiation in τ give

$$V_\tau + (V \cdot \nabla_Y) V + \nabla_Y P - \nu \Delta_Y V = \left(\frac{\mu_\tau}{\mu} + \mu^2 A \right) (V + Y \cdot \nabla_Y V) + \left(\mu B + \mu A q + \frac{q_\tau}{\mu} \right) \cdot \nabla_Y V.$$

This is the displayed equation. Since μ, q, ρ are absolutely continuous and A, B are the raw chart coefficients, a, b are measurable and locally integrable on every finite renormalized window. ☐

Theorem A.17 (Minimal representation theorem)**.** *If a Type II branch satisfying the analytic data has the minimal data of definition A.13, then it has a repaired-gauge representation in the sense of definition A.9.*

Proof. Lemma A.14 gives the raw orbit. The gauge correction places the profile on the repaired scale and centering surface, hence gives the repaired-gauge realization. Lemma A.15 gives pressure reconstruction modulo functions of time. Lemma A.16 gives the final modulation coefficients. These are the four inputs in definition A.9. ☐

Definition A.18 (Minimal representation failures)**.** Outside the admissible repaired-gauge representation class, the ordered failures are:

(i) no absolutely continuous concentration chart is available;
(ii) the chart exists but the repaired scale and centering gauges cannot be solved admissibly and absolutely continuously.

Pressure-pullback and modulation-coefficient failures are not independent alternatives when the chart and the absolutely continuous gauge solve exist.

Corollary A.19 (Ordered minimal representation classification)**.** *Every Type II branch satisfying the analytic data has exactly one ordered alternative: a repaired-gauge representation, failure of the raw chart, or failure of the repaired-gauge solve.*

Proof. If the chart is absent, the first failure occurs. If the chart exists but the admissible absolutely continuous gauge solve is absent, the second failure occurs. If both are present, theorem A.17 gives the repaired-gauge representation. The alternatives are disjoint by their order and exhaustive for the representation row. □

A.4. **Local Retained Mass Cost Divergence.**

Lemma A.20 (Spatially constant retained limits are bounded windows)**.** *Let $0 < r < R_*$, let $J \Subset (-\theta,\theta)$, and let*

$$V_j(y,s) = V(y,\sigma_j + s)$$

be selected retained windows on $B_{R_*} \times (-\theta,\theta)$. Assume

$$V_j \to c(s) \quad \textit{strongly in } L^3(B_{R_*} \times (-\theta,\theta)), \qquad \nabla_y c = 0,$$

and assume the retained time-derivative estimate

$$\sup_j \|\partial_s V_j\|_{L^1(J;H^{-m}(B_r))} < \infty$$

for some m with $C_c^\infty(B_r) \subset H_0^m(B_r)$. Then c has a representative in $BV(J;\mathbb{R}^3)$, hence $c \in L^\infty(J)$. If $c \not\equiv 0$ on $(-\theta,\theta)$, then after restricting to a smaller selected parabolic subwindow and recentering its time coordinate, the sequence has a nonzero bounded selected-window limit in the sense of definition 2.1. Consequently the branch exits the local Type II class by lemma 4.25.

Proof. Choose $\zeta \in C_c^\infty(B_r)$ with $\int_{B_r} \zeta\, dy = 1$. For each component k, set

$$C_{j,k}(s) = \int_{B_r} V_{j,k}(y,s)\zeta(y)\, dy.$$

The strong L^3 convergence implies $C_{j,k} \to c_k$ in $L^1(J)$. In distributions on J,

$$\frac{d}{ds} C_{j,k}(s) = \langle \partial_s V_{j,k}(s), \zeta \rangle,$$

and therefore

$$\operatorname{Var}_J(C_{j,k}) \le \|\zeta\|_{H_0^m(B_r)} \|\partial_s V_j\|_{L^1(J;H^{-m}(B_r))} \le C_{J,r}.$$

Lower semicontinuity of total variation under L^1 convergence gives $\operatorname{Var}_J(c_k) \le C_{J,r}$. Since $c_k \in L^1(J)$, this proves $c_k \in BV(J)$. In one dimension $BV(J) \subset L^\infty(J)$, with the bound depending only on $\|c_k\|_{L^1(J)}$, $\operatorname{Var}_J(c_k)$, and $|J|$. Thus $c \in L^\infty(J)$.

If $c \not\equiv 0$, choose a density point $s_0 \in (-\theta,\theta)$ of the set where $|c| > 0$. Since the preceding argument applies to every compact subinterval $J \Subset (-\theta,\theta)$, choose $J_0 \Subset (-\theta,\theta)$ containing s_0 and use the $BV(J_0) \subset L^\infty(J_0)$ representative. Then choose a radius $\rho > 0$ such that

$$B_\rho \times (s_0 - \rho^2, s_0) \Subset B_r \times J_0, \qquad \int_{s_0-\rho^2}^{s_0} |c(s)|^3\, ds > 0.$$

Replacing the selected centers by $\sigma_j + s_0$ identifies this subwindow with Q_ρ. This finite time recentering is admissible by the same local selected-window covariance used in lemma 9.68: no lower bound is transferred to a different physical time; the original subwindow is merely rewritten with its own center as the origin. The strong convergence remains valid there and the limit is bounded and nonzero. This is exactly a nonzero bounded selected-window limit, so lemma 4.25 applies. □

Theorem A.21 (Local retained-mass cost divergence)**.** *Let (V, P, a, b) be a repaired-gauge local Type II branch on a terminal tail. Assume there are $R_* > 0$, $\theta > 0$, $\eta_* > 0$, and an unbounded family $\mathcal{S} \subset [\tau_0, \infty)$ of retained selected active-window centers such that every $\sigma \in \mathcal{S}$ carries positive local velocity mass,*

$$\int_{\sigma-\theta}^{\sigma+\theta} \int_{B_{R_*}} |V(y,\tau)|^3 \, dy \, d\tau \geq \eta_*.$$

Assume also the retained compact-window bounds, the retained $L^1 H^{-m}$ time-derivative bounds on compact subwindows, the local Aubin–Lions compactness on $B_{R_*} \times (-\theta, \theta)$, local windowed H^1 control, and the scale-rigid routing of nonzero bounded selected-window limits. These are all local compact-window hypotheses; no whole-space critical bound, global profile theorem, or a priori vanishing dissipation is assumed. Then finite total localized dissipation on the retained core is impossible. In particular, for the canonical identity cost on a cutoff $\phi_{R_0} \equiv 1$ on B_{R_*},

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) \, d\tau = \infty$$

unless the branch has already exited through a bounded selected-window, rough-core, exterior, or adjacent state-space alternative.

Proof. Assume the localized cost is finite on the terminal tail and no listed exit has occurred. Since $\phi_{R_0} \equiv 1$ on B_{R_*}, finite cost gives finite localized dissipation on B_{R_*}. Choose retained selected active-window centers $\sigma_j \in \mathcal{S}$, $\sigma_j \to \infty$, with disjoint enlarged windows and such that

$$\int_{\sigma_j-\theta}^{\sigma_j+\theta} \int_{B_{R_*}} |\nabla V|^2 \, dy \, d\tau \to 0.$$

This selection is possible because the total dissipation on the tail is finite and $\mathcal{S}$ is unbounded; if no unbounded retained active-window family with the stated mass floor remains, the lower active-mass row has failed and the branch has already exited by the terminal critical-mass routing.

Set $V_j(y, s) = V(y, \sigma_j + s)$. The retained compact-window bounds give, after passing to a subsequence,

$$V_j \to V_\infty \quad \text{strongly in } L^3(B_{R_*} \times (-\theta, \theta)),$$

by the local Aubin–Lions compactness in the retained state-space topology. The vanishing dissipation gives $\nabla V_\infty = 0$ distributionally on $B_{R_*} \times (-\theta, \theta)$. Hence $V_\infty(y, s) = c(s)$ on the retained core cylinder.

If $c \not\equiv 0$, lemma A.20 turns the spatially constant compact limit into a nonzero bounded selected-window limit on a smaller selected parabolic subwindow. The scale-rigid exit theorem lemma 4.25 therefore removes the branch from the retained local Type II path. If $c \equiv 0$, strong convergence on the core cylinder contradicts the retained local mass floor:

$$\int_{-\theta}^{\theta} \int_{B_{R_*}} |V_j(y,s)|^3 \, dy \, ds \to 0 \quad \text{while} \quad \int_{-\theta}^{\theta} \int_{B_{R_*}} |V_j(y,s)|^3 \, dy \, ds \geq \eta_*.$$

Both alternatives are impossible on the retained branch. Therefore the localized cost cannot be finite. ∎

A.5. **Localized cost and the Type II scale-exclusion condition.**

Definition A.22 (Localized Type II cost)**.** For a repaired-gauge Type II branch, let

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y) \, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y) \, dy,$$

where $a_+ = \max(a, 0)$, ϕ_{R_0} is the compact Type II cutoff, and $\nu > 0$. A Type II scale-exclusion cost is a measurable nonnegative function $\mathfrak{C}_{\mathrm{II}}$ that is locally integrable on finite τ-intervals and whose divergence on $[\tau_0, \infty)$ gives the Type II scale-exclusion condition.

Definition A.23 (Cost comparison)**.** A cost comparison for a repaired-gauge Type II branch consists of $R_0, \phi_{R_0}, \mathfrak{C}_{\mathrm{II}}, c_0, \tau_1$, with $c_0 > 0$ and $\tau_1 \geq \tau_0$, such that

$$\mathfrak{C}_{\mathrm{II}}(\tau) \geq c_0\, \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \text{for a.e. } \tau \geq \tau_1.$$

The canonical Navier–Stokes choice is

$$\mathfrak{C}_{\mathrm{II}}(\tau) = \tilde{\mathfrak{D}}_{R_0}(\tau), \qquad c_0 = 1, \qquad \tau_1 = \tau_0.$$

Lemma A.24 (Identity-cost comparison)**.** *If $\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ-intervals, then the canonical choice $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$ defines a valid Type II scale-exclusion cost and satisfies the cost comparison with $c_0 = 1$ and $\tau_1 = \tau_0$.*

Proof. The assumed measurability, nonnegativity, and local finite-interval integrability give the cost properties in definition A.22. The identity

$$\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$$

gives the comparison in definition A.23 with $c_0 = 1$ and $\tau_1 = \tau_0$. ☐

Theorem A.25 (Cost comparison theorem)**.** *Let (V, P, a, b) be a repaired-gauge Type II branch. Assume:*

(i) *$\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ-intervals;*
(ii)
$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty;$$
(iii) *either the canonical cost $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$ is used, or a cost comparison as in definition A.23 is available.*

Then the Type II scale-exclusion condition associated with $\mathfrak{C}_{\mathrm{II}}$ holds.

Proof. In the canonical case, lemma A.24 supplies the cost comparison. In the general case, the comparison is assumed. Hence there are $c_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$\mathfrak{C}_{\mathrm{II}}(\tau) \geq c_0 \tilde{\mathfrak{D}}_{R_0}(\tau)$$

for a.e. $\tau \geq \tau_1$. Since $\tilde{\mathfrak{D}}_{R_0} \geq 0$ and is locally integrable on finite intervals, divergence on $[\tau_0, \infty)$ implies

$$\int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Therefore

$$\int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau \geq c_0 \int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Since $\mathfrak{C}_{\mathrm{II}}$ is nonnegative and locally integrable on finite intervals,

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau = \int_{\tau_0}^{\tau_1} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau + \int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau = \infty.$$

By definition of the Type II scale-exclusion cost, this divergence is the Type II scale-exclusion condition. ☐

A.6. **Local compact-cost classification.**

Definition A.26 (Remaining compact cost-divergence tests)**.** For a repaired-gauge Type II orbit, define the critical-tightness condition by

$$\forall \varepsilon > 0\ \exists R_\varepsilon : \quad \sup_{\tau \geq \tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3 \, dy < \varepsilon.$$

Failure of critical tightness is the condition

$$\exists \varepsilon_0 > 0\ \forall R > 0\ \exists \tau_R \geq \tau_0 : \quad \int_{|y|>R} |V(y,\tau_R)|^3 \, dy \geq \varepsilon_0.$$

Define local windowed H^1 control by

$$\forall m \geq 1 : \quad \sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|^2_{H^1(B_m)} \, d\tau < \infty.$$

Failure of local windowed H^1 control is the condition

$$\exists m \geq 1 : \quad \sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|^2_{H^1(B_m)} \, d\tau = \infty.$$

Definition A.27 (Compact Type II data)**.** A Type II branch has compact Type II data if it has:

(i) the Type II analytic data of definition A.2;
(ii) the repaired-gauge representation of definition A.9;
(iii) retained positive finite local critical mass on every compact terminal window, in the sense of lemma 9.73;
(iv) the locally integrable nonnegative cost and cost comparison of definitions A.22 and A.23.

This condition does not include critical tightness or windowed H^1 control; those are the two remaining compact cost-divergence tests.

Definition A.28 (Routed pre-compact-data exits)**.** A routed pre-compact-data exit is the first failed ordered test before the compact cost-divergence tests are reached. Thus it is one of the routed C1 exhaustion failures in definition A.3, one of the ordered representation failures in definition A.11 or definition A.18, one of the local terminal critical-mass failures routed by lemma 9.73, or a failure of local cost well-posedness or cost comparison before definitions A.22 and A.23 is available. The first-failure convention makes these exits disjoint from the compact Type II data of definition A.27.

Lemma A.29 (Pre-compact-data routing)**.** *Every admissible Type II branch has exactly one of the following ordered outcomes:*

(i) *it has compact Type II data in the sense of definition A.27;*
(ii) *it has a routed pre-compact-data exit in the sense of definition A.28.*

Proof. Fix an admissible branch. The ordered C1 routing hypothesis of theorem A.4 is supplied by theorem 9.103. Applying theorem A.4, if the concentration, scale, and local profile data are not all present, the first failed C1 test is a routed pre-compact-data exit. Otherwise the first component of definition A.27 holds.

Next run the ordered representation classification corollary A.12, equivalently the minimal representation classification corollary A.19 when the minimal chart package is used. A first representation, repaired-gauge, pressure, or modulation failure is a routed pre-compact-data exit. If no such failure occurs, the branch has the repaired-gauge representation of definition A.9.

On that repaired-gauge branch, apply the local terminal critical-mass routing lemma 9.73. A lower-mass failure means the candidate was not a retained active core; an upper local L^3-failure produces a comparable retained class or an adjacent active label. These are routed pre-compact-data exits unless they are grouped into the retained terminal component. In the retained case the positive finite local critical-mass row holds on every compact terminal window.

Finally test whether the localized cost is well posed and compatible with the Navier–Stokes identity cost. Availability of the local cost and comparison is the fourth component of definition A.27. If it is absent, the first absence is the local cost-well-posedness or cost-comparison failure listed in definition A.28; if it is present, all four components of compact Type II data hold. The first-failure convention makes the two outcomes disjoint and exhaustive. □

Lemma A.30 (Uniform retained active-window mass floor)**.** *Assume an admissible Type II branch has compact Type II data in the sense of definition A.27 and remains in the retained compact cost-divergence stratum after the ordered validation tests of lemma 7.70. Then there exist*

$$R_* > 0, \qquad \theta > 0, \qquad \eta_* > 0,$$

and an unbounded family $\mathcal{S} \subset [\tau_0, \infty)$ of retained selected active-window centers such that

$$\int_{\sigma-\theta}^{\sigma+\theta} \int_{B_{R_*}} |V(y,\tau)|^3 \, dy \, d\tau \geq \eta_* \qquad \textit{for every } \sigma \in \mathcal{S}.$$

If such a fixed retained active-window family does not exist, the branch has already left the retained compact stratum through a lower-mass, pressure-only, exterior/separated, cascade, or comparable-class grouping row.

Proof. The retained finite critical-mass component of definition A.27 is read through lemma 9.73. Thus the lower active mass row does not fail on the retained branch; if it did, the candidate would not be a retained terminal component.

Apply the CKN detection argument in theorem 8.32 to the retained active component on the selected compact terminal cylinders. Its proof gives a fixed local threshold $\eta_{\mathrm{det}} > 0$. If the velocity part of the detected CKN quantity were not bounded below on a compact subcylinder, then either the component would be perturbative by corollary 8.31, or the detected mass would be pressure-only and would be routed by lemma 8.22 to the pressure/exterior row. Neither row remains in the retained compact cost-divergence stratum. Therefore each retained active witness carries a positive spacetime velocity mass on one member of a finite compact covering of the selected terminal core.

The covering is finite and the terminal selection rule groups all comparable same-point retained classes by lemma 9.72. If the active witnesses escaped every fixed member of this finite compact chart along the tail, the first failure would be an exterior/separated or cascade row in the ordered local state space. Since the branch remains retained, one fixed compact member of the cover occurs along an unbounded family of selected windows. Writing that member as $B_{R_*} \times (-\theta, \theta)$, and absorbing the scale-invariant covering constants into $\eta_* := c\,\eta_{\mathrm{det}} R_*^2 > 0$, gives the displayed lower bound for an unbounded family $\mathcal{S}$ of retained active-window centers. □

Lemma A.31 (Local upper estimates give retained Aubin–Lions compactness)**.** *Let (V, P, a, b) be a retained repaired-gauge branch on a terminal tail. Fix a compact model interval $I \Subset \mathbb{R}$, a radius $R > 0$, and selected centers $\sigma_j \to \infty$. Assume that on the shifted windows*

$$V_j(y,s) = V(y, \sigma_j + s), \qquad P_j(y,s) = P(y, \sigma_j + s), \qquad s \in I,$$

the ordered retained rows supply:

(i) *a uniform local critical upper bound*

$$\sup_j \|V_j\|_{L^\infty(I; L^3(B_{2R}))} < \infty;$$

(ii) *local windowed H^1 control on B_{2R};*
(iii) *local pressure reconstruction modulo functions of time;*
(iv) *the repaired-gauge equation with uniformly bounded selected-window modulation integrals*

$$\sup_j \int_I \big(|a(\sigma_j + s)| + |b(\sigma_j + s)|\big)\, ds < \infty.$$

Then, after subtracting spatial pressure means, the sequence satisfies

$$\sup_j \Big(\|V_j\|_{L^\infty(I;L^2(B_{2R}))} + \|V_j\|_{L^2(I;H^1(B_{2R}))} + \|P_j - c_{j,2R}\|_{L^{3/2}(B_{2R}\times I)} \Big) < \infty,$$

and for some large m,

$$\sup_j \|\partial_s V_j\|_{L^1(I;H^{-m}(B_R))} < \infty.$$

Consequently, after passing to a subsequence,

$$V_j \to V_\infty \quad \textit{strongly in } L^3(B_R \times I).$$

Proof. The $L^\infty_s L^2_y(B_{2R})$ bound follows from the local $L^\infty_s L^3_y(B_{2R})$ bound and finite measure:

$$\|V_j(s)\|_{L^2(B_{2R})} \le |B_{2R}|^{1/6} \|V_j(s)\|_{L^3(B_{2R})}.$$

The $L^2_s H^1_y(B_{2R})$ bound is the local windowed H^1 control; a fixed compact model interval I meets only finitely many unit windows, so the uniform unit-window bound gives a uniform bound on I. The pressure bound is the local Calderón–Zygmund pressure reconstruction, modulo functions of time, applied to $V_j \otimes V_j$; the finite length of I and the uniform $L^\infty_s L^3_y$ bound give the displayed $L^{3/2}_{s,y}$ estimate.

It remains to bound the time derivative. On B_R the repaired-gauge equation with

$$a_j(s) := a(\sigma_j + s), \qquad b_j(s) := b(\sigma_j + s)$$

gives

$$\partial_s V_j = \nu \Delta V_j - \nabla\cdot(V_j \otimes V_j) - \nabla P_j + a_j(V_j + y\cdot\nabla V_j) + b_j \cdot \nabla V_j.$$

Choose m so large that

$$L^1(B_{2R}) + L^{3/2}(B_{2R}) + H^{-1}(B_{2R}) \hookrightarrow H^{-m}(B_R).$$

The diffusion term is bounded in $L^1(I;H^{-m})$ by the L^2H^1 bound. The nonlinear term is bounded because

$$\|V_j \otimes V_j\|_{L^1(I;L^{3/2}(B_{2R}))} \le |I| \|V_j\|^2_{L^\infty(I;L^3(B_{2R}))}.$$

The pressure-gradient term is bounded by the pressure estimate. For the modulation terms, integrate by parts in the distributional pairing:

$$y\cdot\nabla V_j = \nabla\cdot(y\otimes V_j) - 3V_j, \qquad b_j\cdot\nabla V_j = \nabla\cdot(b_j \otimes V_j).$$

Thus their $H^{-m}(B_R)$ norms are bounded by

$$C_R\big(|a_j(s)| + |b_j(s)|\big)\|V_j(s)\|_{L^2(B_{2R})}.$$

The retained modulation row gives $a_j, b_j \in L^1(I)$ with uniform selected window bounds; if this row failed, the ordered validation would route the branch to the modulation/scale-drift exit before the retained compact stratum was declared. Hence $\partial_s V_j$ is bounded in $L^1(I;H^{-m}(B_R))$.

The Aubin–Lions–Simon theorem applied to

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-m}(B_R)$$

gives strong convergence in $L^2(B_R \times I)$, after passing to a subsequence. The same bounds give a uniform $L^{10/3}(B_R \times I)$ bound by the standard parabolic interpolation between $L^\infty L^2$ and $L^2 H^1$.

Interpolating the strong L^2 convergence with this uniform $L^{10/3}$-bound gives strong convergence in L^p for every $2 \le p < 10/3$, in particular in $L^3(B_R \times I)$. □

Lemma A.32 (Retained compact data supply the retained mass and compactness package)**.** *Assume an admissible Type II branch has compact Type II data in the sense of definition A.27, satisfies the two compact tests of definition A.26, and remains in the retained compact cost-divergence stratum after the ordered validation tests of lemma 7.70. Then, on every compact retained terminal window after a harmless terminal restriction, the requirements of theorem A.21 are satisfied. More precisely:*

(i) *there are $R_* > 0$, $\theta > 0$, $\eta_* > 0$, and an unbounded retained selected active-window family $\mathcal{S}$ satisfying the positive spacetime local mass floor of theorem A.21;*

(ii) *the local compact-cylinder bounds*

$$V \in L^\infty(I; L^3(B_{2R})) \cap L^\infty(I; L^2(B_{2R})) \cap L^2(I; H^1(B_{2R})), \qquad P - c_{2R} \in L^{3/2}(B_{2R} \times I)$$

hold for each fixed R and compact terminal interval I;

(iii) *on smaller balls, $\partial_\tau V$ is bounded in $L^1(I; H^{-m}(B_R))$ for some large m, hence the local Aubin–Lions compactness used in theorem A.21 is available;*

(iv) *any nonzero bounded selected-window limit is routed out of the retained Type II branch by lemma 4.25.*

If any item in this list fails before the retained stratum is reached, the first failure is one of the named representation, pressure, modulation, critical-mass, rough-core, exterior, or adjacent compact-cost exits of lemma 7.70; it is not repaired by inserting a global estimate.

Proof. The positive spacetime retained-window mass floor is lemma A.30. Its proof uses the active CKN detection threshold, pressure-only routing, and the terminal maximality rule; if the fixed unbounded active-window family does not exist, the branch has already left the retained compact stratum through a named first-failure row.

The repaired-gauge representation, pressure reconstruction, and modulation data are item (ii) of definition A.27 together with the passed representation rows of lemma 7.70. The two compact tests give critical tightness and local windowed H^1 control. Equivalently, by theorem 9.89 and corollary 9.90, their failure is a named non-retained alternative. Combining the retained local L^3 upper bound, local windowed H^1 control, local pressure reconstruction, and the retained modulation-window bound of lemma 9.87, lemma A.31 gives the displayed local compact-cylinder bounds, the $L^1 H^{-m}$ time-derivative bound, and the strong L^3 Aubin–Lions compactness on selected windows. This compactness lemma uses only local upper bounds and local PDE estimates; it does not require the pointwise-in-time lower bound in theorem 9.112.

Finally, a nonzero bounded selected-window limit is incompatible with the local Type II definition by lemma 4.25. Hence all hypotheses used in theorem A.21 are local consequences of the retained compact data and validation rows. □

Lemma A.33 (Compact positive case is excluded)**.** *Assume an admissible Type II branch has compact Type II data and satisfies both compact cost-divergence tests in definition A.26. If the branch remains in the retained compact cost-divergence stratum after the ordered validation tests of lemma 7.70, then the branch satisfies the Type II exclusion conclusion.*

Proof. The compact Type II data give the supercritical scale alternative, a repaired-gauge orbit (V, P, a, b), retained positive local critical mass, local compact-cylinder critical upper bounds, and the cost comparison. By lemma A.32, the retained branch also has the local compact-window, pressure, H^1, time-derivative, and bounded-limit routing package required by theorem A.21. Thus

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Theorem A.25 converts this divergence into the Type II scale-exclusion condition, hence the validated compact cost-divergence conclusion. Since the branch remains in the retained compact cost-divergence stratum, corollary 7.79 gives the Type II exclusion conclusion. $\square$

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